ETSI TS 104 148-1 V1.1.1 (2026-01)



TECHNICAL SPECIFICATION

Mission Critical (MC) services; Mission Critical Push To Talk (MCPTT) Application Server (AS) Protocol conformance specification for server-to-server interface including InterWorking Function (IWF) to non-3GPP systems; Part 1: Test structure, configurations, conformance requirement and test purposes

Reference
DTS/TCCE-00255

Keywords
LTE

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from the
ETSI Search & Browse Standards application.

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format on ETSI deliver repository.

Users should be aware that the present document may be revised or have its status changed, this information is available in the Milestones listing.

If you find errors in the present document, please send your comments to the relevant service listed under Committee Support Staff.

If you find a security vulnerability in the present document, please report it through our Coordinated Vulnerability Disclosure (CVD) program.

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm except as authorized by written permission of ETSI.
The content of the PDF version shall not be modified without the written authorization of ETSI.
The copyright and the foregoing restriction extend to reproduction in all media.

© ETSI 2026.
All rights reserved.

Contents

Intellectual Property Rights 9
Foreword 9
Modal verbs terminology 9
Introduction 10
1 Scope 11
2 References 11
2.1 Normative references 11
2.2 Informative references 13
3 Definitions of terms, symbols and abbreviations 14
3.1 Terms 14
3.2 Symbols 15
3.3 Abbreviations 15
4 General 16
4.1 Test methodology 16
4.1.1 Testing of optional functions and procedures 16
4.1.2 Test interfaces and facilities 16
4.2 Implicit testing 16
4.3 Repetition of tests 16
4.4 Handling of differences between conformance requirements in different releases of cores specifications 16
4.5 Reference conditions 16
5 MCPTT Server - MCPTT Server operation 17
5.1 Configuration 17
5.1.1 MCPTT Server - MCPTT Server / Configuration / Group Affiliation / De-affiliation / Controlling server 17
5.1.1.1 Test Purpose (TP) 17
5.1.1.2 Conformance requirements 17
5.1.1.3 Test description 21
5.1.1.3.1 Pre-test conditions 21
5.1.1.3.2 Test procedure sequence 21
5.1.1.3.3 Specific message contents 22
5.1.2 MCPTT Server - MCPTT Server / Configuration / Group Affiliation / De-affiliation / Implicit Affiliation / Participating server 27
5.1.2.1 Test Purpose (TP) 27
5.1.2.2 Conformance requirements 28
5.1.2.3 Test description 31
5.1.2.3.1 Pre-test conditions 31
5.1.2.3.2 Test procedure sequence 32
5.1.2.3.3 Specific message contents 33
5.1.3 MCPTT Server - MCPTT Server / Group and GMK Subscription / Group Regroup creation / Group Regroup Teardown / GMS owning temporary group 34
5.1.3.1 Test Purpose (TP) 34
5.1.3.2 Conformance requirements 35
5.1.3.3 Test description 48
5.1.3.3.1 Pre-test conditions 48
5.1.3.3.2 Test procedure sequence 49
5.2.3.3.3 Specific message contents 51
5.1.4 MCPTT Server - MCPTT Server / Group and GMK Subscription / Group Regroup creation / Group Regroup Teardown / GMS owning group to be combined 59
5.1.4.1 Test Purpose (TP) 59
5.1.4.2 Conformance requirements 59
5.1.4.3 Test description 66
5.1.4.3.1 Pre-test conditions 66
5.1.4.3.2 Test procedure sequence 67

5.1.4.3.3 Specific message contents 69
5.1.5 MCPTT Server - MCPTT Server / Configuration / Functional Alias / Functional alias status determination / Activate functional alias / Deactivate functional alias / Controlling server 75
5.1.5.1 Test Purpose (TP) 75
5.1.5.2 Conformance requirements 75
5.1.5.3 Test description 80
5.1.5.3.1 Pre-test conditions 80
5.1.5.3.2 Test procedure sequence 81
5.1.5.3.3 Specific message contents 82
5.1.6 MCPTT Server - MCPTT Server / Configuration / Functional Alias / Functional alias status determination / Activate functional alias / Deactivate functional alias / Participating server 88
5.1.6.1 Test Purpose (TP) 88
5.1.6.2 Conformance requirements 88
5.1.6.3 Test description 97
5.1.6.3.1 Pre-test conditions 97
5.1.6.3.2 Test procedure sequence 98
5.2 Group Calls 100
5.2.1 Pre-arranged Group Call 100
5.2.1.1 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Automatic Commencement Mode / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / Controlling server 100
5.2.1.1.1 Test Purpose (TP) 100
5.2.1.1.2 Conformance requirements 102
5.2.1.1.3 Test description 125
5.2.1.2 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Automatic Commencement Mode / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / Originating Participating server 148
5.2.1.2.1 Test Purpose (TP) 148
5.2.1.2.2 Conformance requirements 150
5.2.1.2.3 Test description 155
5.2.1.3 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Automatic Commencement Mode / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / Terminating Participating server 175
5.2.1.3.1 Test Purpose (TP) 175
5.2.1.3.2 Conformance requirements 176
5.2.1.3.3 Test description 185
5.2.1.4 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Temporary Group / Controlling server / Trusted mode 203
5.2.1.4.1 Test Purpose (TP) 203
5.2.1.4.2 Conformance requirements 203
5.2.1.4.3 Test description 221
5.2.1.5 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Temporary Group / Non-Controlling server / Trusted mode 238
5.2.1.5.1 Test Purpose (TP) 238
5.2.1.5.2 Conformance requirements 239
5.2.1.5.3 Test description 245
5.2.1.6 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Temporary Group / Participating server / Trusted and Untrusted modes 250
5.2.1.6.1 Test Purpose (TP) 250
5.2.1.6.2 Conformance requirements 250
5.2.1.6.3 Test description 258
5.2.1.7.1 Test Purpose (TP) 279
5.2.1.7.2 Conformance requirements 280
5.2.1.7.3 Test description 298
5.2.1.8 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Temporary Group / Non-Controlling server / Untrusted mode 315
5.2.1.8.1 Test Purpose (TP) 315
5.2.1.8.2 Conformance requirements 315
5.2.2 Chat Group Call 341
5.2.2.1 MCPTT Server - MCPTT Server / Chat Group Call / Automatic Commencement Mode / Controlling server 341
5.2.2.1.1 Test Purpose (TP) 341

5.2.2.1.2 Conformance requirements....342
5.2.2.1.3 Test description....350
5.2.2.2 MCPTT Server - MCPTT Server / Chat Group Call / Automatic Commencement Mode / Originating Participating server....357
5.2.2.2.1 Test Purpose (TP)....357
5.2.2.2.2 Conformance requirements....358
5.2.2.2.3 Test description....360
5.2.2.3 MCPTT Server - MCPTT Server / Chat Group Call / Temporary Group / Controlling server / Untrusted mode....371
5.2.2.3.1 Test Purpose (TP)....371
5.2.2.3.2 Conformance requirements....372
5.2.2.3.3 Test description....379
5.2.2.4 MCPTT Server - MCPTT Server / Chat Group Call / Temporary Group / Non-Controlling server / Untrusted mode....392
5.2.2.4.1 Test Purpose (TP)....392
5.2.2.4.2 Conformance requirements....393
5.2.2.4.3 Test description....406

6 IWF - MCPTT Server operation....418
6.1 Group Calls....418
6.1.1 Pre-arranged Group Call....418
6.1.1.1 IWF - MCPTT Server / Pre-arranged Group Call / Automatic Commencement Mode / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / IWF Controlling Role....418
6.1.1.1.1 Test Purpose (TP)....418
6.1.1.1.2 Conformance requirements....419
6.1.1.1.3 Test description....435
6.1.1.2 IWF - MCPTT Server / Pre-arranged Group Call / Automatic Commencement Mode / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / IWF Originating Participating Role....458
6.1.1.2.1 Test Purpose (TP)....458
6.1.1.2.2 Conformance requirements....460
6.1.1.2.3 Test description....473

Annex A (normative): Common Test Environment....487

A.1 Introduction....487

A.2 Generic test procedures for Server-Client MCS operation....487
A.2.1 General....487
A.2.2 Void....487
A.2.3 MCX Server Originated notifications regarding temporary group creation or tear down....487
A.2.3.1 Initial conditions....487
A.2.3.2 Definition of system information messages....487
A.2.3.3 Procedure....488
A.2.3.4 Specific message contents....488
A.2.4 MCX Server Terminated Group Creation....489
A.2.4.1 Initial conditions....489
A.2.4.2 Definition of system information messages....489
A.2.4.3 Procedure....489
A.2.4.4 Specific message contents....489
A.2.5 MCX Server Terminated Temporary Group Creation....489
A.2.5.1 Initial conditions....489
A.2.5.2 Definition of system information messages....489
A.2.5.3 Procedure....490
A.2.5.4 Specific message contents....490
A.2.6 MCX Server Terminated Temporary Group Tear Down....490
A.2.6.1 Initial conditions....490
A.2.6.2 Definition of system information messages....490
A.2.6.3 Procedure....490
A.2.6.4 Specific message contents....491
A.2.7 MCX Subscription and Notification....491
A.2.7.1 Initial conditions....491

A.2.7.2 Definition of system information messages 491
A.2.7.3 Procedure 491
A.2.7.4 Specific message contents 491
A.2.8 MCX Participating Terminated SIP MESSAGE 493
A.2.8.1 Initial conditions 493
A.2.8.2 Definition of system information messages 493
A.2.8.3 Procedure 493
A.2.8.4 Specific message contents 493
A.2.9 MCX Participating Originated SIP MESSAGE 494
A.2.9.1 Initial conditions 494
A.2.9.2 Definition of system information messages 494
A.2.9.3 Procedure 494
A.2.9.4 Specific message contents 494
A.2.10 MCX E2E SIP MESSAGE 494
A.2.10.1 Initial conditions 494
A.2.10.2 Definition of system information messages 494
A.2.10.3 Procedure 495
A.2.10.4 Specific message contents 495
A.2.11 MCX E2E SIP MESSAGE Request - Accept 495
A.2.11.1 Initial conditions 495
A.2.11.2 Definition of system information messages 495
A.2.11.3 Procedure 495
A.2.11.4 Specific message contents 496
A.2.12 MCX Presence Event Status Change 496
A.2.12.1 Initial conditions 496
A.2.12.2 Definition of system information messages 496
A.2.12.3 Procedure 496
A.2.12.4 Specific message contents 496
A.2.13 MCX Participating Terminated pre-established session establishment 497
A.2.13.1 Initial conditions 497
A.2.13.2 Definition of system information messages 498
A.2.13.3 Procedure 498
A.2.13.4 Specific message contents 498
A.2.14 MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying 498
A.2.14.1 Initial conditions 498
A.2.14.2 Definition of system information messages 498
A.2.14.3 Procedure 499
A.2.14.4 Specific message contents 499
A.2.15 MCX Participating Terminated call release 499
A.2.15.1 Initial conditions 499
A.2.15.2 Definition of system information messages 499
A.2.15.3 Procedure 500
A.2.15.4 Specific message contents 500
A.2.16 MCX Participating Originated session establishment/modification without provisional responses other than 100 Trying 500
A.2.16.1 Initial conditions 500
A.2.16.2 Definition of system information messages 500
A.2.16.3 Procedure 500
A.2.16.4 Specific message contents 501
A.2.17 MCX Participating Originated call release 501
A.2.17.1 Initial conditions 501
A.2.17.2 Definition of system information messages 501
A.2.17.3 Procedure 501
A.2.17.4 Specific message contents 501
A.2.18 MCX E2E session establishment/modification without provisional responses other than 100 Trying 502
A.2.18.1 Initial conditions 502
A.2.18.2 Definition of system information messages 502
A.2.18.3 Procedure 502
A.2.18.4 Specific message contents 502
A.2.19 MCX E2E call release 503
A.2.19.1 Initial conditions 503

A.2.19.2 Definition of system information messages 503
A.2.19.3 Procedure 503
A.2.19.4 Specific message contents 503
A.2.20 MCX E2E group call establishment, manual commencement 504
A.2.20.1 Initial conditions 504
A.2.20.2 Definition of system information messages 504
A.2.20.3 Procedure 504
A.2.20.4 Specific message contents 504
A.2.21 MCX E2E private call establishment, manual commencement 505
A.2.21.1 Initial conditions 505
A.2.21.2 Definition of system information messages 505
A.2.21.3 Procedure 505
A.2.21.4 Specific message contents 505
A.3 Generic test procedures for MCPTT Server-Client operation 506
A.3.0 General 506
A.3.1 MCPTT E2E session establishment/modification without provisional responses other than 100 Trying 506
A.3.1.1 Initial conditions 506
A.3.1.2 Definition of system information messages 506
A.3.1.3 Procedure 507
A.3.1.4 Specific message contents 507
A.3.2 MCPTT E2E group call establishment, manual commencement 516
A.3.2.1 Initial conditions 516
A.3.2.2 Definition of system information messages 516
A.3.2.3 Procedure 517
A.3.2.4 Specific message contents 517
A.3.3 MCPTT E2E private call establishment, manual commencement 524
A.3.3.1 Initial conditions 524
A.3.3.2 Definition of system information messages 524
A.3.3.3 Procedure 525
A.3.3.4 Specific message contents 525
A.3.4 MCPTT E2E call establishment using a pre-established session, manual commencement 534
A.3.4.1 Initial conditions 534
A.3.4.2 Definition of system information messages 534
A.3.4.3 Procedure 535
A.3.4.4 Specific message contents 535
A.3.5 MCPTT E2E call establishment using a pre-established session, automatic commencement 547
A.3.5.1 Initial conditions 547
A.3.5.2 Definition of system information messages 547
A.3.5.3 Procedure 548
A.3.5.4 Specific message contents 548
A.3.6 MCPTT E2E call release keeping the pre-established session 555
A.3.6.1 Initial conditions 555
A.3.6.2 Definition of system information messages 555
A.3.6.3 Procedure 556
A.3.6.4 Specific message contents 556
A.3.7 MCPTT Group Affiliation Status Determination and Subscription 556
A.3.7.1 Initial conditions 556
A.3.7.2 Definition of system information messages 556
A.3.7.3 Procedure 557
A.3.7.4 Specific message contents 557
A.3.8 MCPTT Group Affiliation Status Change 557
A.3.8.1 Initial conditions 557
A.3.8.2 Definition of system information messages 557
A.3.8.3 Procedure 557
A.3.8.4 Specific message contents 558
A.3.9 MCPTT Functional Alias Status Determination and Subscription 559
A.3.9.1 Initial conditions 559
A.3.9.2 Definition of system information messages 559
A.3.9.3 Procedure 559
A.3.9.4 Specific message contents 559
A.3.10 MCPTT Functional Alias Status Change 560

A.3.10.1 Initial conditions 560
A.3.10.2 Definition of system information messages 560
A.3.10.3 Procedure 560
A.3.10.4 Specific message contents 561
A.3.11 MCPTT Floor Request - Floor Granted 562
A.3.11.1 Initial conditions 562
A.3.11.2 Definition of system information messages 562
A.3.11.3 Procedure 563
A.3.11.4 Specific message contents 563
A.3.12 MCPTT Floor Request - Floor Queue Position Info 563
A.3.12.1 Initial conditions 563
A.3.12.2 Definition of system information messages 563
A.3.12.3 Procedure 564
A.3.12.4 Specific message contents 564
A.3.13 MCPTT Queuing Position Request 564
A.3.13.1 Initial conditions 564
A.3.13.2 Definition of system information messages 564
A.3.13.3 Procedure 565
A.3.13.4 Specific message contents 565
A.3.14 MCPTT Floor Request - Floor Deny 565
A.3.14.1 Initial conditions 565
A.3.14.2 Definition of system information messages 565
A.3.14.3 Procedure 566
A.3.14.4 Specific message contents 566
A.3.15 MCPTT Floor Release - Floor Idle 566
A.3.15.1 Initial conditions 566
A.3.15.2 Definition of system information messages 566
A.3.15.3 Procedure 567
A.3.15.4 Specific message contents 567
A.3.16 MCPTT Floor Release - Floor Taken 567
A.3.16.1 Initial conditions 567
A.3.16.2 Definition of system information messages 568
A.3.16.3 Procedure 568
A.3.16.4 Specific message contents 568
A.4 MCPTT Conformance testing test point models 569
Annex B (informative): Bibliography 572
History 573

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee TETRA and Critical Communications Evolution (TCCE).

The present document is based on ETSI TS 136 579-3 [i.2] (3GPP TS 36.579-3) produced by the 3rd Generation Partnership Project (3GPP).

The present document is part 1 of a multi-part deliverable covering the Mission Critical (MC) services; Mission Critical Push To Talk (MCPTT) Application Server (AS) Protocol conformance specification for server-to-server interface including InterWorking Function (IWF) to non-3GPP systems, as identified below:

Part 1: "Test structure, configurations, conformance requirement and test purposes";

Part 2: "Test Applicability and Implementation Conformance Statement (ICS) proforma specification".

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Introduction

The ETSI Technical Specification used as the reference document for the present document is part 3 of a multi-part deliverable covering conformance test specification for Mission Critical Services over LTE consisting of:

- ETSI TS 136 579-1 [1]: "LTE; Mission Critical (MC) services over LTE; Part 1: Common test environment (3GPP TS 36.579-1)";
- ETSI TS 136 579-2 [i.6]: "LTE; Mission Critical (MC) services over LTE; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-2)";
- **ETSI TS 136 579-3 [i.2]: "LTE; Mission Critical (MC) services over LTE; Part 3: Mission Critical Push To Talk (MCPTT) Server Application conformance specification (3GPP TS 36.579-3)";**
- ETSI TS 136 579-4 [2]: "LTE; Mission Critical (MC) services over LTE; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification (3GPP TS 36.579-4)";
- ETSI TS 136 579-5 [i.7]: "LTE; Mission Critical (MC) services over LTE; Part 5: Abstract test suite (ATS) (3GPP TS 36.579-5)";
- ETSI TS 136 579-6 [i.8]: "LTE; Mission Critical (MC) services over LTE; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-6)";
- ETSI TS 136 579-7 [i.9]: "LTE; Mission Critical (MC) services over LTE; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-7)";
- 3GPP TS 36.579-8 [i.4]: "Mission Critical (MC) services over LTE; Part 8: Mission Critical Video (MCVideo) Server Application conformance specification";
- 3GPP TS 36.579-9 [i.3]: "Mission Critical (MC) services over LTE; Part 9: Mission Critical Data (MCData) Server Application conformance specification".

1 Scope

The present document specifies the protocol conformance testing for testing a MCPTT Server for compliance to the Mission Critical Push To Talk (MCPTT) over LTE protocol requirements defined by 3GPP. The present document addresses only MCPTT Server-Server communication scenarios. It does not cover e.g. MCPTT Server-EPS, MCPTT Server-SIP Core, etc. scenarios which involve interfaces which implementation may widely vary.

In particular the present specification contains:

- the overall test structure;
- the test configurations;
- the conformance requirement and reference to the core specifications;
- the test purposes; and
- a brief description of the test procedure, the specific test requirements and short message exchange table.

The present document is valid for MCPTT Servers implemented according to 3GPP releases starting from Release 14 up to the Release indicated on the cover page of the present document.

The following information relevant to testing specified in the present document could be found in accompanying specifications:

- default setting of the test parameters ETSI TS 136 579-1 [1] (3GPP TS 36.579-1);
- Implementation Conformance Statement (ICS) ETSI TS 136 579-4 [2] (3GPP TS 36.579-4) and Implementation extra Information for Testing (IXIT) ETSI TS 136 579-5 [i.7] (3GPP TS 36.579-5);
- the applicability of each test case ETSI TS 136 579-4 [2] (3GPP TS 36.579-4).

The present document does not specify the protocol conformance testing for the EPS (LTE) bearers which carry the MCPTT data sent or received by the MCPTT Server. The specification of such testing is out of the scope of RAN5.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 136 579-1: "LTE; Mission Critical (MC) services over LTE; Part 1: Common test environment (3GPP TS 36.579-1)".

[2] ETSI TS 136 579-4: "LTE; Mission Critical (MC) services over LTE; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification (3GPP TS 36.579-4)".

[3] ETSI TS 123 179: "LTE; Functional architecture and information flows to support mission critical communication services; Stage 2 (3GPP TS 23.179)".

[4] ETSI TS 124 379: "LTE; Mission Critical Push To Talk (MCPTT) call control; Protocol specification (3GPP TS 24.379)".

[5] ETSI TS 124 380: "LTE; Mission Critical Push To Talk (MCPTT) media plane control; Protocol specification (3GPP TS 24.380)".

[6] ETSI TS 124 481: "LTE; Mission Critical Services (MCS) group management; Protocol specification (3GPP TS 24.481)".

[7] IETF RFC 5245 (April 2010): "Interactive Connectivity Establishment (ICE): A Protocol for Network Address Translator (NAT) Traversal for Offer/Answer Protocols".

[8] IETF RFC 3863 (August 2004): "Presence Information Data Format (PIDF)".

[9] IETF RFC 3556 (July 2003): "Session Description Protocol (SDP) Bandwidth Modifiers for RTP Control Protocol (RTCP) Bandwidth".

[10] ETSI TS 124 282: "LTE; Mission Critical Data (MCData) signalling control; Protocol specification (3GPP TS 24.282)".

[11] ETSI TS 124 281: "LTE; Mission Critical Video (MCVideo) signalling control; Protocol specification (3GPP TS 24.281)".

[12] ETSI TS 126 114: "Universal Mobile Telecommunications System (UMTS); LTE; 5G; IP Multimedia Subsystem (IMS); Multimedia Telephony; Media handling and interaction (3GPP TS 26.114)".

[13] IETF RFC 6335 (August 2011): "Internet Assigned Numbers Authority (IANA) Procedures for the Management of the Service Name and Transport Protocol Port Number Registry".

[14] IETF RFC 3842 (August 2004): "A Message Summary and Message Waiting Indication Event Package for the Session Initiation Protocol (SIP)".

[15] IETF RFC 4867 (April 2007): "RTP Payload Format and File Storage Format for the Adaptive Multi-Rate (AMR) and Adaptive Multi-Rate Wideband (AMR-WB) Audio Codecs".

[16] IETF RFC 5621 (September 2009): "Message Body Handling in the Session Initiation Protocol (SIP)".

[17] IETF RFC 3261 (June 2002): "SIP: Session Initiation Protocol".

[18] IETF RFC 7134 (March 2014): "The Management Policy of the Resource Priority Header (RPH) Registry Changed to "IETF Review"".

[19] IETF RFC 5031 (January 2008): "A Uniform Resource Name (URN) for Emergency and Other Well-Known Services".

[20] IETF RFC 4661 (September 2006): "An Extensible Markup Language (XML)-Based Format for Event Notification Filtering".

[21] IETF RFC 8101 (March 2017): "IANA Registration of New Session Initiation Protocol (SIP) Resource-Priority Namespace for Mission Critical Push To Talk Service".

[22] IETF RFC 3903 (October 2004): "Session Initiation Protocol (SIP) Extension for Event State Publication".

[23] IETF RFC 4412 (February 2006): "Communications Resource Priority for the Session Initiation Protocol (SIP)".

[24] IETF RFC 6665 (July 2012): "SIP-Specific Event Notification".

[25] IETF RFC 4028 (April 2005): "Session Timers in the Session Initiation Protocol (SIP)".

[26] IETF RFC 6050 (November 2010): "A Session Initiation Protocol (SIP) Extension for the Identification of Services".

[27] IETF RFC 3325 (November 2002): "Private Extensions to the Session Initiation Protocol (SIP) for Asserted Identity within Trusted Networks".

[28] IETF RFC 5366 (October 2008): "Conference Establishment Using Request-Contained Lists in the Session Initiation Protocol (SIP)".

[29] ETSI TS 123 379: "LTE; Functional architecture and information flows to support Mission Critical Push To Talk (MCPTT); Stage 2 (3GPP TS 23.379)".

[30] ETSI TS 129 379: "LTE; 5G; Mission Critical Push To Talk (MCPTT) call control interworking with Land Mobile Radio (LMR) systems; Stage-3 (3GPP TS 29.379)".

[31] ETSI TS 129 380: "LTE; 5G; Mission Critical Push To Talk (MCPTT) media plane control interworking with Land Mobile Radio (LMR) systems; Stage 3 (3GPP TS 29.380)".

[32] ETSI TS 123 280: "LTE; Common functional architecture to support mission critical services; Stage 2 (3GPP TS 23.280)".

[33] ETSI TS 123 283: "LTE; Mission Critical Communication Interworking with Land Mobile Radio Systems (3GPP TS 23.283)".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI TR 121 905: "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; Vocabulary for 3GPP Specifications (3GPP TR 21.905)".

[i.2] ETSI TS 136 579-3: "LTE; Mission Critical (MC) services over LTE; Part 3: Mission Critical Push To Talk (MCPTT) Server Application conformance specification (3GPP TS 36.579-3)".

[i.3] 3GPP TS 36.579-9: "Mission Critical (MC) services over LTE; Part 9: Mission Critical Data (MCData) Server Application conformance specification".

[i.4] 3GPP TS 36.579-8: "Mission Critical (MC) services over LTE; Part 8: Mission Critical Video (MCVideo) Server Application conformance specification".

[i.5] ETSI TS 122 179: "LTE; 5G; Mission Critical Push to Talk (MCPTT); Stage 1 (3GPP TS 22.179)".

[i.6] ETSI TS 136 579-2: "LTE; Mission Critical (MC) services over LTE; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-2)".

[i.7] ETSI TS 136 579-5: "LTE; Mission Critical (MC) services over LTE; Part 5: Abstract test suite (ATS) (3GPP TS 36.579-5)".

[i.8] ETSI TS 136 579-6: "LTE; Mission Critical (MC) services over LTE; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-6)".

[i.9] ETSI TS 136 579-7: "LTE; Mission Critical (MC) services over LTE; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-7)".

[i.10] ETSI TS 124 229: "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3 (3GPP TS 24.229)".

[i.11] IETF RFC 3856: "A Presence Event Package for the Session Initiation Protocol (SIP)".

[i.12] ETSI TS 104 148-2: "Mission Critical (MC) services; Mission Critical Push To Talk (MCPTT) Application Server (AS) Protocol conformance specification for server-to-server interface including InterWorking Function (IWF) to non-3GPP systems; Part 2: Test Applicability and Implementation Conformance Statement (ICS) proforma specification".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI TR 121 905 [i.1] (3GPP TR 21.905) apply.

NOTE: A term defined in the present document takes precedence over the definition of the same term, if any, in ETSI TR 121 905 [i.1].

For the purpose of the present document, the following terms given in ETSI TS 124 379 [4] (3GPP TS 24.379) apply:

- An MCPTT user is affiliated to an MCPTT group
- An MCPTT user is affiliated to an MCPTT group at an MCPTT client
- Affiliation status
- Group identity
- In-progress emergency private call state
- In-progress imminent peril group state
- MCPTT client ID
- MCPTT emergency alert state
- MCPTT emergency group state
- MCPTT emergency group call state
- MCPTT emergency private call state
- MCPTT emergency private priority state
- MCPTT imminent peril group call state
- MCPTT imminent peril group state
- MCPTT private emergency alert state
- MCPTT speech
- Media-floor control entity
- Temporary MCPTT group identity
- Trusted mutual aid
- Untrusted mutual aid

For the purposes of the present document, the following terms given in ETSI TS 122 179 [i.5] (3GPP TS 22.179) apply:

- In-progress emergency
- MCPTT emergency alert
- MCPTT emergency group call
- MCPTT emergency state
- Partner MCPTT system
- Primary MCPTT system

For the purpose of the present document, the following terms given in ETSI TS 124 380 [5] (3GPP TS 24.380) apply:

- MBMS subchannel

For the purpose of the present document, the following terms given in ETSI TS 123 179 [3] (3GPP TS 23.179) apply:

- Pre-selected MCPTT user profile

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TR 121 905 [i.1] (3GPP TR 21.905) and the following apply.

NOTE: An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in ETSI TR 121 905 [i.1] (3GPP TR 21.905).

FFS	For Further Study
GKTP	Group Key Transport Payload
ICS	Implementation Conformance Statement
IUT	Implementation Under Test
IWF	Inter-Working Function
IXIT	Implementation eXtra Information for Testing
LMR	Land Mobile Radio
MBMS	Multimedia Broadcast and Multicast Service
MCPTT group ID	MCPTT group IDentity
MCPTT	Mission Critical Push To Talk
MEA	MCPTT Emergency Alert
MEG	MCPTT Emergency Group
MEGC	MCPTT Emergency Group Call
MIG	MCPTT Imminent peril Group
MIGC	MCPTT Imminent peril Group Call
MIME	Multipurpose Internet Mail Extensions
PIDF	Presence Information Data Format
RTP	Real-time Transport Protocol
SDP	Session Description Protocol
SIP	Session Initiation Protocol
SS	System Simulator
SSRC	Synchronization SouRCe
TGI	Temporary MCPTT Group Identity
TP	Test Purpose
TP	Transmission Point
URI	Uniform Resource Identifier
XCAP	eXtensible Markup Language (XML) Access Configuration Protocol

4 General

4.1 Test methodology

4.1.1 Testing of optional functions and procedures

Any function or procedure which is optional may be subject to a conformance test if it is implemented in the MCPTT Server.

A declaration by the Client supplier (to use the Implementation Conformance Statement (ICS) proforma specified in ETSI TS 136 579-4 [2] (3GPP TS 36.579-4)) is used to determine whether an optional function/procedure has been implemented.

4.1.2 Test interfaces and facilities

Descriptions of the MCPTT Server to MCPTT Server interface is specified in clause A.4 of the present document.

4.2 Implicit testing

For some 3GPP MCPTT protocol features conformance is not verified explicitly in the present document. This does not imply that correct functioning of these features is not essential, but that these are implicitly tested to a sufficient degree in tests which are not explicitly dedicated to test the feature.

4.3 Repetition of tests

As a general rule, the test cases specified in the present document are highly reproducible and do not need to be repeated unless otherwise stated.

4.4 Handling of differences between conformance requirements in different releases of cores specifications

The conformance requirements which determine the scope of each test case are explicitly copy-pasted from relevant core specifications in the especially dedicated for this clause of each test with the title 'Conformance requirements'.

NOTE: When in the copy/pasted text there are references to other specifications the reference numbers will not match the reference numbers used in the present specification. This approach has been taken in order to allow easy copy and then search for conformance requirements in those specifications.

When differences between conformance requirements in different releases of the cores specifications have impact on the Pre-test conditions, Test procedure sequence or/and the Specific message contents, the Conformance requirements related to different releases are specified separately with clear indication of the Release of the spec from which they were copied.

When there is no Release indicated for a conformance requirement text, this should be understood either as the Conformance requirements in the latest version of the spec with release = the TC Applicability release (which can be found in ETSI TS 104 148-2 [i.12], Table 4-1 and Table 4-2: Applicability of tests and additional information for MCPTT Server and MCPTT IWF testing, column 'Release'), or, as the Conformance requirements in the latest version of the spec of the release when the feature was introduced to the core specs.

4.5 Reference conditions

The reference environments used by all signalling and protocol tests are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1). Where a test requires an environment that is different, this will be specified in the test itself.

5 MCPTT Server - MCPTT Server operation

5.1 Configuration

5.1.1 MCPTT Server - MCPTT Server / Configuration / Group Affiliation / De-affiliation / Controlling server

5.1.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS (MCPTT server) requests IUT (MCPTT Server) to determine the affiliation status of
the user with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) with a SIP 200 (OK) message and
sends a SIP NOTIFY message informing user not affiliated to any group }
            }
```

(2)

```
with { IUT (MCPTT Server) having provided affiliation status }
ensure that {
  when { the SS (MCPTT server) requests IUT (MCPTT Server) to affiliate user to an MCPTT group with
SIP PUBLISH message }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) with a SIP 200 (OK) message and
sends SIP NOTIFY message to inform the affiliation change }
            }
```

(3)

```
with { IUT (MCPTT Server) having provided affiliation change }
ensure that {
  when { the SS (MCPTT server) requests IUT (MCPTT Server) to de-affiliate user from an MCPTT group
with SIP PUBLISH message }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) with a SIP 200 (OK) message and
sends SIP NOTIFY message to inform the affiliation change }
            }
```

(4)

```
with { IUT (MCPTT Server) having provided affiliation change }
ensure that {
  when { the SS (MCPTT server) requests IUT (MCPTT Server) to de-subscribe from the affiliation
status for user with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) with a SIP 200 (OK) message and
sends a SIP NOTIFY message indicating the termination of the subscription }
            }
```

5.1.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9.2.2.2.7, 9.2.2.3.3, 9.2.2.3.4 and 9.2.2.3.5. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.2.7]

> *"NOTE 1: Usage of one SIP SUBSCRIBE request to subscribe for notification about change of affiliation state of several MCPTT users served by the same MCPTT server is not supported in this version of the specification.*

In order to discover whether a served MCPTT user was successfully affiliated to a handled MCPTT group in the MCPTT server owning the handled MCPTT group, the MCPTT server shall generate an initial SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 3856 [51], and IETF RFC 6665 [26].

In the SIP SUBSCRIBE request, the MCPTT server:

1) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the handled MCPTT group ID;

2) shall include an application/vnd.3gpp.mcptt-info+xml MIME body. In the application/vnd.3gpp.mcptt-info+xml MIME body, the MCPTT server:

a) shall include the <mcptt-request-uri> element set to the handled MCPTT group ID; and

b) shall include the <mcptt-calling-user-id> element set to the served MCPTT ID;

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];

4) if the MCPTT server wants to receive the current status and later notification, shall set the Expires header field according to IETF RFC 6665 [26], to 4294967295;

NOTE 2: 4294967295, which is equal to 2^{32}-1, is the highest value defined for Expires header field in IETF RFC 3261 [24].

5) if the MCPTT server wants to fetch the current state only, shall set the Expires header field according to IETF RFC 6665 [26], to zero;

6) shall include an Accept header field containing the application/pidf+xml MIME type; and

7) shall include an application/simple-filter+xml MIME body indicating per-user restrictions of presence event package notification information according to clause 9.3.2, indicating the served MCPTT ID.

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.3.3]

"Upon receiving a SIP PUBLISH request such that:

1) Request-URI of the SIP PUBLISH request contains the public service identity of the controlling MCPTT function associated with the served MCPTT group;

2) the SIP PUBLISH request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the <mcptt-request-uri> element and the <mcptt-calling-user-id> element;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];

4) the Event header field of the SIP PUBLISH request contains the "presence" event type; and

5) SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-group affiliation information constructed according to clause 9.3.1.2;

then the MCPTT server:

1) shall identify the served MCPTT group ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP PUBLISH request;

2) shall identify the handled MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP PUBLISH request;

…

6) shall respond with SIP 200 (OK) response to the SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37]. In the SIP 200 (OK) response, the MCPTT server:

a) shall set the Expires header field according to IETF RFC 3903 [37], to the selected expiration time;

7) if the "entity" attribute of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request is different than the served MCPTT group ID, shall not continue with the rest of the steps;

8) if the handled MCPTT ID is different from the MCPTT ID in the "id" attribute of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request, shall not continue with the rest of the steps;

9) shall consider an MCPTT group information entry such that:

a) the MCPTT group information entry is in the list of MCPTT group information entries described in clause 9.2.2.3.2; and

b) the MCPTT group ID of the MCPTT group information entry is equal to the served MCPTT group ID;

as the served MCPTT group information entry;

10) if the selected expiration time is zero:

a) shall remove the MCPTT user information entry such that:

i) the MCPTT user information entry is in the list of the MCPTT user information entries of the served MCPTT group information entry; and

ii) the MCPTT user information entry has the MCPTT ID set to the served MCPTT ID;

11) if the selected expiration time is not zero:

a) shall consider an MCPTT user information entry such that:

i) the MCPTT user information entry is in the list of the MCPTT user information entries of the served MCPTT group information entry; and

ii) the MCPTT ID of the MCPTT user information entry is equal to the handled MCPTT ID;

as the served MCPTT user information entry;

b) if the MCPTT user information entry does not exist:

i) shall insert an MCPTT user information entry with the MCPTT ID set to the handled MCPTT ID into the list of the MCPTT user information entries of the served MCPTT group information entry; and

ii) shall consider the inserted MCPTT user information entry as the served MCPTT user information entry; and

c) shall set the following information in the served MCPTT user information entry:

i) set the MCPTT client ID list according to the "client" attributes of the <affiliation> elements of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

ii) set the expiration time according to the selected expiration time;

12) shall identify the handled p-id in the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

13) shall perform the procedures specified in clause 9.2.2.3.5 for the served MCPTT group ID."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.3.4]

"NOTE: Usage of one SIP SUBSCRIBE request to subscribe for notification about change of affiliation state of several MCPTT users served by the same MCPTT server is not supported in this version of the specification.

Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains the public service identity of the controlling MCPTT function associated with the served MCPTT group;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the<mcptt-request-uri> element and the <mcptt-calling-user-id> element;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type; and

5) the SIP SUBSCRIBE request contains an application/simple-filter+xml MIME body indicating per-user restrictions of presence event package notification information according to clause 9.3.2 indicating the same MCPTT ID as in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

then the MCPTT server:

1) shall identify the served MCPTT group ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

2) shall identify the handled MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

…

6) shall generate a SIP 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 6665 [26].

For the duration of the subscription, the MCPTT server shall notify subscriber about changes of the information of the served MCPTT ID, as described in clause 9.2.2.3.5."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.3.5]

"In order to notify the subscriber identified by the handled MCPTT ID about changes of the affiliation status of the served MCPTT group ID, the MCPTT server:

1) shall consider an MCPTT group information entry such that:

a) the MCPTT group information entry is in the list of MCPTT group information entries described in clause 9.2.2.3.2; and

b) the MCPTT group ID of the MCPTT group information entry is equal to the served MCPTT group ID;

2) shall consider an MCPTT user information entry such:

a) the MCPTT user information entry is in the list of the MCPTT user information entries of the served MCPTT group information entry; and

b) the MCPTT ID of the MCPTT user information entry is equal to the handled MCPTT ID;

as the served MCPTT user information entry;

3) shall generate an application/pidf+xml MIME body indicating per-group affiliation information according to clause 9.3.1 and the served list of the served MCPTT user information entry of the MCPTT group information entry with following clarifications:

a) the MCPTT server shall include the "expires" attribute in the <affiliation> element; and

b) if this procedures is invoked by procedure in clause 9.2.2.3.3 where the handled p-id was identified, the MCPTT server shall set the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id value; and

4) send a SIP NOTIFY request according to 3GPP TS 24.229 [4], and IETF RFC 6665 [26] for the subscription created in clause 9.2.2.3.4. In the SIP NOTIFY request, the MCPTT server shall include the generated application/pidf+xml MIME body indicating per-group affiliation information."

5.1.1.3 Test description

5.1.1.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server):
  - SS (MCPTT Server) provides the participating MCPTT functions.
- SS-UE1 (MCPTT client):
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.1.1.3.1-1.

IUT:

- IUT (MCPTT Server):
  - IUT (MCPTT Server) provides the controlling MCPTT function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2.
- Group A is controlled by the controlling MCPTT function.
- User A is not configured to be implicitly affiliated to Group A.

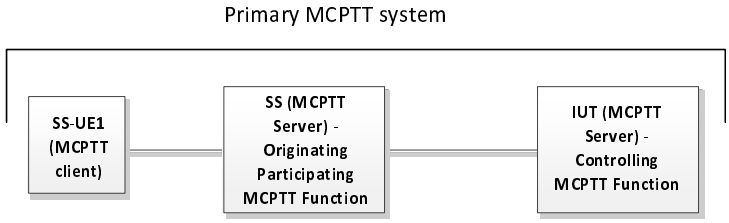


Figure 5.1.1.3.1-1: Functions of the testing components

5.1.1.3.2 Test procedure sequence

Table 5.1.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) sends a SIP SUBSCRIBE message request to determine the affiliation status of user A for group A.	<--	SIP SUBSCRIBE	-	-
2	Check: Does the IUT (MCPTT Server) respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	1	P
3	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message informing user A that is not affiliated to any group?	-->	SIP NOTIFY	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
4	The SS (MCPTT Server) responds with a SIP 200 (OK) message.	<--	SIP 200 (OK)		
5	The SS (MCPTT server) sends a SIP PUBLISH message to affiliate user A to group A.	<--	SIP PUBLISH	-	-
6	Check: Does the IUT (MCPTT Server) respond to the SIP PUBLISH message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	2	P
7	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY informing about the affiliation change?	-->	SIP NOTIFY	2	P
8	The SS responds with a SIP 200 (OK).	<--	SIP 200 (OK)	-	-
9	The SS (MCPTT server) sends a SIP PUBLISH message to de-affiliate user A from group A.	<--	SIP PUBLISH	-	-
10	Check: Does the IUT (MCPTT Server) respond to the SIP PUBLISH message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	3	P
11	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message informing about the affiliation change?	-->	SIP NOTIFY	3	P
12	The SS (MCPTT Client) responds with a SIP 200 (OK)	<--	SIP 200 (OK)	-	-
13	The SS (MCPTT Server) sends a SIP SUBSCRIBE message request to de-subscribe from affiliation status changes for user A and group A.	<--	SIP SUBSCRIBE	-	-
14	Check: Does the IUT (MCPTT Server) respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	4	P
15	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message indicating the termination of the subscription?	-->	SIP NOTIFY	4	P
16	The SS (MCPTT Server) responds with a SIP 200 (OK) message.	<--	SIP 200 (OK)		

5.1.1.3.3 Specific message contents

Table 5.1.1.3.3-1: SIP SUBSCRIBE from the SS (MCPTT Server) (Step 1, Table 5.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	tsc_MCPTT_PublicServiceId_A	The public service identity identifying the controlling MCPTT function owning the MCPTT group		
From				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B	The public service identity identifying the originating participating MCPTT function serving the MCPTT user		
To				
addr-spec				

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
user-info and host	tsc_MCPTT_PublicServiceId_A	The public service identity identifying the controlling MCPTT function owning the MCPTT group		
Expires				
value	"4294967295"	to receive the current status and later notification		
P-Asserted-Service			IETF RFC 6050 [26]	
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.1.1.3.3-1A			
MIME body part		**SIMPLE-FILTER**		
MIME-part-body	Simple-Filter as described in Table 5.1.1.3.3-1B			

Table 5.1.1.3.3-1A: MCPTT-Info in SIP SUBSCRIBE (Table 5.1.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_A_ID			
mcptt-calling-user-id	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 5.1.1.3.3-1B: Simple-Filter in SIP SUBSCRIBE (Table 5.1.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.6-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
filter-set				
filter[1]				
what			IETF RFC 4661 [20]	
include[1]				
type	xpath if present	"xpath" per default		
base	"//pidf:presence/pidf:tuple[@id=" & px_MCPTT_ID_User_A & "]"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.2.2	

Table 5.1.1.3.3-2: SIP 200 (OK) to SIP SUSCRIBE (Steps 2 and 14, Table 5.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition SUBSCRIBE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route				Not present

Table 5.1.1.3.3-3: SIP NOTIFY from the IUT (MCPTT Server) (Steps 3 and 11, Table 5.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions MCPTT and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Service			IETF RFC 6050 [26]	
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
P-Asserted-Identity			IETF RFC 3325 [27]	
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_A	The public service identity identifying the controlling MCPTT function owning the MCPTT group		
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.3.5	
MIME-part-body	PIDF as described in Table 5.1.1.3.3-3A			

Table 5.1.1.3.3-3A: PIDF in SIP NOTIFY (Table 5.1.1.3.3-3)

Derivation Path: IETF RFC 3863 [8]				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence			IETF RFC 3863 [8]	
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_Group_A_ID			
tuple				
id attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_A			
status				
affiliation	not present	MCPTT extension	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
contact	not present			
note	not present			
timestamp	not present			
note	not present			
p-id	same value as received in the SIP PUBLISH message		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.2.5 and 9.2.2.3.5	
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.13.3-1.				

Table 5.1.1.3.3-4: SIP NOTIFY from the IUT (MCPTT Server) (Steps 15, Table 5.1.1.3.2-1)

Derivation Path: Table 5.1.1.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subscription-State			IETF RFC 6665 [24]	
substate-value	"terminated"			
expires	not present			

Table 5.1.1.3.3-5: SIP NOTIFY from the IUT (MCPTT Server) (Steps 7, Table 5.1.1.3.2-1)

Derivation Path: Table 5.1.1.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.3.5	
MIME-part-body	PIDF as described in Table 5.1.1.3.3-5A			

Table 5.1.1.3.3-5A: PIDF in SIP NOTIFY (Table 5.1.1.3.3-5)

Derivation Path: Table 5.1.1.3.3-3A				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence			IETF RFC 3863 [8]	
tuple				
status				
affiliation				
client	Encrypted URI (see note) with value set to px_MCX_Client_B_ID			
group	not present			
status	not present			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.3.5	
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS Table 5.5.13.3-1.				

Table 5.1.1.3.3-6: SIP 200 (OK) to SIP NOTIFY (Steps 4, 8, 12, 16, Table 5.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route				Not present

Table 5.1.1.3.3-7: SIP SUBSCRIBE from the SS (MCPTT Server) (Step 13, Table 5.1.1.3.2-1)

Derivation Path: Table 5.1.1.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	same URI as the IUT (MCPTT Server) has sent earlier in the Contact header of a message within the same dialog	Contact URI of the recipient of the previous 200 OK		
Expires				
value	"0"			

Table 5.1.1.3.3-8: SIP PUBLISH from the SS (MCPTT Server) (Step 5, Table 5.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.11-1, condition PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	tsc_MCPTT_PublicServiceId_A	The public service identity identifying the controlling MCPTT function owning the MCPTT group		
From				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B	The public service identity identifying the originating participating MCPTT function serving the MCPTT user		
To				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_A	The public service identity identifying the controlling MCPTT function owning the MCPTT group		
Expires				
value	"4294967295"			
P-Asserted-Service			IETF RFC 6050 [26]	
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
P-Asserted-Identity			IETF RFC 3325 [27]	
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B	The public service identity identifying the originating participating MCPTT function serving the MCPTT user		
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.1.1.3.3-1A			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.1.1.3.3-8A			

Table 5.1.1.3.3-8A: PIDF in SIP PUBLISH (Table 5.1.1.3.3-8)

Derivation Path: Table 5.1.1.3.3-5A				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence			IETF RFC 3863 [8]	
tuple				
status				
affiliation		MCPTT extension	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	
expires	not present			
p-id	any allowed value		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.2.5 and 9.2.2.3.5	

Table 5.1.1.3.3-9: SIP PUBLISH from the SS (MCPTT Server) (Step 9, Table 5.1.1.3.2-1)

Derivation Path: Table 5.1.1.3.3-8				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
value	"0"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.1.1.3.3-1A			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.1.1.3.3-9A			

Table 5.1.1.3.3-9A: PIDF in SIP PUBLISH (Table 5.1.1.3.3-9)

Derivation Path: Table 5.1.1.3.3-8A				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence			IETF RFC 3863 [8]	
tuple				
status				
affiliation	not present	MCPTT extension	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.3.1	

Table 5.1.1.3.3-10: SIP 200 (OK) to SIP PUBLISH (Steps 6 and 10, Table 5.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition PUBLISH-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route				Not present

5.1.2 MCPTT Server - MCPTT Server / Configuration / Group Affiliation / De-affiliation / Implicit Affiliation / Participating server

5.1.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the IUT (MCPTT server) subscribes to the affiliation status determination }
    then { IUT (MCPTT Server) sends a SIP SUBSCRIBE message to determine the affiliation status of
the user and sends SIP 200 OK after receipt of SIP NOTIFY informing about not affiliated user  }
           }
```

(2)

```
with { SS (MCPTT Server) having established subscription for SS-UE1 }
ensure that {
  when { the IUT (MCPTT server) trigger implicit affiliate for SS-UE1 }
    then { IUT (MCPTT Server) sends SIP PUBLISH message and sends SIP 200 OK after receipt of SIP
NOTIFY informing about SS-UE1 affiliation change }
           }
```

(3)

```
with { SS (MCPTT Server) having affiliated user }
ensure that {
  when { the SS-UE1 requests IUT (MCPTT Server) to de-affiliate user from group with SIP PUBLISH
message }
    then { IUT (MCPTT Server) forwards SIP PUBLISH message to SS (MCPTT Server) and replies with SIP
200 OK to SS-UE1 and sends SIP 200 OK after receipt of SIP NOTIFY informing about SS-UE1 de-
affiliation change  }
            }
```

(4)

```
with { SS (MCPTT Server) having de-affiliated user }
ensure that {
  when { the SS-UE1 requests IUT (MCPTT Server) to affiliate user to group with SIP PUBLISH message
}
    then { IUT (MCPTT Server) forwards SIP PUBLISH message to SS (MCPTT Server) and replies with SIP
200 OK to SS-UE1 and sends SIP 200 OK after receipt of SIP NOTIFY informing about SS-UE1 affiliation
change  }
            }
```

(5)

```
with { SS (MCPTT Server) having affiliated user }
ensure that {
  when { the IUT (MCPTT server) de-subscribes from the affiliation status determination }
    then { IUT (MCPTT Server) sends a SIP SUBSCRIBE message and sends SIP 200 OK after receipt of
SIP NOTIFY for subscription termination  }
            }
```

5.1.2.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9.2.2.2.6, 9.2.2.2.7, 9.2.2.2.13 and 9.2.2.3.5. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.2.6]

"In order:

- *to send an affiliation request of a served MCPTT ID to a handled MCPTT group ID;*
- *to send an de-affiliation request of a served MCPTT ID from a handled MCPTT group ID; or*
- *to send an affiliation request of a served MCPTT ID to a handled MCPTT group ID due to near expiration of the previously published information;*

the MCPTT server shall generate a SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37] and IETF RFC 3856 [51]. In the SIP PUBLISH request, the MCPTT server:

1) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the handled MCPTT group ID;

2) shall include an application/vnd.3gpp.mcptt-info+xml MIME body. In the application/vnd.3gpp.mcptt-info+xml MIME body, the MCPTT server:

   a) shall include the <mcptt-request-uri> element set to the handled MCPTT group ID; and

   b) shall include the <mcptt-calling-user-id> element set to the served MCPTT ID;

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9] ;

4) if sending an affiliation request, shall set the Expires header field according to IETF RFC 3903 [37] , to 4294967295;

NOTE 1: 4294967295, which is equal to 232-1, is the highest value defined for Expires header field in IETF RFC 3261 [24] .

5) if sending an de-affiliation request, shall set the Expires header field according to IETF RFC 3903 [37] , to zero;

6) shall include an P-Asserted-Identity header field set to the public service identity of the MCPTT server according to 3GPP TS 24.229 [4] ;7) shall set the current p-id to a globally unique value;

8) shall consider an MCPTT user information entry such that:

a) the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9.2.2.2.2; and

b) the MCPTT ID of the MCPTT user information entry is equal to the served MCPTT ID;

as the served MCPTT user information entry;

9) for each MCPTT group information entry such that:

a) the MCPTT group information entry has the "affiliating" affiliation status, the MCPTT group ID set to the handled MCPTT group ID, the expiration time has not expired yet and the affiliating p-id is not set;

b) the MCPTT group information entry is in the list of the MCPTT group information entries of an MCPTT client information entry; and

c) the MCPTT client information entry is in the list of the MCPTT client information entries of the served MCPTT user information entry;

shall set the affiliating p-id of the MCPTT group information entry to the current p-id; and

10) shall include an application/pidf+xml MIME body indicating per-group affiliation information constructed according to clause 9.3.1.2. The MCPTT server shall indicate all served MCPTT client IDs, such that:

a) the affiliation status is set to "affiliating" or "affiliated", and the expiration time has not expired yet in an MCPTT group information entry with the MCPTT group ID set to the handled MCPTT group;

b) the MCPTT group information entry is in the list of the MCPTT group information entries of an MCPTT client information entry;

c) the MCPTT client information entry has the MCPTT client ID set to the served MCPTT client ID; and

d) the MCPTT client information entry is in the list of the MCPTT client information entries of the served MCPTT user information entry.

The MCPTT server shall set the <p-id> child element of the <presence> root element to the current p-id.

The MCPTT server shall not include the "expires" attribute in the <affiliation> element.

The MCPTT server shall send the SIP PUBLISH request according to 3GPP TS 24.229 [4].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.2.7]

"NOTE 1: Usage of one SIP SUBSCRIBE request to subscribe for notification about change of affiliation state of several MCPTT users served by the same MCPTT server is not supported in this version of the specification.

In order to discover whether a served MCPTT user was successfully affiliated to a handled MCPTT group in the MCPTT server owning the handled MCPTT group, the MCPTT server shall generate an initial SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 3856 [51], and IETF RFC 6665 [26]. In the SIP SUBSCRIBE request, the MCPTT server:

1) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the handled MCPTT group ID;

2) shall include an application/vnd.3gpp.mcptt-info+xml MIME body. In the application/vnd.3gpp.mcptt-info+xml MIME body, the MCPTT server:

a) shall include the <mcptt-request-uri> element set to the handled MCPTT group ID; and

b) shall include the <mcptt-calling-user-id> element set to the served MCPTT ID;

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];

4) if the MCPTT server wants to receive the current status and later notification, shall set the Expires header field according to IETF RFC 6665 [26] , to 4294967295;

NOTE 2: 4294967295, which is equal to 232-1, is the highest value defined for Expires header field in IETF RFC 3261 [24].

5) if the MCPTT server wants to fetch the current state only, shall set the Expires header field according to IETF RFC 6665 [26], to zero;6) shall include an Accept header field containing the application/pidf+xml MIME type; and

7) shall include an application/simple-filter+xml MIME body indicating per-user restrictions of presence event package notification information according to clause 9.3.2, indicating the served MCPTT ID.

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.2.13]

"This clause is referenced from other procedures.

If the participating MCPTT function has received a SIP 2xx response from the controlling MCPTT function to a SIP request that had triggered performing the procedures of clause 9.2.2.2.12, the participating MCPTT function:

1) shall set the affiliation status of the MCPTT group information entry added to the candidate list of the MCPTT group information entries by the procedures of clause 9.2.2.2.12 to "affiliated"; and

2) shall perform the procedures specified in clause 9.2.2.2.5 for the served MCPTT ID."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9.2.2.3.5]

"In order to notify the subscriber identified by the handled MCPTT ID about changes of the affiliation status of the served MCPTT group ID, the MCPTT server:

1) shall consider an MCPTT group information entry such that:

a) the MCPTT group information entry is in the list of MCPTT group information entries described in clause 9.2.2.3.2; and

b) the MCPTT group ID of the MCPTT group information entry is equal to the served MCPTT group ID;

2) shall consider an MCPTT user information entry such:

a) the MCPTT user information entry is in the list of the MCPTT user information entries of the served MCPTT group information entry; and

b) the MCPTT ID of the MCPTT user information entry is equal to the handled MCPTT ID;

as the served MCPTT user information entry;

3) shall generate an application/pidf+xml MIME body indicating per-group affiliation information according to clause 9.3.1 and the served list of the served MCPTT user information entry of the MCPTT group information entry with following clarifications:

a) the MCPTT server shall include the "expires" attribute in the <affiliation> element; and

b) if this procedures is invoked by procedure in clause 9.2.2.3.3 where the handled p-id was identified, the MCPTT server shall set the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id value; and

4) send a SIP NOTIFY request according to 3GPP TS 24.229 [4], and IETF RFC 6665 [26] for the subscription created in clause 9.2.2.3.4. In the SIP NOTIFY request, the MCPTT server shall include the generated application/pidf+xml MIME body indicating per-group affiliation information."

5.1.2.3 Test description

5.1.2.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the controlling MCPTT function.
- SS-UE1 (MCPTT client)

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the participating MCPTT function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2, and to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- Group A is controlled by the controlling MCPTT function.
- User A is not configured to be implicitly affiliated to any group.

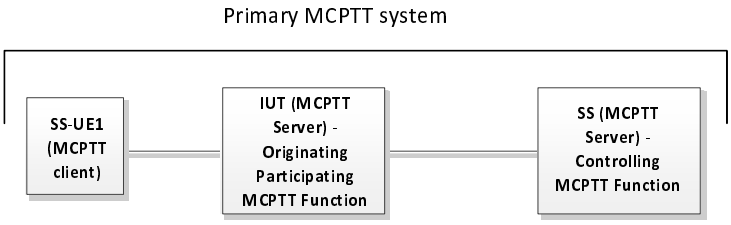


Figure 5.1.2.3.1-1: Functions of the testing components

5.1.2.3.2 Test procedure sequence

Table 5.1.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Make the IUT (MCPTT Server) subscribe to affiliation status determination and later notifications for user A in group A.	-	-	-	-
2	Check: Does the IUT (MCPTT Server) send a SIP SUBSCRIBE message request to determine the affiliation status of user A for group A?	-->	SIP SUBSCRIBE	1	P
3	The SS (MCPTT Server) responds to the SIP SUBSCRIBE message with a SIP 200 (OK) message	<--	SIP 200 (OK)	-	-
4	The SS (MCPTT Server) sends a SIP NOTIFY message informing that user A is not affiliated to group A.	<--	SIP NOTIFY	-	-
5	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK) message?	-->	SIP 200 (OK)	1	P
6	Check: Does the IUT (MCPTT Server) send a SIP PUBLISH message to implicitly affiliate user A to group A?	-->	SIP PUBLISH	2	P
7	The SS (MCPTT Server) responds to the SIP PUBLISH message with a SIP 200 (OK) message.	<--	SIP 200 (OK)	-	-
8	The SS (MCPTT Server) sends a SIP NOTIFY informing about the affiliation change.	<--	SIP NOTIFY	-	-
9	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK)?	-->	SIP 200 (OK)	2	P
10	The SS-UE1 (MCPTT client) sends a SIP PUBLISH message to de-affiliate user A from group A.	<--	SIP PUBLISH	-	-
11	Check: Does the IUT (MCPTT Server) send a SIP PUBLISH message to the SS to de-affiliate user A from group A?	-->	SIP PUBLISH	3	P
12	The SS (MCPTT Server) responds to the SIP PUBLISH message with a SIP 200 (OK) message.	<--	SIP 200 (OK)		
13	Check: Does the IUT (MCPTT Server) respond to the SIP PUBLISH message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	3	P
14	The SS (MCPTT Server) sends a SIP NOTIFY informing about the affiliation change.	<--	SIP NOTIFY	-	-
15	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK)?	-->	SIP 200 (OK)	3	P
16	The SS-UE1 (MCPTT client) sends a SIP PUBLISH message to affiliate user A from group A.	<--	SIP PUBLISH	-	-
17	Check: Does the IUT (MCPTT Server) send a SIP PUBLISH message to the SS to affiliate user A from group A?	-->	SIP PUBLISH	4	P
18	The SS (MCPTT Server) responds to the SIP PUBLISH message with a SIP 200 (OK) message.	<--	SIP 200 (OK)		
19	Check: Does the IUT (MCPTT Server) respond to the SIP PUBLISH message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	4	P
20	The SS (MCPTT Server) sends a SIP NOTIFY message informing about the affiliation change	<--	SIP NOTIFY	-	-
21	Check: Does the MCPTT Client respond with a SIP 200 (OK)?	-->	SIP 200 (OK)	4	P
22	Make the IUT (MCPTT Server) de-subscribe from getting affiliation status notifications for user A and group A.	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
23	Check: Does the IUT (MCPTT Server) send a SIP SUBSCRIBE message request to de-subscribe from affiliation status changes for user A and group A?	-->	SIP SUBSCRIBE	5	P
24	The SS (MCPTT Server) responds to the SIP SUBSCRIBE message with a SIP 200 (OK) message	<--	SIP 200 (OK)	-	-
25	The SS (MCPTT Server) sends a SIP NOTIFY message indicating the termination of the subscription	<--	SIP NOTIFY	-	-
26	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK) message?	-->	SIP 200 (OK)	5	P

5.1.2.3.3 Specific message contents

Table 5.1.2.3.3-1: SIP SUBSCRIBE from the IUT (MCPTT Server) (Step 2, Table 5.1.2.3.2-1)

Derivation Path: Table 5.1.1.3.3-1

Table 5.1.3.3.3-2: SIP 200 (OK) to SUBSCRIBE (Steps 3 and 24, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.1.3.3-2

Table 5.1.2.3.3-3: SIP NOTIFY from the SS (MCPTT Server) (Steps 8 and 20, Table 5.1.2.3.2-1)

Derivation Path: Table 5.1.1.3.3-3

Table 5.1.2.3.3-4: SIP NOTIFY from the SS (MCPTT Server) (Step 25, Table 5.1.2.3.2-1)

Derivation Path: Table 5.1.1.3.3-4

Table 5.1.2.3.3-5: SIP NOTIFY from the SS (MCPTT Server) (Steps 4 and 14, Table 5.1.2.3.2-1)

Derivation Path: Table 5.1.1.3.3-5

Table 5.1.3.3.3-6: SIP 200 (OK) to NOTIFY (Steps 5, 9, 15, 21 and 26, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.1.3.3-6

Table 5.1.2.3.3-7: SIP SUBSCRIBE from the IUT (MCPTT Server) (Step 23, Table 5.1.2.3.2-1)

Derivation Path: Table 5.1.1.3.3-7

Table 5.1.2.3.3-8: SIP PUBLISH from the IUT (MCPTT Server) (Steps 6 and 17, Table 5.1.2.3.2-1)

Derivation Path: Table 5.1.1.3.3-8

Table 5.1.2.3.3-9: SIP PUBLISH from the IUT (MCPTT Server) (Step 11, Table 5.1.2.3.2-1)

Derivation Path: Table 5.1.1.3.3-9

Table 5.1.2.3.3-10: SIP PUBLISH from the SS-UE1 (MCPTT Client) (Step 10, Table 5.1.2.3.2-1)

Derivation Path: Table 5.3.3.3-7

Table 5.1.2.3.3-11: SIP PUBLISH from the SS-UE1 (MCPTT Client) (Steps 16, Table 5.1.2.3.2-1)

Derivation Path: Table 5.3.3.3-8

Table 5.1.2.3.3-12: SIP 200 (OK) to PUBLISH (Steps 7, 12 and 18, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.1.3.3-10

Table 5.1.2.3.3-13: SIP 200 (OK) to PUBLISH (Steps 13 and 19, Table 5.1.3.3.2-1)

Derivation Path: Annex A, Table A.3.8.4-3

5.1.3 MCPTT Server - MCPTT Server / Group and GMK Subscription / Group Regroup creation / Group Regroup Teardown / GMS owning temporary group

5.1.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE1 (MCPTT client) initiates creation of a temporary group }
    then { IUT (MCPTT Server) sends a HTTP POST message to SS (MCPTT Server) to request a GMOP group
regroup check and sends a HTTP POST message to SS (MCPTT Server) to request a GMOP group regroup
notification and replies with HTTP 200 OK message to SS-UE1 (MCPTT client }
            }
```

(2)

```
with { SS (MCPTT Server) having provided the GMOP group regroup creation response for SS-UE1 }
ensure that {
  when { the SS (MCPTT Server) sends a SIP SUBSCRIBE message to get the group key transport payloads
(GKTP) document for both groups }
    then { IUT (MCPTT Server) sends a SIP 200 OK message to the SS (MCPTT Server) and sends SIP
NOTIFY message containing the group key transport payloads (GKTP) document for both groups}
            }
```

(3)

```
with { SS (MCPTT Server) having received the group key transport payloads (GKTP) document }
ensure that {
  when { the SS-UE1 (MCPTT client) sends a HTTP DELETE message to tear down the temporary group }
    then { IUT (MCPTT Server) sends a HTTP DELETE message to the SS (MCPTT Server) to remove the
parameters associated to the temporary group and sends HTTP 200 OK message to the SS-UE1 (MCPTT
client) }
            }
```

(4)

```
with { SS (MCPTT Server) having removed the parameters associated to the temporary group }
ensure that {
  when { the SS (MCPTT Server) sends a SIP SUBSCRIBE message to de-subscribe from receiving
notifications of group key transport payloads (GKTP) for both groups }
    then { IUT (MCPTT Server) sends a SIP 200 OK message to the SS (MCPTT Server) and sends SIP
NOTIFY message indicating the termination of the subscription for both groups }
            }
```

5.1.3.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 481 [6] (3GPP TS 24.481) clauses 6.3.14.3.1, 6.3.14.3.2, 6.3.15.3.1, 6.3.15.3.2, 6.3.13.3.2.2, 6.3.13.3.2.3, 6.3.13.3.2.4, 6.3.13.3.3, 6.2.5.3, 7.3.4.3, 7.3.4.4, 7.3.4.5, 7.3.4.6, 7.7.8, 7.4.2 and A.2.3. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.14.3.1]

"Upon reception of an HTTP POST request:

a) with a Request-URI with an XCAP URI identifying a non-existing group document; and

b) with an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document requesting group regroup creation specified in subclause 7.3.4.3;

then the GMS:

…

d) for each MCS group ID of an MCS group to be combined indicated in content of a <constituent-MCPTT-group-ID> element of the <constituent-MCPTT-group-IDs> element of the <on-network-temporary> element of the group document of the <group> element of the GMOP document requesting group regroup creation specified in subclause 7.3.4.3 of the received HTTP POST request:

1) shall send a HTTP POST request according to procedures specified in IETF RFC 2616 [21] and subclause 6.2.5. In the HTTP POST request, the GMS:

A) shall set the Request-URI to an XCAP URI:

i) with the document selector identifying a group document addressed by a group ID as specified in subclause 7.2.10.2, where the group ID is set to the MCS group ID of the MCS group to be combined;

ii) with the node selector identifying a <on-network-regrouped> element of the constituent MCS group such that the <on-network-regrouped> element has the "temporary-MCPTT-group-ID" attribute set to the content of the "uri" attribute of the <list-service> element of the group document included in the received HTTP POST request;

iii) if the MCS group ID of the MCS group to be combined is owned by the MCS provider of the GMS, with the XCAP root URI of the MCS provider of the GMS; and

iv) if the MCS group ID of the MCS group to be combined is owned by an MCS provider other than the MCS provider of the GMS, with XCAP root URI derived using the group ID routing database as specified in subclause 6.2.5.2 and the MCS group ID of the MCS group to be combined;

B) shall set the X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.482 [5] to a public service identity of the GMS; and

C) shall include an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document requesting group regroup check specified in subclause 7.3.4.4 with a <on-network-regrouped> element. In the <on-network-regrouped> element, the GMS:

i) shall set the "temporary-MCPTT-group-ID" attribute to the content of the "uri" attribute of the <list-service> element of the group document included in the received HTTP POST request;

ii) shall set the "temporary-MCPTT-group-requestor" attribute to the identity of the sender of the received HTTP POST request;

iii) shall include the <constituent-MCPTT-group-IDs> element set to content of the <constituent-MCPTT-group-IDs> element of <on-network-temporary> element of the <list-service> element of the group document of the MCS group included in the received HTTP POST request;

iv) if an <on-network-group-priority> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <on-network-group-priority> element set to content of the <on-network-group-priority> element of the <list-service> element of the group document included in the received HTTP POST request;

v) if a <protect-media> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <protect-media> element set to content of the <protect-media> element of the <list-service> element of the group document included in the received HTTP POST request;

vi) if a <protect-floor-control-signalling> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <protect-floor-control-signalling> element set to content of the <protect-floor-control-signalling> element of the <list-service> element of the group document included in the received HTTP POST request; and

vii) if a <require-multicast-floor-control-signalling> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <require-multicast-floor-control-signalling> element.

Upon reception of a HTTP 3xx, 4xx, or 5xx responses to a sent HTTP POST request or upon timeout, the GMS shall send a HTTP 403 (Forbidden) response to the received HTTP request and shall not continue with rest of the steps.

Upon reception of HTTP 2xx responses to all sent HTTP POST requests, the GMS:

a) for each MCS group ID of an MCS group to be combined indicated in content of a <constituent-MCPTT-group-ID> element of the <constituent-MCPTT-group-IDs> element of the <on-network-temporary> element of the <group> element of the GMOP document requesting group regroup creation specified in subclause 7.3.4.3 of the received HTTP POST request:

1) shall send an HTTP POST request according to procedures specified in IETF RFC 2616 [21] and subclause 6.2.5. In the HTTP POST request, the GMS:

A) shall set the Request-URI to an XCAP URI:

i) with the document selector identifying a group document addressed by a group ID as specified in subclause 7.2.10.2, where the group ID is set to the MCS group ID of the MCS group to be combined;

ii) with the node selector identifying a <on-network-regrouped> element of the constituent MCS group such that the <on-network-regrouped> element has the "temporary-MCPTT-group-ID" attribute set to the content of the "uri" attribute of the <list-service> element of the group document included in the received HTTP POST request;

iii) if the MCS group ID of the MCS group to be combined is owned by the MCS provider of the GMS, with the XCAP root URI of the MCS provider of the GMS; and

iv) if the MCS group ID of the MCS group to be combined is owned by an MCS provider other than the MCS provider of the GMS, with XCAP root URI derived using the group ID routing database as specified in subclause 6.2.5.2 and the MCS group ID of the MCS group to be combined;

B) shall set the X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.482 [5] to a public service identity of the GMS; and

C) shall include an application/vnd.3gpp.GMOP+xml MIME body with a GMOP document requesting group regroup notification specified in subclause 7.3.4.5. In the GMOP document requesting group regroup notification, the GMS:

i) shall include a <on-network-regrouped> element. In the <on-network-regrouped> element, the GMS:

- *shall set the "temporary-MCPTT-group-ID" attribute to the content of the "uri" attribute of the <list-service> element of the group document included in the received HTTP POST request;*
- *shall set the "temporary-MCPTT-group-requestor" attribute to the identity of the sender of the received HTTP POST request;*
- *shall include the <constituent-MCPTT-group-IDs> element set to content of the <constituent-MCPTT-group-IDs> element of <on-network-temporary> element of the <list-service> element of the group document of the MCS group included in the received HTTP POST request;*
- *if an <on-network-group-priority> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <on-network-group-priority> element set to content of the <on-network-group-priority> element of the <list-service> element of the group document included in the received HTTP POST request;*
- *if a <protect-media> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <protect-media> element set to content of the <protect-media> element of the <list-service> element of the group document included in the received HTTP POST request;*
- *if a <protect-floor-control-signalling> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <protect-floor-control-signalling> element set to content of the <protect-floor-control-signalling> element of the <list-service> element of the group document included in the received HTTP POST request; and*
- *if a <require-multicast-floor-control-signalling> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <require-multicast-floor-control-signalling> element.*

NOTE: GMK is not included in the GMOP document requesting group regroup notification as GMK is provided only using SIP.

Upon reception of HTTP 2xx responses to all sent HTTP POST requests, the GMS shall create the group document of the temporary MCS group at the location specified by the Request-URI of the received HTTP POST request and shall send an HTTP 2xx response to the received HTTP request. In the HTTP 2xx response, the GMS shall include an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document with group regroup creation response specified in subclause 7.3.4.6 with a <temporary-group-document-ETag> element set to the current value of the entity tag for the created group document of the temporary MCS group."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.14.3.2]

"Upon reception of an HTTP POST request:

a) with the Request-URI set to an XCAP URI identifying an existing or a non-existing <on-network-regrouped> element of an existing group document of an MCS group; and

b) with an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document requesting group regroup check specified in subclause 7.3.4.4;

the GMS:

a) if:

1) the Request-URI identifies an existing <on-network-regrouped> element of an existing group document of an MCS group;

2) the Request-URI identifies a non-existing <on-network-regrouped> element of an existing group document defining a temporary MCS group;

3) identity indicated in the X-3GPP-Asserted-Identity header field is not listed in the authorized GMS list specified in subclause 6.2.5.1; or

4) the MIME body of the HTTP POST request is not acceptable;

then shall respond with HTTP 403 (Forbidden) response to the HTTP POST request and shall not continue with rest of the steps; and

b) shall respond with HTTP 200 (OK) response to the HTTP POST request.

Upon reception of an HTTP POST request:

a) with the Request-URI set to an XCAP URI identifying an existing or a non-existing <on-network-regrouped> element of an existing group document of an MCS group; and

b) with an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document requesting group regroup notification specified in subclause 7.3.4.5;

the GMS:

a) if:

1) the Request-URI identifies an existing <on-network-regrouped> element of an existing group document of an MCS group;

2) the Request-URI identifies a non-existing <on-network-regrouped> element of an existing group document defining a temporary MCS group;

3) identity indicated in the X-3GPP-Asserted-Identity header field is not listed in the authorized GMS list specified in subclause 6.2.5.1; or

4) the MIME body of the HTTP POST request is not acceptable;

then shall respond with HTTP 403 (Forbidden) response to the HTTP POST request and shall not continue with rest of the steps;

b) shall place the <on-network-regrouped> element of the GMOP document requesting group regroup notification of the HTTP POST request at the location identified by the Request-URI of the received HTTP POST request;

c) shall respond with HTTP 200 (OK) response to the HTTP POST request; and

NOTE: GMKis not included in the GMOP document requesting group regroup notification as GMK is provided only using SIP.

d) shall subscribe for changes of the MCS GKTP document of the temporary MCS Group ID indicated in the "temporary-MCPTT-group-ID" attribute of the <on-network-regrouped> element of the GMOP document requesting group regroup notification according to subclause 6.3.13.2.3."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.15.3.1]

"Upon reception of an HTTP DELETE request with Request-URI with an XCAP URI identifying a group document of a temporary MCS group, the GMS:

a) for each constituent MCS group indicated in the group document indicated by Request-URI:

1) shall send an HTTP DELETE request. In the HTTP DELETE request, the GMS:

A) shall set the Request-URI to an XCAP URI:

i) with the document selector identifying a group document addressed by a group ID as specified in subclause 7.2.10.2, where the group ID is set to the MCS group ID of the constituent MCS group;

ii) with the node selector identifying a <on-network-regrouped> element of the constituent MCS group, such that the "temporary-MCPTT-group-ID" attribute of the <on-network-regrouped> element contains the temporary MCS group ID of the temporary MCS group;

iii) if the MCS group ID of the constituent MCS group is owned by the MCS provider of the GMS, with the XCAP root URI of the MCS provider of the GMS; and

iv) if the MCS group ID of the constituent MCS group is owned by an MCS provider other than the MCS provider of the GMS, with XCAP root URI derived using the group ID routing database as specified in subclause 6.2.5.2 and the MCS group ID of the constituent MCS group; and

B) shall set the X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.482 [5] to a public service identity of the GMS.

Upon reception of an HTTP response to all sent HTTP DELETE requests, the GMS shall remove the group document of the temporary MCS group and shall send an HTTP 2xx response to the received HTTP request."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.15.3.2]

"Upon reception of an HTTP DELETE request with Request-URI identifying a <on-network-regrouped> element of an MCS group document, the GMS:

a) if:

1) the Request-URI identifies an existing <on-network-regrouped> element of a non-existing group document; or

2) identity indicated in the X-3GPP-Asserted-Identity header field is not listed in the authorized GMS list specified in subclause 6.2.5.1;

then shall respond with HTTP 403 (Forbidden) response to the HTTP DELETE request and shall not continue with rest of the steps; and

b) shall act according to according to procedures specified in IETF RFC 4825 [22] "DELETE Handling"."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.13.3.2.2]

"Upon reception of an initial SIP SUBSCRIBE request:

a) with the Event header field set to xcap-diff;

b) with the Request-URI set to own public service identity for performing subscription proxy function of the GMS;

c) with a P-Asserted-Identity header field not containing an identity listed in the authorized MCS server list specified in subclause 6.2.5.1 and not containing an identity listed in the authorized GMS list as specified in subclause 6.2.5.1;

d) with an application/vnd.3gpp.mcptt-info+xml MIME body containing the <mcptt-access-token> element;

e) with an application/resource-lists+xml MIME body; and

f) with the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24 229 [12]), in a P-Asserted-Service header field according to IETF RFC 6050 [14];

the GMS:

a) if an <EncryptedData> XML tag is included in the application/vnd.3gpp.mcptt-info+xml MIME body and the CSK is received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, shall decrypt the application/vnd.3gpp.mcptt-info+xml MIME body;

b) if an <EncryptedData> XML tag is included in the application/resource-lists+xml MIME body and the CSK is received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, shall decrypt the application/resource-lists+xml MIME body;

c) shall identify the originating MCS ID from <mcptt-access-token> element received in the application/vnd.3gpp.mcptt-info+xml MIME body and shall use the originating MCS ID as an authenticated identity when performing the authorization;

d) if the authenticated identity is not authorized to subscribe to notification of changes of any resource in the application/resource-lists+xml MIME body, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps;

e) act as a notifier according to IETF RFC 5875 [13]. Additionally, if an XCAP URI in the "uri" attribute of the <entry> element of the application/resource-lists+xml MIME body of the initial SIP SUBSCRIBE request identifies:

1) a group document addressed by a group ID as described in subclause 7.2.10.2 where the group ID is an MCS group ID owned by an MCS provider other than the MCS provider of the GMS; or

2) a element of an MCS GKTP document as described in subclause 7.7.10 where the group ID is an MCS group ID owned by an MCS provider other than the MCS provider of GMS;

shall perform the procedure in subclause 6.3.13.3.2.4 for each such MCS group ID and shall interwork information of received SIP NOTIFY requests in subclause 6.3.13.3.2.4 in SIP NOTIFY requests associated with a subscription created as result of the received initial SIP SUBSCRIBE request.

Upon sending a SIP NOTIFY request associated with a subscription created as result of the received initial SIP SUBSCRIBE request, if the CSK is received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, the GMS shall perform the confidentiality protection procedures and integrity protection procedures defined in 3GPP TS 24.379 [5] for MCS server.

Upon reception of a SIP re-SUBSCRIBE request:

a) with the Event header field set to xcap-diff; and

b) with an application/resource-lists+xml MIME body;

the GMS:

a) if an <EncryptedData> XML tag is included in the application/resource-lists+xml MIME body of the received SIP re-SUBSCRIBE request and the CSK was received in an application/mikey MIME body of the initial SIP SUBSCRIBE request, shall decrypt the application/resource-lists+xml MIME body; and

b) act as a notifier according to IETF RFC 5875 [13]. Additionally, if an XCAP URI in the "uri" attribute of the <entry> element of the application/resource-lists+xml MIME body of the SIP re-SUBSCRIBE request identifies:

1) a group document addressed by a group ID as described in subclause 7.2.10.2 where the group ID is an MCS group ID owned by an MCS provider other than the MCS provider of the GMS; or

2) a element of an MCS GKTP document as described in subclause 7.7.10 where the group ID is an MCS group ID owned by an MCS provider other than the MCS provider of GMS;

and for which there is no related subscription established according to the subclause 6.3.13.3.2.4, shall perform the procedure in subclause 6.3.13.3.2.4 for each such MCS group ID and shall interwork information of received SIP NOTIFY requests in subclause 6.3.13.3.2.4 in SIP NOTIFY requests associated with a subscription created as result of the received initial SIP SUBSCRIBE request."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.13.3.2.3]

"Upon reception of an initial SIP SUBSCRIBE request:

a) with the Event header field set to xcap-diff;

b) with the Request-URI set to own public service identity for performing subscription proxy function of the GMS;

c) with a P-Asserted-Identity header field containing an identity listed in the authorized MCS server list specified in subclause 6.2.5.1;

d) with an application/resource-lists+xml MIME body; and

e) with the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24 229 [12]), in a P-Asserted-Service header field according to IETF RFC 6050 [14];

the GMS shall act as a notifier according to IETF RFC 5875 [13].

Upon reception of a SIP re-SUBSCRIBE request:

a) with the Event header field set to xcap-diff; and

b) with an application/resource-lists+xml MIME body;

the GMS:

a) shall use URI of the P-Asserted-Identity header field as an authenticated identity when performing the authorization;

b) if the authenticated identity is not authorized to subscribe to notification of changes of any document or element in the application/resource-lists+xml MIME body, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps; and

c) shall act as a notifier according to IETF RFC 5875 [13]."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.13.3.2.4]

"In order to subscribe to notification of changes of:

a) an MCS group document of an MCS group;

b) a <GKTPs> element of an MCS GKTP document for an MCS group; or

c) both;

such that the MCS group is identified by a MCS group ID owned by an MCS provider other than the MCS provider of the GMS, the GMS shall send an initial SIP SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [12] and IETF RFC 5875 [13]. In the initial SIP SUBSCRIBE request, the GMS:

a) shall include an application/resource-lists+xml MIME body. In the application/resource-lists+xml MIME body, the GMS shall include one <entry> element for each document or element to be subscribed to, such that the "uri" attribute of the <entry> element contains XCAP URI:

1) contains a relative path reference:

A) with the base URI being derived from the group ID routing database specified in subclause 6.2.5.2 and the MCS group ID; and

B) identifying a group document addressed by a group ID as described in subclause 7.2.10.2 where the group ID is set to the MCS group ID; or

2) contains a relative path reference

A) with the base URI being equal to the XCAP root URI of the MCS provider of the GMS;

B) with the document selector identifying the MCS GKTP document as described in subclause 7.7.10 where the group ID is set to the MCS group ID; and

C) with the node selector identifying a <GKTPs> element of the MCS GKTP document;

b) shall set the Request-URI to the public service identity for accessing documents of the other MCS provider derived from the group ID routing database specified in subclause 6.2.5.2; and

c) shall include a P-Asserted-Identity header field containing the own public service identity for accessing documents;

d) shall include an application/vnd.3gpp.mcptt-info+xml MIME body. In the application/vnd.3gpp.mcptt-info+xml MIME body, the GMS shall include the <mcptt-calling-user-id> element set to the originating MCS ID;

e) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [12]), in a P-Asserted-Service header field according to IETF RFC 6050 [14]; and

f) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field.

Upon receiving a SIP NOTIFY request associated with a subscription created as result of the sent initial SIP SUBSCRIBE request, the GMS shall handle the SIP NOTIFY request according to IETF RFC 5875 [13] and shall interwork the received information in subclause 6.3.13.3.2.2."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.13.3.3]

"Upon reception of an initial SIP SUBSCRIBE request:

a) with the Event header field set to xcap-diff;

b) with the Request-URI set to own public service identity for accessing documents;

c) with a P-Asserted-Identity header field containing an identity listed in the authorized GMS list as specified in subclause 6.2.5.1;

d) with an application/resource-lists+xml MIME body;

e) with an application/vnd.3gpp.mcptt-info+xml MIME body; and

f) with the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24 229 [12]), in a P-Asserted-Service header field according to IETF RFC 6050 [14];

the GMS:

a) if the <mcptt-calling-user-id> element is included in the application/vnd.3gpp.mcptt-info+xml MIME body:

1) shall use the <mcptt-calling-user-id> element value as an authenticated identity when performing the authorization; and

2) if the authenticated identity is not authorized to subscribe to notification of changes of any document or element in the application/resource-lists+xml MIME body, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps;

b) if the <mcptt-calling-group-id> element is included in the application/vnd.3gpp.mcptt-info+xml MIME body:

1) when performing the authorization, shall use URI of the P-Asserted-Identity header field as an authenticated identity and shall determine that the subscription is on behalf of the identity in the <mcptt-calling-group-id> element;

2) if the authenticated identity is not authorized to subscribe to notification of changes of any document or element in the application/resource-lists+xml MIME body, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps; and

3) *if the MCS group ID indicated in the <mcptt-calling-group-id> element is a constituent MCS group of a temporary MCS group and the documents or elements indicated in the application/resource-lists+xml MIME body are not associated with the temporary MCS group ID, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps; and*

c) shall act as a notifier according to IETF RFC 5875 [13]."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.2.5.1]

*"The GMS shall handle the HTTP request as specified for the HTTP server in annex A of 3GPP TS 24.482 [*5*].*

*The GMS server shall send the HTTP request as specified for the HTTP client in the network entity in annex A of 3GPP TS 24.482 [*5*].*

The GMS shall be configured with own public service identity for accessing documents.

The GMS shall be configured with an authorized GMS list, containing:

a) own public service identity for accessing documents; and

b) public service identities for accessing documents of GMSs of MCS providers which are partners of the MCS provider of the GMS.

The GMS shall be configured with an authorized MCS server list, containing public service identities of MCS servers of the MCS provider of the GMS.

The GMS shall handle SIP requests and SIP responses as specified in 3GPP TS 24.229 [12]."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.2.5.3]

"If the GMS receives an HTTP request with Request-URI identifying a group document addressed by a group ID as described in subclause 7.2.10.2 and the group ID in the Request-URI identifies a group of another MCS provider, then GMS:

a) shall derive XCAP root URI of the other MCS provider using the group ID routing database as specified in subclause 6.2.5.2 and the group ID in the Request-URI;

b) shall replace the XCAP root URI of the Request URI with the derived XCAP root URI of the other MCS provider;

*c) if the X-3GPP-Asserted-Identity header field is not present in the received HTTP request, shall insert an X-3GPP-Asserted-Identity header field with the identity of the sender of the HTTP request determined as specified in 3GPP TS 24.482 [*5*];*

d) if the Authorization header field is present in the received HTTP request, shall remove the Authorization header field from the HTTP request; and

e) shall forward the HTTP request."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.3.4.3]

"The GMOP document requesting group regroup creation is a GMOP document according to subclause 7.3.4.1, where the <request> element includes a <group-regroup-creation> element specified in subclause 7.3.3.

The <group-regroup-creation> element:

a) shall include an <group> element specified in subclause 7.2.4; and

b) may include an <anyExt> element specified in subclause 7.3.3."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.3.4.4]

"The GMOP document requesting group regroup check is a GMOP document according to subclause 7.3.4.1, where the <request> element includes a <group-regroup-check> element specified in subclause 7.3.3.

The <group-regroup-check> element:

a) shall include an <on-network-regrouped> element specified in subclause 7.2.4; and

b) may include an <anyExt> element specified in subclause 7.3.3."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.3.4.5]

"The GMOP document requesting group regroup notification is a GMOP document according to subclause 7.3.4.1, where the <request> element includes a <group-regroup-notification> element specified in subclause 7.3.3.

The <group-regroup-notification> element:

a) shall include an <on-network-regrouped> element specified in subclause 7.2.4; and

c) may include an <anyExt> element specified in subclause 7.3.3."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.3.4.6]

"The GMOP document with group regroup creation response is a GMOP document according to subclause 7.3.4.1, where the <response> element includes a <group-regroup-creation-response> element specified in subclause 7.3.3.

The <group-regroup-creation-response> element:

a) shall include an <temporary-group-document-ETag> element specified in subclause 7.3.3 containing the ETag of the group document of the temporary MCS group; and

b) may include an <anyExt> element specified in subclause 7.3.3."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.7.8]

"…

Value of a <GKTP> element of an <on-network-regrouped-GKTPs> element of a <GMK-GKTPs> element of a <GKTPs> element of the <list-service> element of the MCS GKTP document contains a group key transport payload carrying GMK as described in subclause 7.4, indicating the GMK for protection of:

a) media; and

b) floor control signalling when the UE operates off the network;

to be used when the MCPTT group is a constituent MCS group of the temporary MCS group with the MCS Group ID indicated in the "temporary-MCPTT-group-ID" attribute of the <on-network-regrouped-GKTPs> element, encrypted to the MCPTT ID of the MCPTT user accessing the MCS GKTP document, according to 3GPP TS 33.180 [29], signed using the identity of the GMS.

Value of a <GKTP> element of an <on-network-regrouped-GKTPs> element of a <MKFC-GKTPs> element of a <GKTPs> element of the <list-service> element of the MCS GKTP document contains a group key transport payload carrying MKFC as described in subclause 7.4, indicating the MKFC for protection of multicast floor control signalling when the UE operates on the network, to be used when the MCPTT group is a constituent MCS group of the temporary MCS group with the MCS Group ID indicated in the "temporary-MCPTT-group-ID" attribute of the <on-network-regrouped-GKTPs> element, encrypted to:

- *MCPTT ID of the MCPTT user accessing the MCS GKTP document; or*
- *the identity of MCPTT server accessing the MCS GKTP document;*

according to 3GPP TS 33.180 [29], signed using the identity of the GMS."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.4.2]

"The group key transport payload is an I_MESSAGE as specified in IETF RFC 3830 [16] containing a GMK or an MKFC, with additional fields as specified in IETF RFC 6509 [18], composed with the following clarification:

a) the common header payload specified in IETF RFC 3830 [16] is included and the CSB_ID field of the common header payload:

i) for transport of the GMK, contains GUK-ID as specified in 3GPP TS 33.180 [29]; and

ii) for transport of the MKFC, contains MKFC-ID as specified in 3GPP TS 33.180 [29];

b) the timestamp payload specified in IETF RFC 3830 [16] is included and the TS type field of the timestamp payload is set to 'NTP-UTC';

c) the RAND payload specified in IETF RFC 3830 [16] is included;

d) if MCS identifiers are not protected, the IDRi payload specified in IETF RFC 6509 [18] is included and:

1) the ID type field of the IDRi payload is set to the 'URI';

2) if the originator of the I_MESSAGE is the GMS, the ID data field of the IDRi payload is set to the GMS's URI, consisting of the HTTP URI identifying the directory of the application unique ID as specified in subclause 7.2.3 under the XCAP root URI; and

3) if the originator of the I_MESSAGE is an MCS server, the ID data field of the IDRi payload is set to the MCS server's URI;

e) if MCS identifiers are protected, the ID payload with role indicator specified in IETF RFC 6043 [17] is included and:

1) the ID role field of the ID payload is set to the 'IDRuidi' as specified in subclause 7.5.2;

2) the ID type field of the ID payload is set to the 'Byte string';

3) if the originator of the I_MESSAGE is the GMS, the ID data field of the ID payload is set to the UID generated from the GMS's URI as specified in 3GPP TS 33.180 [29], consisting of the HTTP URI identifying the directory of the application unique ID as specified in subclause 7.2.3 under the XCAP root URI; and

4) if the originator of the I_MESSAGE is an MCS server, the ID data field of the ID payload is set to the MCS server's URI;

f) if MCS identifiers are not protected, the IDRr payload specified in IETF RFC 6509 [18] is included and:

1) the ID type field of the IDRr payload is set to the 'URI';

2) if an MCS user is targeted, the ID data field of the IDRr payload is set to the MCS ID of the targeted MCS user;

3) if an constituent MCS group is targeted, the ID data field of the IDRr payload is set to the MCS Group ID of the targeted constituent MCS group; and

4) if an MCS server is targeted, the ID data field of the IDRr payload is set to the MCS server's URI;

g) if MCS identifiers are protected, the ID payload with role indicator specified in IETF RFC 6043 [17] is included and:

1) the ID role field of the ID payload is set to the 'IDRuidr' as specified in subclause 7.5.2;

2) the ID type field of the ID payload is set to the 'Byte string';

3) if an MCS user is targeted, the ID data field of the ID payload is set to the UID generated from the MCS ID of the targeted MCS user;

4) if an constituent MCS group is targeted, the ID data field of the ID payload is set to the UID generated from the MCS Group ID of the targeted constituent MCS group; and

5) *if an MCS server is targeted, the ID data field of the ID payload is set to the MCS's server URI;*

h) *the IDRkmsi payload specified in IETF RFC 6509 [18] is included and:*

1) *the ID type field of the IDRkmsi payload is set to the 'URI';*

2) *if the originator of the I_MESSAGE is the GMS, the ID data field of the IDRkmsi payload is set to the URI of the MCS KMS used by the group management server; and*

3) *if the originator of the I_MESSAGE is an MCS server, the ID data field of the IDRkmsi payload is set to the URI of the MCS KMS used by MCS server;*

i) *the IDRkmsr payload specified in IETF RFC 6509 [18] is included and:*

1) *the ID type field of the IDRkmsr payload is set to the 'URI';*

2) *if an MCS user is targeted, the ID data field of the IDRkmsr payload is set to the URI of the MCS KMS used by targeted MCS user;*

NOTE: The KMS URI is stored in the group configuration data.

Editor's Note:The XML element(s) (or XML attribute(s)) in the MCPTT Group for carrying the KMS URI need to be defined.

3) *if an constituent MCS group is targeted, the ID data field of the IDRkmsr payload is set to the URI of the MCS KMS used by targeted constituent MCS group; and*

4) *if an MCS server is targeted, the ID data field of the IDRkmsr payload is set to the URI of the MCS KMS used by the MCS server;*

j) *the SAKKE payload specified in IETF RFC 6509 [18] is included and:*

1) *the SAKKE params field of the SAKKE payload is set to 'Parameter Set 1';*

2) *the ID scheme field of the SAKKE payload is set to 'MCPTT-ID-scheme' or '3GPP MCX hashed UID' as specified in subclause 7.5.3; and*

3) *the SAKKE data field of the SAKKE payload contains the GMK or MKFC specified in 3GPP TS 33.180 [29];*

k) *the SIGN payload specified in IETF RFC 6509 [18] is included and the S type field of the SIGN payload is set to the 'ECCSI';*

l) *the security policy payload specified in IETF RFC 3830 [16] can be included;*

m) *the general extension payload specified in IETF RFC 3830 [16] with the type field set to 'GMK-or-MKFC-associated-parameters' or to '3GPP key parameters' as specified in subclause 7.5.4 is included and the data field of the general extension payload contains the associated parameters of GMK or MKFC as specified in 3GPP TS 33.180 [29] figure E.6.1-1; and*

n) *the General extension payload specified in IETF RFC 3830 [16] with the type field set to the 'SAKKE-to-self' as specified in subclause 7.5.4 can be included and the data field of the general extension payload contains a SAKKE payload specified in IETF RFC 6509 [18]. In the SAKKE payload included in the general extension payload:*

1) *the SAKKE params field of the SAKKE payload is set to 'Parameter Set 1';*

2) *the ID scheme field of the SAKKE payload is set to 'MCPTT-SAKKE-to-self-ID-scheme' or '3GPP MCX hashed UID' as specified in subclause 7.5.3; and*

3) *the SAKKE data field of the SAKKE payload contains the GMK or MKFC specified in 3GPP TS 33.180 [29]."*

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause A.2.3]

"…

The temporary MCPTT group is identified by the temporary MCPTT Group ID sip:groupT@MCPTTSP1.example.com and is being created in GMS-1 under the groupdocumentT.xml document name.

The temporary MCPTT group is a pre-arranged MCPTT group.

The GMK is required to confidentiality and integrity protect media for calls on the temporary MCPTT group. Both confidentiality protection and integrity protection of floor control signalling are required for calls on the temporary MCPTT group. Multicast bearers are used for floor controlling signalling for the temporary MCPTT group.

One MCPTT group to be combined is identified by the MCPTT Group ID sip:groupA@MCPTTSP1.example.com and is owned by GMS-1.

The other MCPTT group to be combined is identified by the MCPTT Group ID sip:groupB@MCPTTSP2.example.com and is owned by GMS-2.

The GMS-1 is configured with the group ID routing database stating:

- *the XCAP Root URI of sip:groupB@MCPTTSP2.example.com is http://xcap.MCPTTSP2.example.com/path2; and*
- *the public service identity for accessing documents of the MCPTT provider of sip:groupB@MCPTTSP2.example.com is sip:gms.MCPTTSP2.example.com.*

The GMS-2 is configured with the group ID routing database stating:

- *the XCAP Root URI of sip:groupT@MCPTTSP1.example.com is http://xcap.MCPTTSP1.example.com/path1; and*
- *the public service identity for accessing documents of the MCPTT provider of sip:groupT@MCPTTSP1.example.com is sip:gms.MCPTTSP1.example.com.*

…

The details of the flows are as follows:

1) GMC-1 sends an HTTP POST request with the GMOP document requesting group regroup creation shown in table A.2.3-1 to the GMS-1.

...

2) GMS-1 authorizes the request as specified in subclause 6.3.14.3.1 and sends HTTP POST request with the GMOP document requesting group regroup check.

3) GMS-1 sends HTTP POST request with the GMOP document requesting group regroup check to GMS-2 as shown in table A.2.3-3.

…

4) GMS-1 authorizes the request as specified in subclause 6.3.14.3.2 and responds with a HTTP 200 (OK) response to the HTTP POST request.

5) GMS-2 authorizes the request as specified in subclause 6.3.14.3.2 and responds with a HTTP 200 (OK) response to the HTTP POST request.

6) GMS-1 sends HTTP POST request with the GMOP document requesting group regroup notification.

7) GMS-1 sends HTTP POST request with the GMOP document requesting group regroup notification to GMS-2 as shown in table A.2.3-7.

…

8) GMS-1 authorizes the request as specified in subclause 6.3.14.3.2, place the <on-network-regrouped> element of the GMOP document requesting group regroup notification of the HTTP POST request at the location identified by the Request-URI of the received HTTP POST request and responds with a HTTP 200 (OK) response to the HTTP POST request.

9) GMS-2 authorizes the request as specified in subclause 6.3.14.3.2, place the <on-network-regrouped> element of the GMOP document requesting group regroup notification of the HTTP POST request at the location identified by the Request-URI of the received HTTP POST request and responds with a HTTP 200 (OK) response to the HTTP POST request.

10) GMS-1 creates the temporary MCPTT group document so that it is accessible using the XCAP URIs

…

GMS-1 also creates the MCPTT GKTP document …

GMS-1 accepts the HTTP POST request with HTTP 200 (OK) response. The HTTP 200 (OK) response includes the GMOP document with group regroup creation response.

10) GMS-1 subscribes to changes of the <GKTPs> element of MCPTT-GKTP document of the temporary MCPTT group, is notified about the current state and updates the MCPTT-GKTP document for the MCPTT group with sip:groupA@MCPTTSP1.example.com MCPTT Group ID.

11) GMS-2 subscribes to changes of the <GKTPs> element of MCPTT-GKTP document of the temporary MCPTT group as shown in table A.2.3-11.

…

12) GMS-1 authorizes the SIP SUBSCRIBE request and accepts it with SIP 200 (OK) response.

13) GMS-1 sends a SIP NOTIFY request with the content of the <GKTPs> element of MCPTT-GKTP document of the temporary MCPTT group as shown in table A.2.3-13.

…

14) GMS-2 accepts the SIP NOTIFY request with SIP 200 (OK) response and updates the MCPTT-GKTP document for the MCPTT group with sip:groupB@MCPTTSP2.example.coms MCPTT Group ID."

5.1.3.3 Test description

5.1.3.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the Group Management function.
- SS-UE1 (MCPTT client)

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the Group Management function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.6, and to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in subclause 5.1.
- Group A and Group B are owned by the SS (MCPTT Server).
- Group T will be owned by the IUT (MCPTT Server).
- The SS-UE1 (MCPTT client) is member of group A.

5.1.3.3.2 Test procedure sequence

Table 5.1.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT Client) sends a HTTP POST to the IUT (MCPTT Server) to request for creation of a temporary group.	<--	HTTP POST		-
	- EXCEPTION: The IUT (MCPTT Server) sends one GMOP group regroup check requests for each of the groups being part of the temporary group. The following sequence of steps identified with "lower case letters" can happen in parallel.				
2a	Check: Does the IUT (MCPTT Server) send a HTTP POST to the SS (MCPTT Server) to request a GMOP group regroup check for group A?	-->	HTTP POST	1	P
3a	The SS (MCPTT Server) responds with a HTTP 200 (OK) response message	<--	HTTP 200 (OK)		-
2b	Check: Does the IUT (MCPTT Server) send a HTTP POST to the SS (MCPTT Server) to request a GMOP group regroup check for group B?	-->	HTTP POST	1	P
3b	The SS (MCPTT Server) responds with a HTTP 200 (OK) response message	<--	HTTP 200 (OK)		-
	- EXCEPTION: The IUT (MCPTT Server) sends one GMOP group regroup notification for each of the groups being part of the temporary group. The following sequence of steps identified with "lower case letters" can happen in parallel.				
4a	Check: Does the IUT (MCPTT Server) send a HTTP POST to the SS (MCPTT Server) for notifying a GMOP group regroup notification to group A?	-->	HTTP POST	1	P
5a	The SS (MCPTT Server) responds with a HTTP 200 (OK) response message	<--	HTTP 200 (OK)		-
4b	Check: Does the IUT (MCPTT Server) send a HTTP POST to the SS (MCPTT Server) for notifying a GMOP group regroup notification to group B?	-->	HTTP POST	1	P
5b	The SS (MCPTT Server) responds with a HTTP 200 (OK) response message	<--	HTTP 200 (OK)		-
6	Check: Does the IUT (MCPTT Server) send a HTTP 200 (OK) containing the GMOP group-regroup-creation-response?	-->	HTTP 200 (OK)	1	P
7a	The SS (MCPTT Server) sends a SIP SUBSCRIBE message request to get the group key transport payloads (GKTP) document including the group keys for the temporary group, related to group A.	<--	SIP SUBSCRIBE		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
8a	Check: Does the IUT (MCPTT Server) respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	2	P
9a	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message to the SS (MCPTT Server) containing the group key transport payloads (GKTP) document including the group keys?	-->	SIP NOTIFY		P
10a	The SS (MCTT Server) sends a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
7b	The SS (MCPTT Server) sends a SIP SUBSCRIBE message request to get the group key transport payloads (GKTP) document including the group keys for the temporary group, related to group B.	<--	SIP SUBSCRIBE		-
8b	Check: Does the IUT (MCPTT Server) respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	2	P
9b	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message to the SS (MCPTT Server) containing the group key transport payloads (GKTP) document including the group keys?	-->	SIP NOTIFY	2	P
10b	The SS (MCTT Server) sends a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
11	The SS-UE1 (MCPTT Client) sends a HTTP DELETE to request for tear down of the temporary group.	<--	HTTP DELETE		-
	- EXCEPTION: The IUT (MCPTT Server) sends one GMOP group regroup notification for each of the groups being part of the temporary group. The following sequence of steps identified with "lower case letters" can happen in parallel.				
12a	Check: Does the IUT (MCPTT Server) send a HTTP DELETE to request the removal of the parameters associated to the temporary group from the group configuration file of group A?	-->	HTTP DELETE	3	P
13a	The SS (MCPTT Server) sends a HTTP 200 (OK).	<--	HTTP 200 (OK)		
12b	Check: Does the IUT (MCPTT Server) send a HTTP DELETE to request the removal of the parameters associated to the temporary group from the group configuration file of group B?	-->	HTTP DELETE	3	P
13b	The SS (MCPTT Server) sends a HTTP 200 (OK).	<--	HTTP 200 (OK)		
14	Check: Does the IUT (MCPTT Server) send a HTTP 200 (OK) to the SS-UE1 (MCPTT Client)?	-->	HTTP 200 (OK)	3	P
15a	The SS (MCPTT Server) sends a SIP SUBSCRIBE message request to de-subscribe from receiving notifications of group key transport payloads (GKTP) for the temporary group, related to group A.	<--	SIP SUBSCRIBE		-
16a	Check: Does the IUT (MCPTT Server) respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	4	P
17a	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message to the SS (MCPTT Server) indicating the termination of the subscription?	-->	SIP NOTIFY	4	P
18a	The SS (MCTT Server) sends a SIP 200 (OK) message.	<--	SIP 200 (OK)		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
15b	The SS (MCPTT Server) sends a SIP SUBSCRIBE message request to de-subscribe from receiving notifications of group key transport payloads (GKTP) for the temporary group, related to group B.	<--	SIP SUBSCRIBE		-
16b	Check: Does the IUT (MCPTT Server) respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	4	P
17b	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message to the SS (MCPTT Server) indicating the termination of the subscription?	-->	SIP NOTIFY	4	P
18b	The SS (MCTT Server) sends a SIP 200 (OK) message.	<--	SIP 200 (OK)		-

5.2.3.3.3 Specific message contents

Table 5.1.3.3.3-1: HTTP POST from the SS-UE1 (MCPTT Client) (Step 1, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, conditions MCPTT and TEMPGROUP

Table 5.1.3.3.3-2: HTTP POST from the IUT (MCPTT Server) (Step 2a, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, conditions TEMPGROUP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI				
uri	tsc_MCX_GMSXCAPRootURI_B & "/" & "org.openmobilealliance.groups/global/byGroupID/" & px_MCPTT_Group_A_ID & "/~~/group/list-service/mcpttgi:on-network-regrouped%5b@temporary-MCPTT-group-ID=%22" & px_MCPTT_Group_T_ID & "%22%5d?xmlns(mcpttgi=urn:3gpp:ns:mcpttGroupInfo:1.0)"	tsc_MCX_GMSXCAPRootURI_B contains the XCAP Root URI of the GMS owning an MCS group to be combined.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.14.3.1	
Authorization	Not present			
X-3GPP-Asserted-Identity	tsc_MCPTT_PublicServiceId_A	The URI of the GMS owning the temporary group.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.14.3.1	
Message-body				
GMOP document requesting group regroup check	As described in Table 5.1.3.3.3-2A			

Table 5.1.3.3.3-2A: GMOP group regroup check in HTTP POST (Table 5.1.3.3.3-2)

Derivation Path: ETSI TS 124 481 [6] (3GPP TS 24.481) clause 5.1.4.4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
gmop:document				
gmop:request				
gmop:group-regroup-check				
mcpttgi:on-network-regrouped				
temporary-MCPTT-group-ID	px_MCPTT_Group_T_ID	MCPTT temporary group identity		
temporary-MCPTT-group-requestor	px_MCPTT_ID_User_A	The URI of the MCPTT user requesting the creation of the temporary group.		
mcpttgi:constituent-MCPTT-group-IDs				
constituent-MCPTT-group-ID[1]	px_MCPTT_Group_A_ID	MCS group ID of a constituent MCS group of the temporary MCS group		
constituent-MCPTT-group-ID[2]	px_MCPTT_Group_B_ID	MCS group ID of a constituent MCS group of the temporary MCS group		

Table 5.1.3.3.3-3: HTTP POST from the IUT (MCPTT Server) (Step 2b, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, conditions TEMPGROUP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI				
uri	tsc_MCX_GMSXCAPRootURI_B & "/" & "org.openmobilealliance.groups/global/byGroupID/" & px_MCPTT_Group_B_ID & "/~~/group/list-service/mcpttgi:on-network-regrouped%5b@temporary-MCPTT-group-ID=%22" & px_MCPTT_Group_T_ID & "%22%5d?xmlns(mcpttgi=urn:3gpp:ns:mcpttGroupInfo:1.0)"	tsc_MCX_GMSXCAPRootURI_B contains the XCAP Root URI of the GMS owning an MCS group to be combined.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.14.3.1	
Authorization	Not present			
X-3GPP-Asserted-Identity	tsc_MCPTT_PublicServiceId_A	The URI of the GMS owning the temporary group.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.14.3.1	
Message-body				
GMOP document requesting group regroup check	As described in Table 5.1.3.3.3-2A			

Table 5.1.3.3.3-4: HTTP POST from the IUT (MCPTT Server) (Step 4a, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.3.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
GMOP document requesting group regroup notification	As described in Table 5.1.3.3.3-4A			

Table 5.1.3.3.3-4A: GMOP group regroup notification in HTTP POST (Table 5.1.3.3.3-4)

Derivation Path: ETSI TS 124 481 [6] (3GPP TS 24.481) clause 5.1.4.4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
gmop:document				
gmop:request				
gmop:group-regroup-notification				
mcpttgi:on-network-regrouped				
temporary-MCPTT-group-ID	px_MCPTT_Group_T_ID	MCPTT temporary group identity		
temporary-MCPTT-group-requestor	px_MCPTT_ID_User_A	The URI of the MCPTT user requesting the creation of the temporary group.		
mcpttgi:constituent-MCPTT-group-IDs				
constituent-MCPTT-group-ID[1]	px_MCPTT_Group_A_ID	MCS group ID of a constituent MCS group of the temporary MCS group		
constituent-MCPTT-group-ID[2]	px_MCPTT_Group_B_ID	MCS group ID of a constituent MCS group of the temporary MCS group		

Table 5.1.3.3.3-5: HTTP POST from the IUT (MCPTT Server) (Step 4b, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.3.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
GMOP document requesting group regroup notification	As described in Table 5.1.3.3.3-4A			

Table 5.1.3.3.3-6: HTTP 200 (OK) from the IUT (MCPTT Server) (Step 6, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition TEMPGROUP

Table 5.1.3.3.3-6A: HTTP 200 (OK) from the IUT (MCPTT Server) (Step 14, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1

Table 5.1.3.3.3-7: HTTP 200 (OK) from the SS (MCPTT Server) (Steps 3a, 3b, 5a, 5b, 13a and 13b, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1

Table 5.1.3.3.3-8: HTTP DELETE from the SS-UE1 (MCPTT Client) (Step 11, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.5-1, condition TEMPGROUP

Table 5.1.3.3.3-9: HTTP DELETE from the IUT (MCPTT Server) (Step 12a, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.5-1, conditions TEMPGROUP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI				
uri	tsc_MCX_GMSXCAPRootURI_B & "/" & "org.openmobilealliance.groups/global/byGroupID/" & px_MCPTT_Group_A_ID & "/~~/group/list-service/mcpttgi:on-network-regrouped%5b@temporary-MCPTT-group-ID=%22" & px_MCPTT_Group_T_ID & "%22%5d?xmlns(mcpttgi=urn:3gpp:ns:mcpttGroupInfo:1.0)"	tsc_MCX_GMSXCAPRootURI_B contains the XCAP Root URI of the GMS owning an MCS group to be combined.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.15.3.1	
Authorization	Not present			
X-3GPP-Asserted-Identity	tsc_MCPTT_PublicServiceId_A	The URI of the GMS owning the temporary group.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.15.3.1	

Table 5.1.3.3.3-10: HTTP DELETE from the IUT (MCPTT Server) (Step 12b, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.5-1, conditions TEMPGROUP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Request-URI				
uri	tsc_MCX_GMSXCAPRootURI_B & "/" & "org.openmobilealliance.groups/global/byGroupID/" & px_MCPTT_Group_B_ID & "/~~/group/list-service/mcpttgi:on-network-regrouped%5b@temporary-MCPTT-group-ID=%22" & px_MCPTT_Group_T_ID & "%22%5d?xmlns(mcpttgi=urn:3gpp:ns:mcpttGroupInfo:1.0)"	tsc_MCX_GMSXCAPRootURI_B contains the XCAP Root URI of the GMS owning an MCS group to be combined.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.15.3.1	
Authorization	Not present			
X-3GPP-Asserted-Identity	tsc_MCPTT_PublicServiceId_A	The URI of the GMS owning the temporary group.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.15.3.1	

Table 5.1.3.3.3-11: SIP SUBSCRIBE from the SS (MCPTT Server) (Step 7a, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	tsc_MCPTT_PublicServiceId_A	The URI of the GMS owning the temporary group.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the GMS owning the groups being combined into the temporary group.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the GMS owning the temporary group.		
Contact				
addr-spec	SIP URI			
user-info and host	IP address or FQDN			
port	protected server port of SS (MCPTT Server)			
Expires			IETF RFC 3261 [17] IETF RFC 3903 [22]	
value	3600			
Require	not present			
Proxy-Require	not present			
Security-Verify	not present			
Max-Forwards			IETF RFC 3261 [17]	
value	70	The recommended initial value is 70 in IETF RFC 3261 [17].		
P-Access-Network-Info	not present			
P-Asserted-Identity	tsc_MCPTT_PublicServiceId_B	The URI of the GMS owning the groups being combined into the temporary group.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.13.3.2	
Content-Type			IETF RFC 5621 [16]	
media-type	"multipart/mixed"			
Message-body				
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 5.1.3.3.3-11A			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME body part		**MCPTT Info**		
MIME-part-body	Resource-lists as described in Table 5.1.3.3.3-11B			

Table 5.1.3.3.3-11A: Resource-Lists in SIP SUBSCRIBE (Table 5.1.3.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1A-1, condition GROUPKEY				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list[1]				
entry[1]			ETSI TS 124 481 [6] (3GPP TS 24.481)	GROUPKEY
uri attribute	"org.3gpp.MCPTT-GKTP/global/byGroupID/" & px_MCPTT_Group_T_ID & "/" & "~~" & "/group/list-service/mgktp:GKTPs?xmlns(mgktp=urn:3gpp:ns:mcpttGKTP:1.0)"			
display-name	Not present			

Table 5.1.3.3.3-11B: MCPTT-Info in SIP SUBSCRIBE (Table 5.1.3.3.3-11)

Derivation Path: ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_A_ID			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.1.3.3.3-12: SIP NOTIFY from the IUT (MCPTT Server) (Steps 9a and 9b, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	The recommended initial value is 70 in IETF RFC 3261 [17].		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 3261 [17] IETF RFC 3842 [14]	
media-type	"application/xcap-diff+xml"			
Message-body				
xcap-diff document	xcap-diff document as described in Table 5.1.3.3.3-12A			

Table 5.1.3.3.3-12A: Xcap-Diff Document (Table 5.1.3.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.12-2, condition GROUPKEY				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
xcap-diff				
xcap-root	tsc_MCX_GMSXCAPRootURI	The URI of the GMS owning the temporary group.		
element[1]				
sel attribute	"org.3gpp.MCPTT-GKTP/global/byGroupID/" & px_MCPTT_Group_T_ID & "/~~" & "/group/list-service/mgktp:GKTPs?xmlns(mgktp=urn:3gpp:ns:mcpttGKTP:1.0) "			
GKTPs	group key transport payloads (GKTP) document as described in Table 5.1.3.3.3-12B			

Table 5.1.3.3.3-12B: group key transport payloads (GKTP) document (Table 5.1.3.3.3-12A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.14-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
GKTPs				
GMK-GKTPs				
GKTP[1]	MIKEY message as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-9			

Table 5.1.3.3.3-13: SIP 200 (OK) to SIP SUBSCRIBE (Steps 8a, 8b, 16a and 16b, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition SUBSCRIBE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			

Table 5.1.3.3.3-14: SIP SUBSCRIBE from the SS (MCPTT Server) (Step 15a, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.3.3.3-11				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	same URI as the IUT (MCPTT Server) has sent earlier in the Contact header of a message within the same dialog	Contact URI of the recipient of the previous 200 OK		
Expires				
value	"0"			

Table 5.1.3.3.3-15: SIP NOTIFY from the IUT (MCPTT Server) (Steps 17a and 17b, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.3.3.3-12				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subscription-State			IETF RFC 6665 [24]	
substate-value	"terminated"			
expires	not present			

Table 5.1.3.3.3-16: SIP 200 (OK) to SIP NOTIFY (Steps 10a, 10b, 18a, 18b, Table 5.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			

Table 5.1.3.3.3-17: SIP SUBSCRIBE from the SS (MCPTT Server) (Step 7b, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.3.3.3-11				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 5.1.3.3.3-11A			
MIME body part		**MCPTT Info**		
MIME-part-body	Resource-lists as described in Table 5.1.3.3.3-17A			

Table 5.1.3.3.3-17A: MCPTT-Info in SIP SUBSCRIBE (Table 5.1.3.3.3-17)

Derivation Path: ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_B_ID			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A				

Table 5.1.3.3.3-18: SIP SUBSCRIBE from the SS (MCPTT Server) (Step 15b, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.3.3.3-17				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	same URI as the IUT (MCPTT Server) has sent earlier in the Contact header of a message within the same dialog	Contact URI of the recipient of the previous 200 OK		
Expires				
value	"0"			

5.1.4 MCPTT Server - MCPTT Server / Group and GMK Subscription / Group Regroup creation / Group Regroup Teardown / GMS owning group to be combined

5.1.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { SS (MCPTT Server) sends HTTP POST message to IUT (MCPTT Server to request a GMOP group
regroup check for both groups }
    then { IUT (MCPTT Server) sends a HTTP 200 OK message to SS (MCPTT Server) }
            }
```

(2)

```
with { SS (MCPTT Server) having requested a GMOP group regroup check }
ensure that {
  when { the IUT (MCPTT Server) requests the group key transport payloads (GKTP) document }
    then { IUT (MCPTT Server) sends a SIP SUBSCRIBE message to the SS (MCPTT Server) and replies
with SIP 200 OK message to the received notification }
            }
```

(3)

```
with { IUT (MCPTT Server) having establish a temporary group }
ensure that {
  when { the SS (MCPTT Server) initiates tear down of the temporary group by sending a HTTP DELETE
request }
    then { IUT (MCPTT Server) responds to the SS (MCPTT Server) with HTTP 200 (OK) message }
            }
```

(4)

```
with { IUT (MCPTT Server) having establish a temporary group }
ensure that {
  when { the IUT (MCPTT Server) requests to de-subscribe from receiving notifications }
    then { IUT (MCPTT Server) sends a SIP SUBSCRIBE message to the SS (MCPTT Server) and replies
with SIP 200 OK message to the received notification }
            }
```

5.1.4.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 481 [6] (3GPP TS 24.481) clauses 6.3.14.3.2, 6.3.15.3.2, 6.3.13.3.3, 7.3.4.4, 7.3.4.5, 7.7.8, 7.4.2 and A.2.3. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.14.3.2]

"Upon reception of an HTTP POST request:

a) with the Request-URI set to an XCAP URI identifying an existing or a non-existing <on-network-regrouped> element of an existing group document of an MCS group; and

b) with an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document requesting group regroup check specified in subclause 7.3.4.4;

the GMS:

a) if:

1) the Request-URI identifies an existing <on-network-regrouped> element of an existing group document of an MCS group;

2) the Request-URI identifies a non-existing <on-network-regrouped> element of an existing group document defining a temporary MCS group;

3) identity indicated in the X-3GPP-Asserted-Identity header field is not listed in the authorized GMS list specified in subclause 6.2.5.1; or

4) the MIME body of the HTTP POST request is not acceptable;

then shall respond with HTTP 403 (Forbidden) response to the HTTP POST request and shall not continue with rest of the steps; and

b) shall respond with HTTP 200 (OK) response to the HTTP POST request.

Upon reception of an HTTP POST request:

a) with the Request-URI set to an XCAP URI identifying an existing or a non-existing <on-network-regrouped> element of an existing group document of an MCS group; and

b) with an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document requesting group regroup notification specified in subclause 7.3.4.5;

the GMS:

a) if:

1) the Request-URI identifies an existing <on-network-regrouped> element of an existing group document of an MCS group;

2) the Request-URI identifies a non-existing <on-network-regrouped> element of an existing group document defining a temporary MCS group;

3) identity indicated in the X-3GPP-Asserted-Identity header field is not listed in the authorized GMS list specified in subclause 6.2.5.1; or

4) the MIME body of the HTTP POST request is not acceptable;

then shall respond with HTTP 403 (Forbidden) response to the HTTP POST request and shall not continue with rest of the steps;

b) shall place the <on-network-regrouped> element of the GMOP document requesting group regroup notification of the HTTP POST request at the location identified by the Request-URI of the received HTTP POST request;

c) shall respond with HTTP 200 (OK) response to the HTTP POST request; and

NOTE: GMKis not included in the GMOP document requesting group regroup notification as GMK is provided only using SIP.

d) shall subscribe for changes of the MCS GKTP document of the temporary MCS Group ID indicated in the "temporary-MCPTT-group-ID" attribute of the <on-network-regrouped> element of the GMOP document requesting group regroup notification according to subclause 6.3.13.2.3."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.15.3.2]

"Upon reception of an HTTP DELETE request with Request-URI identifying a <on-network-regrouped> element of an MCS group document, the GMS:

a) if:

1) the Request-URI identifies an existing <on-network-regrouped> element of a non-existing group document; or

2) identity indicated in the X-3GPP-Asserted-Identity header field is not listed in the authorized GMS list specified in subclause 6.2.5.1;

then shall respond with HTTP 403 (Forbidden) response to the HTTP DELETE request and shall not continue with rest of the steps; and

b) shall act according to according to procedures specified in IETF RFC 4825 [22] "DELETE Handling"."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.13.3.3]

"Upon reception of an initial SIP SUBSCRIBE request:

a) with the Event header field set to xcap-diff;

b) with the Request-URI set to own public service identity for accessing documents;

c) with a P-Asserted-Identity header field containing an identity listed in the authorized GMS list as specified in subclause 6.2.5.1;

d) with an application/resource-lists+xml MIME body;

e) with an application/vnd.3gpp.mcptt-info+xml MIME body; and

f) with the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24 229 [12]), in a P-Asserted-Service header field according to IETF RFC 6050 [14];

the GMS:

a) if the <mcptt-calling-user-id> element is included in the application/vnd.3gpp.mcptt-info+xml MIME body:

1) shall use the <mcptt-calling-user-id> element value as an authenticated identity when performing the authorization; and

2) if the authenticated identity is not authorized to subscribe to notification of changes of any document or element in the application/resource-lists+xml MIME body, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps;

b) if the <mcptt-calling-group-id> element is included in the application/vnd.3gpp.mcptt-info+xml MIME body:

1) when performing the authorization, shall use URI of the P-Asserted-Identity header field as an authenticated identity and shall determine that the subscription is on behalf of the identity in the <mcptt-calling-group-id> element;

2) if the authenticated identity is not authorized to subscribe to notification of changes of any document or element in the application/resource-lists+xml MIME body, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps; and

3) if the MCS group ID indicated in the <mcptt-calling-group-id> element is a constituent MCS group of a temporary MCS group and the documents or elements indicated in the application/resource-lists+xml MIME body are not associated with the temporary MCS group ID, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps; and

c) shall act as a notifier according to IETF RFC 5875 [13]."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.2.5.1]

"The GMS shall handle the HTTP request as specified for the HTTP server in annex A of 3GPP TS 24.482 [5].

The GMS server shall send the HTTP request as specified for the HTTP client in the network entity in annex A of 3GPP TS 24.482 [5].

The GMS shall be configured with own public service identity for accessing documents.

The GMS shall be configured with an authorized GMS list, containing:

a) own public service identity for accessing documents; and

b) public service identities for accessing documents of GMSs of MCS providers which are partners of the MCS provider of the GMS.

The GMS shall be configured with an authorized MCS server list, containing public service identities of MCS servers of the MCS provider of the GMS.

The GMS shall handle SIP requests and SIP responses as specified in 3GPP TS 24.229 [12]."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.3.4.4]

"The GMOP document requesting group regroup check is a GMOP document according to subclause 7.3.4.1, where the <request> element includes a <group-regroup-check> element specified in subclause 7.3.3.

The <group-regroup-check> element:

a) shall include an <on-network-regrouped> element specified in subclause 7.2.4; and

b) may include an <anyExt> element specified in subclause 7.3.3."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.3.4.5]

"The GMOP document requesting group regroup notification is a GMOP document according to subclause 7.3.4.1, where the <request> element includes a <group-regroup-notification> element specified in subclause 7.3.3.

The <group-regroup-notification> element:

a) shall include an <on-network-regrouped> element specified in subclause 7.2.4; and

c) may include an <anyExt> element specified in subclause 7.3.3."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.7.8]

"…

Value of a <GKTP> element of an <on-network-regrouped-GKTPs> element of a <GMK-GKTPs> element of a <GKTPs> element of the <list-service> element of the MCS GKTP document contains a group key transport payload carrying GMK as described in subclause 7.4, indicating the GMK for protection of:

a) media; and

b) floor control signalling when the UE operates off the network;

to be used when the MCPTT group is a constituent MCS group of the temporary MCS group with the MCS Group ID indicated in the "temporary-MCPTT-group-ID" attribute of the <on-network-regrouped-GKTPs> element, encrypted to the MCPTT ID of the MCPTT user accessing the MCS GKTP document, according to 3GPP TS 33.180 [29], signed using the identity of the GMS.

Value of a <GKTP> element of an <on-network-regrouped-GKTPs> element of a <MKFC-GKTPs> element of a <GKTPs> element of the <list-service> element of the MCS GKTP document contains a group key transport payload carrying MKFC as described in subclause 7.4, indicating the MKFC for protection of multicast floor control signalling when the UE operates on the network, to be used when the MCPTT group is a constituent MCS group of the temporary MCS group with the MCS Group ID indicated in the "temporary-MCPTT-group-ID" attribute of the <on-network-regrouped-GKTPs> element, encrypted to:

- *MCPTT ID of the MCPTT user accessing the MCS GKTP document; or*
- *the identity of MCPTT server accessing the MCS GKTP document;*

according to 3GPP TS 33.180 [29], signed using the identity of the GMS."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.4.2]

"The group key transport payload is an I_MESSAGE as specified in IETF RFC 3830 [16] containing a GMK or an MKFC, with additional fields as specified in IETF RFC 6509 [18], composed with the following clarification:

a) the common header payload specified in IETF RFC 3830 [16] is included and the CSB_ID field of the common header payload:

i) for transport of the GMK, contains GUK-ID as specified in 3GPP TS 33.180 [29]; and

ii) for transport of the MKFC, contains MKFC-ID as specified in 3GPP TS 33.180 [29];

b) the timestamp payload specified in IETF RFC 3830 [16] is included and the TS type field of the timestamp payload is set to 'NTP-UTC';

c) the RAND payload specified in IETF RFC 3830 [16] is included;

d) if MCS identifiers are not protected, the IDRi payload specified in IETF RFC 6509 [18] is included and:

1) the ID type field of the IDRi payload is set to the 'URI';

2) if the originator of the I_MESSAGE is the GMS, the ID data field of the IDRi payload is set to the GMS's URI, consisting of the HTTP URI identifying the directory of the application unique ID as specified in subclause 7.2.3 under the XCAP root URI; and

3) if the originator of the I_MESSAGE is an MCS server, the ID data field of the IDRi payload is set to the MCS server's URI;

e) if MCS identifiers are protected, the ID payload with role indicator specified in IETF RFC 6043 [17] is included and:

1) the ID role field of the ID payload is set to the 'IDRuidi' as specified in subclause 7.5.2;

2) the ID type field of the ID payload is set to the 'Byte string';

3) if the originator of the I_MESSAGE is the GMS, the ID data field of the ID payload is set to the UID generated from the GMS's URI as specified in 3GPP TS 33.180 [29], consisting of the HTTP URI identifying the directory of the application unique ID as specified in subclause 7.2.3 under the XCAP root URI; and

4) if the originator of the I_MESSAGE is an MCS server, the ID data field of the ID payload is set to the MCS server's URI;

f) if MCS identifiers are not protected, the IDRr payload specified in IETF RFC 6509 [18] is included and:

1) the ID type field of the IDRr payload is set to the 'URI';

2) if an MCS user is targeted, the ID data field of the IDRr payload is set to the MCS ID of the targeted MCS user;

3) if an constituent MCS group is targeted, the ID data field of the IDRr payload is set to the MCS Group ID of the targeted constituent MCS group; and

4) if an MCS server is targeted, the ID data field of the IDRr payload is set to the MCS server's URI;

g) if MCS identifiers are protected, the ID payload with role indicator specified in IETF RFC 6043 [17] is included and:

1) the ID role field of the ID payload is set to the 'IDRuidr' as specified in subclause 7.5.2;

2) the ID type field of the ID payload is set to the 'Byte string';

3) if an MCS user is targeted, the ID data field of the ID payload is set to the UID generated from the MCS ID of the targeted MCS user;

4) if an constituent MCS group is targeted, the ID data field of the ID payload is set to the UID generated from the MCS Group ID of the targeted constituent MCS group; and

5) if an MCS server is targeted, the ID data field of the ID payload is set to the MCS's server URI;

h) the IDRkmsi payload specified in IETF RFC 6509 [18] is included and:

1) the ID type field of the IDRkmsi payload is set to the 'URI';

2) if the originator of the I_MESSAGE is the GMS, the ID data field of the IDRkmsi payload is set to the URI of the MCS KMS used by the group management server; and

3) if the originator of the I_MESSAGE is an MCS server, the ID data field of the IDRkmsi payload is set to the URI of the MCS KMS used by MCS server;

i) the IDRkmsr payload specified in IETF RFC 6509 [18] is included and:

1) the ID type field of the IDRkmsr payload is set to the 'URI';

2) if an MCS user is targeted, the ID data field of the IDRkmsr payload is set to the URI of the MCS KMS used by targeted MCS user;

NOTE: The KMS URI is stored in the group configuration data.

Editor's Note:The XML element(s) (or XML attribute(s)) in the MCPTT Group for carrying the KMS URI need to be defined.

3) if an constituent MCS group is targeted, the ID data field of the IDRkmsr payload is set to the URI of the MCS KMS used by targeted constituent MCS group; and

4) if an MCS server is targeted, the ID data field of the IDRkmsr payload is set to the URI of the MCS KMS used by the MCS server;

j) the SAKKE payload specified in IETF RFC 6509 [18] is included and:

1) the SAKKE params field of the SAKKE payload is set to 'Parameter Set 1';

2) the ID scheme field of the SAKKE payload is set to 'MCPTT-ID-scheme' or '3GPP MCX hashed UID' as specified in subclause 7.5.3; and

3) the SAKKE data field of the SAKKE payload contains the GMK or MKFC specified in 3GPP TS 33.180 [29];

k) the SIGN payload specified in IETF RFC 6509 [18] is included and the S type field of the SIGN payload is set to the 'ECCSI';

l) the security policy payload specified in IETF RFC 3830 [16] can be included;

m) the general extension payload specified in IETF RFC 3830 [16] with the type field set to 'GMK-or-MKFC-associated-parameters' or to '3GPP key parameters' as specified in subclause 7.5.4 is included and the data field of the general extension payload contains the associated parameters of GMK or MKFC as specified in 3GPP TS 33.180 [29] figure E.6.1-1; and

n) the General extension payload specified in IETF RFC 3830 [16] with the type field set to the 'SAKKE-to-self' as specified in subclause 7.5.4 can be included and the data field of the general extension payload contains a SAKKE payload specified in IETF RFC 6509 [18]. In the SAKKE payload included in the general extension payload:

1) the SAKKE params field of the SAKKE payload is set to 'Parameter Set 1';

2) the ID scheme field of the SAKKE payload is set to 'MCPTT-SAKKE-to-self-ID-scheme' or '3GPP MCX hashed UID' as specified in subclause 7.5.3; and

3) the SAKKE data field of the SAKKE payload contains the GMK or MKFC specified in 3GPP TS 33.180 [29]."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause A.2.3]

"…

The temporary MCPTT group is identified by the temporary MCPTT Group ID sip:groupT@MCPTTSP1.example.com and is being created in GMS-1 under the groupdocumentT.xml document name.

The temporary MCPTT group is a pre-arranged MCPTT group.

The GMK is required to confidentiality and integrity protect media for calls on the temporary MCPTT group. Both confidentiality protection and integrity protection of floor control signalling are required for calls on the temporary MCPTT group. Multicast bearers are used for floor controlling signalling for the temporary MCPTT group.

One MCPTT group to be combined is identified by the MCPTT Group ID sip:groupA@MCPTTSP1.example.com and is owned by GMS-1.

The other MCPTT group to be combined is identified by the MCPTT Group ID sip:groupB@MCPTTSP2.example.com and is owned by GMS-2.

The GMS-1 is configured with the group ID routing database stating:

- *the XCAP Root URI of sip:groupB@MCPTTSP2.example.com is http://xcap.MCPTTSP2.example.com/path2; and*
- *the public service identity for accessing documents of the MCPTT provider of sip:groupB@MCPTTSP2.example.com is sip:gms.MCPTTSP2.example.com.*

The GMS-2 is configured with the group ID routing database stating:

- *the XCAP Root URI of sip:groupT@MCPTTSP1.example.com is http://xcap.MCPTTSP1.example.com/path1; and*
- *the public service identity for accessing documents of the MCPTT provider of sip:groupT@MCPTTSP1.example.com is sip:gms.MCPTTSP1.example.com.*

…

The details of the flows are as follows:

1) GMC-1 sends an HTTP POST request with the GMOP document requesting group regroup creation shown in table A.2.3-1 to the GMS-1.

…

2) GMS-1 authorizes the request as specified in subclause 6.3.14.3.1 and sends HTTP POST request with the GMOP document requesting group regroup check.

3) GMS-1 sends HTTP POST request with the GMOP document requesting group regroup check to GMS-2 as shown in table A.2.3-3.

…

4) GMS-1 authorizes the request as specified in subclause 6.3.14.3.2 and responds with a HTTP 200 (OK) response to the HTTP POST request.

5) GMS-2 authorizes the request as specified in subclause 6.3.14.3.2 and responds with a HTTP 200 (OK) response to the HTTP POST request.

6) GMS-1 sends HTTP POST request with the GMOP document requesting group regroup notification.

7) GMS-1 sends HTTP POST request with the GMOP document requesting group regroup notification to GMS-2 as shown in table A.2.3-7.

…

8) GMS-1 authorizes the request as specified in subclause 6.3.14.3.2, place the <on-network-regrouped> element of the GMOP document requesting group regroup notification of the HTTP POST request at the location identified by the Request-URI of the received HTTP POST request and responds with a HTTP 200 (OK) response to the HTTP POST request.

9) GMS-2 authorizes the request as specified in subclause 6.3.14.3.2, place the <on-network-regrouped> element of the GMOP document requesting group regroup notification of the HTTP POST request at the location identified by the Request-URI of the received HTTP POST request and responds with a HTTP 200 (OK) response to the HTTP POST request.

10) GMS-1 creates the temporary MCPTT group document so that it is accessible using the XCAP URIs

…

GMS-1 also creates the MCPTT GKTP document …

GMS-1 accepts the HTTP POST request with HTTP 200 (OK) response. The HTTP 200 (OK) response includes the GMOP document with group regroup creation response.

10) GMS-1 subscribes to changes of the <GKTPs> element of MCPTT-GKTP document of the temporary MCPTT group, is notified about the current state and updates the MCPTT-GKTP document for the MCPTT group with sip:groupA@MCPTTSP1.example.com MCPTT Group ID.

11) GMS-2 subscribes to changes of the <GKTPs> element of MCPTT-GKTP document of the temporary MCPTT group as shown in table A.2.3-11.

…

12) GMS-1 authorizes the SIP SUBSCRIBE request and accepts it with SIP 200 (OK) response.

13) GMS-1 sends a SIP NOTIFY request with the content of the <GKTPs> element of MCPTT-GKTP document of the temporary MCPTT group as shown in table A.2.3-13.

…

14) GMS-2 accepts the SIP NOTIFY request with SIP 200 (OK) response and updates the MCPTT-GKTP document for the MCPTT group with sip:groupB@MCPTTSP2.example.comsip:groupB@MCPTTSP2.example.com MCPTT Group ID."

5.1.4.3 Test description

5.1.4.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the Group Management function.
- SS-UE1 (MCPTT client)

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the Group Management function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.6, and to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- Group A is owned by the IUT (MCPTT Server).
- Group T will be owned by the SS (MCPTT Server).
- The SS-UE1 (MCPTT client) is member of group A.

5.1.4.3.2 Test procedure sequence

Table 5.1.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) sends a HTTP POST to the IUT (MCPTT Server) to request a GMOP group regroup check for group A.	<--	HTTP POST		-
2	Check: Does the IUT (MCPTT Server) respond with a HTTP 200 (OK) response message?	-->	HTTP 200 (OK)	1	P
3	The SS (MCPTT Server) sends a HTTP POST to the IUT (MCPTT Server) for notifying a GMOP group regroup notification to group A.	<--	HTTP POST		-
4	Check: Does the IUT (MCPTT Server) respond with a HTTP 200 (OK) response message?	-->	HTTP 200 (OK)	1	P
5	Check: Does the IUT (MCPTT Server) send a SIP SUBSCRIBE message request to get the group key transport payloads (GKTP) document including the group keys for the temporary group, related to group A?	-->	SIP SUBSCRIBE	2	P
6	The SS (MCPTT Server) responds to the SIP SUBSCRIBE message with a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
	- EXCEPTION: Depending on the IUT (MCPTT Server) implementation, steps 8 and 9 can happen in parallel.				
7	The SS (MCPTT Server) sends a SIP NOTIFY message to the IUT (MCPTT Server) containing the group key transport payloads (GKTP) document including the group keys.	<--	SIP NOTIFY		-
8	Check: Does the IUT (MCTT Server) send a SIP 200 (OK) message?	-->	SIP 200 (OK)	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
9	Check: Does the IUT (MCPTT Server) correctly perform procedure 'MCX Server Originated notifications regarding temporary group creation or tear down' as described in Annex A, Table A.2.3.3-1 to update the group document of group A and retrieve the GMK for the temporary group T?	-	-	2	P
10	The SS (MCPTT Server) sends a HTTP DELETE to request the removal of the parameters associated to the temporary group from the group configuration file of group A.	<--	HTTP DELETE		-
11	Check: Does the IUT (MCPTT Server) respond with a HTTP 200 (OK) response message?	-->	HTTP 200 (OK)	3	P
12	Check: Does the IUT (MCPTT Server) send a SIP SUBSCRIBE message request to de-subscribe from receiving notifications of group key transport payloads (GKTP) for the temporary group, related to group A?	-->	SIP SUBSCRIBE	4	P
13	The SS (MCPTT Server) responds to the SIP SUBSCRIBE message with a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
	- EXCEPTION: Depending on the IUT (MCPTT Server) implementation, steps 15 and 16 can happen in parallel.				
14	The SS (MCPTT Server) sends a SIP NOTIFY message to the IUT (MCPTT Server) containing the group key transport payloads (GKTP) document including the group keys.	<--	SIP NOTIFY		-
15	Check: Does the IUT (MCTT Server) send a SIP 200 (OK) message?	-->	SIP 200 (OK)	4	P
16	Check: Does the IUT (MCPTT Server) correctly perform procedure 'MCX Server Originated notifications regarding temporary group creation or tear down' as described in Annex A, Table A.2.3.3-1 to update the group document and the GKTP document of group A?	-	-	4	P

5.1.4.3.3 Specific message contents

Table 5.1.4.3.3-1: HTTP POST from the SS (MCPTT Server) (Step 1, Table 5.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, conditions TEMPGROUP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI				
uri	tsc_MCX_GMSXCAPRootURI & "/" & "org.openmobilealliance.groups/global/byGroupID/" & px_MCPTT_Group_A_ID & "/~~/group/list-service/mcpttgi:on-network-regrouped%5b@temporary-MCPTT-group-ID=%22" & px_MCPTT_Group_T_ID & "%22%5d?xmlns(mcpttgi=urn:3gpp:ns:mcpttGroupInfo:1.0)"	tsc_MCX_GMSXCAPRootURI contains the XCAP Root URI of the GMS owning an MCS group to be combined.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.14.3.1	
Authorization	Not present			
X-3GPP-Asserted-Identity	tsc_MCPTT_PublicServiceId_B	The URI of the GMS owning the temporary group.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.14.3.1	
Message-body				
GMOP document requesting group regroup check	As described in Table 5.1.4.3.3-1A			

Table 5.1.4.3.3-1A: GMOP group regroup check in HTTP POST (Table 5.1.4.3.3-1)

Derivation Path: ETSI TS 124 481 [6] (3GPP TS 24.481) clause 5.1.4.4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
gmop:document				
gmop:request				
gmop:group-regroup-check				
mcpttgi:on-network-regrouped				
temporary-MCPTT-group-ID	px_MCPTT_Group_T_ID	MCPTT temporary group identity		
temporary-MCPTT-group-requestor	px_MCPTT_ID_User_A	The URI of the MCPTT user requesting the creation of the temporary group.		
mcpttgi:constituent-MCPTT-group-IDs				
constituent-MCPTT-group-ID[1]	px_MCPTT_Group_A_ID	MCS group ID of a constituent MCS group of the temporary MCS group		
constituent-MCPTT-group-ID[2]	px_MCPTT_Group_B_ID	MCS group ID of a constituent MCS group of the temporary MCS group		

Table 5.1.4.3.3-2: HTTP POST from the SS (MCPTT Server) (Step 3, Table 5.1.4.3.2-1)

Derivation Path: Table 5.1.4.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
GMOP document requesting group regroup notification	As described in Table 5.1.4.3.3-2A			

Table 5.1.4.3.3-2A: GMOP group regroup notification in HTTP POST (Table 5.1.4.3.3-2)

Derivation Path: ETSI TS 124 481 [6] (3GPP TS 24.481) clause 5.1.4.4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
gmop:document				
gmop:request				
gmop:group-regroup-notification				
mcpttgi:on-network-regrouped				
temporary-MCPTT-group-ID	px_MCPTT_Group_T_ID	MCPTT temporary group identity		
temporary-MCPTT-group-requestor	px_MCPTT_ID_User_A	The URI of the MCPTT user requesting the creation of the temporary group.		
mcpttgi:constituent-MCPTT-group-IDs				
constituent-MCPTT-group-ID[1]	px_MCPTT_Group_A_ID	MCS group ID of a constituent MCS group of the temporary MCS group		
constituent-MCPTT-group-ID[2]	px_MCPTT_Group_B_ID	MCS group ID of a constituent MCS group of the temporary MCS group		

Table 5.1.4.3.3-3: HTTP 200 (OK) from the IUT (MCPTT Server) (Steps 2, 4 and 11, Table 5.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1

Table 5.1.4.3.3-4: HTTP DELETE from the SS (MCPTT Server) (Step 10, Table 5.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.5-1, conditions TEMPGROUP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI				
uri	tsc_MCX_GMSXCAPRootURI & "/" & "org.openmobilealliance.groups/global/byGroupID/" & px_MCPTT_Group_A_ID & "/~~/group/list-service/mcpttgi:on-network-regrouped%5b@temporary-MCPTT-group-ID=%22" & px_MCPTT_Group_T_ID & "%22%5d?xmlns(mcpttgi=urn:3gpp:ns:mcpttGroupInfo:1.0)"	tsc_MCX_GMSXCAPRootURI contains the XCAP Root URI of the GMS owning an MCS group to be combined.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.15.3.1	
Authorization	Not present			
X-3GPP-Asserted-Identity	tsc_MCPTT_PublicServiceId_B	The URI of the GMS owning the temporary group.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.15.3.1	

Table 5.1.4.3.3-5: SIP SUBSCRIBE from the IUT (MCPTT Server) (Step 5, Table 5.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the GMS owning the temporary group.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the GMS owning the groups being combined into the temporary group.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the GMS owning the temporary group.		
Contact				
addr-spec	SIP URI			
user-info and host	IP address or FQDN			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
port	protected server port of IUT (MCPTT Server)			
Expires			IETF RFC 3261 [17] IETF RFC 3903 [22]	
value	any value			
Require	not present			
Proxy-Require	not present			
Security-Verify	not present			
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value			
P-Access-Network-Info	not present			
P-Asserted-Identity	tsc_MCPTT_PublicServiceId_A	The URI of the GMS owning the groups being combined into the temporary group.	ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.13.3.2	
Content-Type			IETF RFC 5621 [16]	
media-type	"multipart/mixed"			
Message-body				
MIME body part		**Resource-lists**		
MIME-part-body	Resource-lists as described in Table 5.1.4.3.3-5A			
MIME body part		**MCPTT Info**		
MIME-part-body	Resource-lists as described in Table 5.1.4.3.3-5B			

Table 5.1.4.3.3-5A: Resource-Lists in SIP SUBSCRIBE (Table 5.1.4.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1A-1, condition GROUPKEY				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list[1]				
entry[1]			ETSI TS 124 481 [6] (3GPP TS 24.481)	GROUPKEY
uri attribute	"org.3gpp.MCPTT-GKTP/global/byGroupID/" & px_MCPTT_Group_T_ID & "/" & "~~" & "/group/list-service/mgktp:GKTPs?xmlns(mgktp=urn:3gpp:ns:mcpttGKTP:1.0)"			
display-name	Not present			

Table 5.1.4.3.3-5B: MCPTT-Info in SIP SUBSCRIBE (Table 5.1.4.3.3-5)

Derivation Path: ETSI TS 124 379 [4] (3GPP TS 24.379) clause F.1.2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_A_ID			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A				

Table 5.1.4.3.3-6: SIP NOTIFY from the SS (MCPTT Server) (Step 7, Table 5.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Max-Forwards			IETF RFC 3261 [17]	
value	70	The recommended initial value is 70 in IETF RFC 3261 [17].		
Content-Type			IETF RFC 3261 [17] IETF RFC 3842 [14]	
media-type	"application/xcap-diff+xml"			
Message-body				
xcap-diff document	xcap-diff document as described in Table 5.1.4.3.3-6A			

Table 5.1.4.3.3-6A: Xcap-Diff Document (Table 5.1.4.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.12-2, condition GROUPKEY				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
xcap-diff				
xcap-root	tsc_MCX_GMSXCAPRootURI	The URI of the GMS owning the temporary group.		
element[1]				
sel attribute	"org.3gpp.MCPTT-GKTP/global/byGroupID/" & px_MCPTT_Group_T_ID & "/~~" & "/group/list-service/mgktp:GKTPs?xmlns(mgktp=urn:3gpp:ns:mcpttGKTP:1.0)"			
GKTPs	group key transport payloads (GKTP) document as described in Table 5.1.4.3.3-6B			

Table 5.1.4.3.3-6B: group key transport payloads (GKTP) document (Table 5.1.4.3.3-6A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.14-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
GKTPs				
GMK-GKTPs				
GKTP[1]	MIKEY message as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-9			

Table 5.1.4.3.3-7: SIP 200 (OK) to SIP SUSCRIBE (Steps 6 and 13, Table 5.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition SUBSCRIBE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			

Table 5.1.4.3.3-8: SIP SUBSCRIBE from the IUT (MCPTT Server) (Step 12, Table 5.1.4.3.2-1)

Derivation Path: Table 5.1.4.3.3-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a message within the same dialog	Contact URI of the recipient of the previous 200 OK		
Expires				
value	"0"			

Table 5.1.4.3.3-9: SIP NOTIFY from the SS (MCPTT Server) (Step 14, Table 5.1.4.3.2-1)

Derivation Path: Table 5.1.4.3.3-6				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subscription-State			IETF RFC 6665 [24]	
substate-value	"terminated"			
expires	not present			

Table 5.1.4.3.3-10: SIP 200 (OK) to SIP NOTIFY (Steps 8 and 15, Table 5.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			

5.1.5 MCPTT Server - MCPTT Server / Configuration / Functional Alias / Functional alias status determination / Activate functional alias / Deactivate functional alias / Controlling server

5.1.5.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS (MCPTT server) requests IUT (MCPTT Server) to determine the current status of a
functional alias and to get later notification of status changes of a functional alias with SIP
SUBSCRIBE message }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) with a SIP 200 (OK) message and
sends a SIP NOTIFY message informing about the status of the functional alias }
            }
```

(2)

```
with { IUT (MCPTT Server) having provided the status of the functional alias }
ensure that {
  when { the SS (MCPTT server) requests IUT (MCPTT Server) activate the functional alias for user
with SIP PUBLISH message }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) with a SIP 200 (OK) message and
sends SIP NOTIFY message to inform successful activation of the functional alias }
            }
```

(3)

```
with { IUT (MCPTT Server) having provided activation of the functional alias }
ensure that {
  when { the SS (MCPTT server) requests IUT (MCPTT Server) to deactivate the functional alias with
SIP PUBLISH message }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) with a SIP 200 (OK) message and
sends SIP NOTIFY message to inform successful deactivation of the functional alias }
            }
```

(4)

```
with { IUT (MCPTT Server) having provided deactivation of the functional alias }
ensure that {
  when { the SS (MCPTT server) requests IUT (MCPTT Server) to de-subscribe from functional alias
status changes of the functional alias with SIP SUBSCRIBE message }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) with a SIP 200 (OK) message and
sends a SIP NOTIFY message indicating the termination of the subscription }
            }
```

5.1.5.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9A.2.2.3.3, 9A.2.2.3.4, 9A.2.2.3.5, 9A.2.2.3.6, 9A.2.2.3.7 and 9A.2.2.3.8. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.3.3]

"Upon receiving a SIP PUBLISH request such that:

1) *Request-URI of the SIP PUBLISH request contains the public service identity of the controlling MCPTT function associated with the served functional alias;*

2) *the SIP PUBLISH request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the <mcptt-request-uri> element and the <mcptt-calling-user-id> element;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];*

4) *the Event header field of the SIP PUBLISH request contains the "presence" event type; and*

5) *SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-functional alias information constructed according to clause 9A.2.3.2;*

then the MCPTT server:

1) shall identify the served functional alias in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP PUBLISH request;

2) shall identify the handled MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP PUBLISH request;

3) if the Expires header field of the SIP PUBLISH request is not included or has nonzero value lower than 4294967295, shall send a SIP 423 (Interval Too Brief) response to the SIP PUBLISH request, where the SIP 423 (Interval Too Brief) response contains a Min-Expires header field set to 4294967295, and shall not continue with the rest of the steps;

4) if the functional alias does not exist in the MCPTT server, shall reject the SIP PUBLISH request with SIP 403 (Forbidden) response to the SIP PUBLISH request according o 3GPP TS 24.229 [4], IETF RFC 3903 [37] and IETF RFC 3856 [51] and skip the rest of the steps;

4a) if SIP PUBLISH request is for activation of a functional alias then:

a) if handled MCPTT ID does not match with any of the entries in the <mcptt-user-list> which contains the MCPTT IDs of MCPTT users which are allowed to activate the functional alias; or

b) if no local policy exists that authorizes the request by the handled MCPTT ID;

shall reject the SIP PUBLISH request with SIP 403 (Forbidden) response according to 3GPP TS 24.229 [4], IETF RFC 3903 [37] and IETF RFC 3856 [51] and skip the rest of the steps;

5) if SIP PUBLISH request is for activation of a functional alias and the number of activations for the handled functional alias is equal <max-simultaneous-activations>, shall reject the SIP PUBLISH request with SIP 403 (Forbidden) response to the SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37] and IETF RFC 3856 [51] and skip the rest of the steps;

6) if SIP PUBLISH request is for take over of a functional alias, the MCPTT server shall use the <allow-takeover> element to determine if take over is possible. If take over is not possible, the MCPTT server shall reject the SIP PUBLISH request with SIP 403 (Forbidden) response to the SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37] and IETF RFC 3856 [51] and skip the rest of the steps;

7) shall respond with SIP 200 (OK) response to the SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37]. In the SIP 200 (OK) response, the MCPTT server:

a) shall set the Expires header field according to IETF RFC 3903 [37], to the selected expiration time;

8) if the "entity" attribute of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request is different than the served functional alias ID, shall not continue with the rest of the steps;

9) if the handled MCPTT ID is different from the MCPTT ID in the "id" attribute of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request, shall not continue with the rest of the steps;

10) shall consider a functional alias information entry such that:

a) the functional alias information entry is in the list of functional alias information entries described in clause 9A.2.2.3.2; and

b) the functional alias ID of the functional alias information entry is equal to the served functional alias ID;

as the served functional alias information entry;

11) if the selected expiration time is zero:

a) shall remove the MCPTT user information entry such that:

i) the MCPTT user information entry is in the list of the MCPTT user information entries of the served functional alias information entry; and

ii) the MCPTT user information entry has the MCPTT ID set to the served MCPTT ID;

12) if the selected expiration time is not zero:

a) shall consider an MCPTT user information entry such that:

i) the MCPTT user information entry is in the list of the MCPTT user information entries of the served functional alias information entry; and

ii) the MCPTT ID of the MCPTT user information entry is equal to the handled MCPTT ID;

as the served MCPTT user information entry;

b) if the MCPTT user information entry does not exist:

i) shall insert an MCPTT user information entry with the MCPTT ID set to the handled MCPTT ID into the list of the MCPTT user information entries of the served functional alias information entry; and

ii) shall consider the inserted MCPTT user information entry as the served MCPTT user information entry; and

c) shall set the expiration time according to the selected expiration time in the served MCPTT user information entry;

13) shall identify the handled p-id-fa in the <p-id-fa> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

14) shall perform the procedures specified in clause 9A.2.2.3.5 for the served functional alias ID."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.3.4]

"NOTE: Usage of one SIP SUBSCRIBE request to subscribe for notification about change of functional alias state of several MCPTT users served by the same MCPTT server is not supported in this version of the specification.

Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains the public service identity of the controlling MCPTT function associated with the served functional alias;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the<mcptt-request-uri> element and the <mcptt-calling-user-id> element;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type; and

5) the SIP SUBSCRIBE request contains an application/simple-filter+xml MIME body indicating per-user restrictions of presence event package notification information according to clause 9A.3.2 indicating the same MCPTT ID as in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

then the MCPTT server:

1) shall identify the served functional alias ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

2) shall identify the handled MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

3) if the Expires header field of the SIP SUBSCRIBE request is not included or has nonzero value lower than 4294967295, shall send a SIP 423 (Interval Too Brief) response to the SIP SUBSCRIBE request, where the SIP 423 (Interval Too Brief) response contains a Min-Expires header field set to 4294967295, and shall not continue with the rest of the steps;

4) if a functional alias does not exist in the MCPTT server, shall reject the SIP PUBLISH request with SIP 403 (Forbidden) response to the SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37] and IETF RFC 3856 [51] and skip the rest of the steps;

5) if the handled MCPTT ID based on local policy is not authorized for notifications of the functional alias identified by the served functional alias ID, shall reject the SIP SUBSCRIBE request with SIP 403 (Forbidden) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37] and IETF RFC 3856 [51] and skip the rest of the steps; and

6) shall generate a SIP 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 6665 [26].

For the duration of the subscription, the MCPTT server shall notify subscriber about changes of the information of the served MCPTT ID, as described in clause 9A.2.2.3.5."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.3.5]

"In order to notify the subscriber identified by the handled MCPTT ID about changes of the functional alias status of the served functional alias ID, the MCPTT server:

1) shall consider a functional alias information entry such that:

a) the functional alias information entry is in the list of functional alias information entries described in clause 9A.2.2.3.2; and

b) the functional alias ID of the functional alias information entry is equal to the served functional alias ID;

2) shall consider an MCPTT user information entry such:

a) the MCPTT user information entry is in the list of the MCPTT user information entries of the served functional alias information entry; and

b) the MCPTT ID of the MCPTT user information entry is equal to the handled MCPTT ID;

as the served MCPTT user information entry;

3) shall generate an application/pidf+xml MIME body indicating per-functional alias information according to clause 9A.3.1 and the served list of the served MCPTT user information entry of the functional alias information entry with following clarifications:

a) the MCPTT server shall include the "expires" attribute in the <functionalAlias> element; and

b) if this procedures is invoked by procedure in clause 9A.2.2.3.3 where the handled p-id-fa was identified, the MCPTT server shall set the <p-id-fa> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id-fa value; and

4) send a SIP NOTIFY request according to 3GPP TS 24.229 [4], and IETF RFC 6665 [26] for the subscription created in clause 9A.2.2.3.4. In the SIP NOTIFY request, the MCPTT server shall include the generated application/pidf+xml MIME body indicating per-functional alias information."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.3.6]

"In order to deactivate a functional alias associated with a target MCPTT ID:

1) externally triggered by an MCPTT administrator by a mechanism outside of the scope of the standard; or

2) directly by the MCPTT function owning the functional alias as a result of an internal trigger like the expiration of the functional alias association;

the MCPTT server

1) *shall consider a functional alias information entry such that:*

   a) *the functional alias information entry is in the list of functional alias information entries described in clause 9A.2.2.3.2; and*

   b) *the functional alias ID of the functional alias information entry is equal to the served functional alias ID;*

   as the served functional alias information entry;

2) *shall remove the MCPTT user information entry such that:*

   a) *the MCPTT user information entry is in the list of the MCPTT user information entries of the served functional alias information entry; and*

   b) *the MCPTT user information entry has the MCPTT ID set to the target MCPTT ID; and*

3) *shall perform the procedures specified in clause 9A.2.2.3.5 for the served functional alias ID."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.3.7]

"Upon receiving a SIP SUBSCRIBE request such that:

1) *Request-URI of the SIP SUBSCRIBE request contains the public service identity of the controlling MCPTT function associated with the requested functional alias;*

2) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];*

3) *the Event header field of the SIP SUBSCRIBE request contains the "presence" event type; and*

4) *the SIP SUBSCRIBE request contains an application/simple-filter+xml MIME body indicating per-functional alias restrictions of presence event package notification information according to clause 9A.3.2;*

then the MCPTT server:

1) *shall identify the requested functional alias ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;*

2) *if the Expires header field of the SIP SUBSCRIBE request is not included or has nonzero value lower than 4294967295, shall send a SIP 423 (Interval Too Brief) response to the SIP SUBSCRIBE request, where the SIP 423 (Interval Too Brief) response contains a Min-Expires header field set to 4294967295, and shall not continue with the rest of the steps;*

3) *if the requested functional alias does not exist in the MCPTT server, shall reject the SIP SUBSCRIBE request with SIP 403 (Forbidden) response to the SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37] and IETF RFC 3856 [51] and skip the rest of the steps; and*

4) *shall generate a SIP 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 6665 [26].*

For the duration of the subscription, the MCPTT server shall notify subscriber about changes of the information of the requested functional alias, as described in clause 9.2.2.3.y."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.3.8]

"In order to notify the subscriber about the MCPTT users that have successfully activated the functional alias corresponding to the requested functional alias ID, the MCPTT server:

1) *shall consider a functional alias information entry such that:*

   a) *the functional alias information entry is in the list of functional alias information entries described in clause 9A.2.2.3.2; and*

   b) *the functional alias ID of the functional alias information entry is equal to the requested functional alias ID;*

2) shall consider any MCPTT user information entry such that the MCPTT user information entry is in the list of the MCPTT user information entries of the served functional alias information entry, as the served MCPTT user information entry;

3) shall generate an application/pidf+xml MIME body indicating per-functional alias information according to clause 9A.3.1 and the served list of the served MCPTT user information entry of the functional alias information entry

4) send a SIP NOTIFY request according to 3GPP TS 24.229 [4], and IETF RFC 6665 [26] for the subscription created in clause 9A.2.2.3.7. In the SIP NOTIFY request, the MCPTT server shall include the generated application/pidf+xml MIME body indicating per-functional alias information."

5.1.5.3 Test description

5.1.5.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the participating MCPTT functions.
- SS-UE1 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.1.5.3.1-1.

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the controlling MCPTT function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2.
- Group A is controlled by the controlling MCPTT function.
- User A is not configured to be implicitly affiliated to Group A.

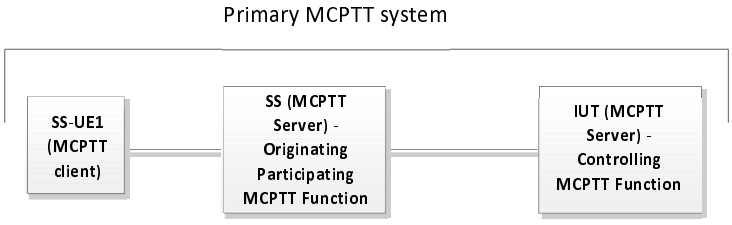


Figure 5.1.5.3.1-1: Functions of the testing components

5.1.5.3.2 Test procedure sequence

Table 5.1.5.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) sends a SIP SUBSCRIBE message request to determine the current status of a functional alias and to get later notification of status changes of a functional alias.	<--	SIP SUBSCRIBE	-	-
2	Check: Does the IUT (MCPTT Server) respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	1	P
3	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message informing about the status of the functional alias?	-->	SIP NOTIFY	1	P
4	The SS (MCPTT Server) responds with a SIP 200 (OK) message.	<--	SIP 200 (OK)		
5	The SS (MCPTT server) sends a SIP PUBLISH message to activate the functional alias for user A.	<--	SIP PUBLISH	-	-
6	Check: Does the IUT (MCPTT Server) respond to the SIP PUBLISH message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	2	P
7	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY informing about the successful activation of the functional alias?	-->	SIP NOTIFY	2	P
8	The SS (MCPTT Server) responds with a SIP 200 (OK) to the SIP NOTIFY.	<--	SIP 200 (OK)	-	-
9	The SS (MCPTT server) sends a SIP PUBLISH message to deactivate the functional alias.	<--	SIP PUBLISH	-	-
10	Check: Does the IUT (MCPTT Server) respond to the SIP PUBLISH message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	3	P
11	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY informing about the successful deactivation of the functional alias?	-->	SIP NOTIFY	3	P
12	The SS (MCPTT Server) responds with a SIP 200 (OK) to the SIP NOTIFY.	<--	SIP 200 (OK)	-	-
13	The SS (MCPTT Server) sends a SIP SUBSCRIBE message request to de-subscribe from functional alias status changes of the functional alias.	<--	SIP SUBSCRIBE	-	-
14	Check: Does the IUT (MCPTT Server) respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	4	P
15	Check: Does the IUT (MCPTT Server) send a SIP NOTIFY message indicating the termination of the subscription?	-->	SIP NOTIFY	4	P
16	The SS (MCPTT Server) responds with a SIP 200 (OK) message.	<--	SIP 200 (OK)		

5.1.5.3.3 Specific message contents

Table 5.1.5.3.3-1: SIP SUBSCRIBE from the SS (MCPTT Server) (Step 1, Table 5.1.5.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	tsc_MCPTT_PublicServiceId_B	The public service identity identifying the controlling MCPTT function owning the MCPTT functional alias		
From				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_A	The public service identity identifying the originating participating MCPTT function serving the MCPTT user		
To				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B	The public service identity identifying the controlling MCPTT function owning the MCPTT functional alias		
Expires				
value	"4294967295"	to receive the current status and later notification		
P-Asserted-Service			IETF RFC 6050 [26]	
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.1.5.3.3-1A			
MIME body part		**SIMPLE-FILTER**		
MIME-part-body	Simple-Filter as described in Table 5.1.5.3.3-1B			

Table 5.1.5.3.3-1A: MCPTT-Info in SIP SUBSCRIBE (Table 5.1.5.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_FA_A			
mcptt-calling-user-id	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A			
anyExt				
request-type	functional-alias-status-determination			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 5.1.5.3.3-1B: Simple-Filter in SIP SUBSCRIBE (Table 5.1.5.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.6-1, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
filter-set				
ns-bindings				
ns-binding urn [2]				
prefix	"mcpttPIFA10"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.3.2.2	
urn	"urn:3gpp:ns:mcpttPresInfoFA:1.0"			
filter[1]				
what			IETF RFC 4661 [20]	
include[1]				
type	Not present			
base	"//pidf:presence/pidf:tuple[@id=" & px_MCPTT_ID_User_A & "]"		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.3.2.2	

Table 5.1.5.3.3-2: SIP 200 (OK) to SIP SUSCRIBE (Steps 2 and 14, Table 5.1.5.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition SUBSCRIBE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route				Not present

Table 5.1.5.3.3-3: SIP NOTIFY from the IUT (MCPTT Server) (Steps 3 and 11, Table 5.1.5.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions MCPTT and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Service			IETF RFC 6050 [26]	
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
P-Asserted-Identity			IETF RFC 3325 [27]	
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B	The public service identity identifying the controlling MCPTT function owning the MCPTT functional alias		
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.3.5	
MIME-part-body	PIDF as described in Table 5.1.5.3.3-3A			

Table 5.1.5.3.3-3A: PIDF in SIP NOTIFY (Table 5.1.5.3.3-3)

Derivation Path: IETF RFC 3863 [8]				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence			IETF RFC 3863 [8]	
entity attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_FA_A			
tuple				
id attribute	Encrypted URI (see note) with value set to px_MCPTT_ID_User_A			
status				
functionalAlias	not present	MCPTT extension	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.3.1	
contact	not present			
note	not present			
timestamp	not present			
note	not present			
p-id-fa	same value as received in the SIP PUBLISH message		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.2.5 and 9A.2.2.3.5	
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.13.3-1.				

Table 5.1.5.3.3-4: SIP NOTIFY from the IUT (MCPTT Server) (Step 15, Table 5.1.5.3.2-1)

Derivation Path: Table 5.1.5.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subscription-State			IETF RFC 6665 [24]	
substate-value	"terminated"			
expires	not present			

Table 5.1.5.3.3-5: SIP NOTIFY from the IUT (MCPTT Server) (Step 7, Table 5.1.5.3.2-1)

Derivation Path: Table 5.1.5.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.3.5	
MIME-part-body	PIDF as described in Table 5.1.5.3.3-5A			

Table 5.1.5.3.3-5A: PIDF in SIP NOTIFY (Table 5.1.5.3.3-5)

Derivation Path: Table 5.1.5.3.3-3A				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence			IETF RFC 3863 [8]	
tuple				
status				
functionalAlias				
user	Encrypted URI (see note) with value set to px_MCX_Client_A_ID			
functionalAliasID	not present			
status	not present			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.3.5	
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.13.3-1.				

Table 5.1.5.3.3-6: SIP 200 (OK) to SIP NOTIFY (Steps 4, 8, 12 and 16, Table 5.1.5.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route				Not present

Table 5.1.5.3.3-7: SIP SUBSCRIBE from the SS (MCPTT Server) (Step 13, Table 5.1.5.3.2-1)

Derivation Path: Table 5.1.5.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	same URI as the IUT (MCPTT Server) has sent earlier in the Contact header of a message within the same dialog	Contact URI of the recipient of the previous 200 OK		
Expires				
value	"0"			

Table 5.1.5.3.3-8: SIP PUBLISH from the SS (MCPTT Server) (Step 5, Table 5.1.5.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.11-1, condition PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [17] IETF RFC 5031 [19]	
Request-URI	tsc_MCPTT_PublicServiceId_B	The public service identity identifying the controlling MCPTT function owning the MCPTT functional alias		
From				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_A	The public service identity identifying the originating participating MCPTT function serving the MCPTT user		
To				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B	The public service identity identifying the controlling MCPTT function owning the MCPTT functional alias		
Expires				
value	"4294967295"			
P-Asserted-Service			IETF RFC 6050 [26]	
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcptt"			
P-Asserted-Identity			IETF RFC 3325 [27]	
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_A	The public service identity identifying the originating participating MCPTT function serving the MCPTT user		
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.1.5.3.3-1A			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.1.5.3.3-8A			

Table 5.1.5.3.3-8A: PIDF in SIP PUBLISH (Table 5.1.5.3.3-8)

Derivation Path: Table 5.1.5.3.3-5A				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence			IETF RFC 3863 [8]	
tuple				
status				
functionalAlias		MCPTT extension	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.3.1	
expires	not present			
p-id-fa	any allowed value		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.2.5 and 9A.2.2.3.5	

Table 5.1.5.3.3-9: SIP PUBLISH from the SS (MCPTT Server) (Step 9, Table 5.1.5.3.2-1)

Derivation Path: Table 5.1.5.3.3-8				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
value	"0"			
Message-body				
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.1.5.3.3-1A			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.1.5.3.3-9A			

Table 5.1.5.3.3-9A: PIDF in SIP PUBLISH (Table 5.1.5.3.3-9)

Derivation Path: Table 5.1.5.3.3-8A				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence			IETF RFC 3863 [8]	
tuple				
status				
functionalAlias	not present	MCPTT extension	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.3.1	

Table 5.1.5.3.3-10: SIP 200 (OK) to SIP PUBLISH (Steps 6 and 10, Table 5.1.5.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition PUBLISH-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route				Not present

5.1.6 MCPTT Server - MCPTT Server / Configuration / Functional Alias / Functional alias status determination / Activate functional alias / Deactivate functional alias / Participating server

5.1.6.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the IUT (MCPTT server) subscribes to functional alias status determination and to get later
notifications }
    then { IUT (MCPTT Server) sends a SIP SUBSCRIBE message to determine the current status of the
functional alias and sends SIP 200 OK after receipt of SIP NOTIFY informing about the status of the
functional alias  }
            }
```

(2)

```
with { SS (MCPTT Server) having established subscription for user }
ensure that {
  when { the SS-UE1 requests IUT (MCPTT Server) to activate the functional alias for user with SIP
PUBLISH message }
    then { IUT (MCPTT Server) forwards SIP PUBLISH message to SS (MCPTT Server) and replies with SIP
200 OK to SS-UE1 and sends SIP 200 OK after receipt of SIP NOTIFY informing about the functional
alias activation   }
            }
```

(3)

```
with { SS (MCPTT Server) having provided functional alias activation }
ensure that {
  when { the SS-UE1 requests IUT (MCPTT Server) to deactivate the functional alias for user with SIP
PUBLISH message }
    then { IUT (MCPTT Server) forwards SIP PUBLISH message to SS (MCPTT Server) and replies with SIP
200 OK to SS-UE1 and sends SIP 200 OK after receipt of SIP NOTIFY informing about the functional
alias deactivation   }
            }
```

(4)

```
with { SS (MCPTT Server) having provided functional alias deactivation }
ensure that {
  when { the IUT (MCPTT server) de-subscribes from functional alias status changes of the functional
alias }
    then { IUT (MCPTT Server) sends a SIP SUBSCRIBE message and sends SIP 200 OK after receipt of
SIP NOTIFY for subscription termination  }
            }
```

5.1.6.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9A.2.2.2.3, 9A.2.2.2.4, 9A.2.2.2.5, 9A.2.2.2.6, 9A.2.2.2.7, 9A.2.2.2.8, 9A.2.2.2.9. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.2.3]

"Upon receiving a SIP PUBLISH request such that:

1) *Request-URI of the SIP PUBLISH request contains either the public service identity identifying the originating participating MCPTT function serving the MCPTT user, or the public service identity identifying the terminating participating MCPTT function serving the MCPTT user;*

2) *the SIP PUBLISH request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing the<mcptt-request-uri> element which identifies an MCPTT ID served by the MCPTT server;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];*

4) *the Event header field of the SIP PUBLISH request contains the "presence" event type; and*

5) SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-user functional alias information according to clause 9A.3.1;

then the MCPTT server:

1) shall identify the served MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP PUBLISH request;

2) if the Request-URI of the SIP PUBLISH request contains the public service identity identifying the originating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID from public user identity in the P-Asserted-Identity header field of the SIP PUBLISH request;

3) if the Request-URI of the SIP PUBLISH request contains the public service identity identifying the terminating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP PUBLISH request;

4) if the originating MCPTT ID is different than the served MCPTT ID or the originating MCPTT ID is not authorized to modify functional alias status of the served MCPTT ID, shall send a SIP 403 (Forbidden) response and shall not continue with the rest of the steps;

5) if the Expires header field of the SIP PUBLISH request is not included or has nonzero value lower than 4294967295, shall send a SIP 423 (Interval Too Brief) response to the SIP PUBLISH request, where the SIP 423 (Interval Too Brief) response contains a Min-Expires header field set to 4294967295, and shall not continue with the rest of the steps;

6) if the Expires header field of the SIP PUBLISH request has nonzero value, shall determine the candidate expiration interval to according to IETF RFC 3903 [37];

7) if the Expires header field of the SIP PUBLISH request has zero value, shall set the candidate expiration interval to zero;

8) shall respond with SIP 200 (OK) response to the SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37]. In the SIP 200 (OK) response, the MCPTT server:

a) shall set the Expires header field according to IETF RFC 3903 [37], to the candidate expiration time;

9) if the "entity" attribute of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request is different than the served MCPTT ID, shall not continue with the rest of the steps;

10) shall consider an MCPTT user information entry such that:

a) the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9A.2.2.2.2; and

b) the MCPTT ID of the MCPTT user information entry is equal to the served MCPTT ID;

as the served MCPTT user information entry;

11) shall consider a copy of the list of the MCPTT functional alias entries of the served MCPTT user information entry as the served list of the MCPTT functional alias information entries;

12) if the candidate expiration interval is nonzero, shall construct the candidate list of the MCPTT functional alias entries as follows:

a) for each functional alias ID which has a functional alias information entry in the served list of the functional alias information entries, such that the expiration time of the functional alias information entry has not expired yet, and which is indicated in a "functionalAliasID" attribute of a <functionalAlias> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

i) shall copy the functional alias information entry into a new functional alias information entry of the candidate list of the functional alias information entries;

ii) if the functional alias status of the functional alias information entry is "deactivating" or "deactivated", shall set the functional alias status of the new functional alias information entry to the "activated" state and shall reset the activating p-id-fa of the new functional alias information entry; and

iii) shall set the expiration time of the new functional alias information entry to the current time increased with the candidate expiration interval;

b) for each functional alias ID which has a functional alias information entry in the served list of the functional alias information entries, such that the expiration time of the functional alias information entry has not expired yet, and which is not indicated in any "functionalAliasID" attribute of the <functionalAlias> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

i) shall copy the functional alias information entry into a new functional alias information entry of the candidate list of the functional alias information entries; and

ii) if the functional alias status of the functional alias information entry is "activated" or "activating":

- *shall set the functional alias status of the new functional alias entry to the "deactivating" state; and*
- *shall set the expiration time of the new functional alias information entry to the current time increased with twice the value of timer F; and*

c) for each functional alias ID:

i) which does not have a functional alias information entry in the served list of the functional alias entries; or

ii) which has a functional alias information entry in the served list of the functional alias information entries, such that the expiration time of the functional alias information entry has already expired;

and which is indicated in a "functionalAliasID" element of the <functionalAlias> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

i) shall add a new functional alias information entry in the candidate list of the functional alias information list for the functional alias ID;

ii) shall set the functional alias status of the new functional alias information entry to the "activating" state;

iii) shall set the expiration time of the new functional alias information entry to the current time increased with the candidate expiration interval; and

iv) shall reset the activating p-id-fa of the new functional alias information entry;

13) if the candidate expiration interval is zero, constructs the candidate list of the functional alias information entries as follows:

a) for each functional alias ID which has an entry in the served list of the functional alias information entries:

i) shall copy the functional alias entry of the served list of the functional alias information into a new functional alias information entry of the candidate list of the functional alias information entries;

ii) shall set the functional alias status of the new functional alias information entry to the "deactivating" state; and

iii) shall set the expiration time of the new functional alias information entry to the current time increased with twice the value of timer F;

14) shall replace the list of the functional alias information entries stored in the served MCPTT user information entry with the candidate list of the functional alias information entries;

15) shall perform the procedures specified in clause 9A.2.2.2.6 for the served MCPTT ID and each functional alias:

a) which does not have a functional alias information entry in the served list of the functional alias information entries and which has a functional alias information entry in the candidate list of the functional alias information entries with the functional alias status set to the "activating" state;

b) which has a functional alias information entry in the served list of the functional alias information entries with the expiration time already expired, and which has a functional alias information entry in the candidate list of the functional alias information entries with the functional alias status set to the "activating" state;

c) which has a functional alias information entry in the served list of the functional alias information entries with the functional alias status set to the "deactivating" state or the "deactivated" state and with the expiration time not expired yet, and which has an functional alias information entry in the candidate list of the functional alias information entries with the functional alias status set to the "activating" state; or

d) which has a functional alias information entry in the served list of the functional alias information entries with the functional alias status set to the "activated" state and with the expiration time not expired yet, and which has an functional alias information entry in the candidate list of the functional alias information entries with the functional alias status set to the "deactivating" state;

16) shall identify the handled p-id-fa in the <p-id-fa> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

17) shall perform the procedures specified in clause 9A.2.2.2.5 for the served MCPTT ID."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.2.4]

"Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains either the public service identity identifying the originating participating MCPTT function serving the MCPTT user, or the public service identity identifying the terminating participating MCPTT function serving the MCPTT user;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcptt-info+xml MIME body containing:

a) the<mcptt-request-uri> element which identifies an MCPTT ID served by the MCPTT server; and

b) the <mcpttinfo> element with the <mcptt-Params> element containing an <anyExt> element with the <request-type> element set to a value of "functional-alias-status-determination";

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9]; and

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type;

the MCPTT server:

1) shall identify the served MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

2) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the originating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;

3) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the terminating participating MCPTT function serving the MCPTT user, shall identify the originating MCPTT ID in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP SUBSCRIBE request;

4) if the originating MCPTT ID is different than the served MCPTT ID and the originating MCPTT ID is not authorized to modify functional alias status of the served MCPTT ID, shall send a SIP 403 (Forbidden) response and shall not continue with the rest of the steps; and

5) shall generate a SIP 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 6665 [26].

For the duration of the subscription, the MCPTT server shall notify the subscriber about changes of the information of the served MCPTT ID, as described in clause 9A.2.2.2.5."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.2.5]

"In order to notify the subscriber about changes of functional aliases of the served MCPTT ID, the MCPTT server:

1) shall consider an MCPTT user information entry such that:

a) the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9A.2.2.2.2; and

b) the MCPTT ID of the MCPTT user information entry is equal to the served MCPTT ID;

as the served MCPTT user information entry;

2) shall generate an application/pidf+xml MIME body indicating per-user functional alias information according to clause 9A.3.1 and the served list of the MCPTT user information entries with the following clarifications:

a) the MCPTT server shall not include information from functional alias information entry with the expiration time already expired;

b) the MCPTT server shall not include information from a functional alias information entry with the functional alias status set to the "deactivated" state;

c) if this procedures is invoked by procedure in clause 9A.2.2.2.3 where the handled p-id-fa value was identified, the MCPTT server shall set the <p-id-fa> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id-fa value; and

3) send a SIP NOTIFY request according to 3GPP TS 24.229 [4], and IETF RFC 6665 [26] for the subscription created in clause 9A.2.2.2.4. In the SIP NOTIFY request, the MCPTT server shall include the generated application/pidf+xml MIME body indicating per-user functional alias information."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.2.6]

"NOTE 1: Usage of one SIP PUBLISH request to carry information about change of functional alias state of several MCPTT users served by the same MCPTT server is not supported in this version of the specification.

In order:

- *to send an activation request of a served MCPTT ID for a handled functional alias ID;*
- *to send an deactivation request of a served MCPTT ID for a handled functional alias ID;*
- *to send a take over request of a served MCPTT ID for a handled functional alias ID due to take over; or*
- *to send an activation request of a served MCPTT ID for a handled functional alias ID due to near expiration of the previously published information;*

the MCPTT server shall generate a SIP PUBLISH request according to 3GPP TS 24.229 [4], IETF RFC 3903 [37] and IETF RFC 3856 [51]. In the SIP PUBLISH request, the MCPTT server:

1) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the handled functional alias ID;

2) shall include an application/vnd.3gpp.mcptt-info+xml MIME body. In the application/vnd.3gpp.mcptt-info+xml MIME body, the MCPTT server:

a) shall include the <mcptt-request-uri> element set to the handled functional alias ID; and

b) shall include the <mcptt-calling-user-id> element set to the served MCPTT ID;

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];

4) if sending an activation request, shall set the Expires header field according to IETF RFC 3903 [37], to 4294967295;

NOTE 1: 4294967295, which is equal to 232-1, is the highest value defined for Expires header field in IETF RFC 3261 [24].

5) if sending an deactivation request, shall set the Expires header field according to IETF RFC 3903 [37], to zero;

6) shall include an P-Asserted-Identity header field set to the public service identity of the MCPTT server according to 3GPP TS 24.229 [4];

7) shall set the current p-id-fa to a globally unique value;

8) shall consider an MCPTT user information entry such that:

a) the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9A.2.2.2.2; and

b) the MCPTT ID of the MCPTT user information entry is equal to the served MCPTT ID;

as the served MCPTT user information entry;

9) for each functional alias information entry such that:

a) the functional alias information entry has the "activating" functional alias status, the functional alias ID set to the handled functional alias ID, the expiration time has not expired yet and the activating p-id-fa is not set; and

b) the functional alias information entry is in the list of the functional alias information entries of the served MCPTT user information entry;

shall set the activating p-id-fa of the functional alias information entry to the current p-id-fa; and

10) shall include an application/pidf+xml MIME body indicating per-functional alias status information constructed according to clause 9A.3.1.2. The MCPTT server shall indicate all served MCPTT user IDs, such that:

a) the functional alias status is set to "activating" with or without "take-over" element or "activated", and the expiration time has not expired yet in a functional alias information entry with the functional alias ID set to the handled functional alias;

b) the functional alias information entry is in the list of the functional alias information entries of an MCPTT user information entry; and

d) the MCPTT user information entry is a served MCPTT user information entry.

The MCPTT server shall set the <p-id-fa> child element of the <presence> root element to the current p-id-fa.

The MCPTT server shall not include the "expires" attribute in the <functionalAlias> element.

The MCPTT server shall send the SIP PUBLISH request according to 3GPP TS 24.229 [4].

If timer F expires for the SIP PUBLISH request sent for a (de)activation request of served MCPTT ID for the functional alias ID or upon receiving a SIP 3xx, 4xx, 5xx or 6xx response to the SIP PUBLISH request, the MCPTT server:

1) shall remove each functional alias ID entry such that:

a) the functional alias information entry has the functional alias ID set to the handled functional alias ID; and

b) the functional alias information entry is in the list of the functional alias information entries of the served MCPTT user information entry."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.2.7]

"NOTE 1: Usage of one SIP SUBSCRIBE request to subscribe for notification about change of functional alias state of several MCPTT users served by the same MCPTT server is not supported in this version of the specification.

In order to discover whether a served MCPTT user successfully activated a handled functional alias in the MCPTT server owning the functional alias, the MCPTT server shall generate an initial SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 3856 [51], and IETF RFC 6665 [26].

In the SIP SUBSCRIBE request, the MCPTT server:

1) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the handled functional alias;

2) shall include an application/vnd.3gpp.mcptt-info+xml MIME body. In the application/vnd.3gpp.mcptt-info+xml MIME body, the MCPTT server:

a) shall include the <mcptt-request-uri> element set to the handled functional alias ID; and

b) shall include the <mcptt-calling-user-id> element set to the served MCPTT ID;

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];

4) if the MCPTT server wants to receive the current status and later notification, shall set the Expires header field according to IETF RFC 6665 [26], to 4294967295;

NOTE 2: 4294967295, which is equal to 2^{32}-1, is the highest value defined for Expires header field in IETF RFC 3261 [24].

5) if the MCPTT server wants to fetch the current state only, shall set the Expires header field according to IETF RFC 6665 [26], to zero;

6) shall include an Accept header field containing the application/pidf+xml MIME type;

7) shall include an Events header field set to "presence"; and

8) shall include an application/simple-filter+xml MIME body indicating per-user restrictions of presence event package notification information according to clause 9A.3.2, indicating the served MCPTT ID.

In order to re-subscribe or de-subscribe, the MCPTT server shall generate an in-dialog SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 3856 [51], and IETF RFC 6665 [26]. In the SIP SUBSCRIBE request, the MCPTT server:

1) if the MCPTT server wants to receive the current status and later notification, shall set the Expires header field according to IETF RFC 6665 [26], to 4294967295;

NOTE 3: 4294967295, which is equal to 2^{32}-1, is the highest value defined for Expires header field in IETF RFC 3261 [24].

2) if the MCPTT server wants to de-subscribe, shall set the Expires header field according to IETF RFC 6665 [26], to zero;

3) shall include an Events header field set to "presence"; and

4) shall include an Accept header field containing the application/pidf+xml MIME type.

Upon receiving a SIP NOTIFY request according to 3GPP TS 24.229 [4], IETF RFC 3856 [51], and IETF RFC 6665 [26], if SIP NOTIFY request contains an application/pidf+xml MIME body indicating per-functional alias information constructed according to clause 9A.3.1, then the MCPTT server:

1) *for each served MCPTT ID such that the application/pidf+xml MIME body of SIP NOTIFY request contains:*

   a) *a <tuple> element of the root <presence> element;*

   b) *the "id" attribute of the <tuple> element indicating the served MCPTT ID;*

   c) *an <functionalAlias> child element of the <status> element of the <tuple> element; and*

   d) *the "expires" attribute of the <functionalAlias> element indicating expiration of activation of functional alias;*

   perform the following:

   a) *if a functional alias information entry exists such that:*

      i) *the functional alias information entry has the "activating" functional alias status, the functional alias ID set to the handled functional alias ID, and the expiration time has not expired yet;*

      ii) *the functional alias information entry is in the list of the functional alias information entries of an MCPTT user information entry with the MCPTT ID set to the served MCPTT ID; and*

      iii) *the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9A.2.2.2.2;*

      shall set the functional alias status of the functional alias information entry to "activated"; and

      shall set the next publishing time of the functional alias information entry to the current time and half of the time between the current time and the expiration of the functional alias; and

2) *for each functional alias information entry such that:*

   a) *the functional alias information entry has the "activated" functional alias status or the "deactivating" functional alias status, the functional alias ID set to the handled functional alias ID, and the expiration time has not expired yet;*

   b) *the functional alias information entry is in the list of the functional alias information entries of an MCPTT user information entry with the MCPTT ID set to a served MCPTT ID; and*

   c) *the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9A.2.2.2.2; and*

   for which the application/pidf+xml MIME body of SIP NOTIFY request does not contain:

   a) *a <tuple> element of the root <presence> element;*

   b) *the "id" attribute of the <tuple> element indicating the served MCPTT ID; and*

   c) *an <functionalAlias> child element of the <status> child element of the <tuple> element.*

   perform the following:

   a) *shall set the functional alias status of the functional alias information entry to "deactivated"; and*

   b) *shall set the expiration time of the functional alias information entry to the current time; and*

3) *if a <p-id-fa> element is included in the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request, then for each functional alias information entry such that:*

   a) *the functional alias information entry has the "activating" functional alias status, the functional alias ID set to the handled functional alias ID, the expiration time has not expired yet and with the activating p-id-fa set to the value of the <p-id-fa> element;*

b) the functional alias information entry is in the list of the functional alias information entries of an MCPTT user information entry with the MCPTT ID set to a served MCPTT ID; and

d) the MCPTT user information entry is in the list of MCPTT user information entries described in clause 9A.2.2.2.2; and

for which the application/pidf+xml MIME body of SIP NOTIFY request does not contain:

a) a <tuple> element of the root <presence> element;

b) the "id" attribute of the <tuple> element indicating the served MCPTT ID; and

c) an <functionalAlias> child element of the <status> child element of the <tuple> element;

perform the following:

a) shall set the functional alias status of the functional alias information entry to "deactivated"; and

b) shall set the expiration time of the functional alias information entry to the current time."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.2.8]

"In order to discover the MCPTT users that have successfully activated a handled functional alias in the MCPTT server owning the functional alias, the MCPTT server shall generate an initial SIP SUBSCRIBE request according to 3GPP TS 24.229 [4], IETF RFC 3856 [51], and IETF RFC 6665 [26].

In the SIP SUBSCRIBE request, the MCPTT server:

1) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the handled functional alias;

2) shall include an application/vnd.3gpp.mcptt-info+xml MIME body. In the application/vnd.3gpp.mcptt-info+xml MIME body, the MCPTT server shall include the <mcptt-request-uri> element set to the handled functional alias ID;

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9];

4) shall set the Expires header field according to IETF RFC 6665 [26] to zero;

NOTE: if the MCPTT server wants to receive the current status and later notification, can set the Expires header field according to IETF RFC 6665 [26], to 4294967295, which is the highest value defined for Expires header field in IETF RFC 3261 [24].

5) shall include an Accept header field containing the application/pidf+xml MIME type;

6) shall include an Events header field set to "presence"; and

7) shall include an application/simple-filter+xml MIME body indicating per-functional alias restrictions of presence event package notification information indicating the served functional alias."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 9A.2.2.2.9]

"Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the originating participating MCPTT function serving the MCPTT user;

2) the SIP SUBCRIBE request contains an application/vnd.3gpp.mcptt-info MIME body containing the<mcptt-request-uri> element which identifies an MCPTT ID not served by MCPTT server;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service header field according to IETF RFC 6050 [9]; and

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type;

then the MCPTT server:

1) *shall identify the target MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP SUBSCRIBE request;*

2) *shall identify the originating MCPTT ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;*

3) *shall generate a SIP SUBSCRIBE request from the received SIP SUBSCRIBE request. In the generated SIP SUBSCRIBE request, the MCPTT server:*

   a) *shall set the Request-URI to the public service identity identifying the terminating participating MCPTT function serving the target MCPTT ID;*

   b) *shall include a P-Asserted-Identity header field containing the public service identity identifying the originating participating MCPTT function serving the MCPTT user;*

   c) *shall include an application/vnd.3gpp.mcptt-info+xml MIME body. In the application/vnd.3gpp.mcptt-info+xml MIME body, the MCPTT server:*

      A) *shall include the <mcptt-request-uri> element set to the target MCPTT ID; and*

      B) *shall include the <mcptt-calling-user-id> element set to the originating MCPTT ID; and*

   d) *shall include other signalling elements from the received SIP SUBSCRIBE request; and*

4) *shall send the generated SIP SUBSCRIBE request according to 3GPP TS 24.229 [4].*

The MCPTT server shall forward to the originating MCPTT ID any received SIP responses to the SIP SUBSCRIBE request, and for the duration of the subscription any received SIP NOTIFY requests and any received SIP responses to the SIP NOTIFY request according to 3GPP TS 24.229 [4]."

5.1.6.3 Test description

5.1.6.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the controlling MCPTT function.
- SS-UE1 (MCPTT client)

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the participating MCPTT function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2, and to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.

- The SS-UE1 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.

- Group A is controlled by the controlling MCPTT function.
- User A is not configured to be implicitly affiliated to any group.

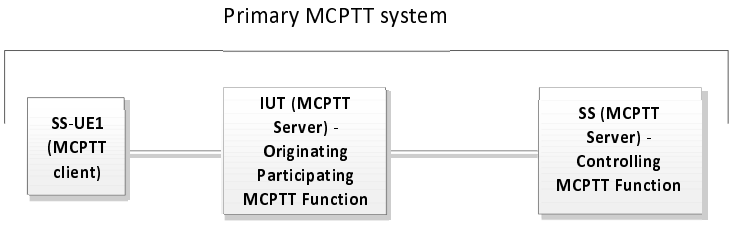


Figure 5.1.6.3.1-1: Functions of the testing components

5.1.6.3.2 Test procedure sequence

Table 5.1.6.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Make the IUT (MCPTT Server) subscribe to functional alias status determination and to get later notification of status changes of a functional alias.	-	-	-	-
2	Check: Does the IUT (MCPTT Server) send a SIP SUBSCRIBE message request to determine the current status of the functional alias?	-->	SIP SUBSCRIBE	1	P
3	The SS (MCPTT Server) responds to the SIP SUBSCRIBE message with a SIP 200 (OK) message	<--	SIP 200 (OK)	-	-
4	The SS (MCPTT Server) sends a SIP NOTIFY message informing about the status of the functional alias.	<--	SIP NOTIFY	-	-
5	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK) message?	-->	SIP 200 (OK)	1	P
6	The SS-UE1 (MCPTT client) sends a SIP PUBLISH message to activate the functional alias for user A.	<--	SIP PUBLISH	-	-
7	Check: Does the IUT (MCPTT Server) send a SIP PUBLISH message to the SS (MCPTT Server) to activate the functional alias for user A?	-->	SIP PUBLISH	2	P
8	The SS (MCPTT Server) responds to the SIP PUBLISH message with a SIP 200 (OK) message.	<--	SIP 200 (OK)		
9	Check: Does the IUT (MCPTT Server) respond to the SIP PUBLISH message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	2	P
10	The SS (MCPTT Server) sends a SIP NOTIFY informing about the functional alias activation.	<--	SIP NOTIFY	-	-
11	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK)?	-->	SIP 200 (OK)	2	P
12	The SS-UE1 (MCPTT client) sends a SIP PUBLISH message to deactivate the functional alias for user A.	<--	SIP PUBLISH	-	-
13	Check: Does the IUT (MCPTT Server) send a SIP PUBLISH message to the SS (MCPTT Server) to deactivate the functional alias for user A?	-->	SIP PUBLISH	3	P
14	The SS (MCPTT Server) responds to the SIP PUBLISH message with a SIP 200 (OK) message.	<--	SIP 200 (OK)		

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
15	Check: Does the IUT (MCPTT Server) respond to the SIP PUBLISH message with a SIP 200 (OK) message?	-->	SIP 200 (OK)	3	P
16	The SS (MCPTT Server) sends a SIP NOTIFY informing about the functional alias deactivation.	<--	SIP NOTIFY	-	-
17	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK)?	-->	SIP 200 (OK)	3	P
18	Make the IUT (MCPTT Server) de-subscribe from functional alias status determination.	-	-	-	-
19	Check: Does the IUT (MCPTT Server) send a SIP SUBSCRIBE message request to de-subscribe from functional alias status changes of the functional alias?	-->	SIP SUBSCRIBE	4	P
20	The SS (MCPTT Server) responds to the SIP SUBSCRIBE message with a SIP 200 (OK) message	<--	SIP 200 (OK)	-	-
21	The SS (MCPTT Server) sends a SIP NOTIFY message indicating the termination of the subscription	<--	SIP NOTIFY	-	-
22	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK) message?	-->	SIP 200 (OK)	4	P

5.1.6.3.3 Specific message contents

Table 5.1.6.3.3-1: SIP SUBSCRIBE from the IUT (MCPTT Server) (Step 2, Table 5.1.6.3.2-1)

Derivation Path: Table 5.1.5.3.3-1

Table 5.1.3.3.3-2: SIP 200 (OK) to SUBSCRIBE (Steps 3, 20, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.5.3.3-2

Table 5.1.6.3.3-3: SIP NOTIFY from the SS (MCPTT Server) (Steps 4 and 16, Table 5.1.6.3.2-1)

Derivation Path: Table 5.1.5.3.3-3

Table 5.1.6.3.3-4: SIP NOTIFY from the SS (MCPTT Server) (Step 21, Table 5.1.6.3.2-1)

Derivation Path: Table 5.1.5.3.3-4

Table 5.1.6.3.3-5: SIP NOTIFY from the SS (MCPTT Server) (Step 10, Table 5.1.6.3.2-1)

Derivation Path: Table 5.1.5.3.3-5

Table 5.1.3.3.3-6: SIP 200 (OK) to NOTIFY (Steps 5, 11, 17 and 22, Table 5.1.3.3.2-1)

Derivation Path: Table 5.1.5.3.3-6

Table 5.1.6.3.3-7: SIP SUBSCRIBE from the IUT (MCPTT Server) (Step 19, Table 5.1.6.3.2-1)

Derivation Path: Table 5.1.5.3.3-7

Table 5.1.6.3.3-8: SIP PUBLISH from the IUT (MCPTT Server) (Step 7, Table 5.1.6.3.2-1)

Derivation Path: Table 5.1.5.3.3-8

Table 5.1.6.3.3-9: SIP PUBLISH from the IUT (MCPTT Server) (Step 13, Table 5.1.6.3.2-1)

Derivation Path: Table 5.1.5.3.3-9

Table 5.1.6.3.3-10: SIP PUBLISH from the SS-UE1 (MCPTT Client) (Step 6, Table 5.1.6.3.2-1)

Derivation Path: Annex A, Table A.3.10.4-1

Table 5.1.6.3.3-11: SIP PUBLISH from the SS-UE1 (MCPTT Client) (Step 12, Table 5.1.6.3.2-1)

Derivation Path: Annex A, Table A.3.10.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
delta-seconds	"0"			

Table 5.1.3.3.3-12: SIP 200 (OK) to PUBLISH (Steps 8 and 14, Table 5.1.3.3.2-1)

Derivation Path: 5.1.5.3.3-10

Table 5.1.3.3.3-13: SIP 200 (OK) to PUBLISH (Steps 9 and 15, Table 5.1.3.3.2-1)

Derivation Path: Annex A, Table A.3.10.4-3

5.2 Group Calls

5.2.1 Pre-arranged Group Call

5.2.1.1 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Automatic Commencement Mode / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / Controlling server

5.2.1.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE1 (MCPTT client) initiates a pre-arranged group call with automatic commencement
mode and implicit floor control }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the call by sending SIP
messages }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
Automatic Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted, Floor Revoke, Floor Deny, Floor Queue Position Info ) by sending messages via the SS (MCPTT
Server) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
Automatic Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) upgrades the call to an emergency group call with implicit floor
request }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up an emergency group call by
sending SIP messages }
            }
```

(4)

```
with { IUT (MCPTT Server) having established an MCPTT emergency group call with implicit floor
request }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted ) by sending messages via the SS (MCPTT Server) }
            }
```

(5)

```
with { IUT (MCPTT Server) having established an MCPTT emergency group call with implicit floor
request }
ensure that {
  when { the SS-UE1 (MCPTT client) downgrades the call to a normal group call without implicit floor
request }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the normal group call by
sending SIP messages }
            }
```

(6)

```
with { IUT (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted ) by sending messages via the SS (MCPTT Server) }
            }
```

(7)

```
with { IUT (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) upgrades the call to imminent-peril group call with implicit
floor request }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the imminent-peril group
call by sending SIP messages }
            }
```

(8)

```
with { IUT (MCPTT Server) having established an MCPTT imminent-peril group call with implicit floor
request }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted ) by sending messages via the SS (MCPTT Server) }
            }
```

(9)

```
with { IUT (MCPTT Server) having established an MCPTT imminent-peril group call with implicit floor
request }
ensure that {
  when { the SS-UE1 (MCPTT client) downgrades the call to a normal group call without implicit floor
request }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the normal group call by
sending SIP messages }
            }
```

(10)

```
with { IUT (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted ) by sending messages via the SS (MCPTT Server) }
            }
```

(11)

```
with { IUT (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the pre-arranged group call }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) by sending a SIP 200 (OK) message
to the client ending the call and sends a SIP BYE message towards SS-UE2 (MCPTT client) }
            }
```

5.2.1.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 10.1.1.4.2, 6.3.3.2.2, 6.3.5.5, 6.3.6, 10.1.1.4.1.1, 6.3.3.1.2, 6.3.3.1.1, 6.3.3.2.3.2, 6.3.3.2.1, 10.1.1.4.7, 10.1.1.4.8, 6.3.3.1.6, 6.3.3.1.9, 6.3.3.1.10, 6.3.3.1.15, 6.3.3.1.19, 10.1.1.4.3, 6.3.3.2.4, 6.3.8.1, 6.3.3.1.5, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 6.3.2.2, 6.3.4.2.2, 6.3.4.3.2, 6.3.4.3.3, 6.3.4.4.2, 6.3.4.4.6, 6.3.4.4.7, 6.3.4.4.12, 6.3.4.5.2, 6.3.4.5.4, 6.3.5.3.3, 6.3.5.3.4, 6.3.5.3.5, 6.3.5.3.10, 6.3.5.4.4, 6.3.5.4.7, 6.3.5.4.8, 6.3.5.5.3, 6.3.5.5.5, 6.3.5.6.2, 6.3.5.6.5. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.2]

"In the procedures in this clause:

1) *MCPTT ID in an incoming SIP INVITE request refers to the MCPTT ID of the originating user from the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;*

2) *group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;*

3) *MCPTT ID in an outgoing SIP INVITE request refers to the MCPTT ID of the called user in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the outgoing SIP INVITE request;*

4) *indication of required group members in a SIP 183 (Session Progress) response refers to the <required> element of the application/vnd.3gpp.mcptt-info+xml MIME body set to "true" in a SIP 183 (Session Progress) sent by the non-controlling MCPTT function of an MCPTT group;*

5) *emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and*

6) *imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.*

Upon receipt of a "SIP INVITE request for controlling MCPTT function of an MCPTT group", the controlling MCPTT function:

...

4) if received SIP INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

...

7) shall perform the actions as described in clause 6.3.3.2.2;

8) shall maintain a local counter of the number of SIP 200 (OK) responses received from invited members and shall initialise this local counter to zero;

9) shall determine if an MCPTT group call for the group identity is already ongoing by determining if an MCPTT session identity has already been allocated for the group call and the MCPTT session is active;

...

14) if the MCPTT group call is not ongoing then:

...

c) if the received SIP INVITE request contains an authorised request for an MCPTT emergency group call as determined by clause 6.3.3.1.13.2 or MCPTT imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCPTT user is eligible to be implicitly affiliated with the MCPTT group as determined in step 14) a) iv) above, shall perform the implicit affiliation as specified in clause 9.2.2.3.7;

d) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [4];

e) shall create a prearranged group session and allocate an MCPTT session identity for the prearranged group call, and shall handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

...

g) if the group identity in the SIP INVITE request for controlling MCPTT function of an MCPTT group is an MCPTT group ID:

i) shall determine the members to invite to the prearranged MCPTT group call as specified in clause 6.3.5.5.

...

iii) if the received SIP INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element set to a value of "true":

A) shall cache the information that the MCPTT user has initiated an MCPTT emergency call;

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCPTT emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that the MCPTT user has initiated an MCPTT emergency alert; and

C) if the in-progress emergency state of the group is set to a value of "false":

I) shall set the value of the in-progress emergency state of the group to "true"; and

II) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iv) if the in-progress emergency state of the group is set to a value of "false" and if the received SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCPTT function shall:

A) shall cache the information that the MCPTT user has initiated an MCPTT imminent peril call; and

B) if the in-progress imminent peril state of the group is set to a value of "false", shall set the in-progress imminent peril state of the group to a value of "true";

v) shall invite each group member determined in step 13)g)i) above, to the group session, as specified in clause 10.1.1.4.1.1; and

vi) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3; and

...

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 10.1.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [34], the timer TNG1 (acknowledged call setup timer) is not running, the controlling MCPTT function supports media buffering and the SIP final response is not yet sent to the inviting MCPTT client:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4,4 in the SIP 200 (OK) response, if the prearranged MCPTT group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [31];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) shall include a P-Answer-State header field with the value "Unconfirmed";

...

7) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 9: Resulting user plane processing is completed before the next step is performed.

8) shall send the SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4];

...

Upon:

1) receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 10.1.1.4.1;

2) the timer TNG1 (acknowledged call setup timer) is not running;

3) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document;

4) the controlling MCPTT function supports media buffering; and

5) the SIP final response has not yet been sent to the inviting MCPTT client;

the controlling MCPTT function according to local policy:

1) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCPTT group has exceeded the <max-participant-count> members as specified in 3GPP TS 24.481 [31];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

...

6) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 13: Resulting media plane processing is completed before the next step is performed.

7) shall send a SIP 200 (OK) response to the inviting MCPTT client according to 3GPP TS 24.229 [4];

..."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.2]

"On receipt of an initial SIP INVITE request the controlling MCPTT function shall cache SIP feature tags, if received in the Contact header field and if the specific feature tags are supported."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.5]

"The MCPTT server shall only invite affiliated group members to a group session.

If the group is not a regroup based on a preconfigured group, the MCPTT server determines the affiliated members from the entries contained in the <list> element of the group document by following the procedures specified in clause 6.3.6.

...

If the number of members of the MCPTT group exceeds the value contained in the <on-network-max-participant-count> element the MCPTT server shall invite only <on-network-max-participant-count> members from the list, but shall prioritise inviting those group members to the group session that have an <entry> element in the <list> element with a <on-network-required> element present.

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the group session. The <on-network-required> element is used to determine which group members need to acknowledge the group call before audio transmission can proceed.

NOTE 3: Other requirements for how the controlling MCPTT function selects which of the <on-network-max-participant-count> members to invite is outside the scope of this specification.

NOTE 4: It is assumed that validation checks are performed at the group management server to ensure that the <on-network-max-participant-count> cannot be less than the number of <on-network-required> users."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.6]

"The MCPTT server checks if an MCPTT user is affiliated to an MCPTT group at an MCPTT client by following the procedures specified below:

1. *the MCPTT server shall find the applicable MCPTT group information entry as an MCPTT group information entry of the list of MCPTT group information entries described in clause 9.2.2.3.2, such that the MCPTT group ID of the MCPTT group information entry is equal to the MCPTT group identity of the MCPTT group. If the applicable MCPTT group information entry cannot be found, then the MCPTT server shall determine that the MCPTT user is not affiliated to the MCPTT group at the MCPTT client and the MCPTT server shall not continue with rest of the steps;*

2. *the MCPTT server shall find the applicable MCPTT user information entry as an MCPTT user information entry of the list of MCPTT user information entries of the applicable MCPTT group information entry, such that the MCPTT ID of the MCPTT user information entry is equal to the MCPTT ID of the MCPTT user. If the applicable MCPTT user information entry cannot be found, then the MCPTT server shall determine that the MCPTT user is not affiliated to the MCPTT group at the MCPTT client and the MCPTT server shall not continue with rest of the steps;*

3. *if the MCPTT client ID of the MCPTT client cannot be found in the list of MCPTT client information entries of the applicable MCPTT user information entry, then the MCPTT server shall determine that the MCPTT user is not affiliated to the MCPTT group at the MCPTT client and the MCPTT server shall not continue with rest of the steps;*

NOTE: the MCPTT client ID of the originating MCPTT client can be found in the <mcptt-client-id> element contained in the application/vnd.3gpp.mcptt-info+xml MIME body of a SIP INVITE request, SIP REFER request or SIP MESSAGE request originated by the MCPTT client.

4. *if the expiration time of the applicable MCPTT user information entry has been reached, then the MCPTT server shall determine that the MCPTT user is not affiliated to the MCPTT group at the MCPTT client and the MCPTT server shall not continue with rest of the steps; and*

5. *the MCPTT server shall determine that the MCPTT user is affiliated to the MCPTT group at the MCPTT client."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.1.1]

"This clause describes the procedures for inviting an MCPTT user to an MCPTT session. The procedure is initiated by the controlling MCPTT function as the result of an action in clause 10.1.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the terminating participating MCPTT function associated to the MCPTT user to be invited.;

NOTE 1: How the controlling MCPTT function finds the address of the terminating MCPTT participating function is out of the scope of the current release.

NOTE 2: If the terminating MCPTT user is part of a partner MCPTT system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCPTT function.

3) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCPTT function;

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcptt-request-uri> element set to the MCPTT ID of the terminating user; and

b) the <mcptt-calling-group-id> element set to the group identity;

NOTE 3: The <mcptt-calling-user-id> is already included in the MIME body as a result of calling clause 6.3.3.1.2 in step 1).

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

6) if the in-progress emergency state of the group is set to a value of "true" the controlling MCPTT function:

a) shall include a Resource-Priority header field populated with the values for an MCPTT emergency group call as specified in clause 6.3.3.1.19;

b) if the received SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "true":

i) shall include in the outgoing SIP INVITE request in the application/vnd.3gpp.mcptt-info+xml MIME body an <emergency-ind> element set to a value of "true"; and

ii) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCPTT user and MCPTT group are authorised for the initiation of MCPTT emergency alerts as determined by the procedures of clause 6.3.3.1.13.1, shall populate the application/vnd.3gpp.mcptt-info+xml MIME body and the application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause 6.3.3.1.12. Otherwise, shall set the <alert-ind> element to a value of "false"; and

c) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCPTT function:

a) shall include a Resource-Priority header field populated with the values for an MCPTT imminent peril group call as specified in clause 6.3.3.1.19; and

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true";

8) void

9) shall send the SIP INVITE request towards the terminating network in accordance with 3GPP TS 24.229 [4].

Upon receiving a SIP 183 (Session Progress) response containing a Require header field with the option tag "100rel" and containing a P-Answer-State header field with the value "Unconfirmed" in response to the SIP INVITE request the controlling MCPTT function:

1) shall send a SIP PRACK request towards the MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

...

3) shall increment the local counter of the number of SIP 200 (OK) responses received from invited members, by 1.

..."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.2]

"This clause is referenced from other procedures.

The controlling MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4].

The controlling MCPTT function:

1) shall include in the Contact header field an MCPTT session identity for the MCPTT session with the g.3gpp.mcptt media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" according to IETF RFC 3840 [16];

2) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [9] in the SIP INVITE request;

4) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

5) shall include a Referred-By header field with the public user identity of the inviting MCPTT client;

6) should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7]. The refresher parameter shall be omitted;

7) shall include the Supported header field set to "timer";

...

10) if the received SIP INVITE request did not contain an application/vnd.3gpp.mcptt-info+xml MIME body containing an <ambient-listening-type> element and did not contain a Priv-Answer-Mode header field, shall include an unmodified Answer-Mode header field if present in the incoming SIP INVITE request; and

11) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing INVITE request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the controlling MCPTT function:

1) when initiating a new MCPTT session the SDP offer;

a) shall contain only one SDP media-level section for MCPTT speech media stream as contained in the received SDP offer; and

b) shall contain one SDP media-level section for media- plane control messages, if present in the received SDP offer; and

2) when adding a new MCPTT user to an existing MCPTT Session, the SDP offer shall contain the media stream currently used in the MCPTT session.

When composing the SDP offer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the controlling MCPTT function;

2) for the MCPTT speech media stream, shall include all media-level attributes from the received SDP offer;

3) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the controlling MCPTT function; and

4) for the offered media plane control channel, shall include the offered media planecontrol channel 'fmtp' attributes as specified in 3GPP TS 24.380 [5] clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.3.2]

"When sending a SIP 200 (OK) response to the initial SIP INVITE request, the controlling MCPTT function:

1) shall generate the SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [4];

2) shall include the Session-Expires header field and start supervising the SIP session according to rules and procedures of IETF RFC 4028 [7], "UAS Behavior". The "refresher" parameter in the Session-Expires header field shall be set to "uac";

3) shall include the option tag "timer" in a Require header field;

4) shall include a P-Asserted-Identity header field with the public service identity of the controlling MCPTT function;

5) shall include a SIP URI for the MCPTT session identity in the Contact header field identifying the MCPTT session at the controlling MCPTT function;

6) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and

c) the isfocus media feature tag;

7) shall include Warning header field(s) received in incoming responses to the SIP INVITE request;

8) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];

9) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [22];

10) shall include the "explicitsub" and "nosub" option tags in a Supported header field according to IETF RFC 7614 [35]; and

11) void

12) shall interact with the media plane as specified in 3GPP TS 24.380 [5]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) for the accepted media stream in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCPTT function; and

2) for the accepted media channel control channel, if present in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCPTT function, for the accepted media plane control channel, if present in the received SDP offer; and

b) shall include 'fmtp' attributes as specified in 3GPP TS 24.380 clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.7]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a SIP re-INVITE request for an MCPTT session identity identifying an on-demand prearranged MCPTT group session, the controlling MCPTT function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP re-INVITE request with a SIP 500 (Server Internal Error) response. The controlling MCPTT function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [24] and skip the rest of the steps;

NOTE 1: If the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "true", the controlling MCPTT function can choose to accept the request.

2) if received SIP re-INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

...

6) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "true" and is an authorised request to initiate an MCPTT emergency group call as determined by clause 6.3.3.1.13.2, the controlling MCPTT function shall:

i) shall cache the MCPTT ID of the MCPTT user that has initiated an MCPTT emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCPTT emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the MCPTT ID of the MCPTT user that has initiated an MCPTT emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCPTT user's emergency indication as specified in clause 6.3.3.1.11, setting the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true";

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4]; and

C) if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false"; and

iv) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

NOTE 2: The interactions of TNG2 with the TNG3 (group call timer) are explained in clause 6.3.3.5.2.

C) shall generate SIP re-INVITE requests for the MCPTT emergency group call to the other participants of the MCPTT group call as specified in clause 6.3.3.1.6;

D) shall send the SIP re-INVITEs towards the other participants of the MCPTT group call; and

E) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5];

…

8) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is determined to be an authorised request for an MCPTT emergency call cancellation as specified in clause 6.3.3.1.16 and the in-progress emergency state of the group to is set to a value of "true" the controlling MCPTT function:

a) shall set the in-progress emergency group state of the group to a value of "false";

b) shall clear the cache of the MCPTT ID of the MCPTT user as having an outstanding MCPTT emergency group call;

c) if an <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body is included and set to "false" and is determined to be an authorised request for an MCPTT emergency alert cancellation as specified in clause 6.3.3.1.13.3 and there is an outstanding MCPTT emergency alert for the MCPTT user shall:

i) if the received SIP re-INVITE request contains an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, clear the cache of the MCPTT ID of the MCPTT user identified by the <originated-by> element as having an outstanding MCPTT emergency alert; or

ii) if the received SIP re-INVITE request does not contain an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, clear the cache of the MCPTT ID of the sender of the SIP re-INVITE request as having an outstanding MCPTT emergency alert;

d) shall generate SIP re-INVITE requests to the participants in the group call as specified in clause 6.3.3.1.6. The MCPTT controlling function:

i) for each of the other participants in the group call shall send the SIP re-INVITE request towards the MCPTT client as specified in 3GPP TS 24.229 [4]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5];

NOTE 3: Clause 6.3.3.1.6 will inform the group call participants of the cancellation of the MCPTT group's in-progress emergency state and the cancellation of the MCPTT emergency alert if applicable.

e) shall stop timer TNG2 (in-progress emergency group call timer); and

NOTE 4: The interactions of TNG2 with the TNG3 (group call timer) are explained in clause 6.3.3.5.2;

f) for each of the affiliated members of the group that are not participating in the call:

i) generate a SIP MESSAGE request notification of the cancellation of the MCPTT user's emergency call as specified in clause 6.3.3.1.11;

ii) set the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "false";

iii) if indicated above in step 8) c), set the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "false"; and

iv) send the SIP MESSAGE request according to 3GPP TS 24.229 [4];

9) if the received SIP re-INVITE request contains an imminent peril indication and the in-progress emergency group state of the group is set to a value of "false", shall perform the procedures specified in clause 10.1.1.4.8 and skip the rest of the steps.

Upon receiving a SIP 200 (OK) response to a SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5];

1) shall generate a SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [4];

2) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.3.3.2.1;

3) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [22];

4) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [23];

…

8) shall interact with media plane as specified in 3GPP TS 24.380 [5]; and

9) shall send the SIP 200 (OK) response towards the MCPTT client according to 3GPP TS 24.229 [4].

...

Upon receipt of a SIP 2xx response for an outgoing SIP MESSAGE request, shall handle according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.8]

"This procedure is initiated by the controlling MCPTT function as the result of an action in clause 10.1.1.4.7.

In the procedures in this clause:

1) imminent peril indication in an incoming SIP re-INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

When the controlling function receives a SIP re-INVITE request with an imminent peril indication set to "true", the controlling function:

1) if the in-progress emergency state of the group is set to a value of "false" and if the SIP re-INVITE request contains an imminent peril indication set to a value of "true" or the in-progress imminent peril state of the group to "true", the controlling MCPTT function shall:

NOTE: 1 The calling procedure has already determined that this is not an unauthorised request for an MCPTT imminent peril call, therefore that check does not need to be repeated in the current procedure.

a) if the in-progress imminent peril state of the group is set to a value of "true" and the MCPTT user is indicating a new imminent peril indication:

i) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCPTT user's imminent peril indication as specified in clause 6.3.3.1.11 with the following clarifications;

A) set the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4];

b) if the in-progress imminent peril state of the group is set to a value of "false";

i) set the value of the in-progress imminent peril state of the group to "true";

ii) generate SIP re-INVITE requests for the MCPTT imminent peril group call to participants in the MCPTT group call as specified in clause 6.3.3.1.15;

iii) send the SIP re-INVITES to all of the other participants in the MCPTT group call;

iv) for each of the affiliated members of the group not participating in the group call, generate a SIP MESSAGE request notification of the MCPTT user's imminent peril indication as specified in clause 6.3.3.1.11 with the following clarifications;

A) set the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4]; and

c) cache the information that the MCPTT user has initiated an MCPTT imminent peril call;

...

3) if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "false" and is determined to be an authorised request for an MCPTT imminent peril call cancellation as specified in clause 6.3.3.1.13.6 and the in-progress imminent peril state of the group to is set to a value of "true" the controlling MCPTT function shall:

a) set the in-progress imminent peril state of the group to a value of "false";

b) cache the information that the MCPTT user no longer has an outstanding MCPTT imminent peril group call;

c) generate SIP re-INVITES requests to the other participants in the MCPTT group call as specified in clause 6.3.3.1.15. The MCPTT controlling function:

i) for each participant shall send the SIP re-INVITE request towards the MCPTT client as specified in 3GPP TS 24.229 [4]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function interact with the media plane as specified in 3GPP TS 24.380 [5]; and

NOTE 2: Clause 6.3.3.1.15 will inform the affiliated and joined members of the cancellation of the MCPTT group's in-progress emergency state and the cancellation of the MCPTT emergency alert if applicable.

d) for each of the affiliated members of the group not participating in the call shall:

i) generate a SIP MESSAGE request notification of the cancellation of the MCPTT user's imminent peril call as specified in clause 6.3.3.1.11;

ii) set the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "false"; and

iii) send the SIP MESSAGE request according to 3GPP TS 24.229 [4];

4) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.3.3.2.1;

5) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [22];

6) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [23];

7) shall interact with media plane as specified in 3GPP TS 24.380 [5]; and

8) shall send the SIP 200 (OK) response towards the MCPTT client according to 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response for an outgoing SIP MESSAGE request, shall handle according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.6]

"This clause is referenced from other procedures.

The controlling MCPTT function shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [4].

The controlling MCPTT function:

1) shall include an SDP offer with the media parameters as currently established with the terminating MCPTT client according to 3GPP TS 24.229 [4];

2) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcptt-calling-user-id> element set to the MCPTT ID of the initiating MCPTT user;

3) if the in-progress emergency group state of the group is set to a value of "true" the controlling MCPTT function:

a) shall include a Resource-Priority header field with the namespace populated with the values for an MCPTT emergency group call as specified in clause 6.3.3.1.19;

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body the <emergency-ind> element set to a value of "true";

c) if the <alert-ind> element is set to "true" in the received SIP re-INVITE request and MCPTT emergency alerts are authorised for this group and MCPTT user as determined by the procedures of clause 6.3.3.1.13.1, shall populate the application/vnd.3gpp.mcptt-info+xml MIME body and application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause 6.3.3.1.12. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcptt-info+xml MIME body; and

d) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <imminentperil-ind> element set to a value of "false"; and

NOTE: If the imminent peril state of the group is true at this point, the controlling function will be setting it to false as part of the calling procedure. This is in effect an upgrade of an MCPTT imminent peril group call to an MCPTT emergency group call.

4) if the in-progress emergency group state of the group is set to a value of "false":

a) shall include a Resource-Priority header field populated with the values for a normal MCPTT group call as specified in clause 6.3.3.1.19; and

b) if the received SIP re-INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "false" and this is an authorised request to cancel an MCPTT emergency group call as determined by the procedures of clause 6.3.3.1.13.4:

i) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "false"; and

ii) if the received SIP re-INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "false" and this is an authorised request to cancel an MCPTT emergency alert as determined by the procedures of clause 6.3.3.1.15, shall:

A) include in the application/vnd.3gpp.mcptt-info+xml MIME body an <alert-ind> element set to a value of "false"; and

B) if the received SIP request contains an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, copy the contents of the received <originated-by> element to an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP re-INVITE request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.9]

"This clause is referenced from other procedures.

This clause describes the procedures for generating a SIP re-INVITE request to be sent by the controlling MCPTT function.

The controlling MCPTT function:

1) shall generate an SIP re-INVITE request according to 3GPP TS 24.229 [4]; and

2) shall include an SDP offer with the media parameters as currently established with the terminating MCPTT client according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.3.3.1.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.10]

"This clause is referenced from other procedures.

This clause describes the procedures for generating a SIP re-INVITE request to cancel the in-progress emergency state of an MCPTT group. The procedure is initiated by the controlling MCPTT function when it determines the cancellation of the in-progress emergency state of an MCPTT group is required.

The controlling MCPTT function:

1) shall generate a SIP re-INVITE request as specified in 3GPP TS 24.229 [4] with the clarifications specified in clause 6.3.3.1.9;

2) shall include a Resource-Priority header field populated with the values for a normal MCPTT group call as specified in clause 6.3.3.1.19; and

3) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "false"."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.15]

"This clause is referenced from other procedures.

The controlling MCPTT function shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [4].

The controlling MCPTT function:

1) shall include in the Contact header field an MCPTT session identity for the MCPTT session with the g.3gpp.mcptt media feature tag and the isfocus media feature tag according to IETF RFC 3840 [16];

2) shall include an SDP offer with the media parameters as currently established with the terminating MCPTT client according to 3GPP TS 24.229 [4];

3) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcptt-calling-user-id> element set to the MCPTT ID of the initiating MCPTT user;

4) if the in-progress imminent peril state of the group is set to a value of "true" the controlling MCPTT function:

a) shall include a Resource-Priority header field populated with the values for an MCPTT imminent peril group call as specified in clause 6.3.3.1.19; and

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <imminentperil-ind> element set to a value of "true"; and

5) if the in-progress imminent peril state of the group is set to a value of "false":

a) shall include a Resource-Priority header field populated with the values for a normal MCPTT group call as specified in clause 6.3.3.1.19; and

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <emergency-ind> element set to a value of "false" and the <imminentperil-ind> element set to a value of "false"."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.19]

"This clause is referenced from other procedures.

When determining the Resource-Priority header field namespace and priority values as specified in IETF RFC 8101 [48] for an MCPTT emergency group call or MCPTT emergency private call the controlling MCPTT function:

1) shall retrieve the value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCPTT service configuration document (see the service configuration document in 3GPP TS 24.484 [50]); and

2) *shall retrieve the value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCPTT service configuration document (see the service configuration document in 3GPP TS 24.484 [50]).*

When determining the Resource-Priority header field namespace and priority values as specified in IETF RFC 8101 [48] for an MCPTT imminent peril group call the controlling MCPTT function:

1) *shall retrieve the value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCPTT service configuration document (see the service configuration document in 3GPP TS 24.484 [50]); and*

2) *shall retrieve the value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCPTT service configuration document (see the service configuration document in 3GPP TS 24.484 [50])*

When determining the Resource-Priority header field namespace and priority values as specified in IETF RFC 8101 [48] for a normal MCPTT group or private call the controlling MCPTT function:

1) *shall retrieve the value of the <resource-priority-namespace> element contained in the <normal-resource-priority> element contained in the <OnNetwork> element of the MCPTT service configuration document (see the service configuration document in 3GPP TS 24.484 [50]); and*

2) *shall retrieve the value of the <resource-priority-priority> element contained in the <normal-resource-priority> element contained in the <OnNetwork> element of the MCPTT service configuration document (see the service configuration document in 3GPP TS 24.484 [50]).*

NOTE: The "normal" Resource-Priority header field value is needed to return to a normal priority value from a priority value adjusted for an MCPTT emergency group or private call or an MCPTT imminent peril group call. The "normal" priority received from the EPS by use of the "normal" Resource-Priority header field value is expected to be the same as the "normal" priority received from the EPS when initiating a call with no Resource-Priority header field included."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.3]

"Upon receiving a SIP BYE request the controlling MCPTT function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCPTT function:

1) *shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing the media plane resource associated with the SIP session towards the MCPTT client;*

…

2) *shall generate a SIP 200 (OK) response and send the SIP response towards the MCPTT client according to 3GPP TS 24.229 [4];*

3) *shall check the MCPTT session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCPTT session needs to be released for each participant of the MCPTT session;*

4) *if release of the MCPTT session is required:*

   a) *shall perform the procedures as specified in the clause 6.3.3.1.5 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and*

...

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT participant."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.8.1]

"If:

1) *the call is a pre-arranged group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 24.380 [5] expired;*

2) *there are only one or no participants in the MCPTT session;*

3) *the call is a pre-arranged group call and if it is according to local policy, the initiator of the group call leaves the MCPTT session;*

4) *the minimum number of affiliated MCPTT group members is not present;*

5) *timer TNG3 (group call timer) expires; or*

6) *the call is a broadcast group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 24.380 [5] expired;*

the controlling MCPTT function shall release the MCPTT session for the group call."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.5]

"When a participant needs to be removed from the MCPTT session, the controlling MCPTT function:

1) *shall interact with the media plane as specified in 3GPP TS 24.380 [5] for the MCPTT session release;*

2) *shall generate a SIP BYE request according to 3GPP TS 24.229 [4]; and*

3) *shall send the SIP BYE request to the MCPTT clients according to 3GPP TS 24.229 [4].*

If timer TNG3 (group call timer) has not expired, then when the last MCPTT client is removed from the MCPTT session, the controlling MCPTT function shall stop timer TNG3 (group call timer).

When the MCPTT group session needs to be released, the controlling MCPTT function shall send SIP BYE requests as described in this clause, to all participants of the group session.

Upon receiving a SIP 200 (OK) response to a SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT clients."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.2.2]

"When an MCPTT call is established a new instance of the floor control server state machine for 'general floor control operation' is created.

For each MCPTT client added to the MCPTT call, a new instance of the floor control server state machine for 'basic floor control operation towards the floor participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the floor control server could queue the implicit floor control request for the media-floor control entity.

...

Before the floor control server sends the first floor control message in the MCPTT call, the floor control server has to assign itself a SSRC identifier to be included in media floor control messages and quality feedback messages if the MCPTT server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The floor participant and the floor control server can negotiate the maximum priority level that the floor participant is permitted to request. The floor control server can pre-empt the current sender based on the negotiated maximum priority level that the floor participant is permitted to request and the priority level included in the Floor Request message.

NOTE: The maximum priority level that a floor participant can use is negotiated as specified in subclause 14.3.3 and is based on group configuration data retrieved by the controlling MCPTT function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCPTT function from the configuration management server as described in 3GPP TS 24.484 [13].

...

The floor participant and the floor control server can negotiate queueing of floor requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the floor control server queues the floor control request if a Floor Request message is received when another floor participant has the floor and the priority of the current speaker is the same or higher. Queueing is not permitted for groups configured for audio cut-in floor control."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.2.2]

"When an MCPTT call is initiated as specified in 3GPP TS 24.379 [2] and

1. *if a confirmed indication is required and at least one invited MCPTT client has accepted the invitation;*

...then the floor control arbitration logic in the floor control server:

1. *shall create an instance of the 'general floor control operation' state machine;*
2. *shall wait for the 'basic floor control operation towards the floor participant' to be initialized before continuing the following steps;*
3. *when the 'basic floor control operation towards the floor participant' state machine is initialized and the initialised session is not a temporary group session:*

...

   b. *if the "mc_granted" fmtp attribute is negotiated as specified in clause 14 and this is not an ambient listening call, shall enter the 'G: Floor Taken' state as specified in subclause 6.3.4.4.2; and*

..."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCPTT client negotiated support of queueing floor requests as described in clause 14, and the state machine specified in subclause 6.3.6 does not exist, the floor control arbitration logic in the floor control server:

1. *if there is a Track Info field associated with the floor control server state transition diagram for 'general floor control operation' stored, shall remove the Track Info field from the storage;*
2. *if the active floor request queue is empty the floor control server:*
   a. *shall send Floor Idle message to all floor participants. The Floor Idle message:*
      i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
      ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications;*
   b. *shall start timer T7 (Floor Idle) and initialise counter C7 (Floor Idle) to 1;*
   c. *shall start timer T4 (Inactivity); and*
   d. *shall set the general state to the 'G: Floor Idle' state; and*
3. *if the active floor request queue is not empty the floor control server:*
   a. *shall select a queued floor request from the top of the active floor request queue;*
   b. *shall remove that queued floor request from the active floor request queue;*

c. if the queued floor request includes a Track Info field, shall store the Track Info field and associate it with the floor control server state transition diagram for 'general floor control operation'; and

d. shall enter the 'G: Floor Taken' state as specified in the subclause 6.3.4.4.2 with respect to that floor participant.

..."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.3.3]

"Upon receiving a floor request message (from a floor participant that is permitted to make a floor request) the floor control arbitration logic in the floor control server:

...

3. if the floor request is granted the floor control server:

...

c. shall store the SSRC of floor participant granted the permission to send media until the floor is released associated to that floor request;

d. if a Track Info field is included in the Floor Request message, shall store the received Track Info field, and

e. shall enter the 'G: Floor Taken' state as specified in the subclause 6.3.4.4.2."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.2]

"When entering this state the floor control arbitration logic in the floor control server:

1. shall send a Floor Granted message to the floor participant to which the floor is granted. The Floor Granted message:

a. shall include the value of timer T2 (Stop talking)in the Duration field;

b. shall include the granted priority in the Floor priority field;

c. if a Track Info field associated with the floor control server state transition diagram for 'general floor control operation' is stored, shall include the stored Track Info field;

...

3. shall send Floor Taken message to all other floor participants. The Floor Taken message:

a. if the floor is currently granted only to one particpant:

i shall include the granted MCPTT user's MCPTT ID in the Granted Party's Identity field, if privacy is not requested;

ii. may include the functional alias of the granted MCPTT user in the Functional Alias field, if privacy is not requested; and

iii. shall include the Location field:

A. if the current location of the talker is not available or is not to be reported according to the MCPTT user profile, then the location type is set to '0' (Not provided); or

B. if the current location of the talker is available and may be reported according to the MCPTT user profile, then the location type and location value are set as specified in table 8.2.3.21-3;

…

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

…

e. if the session is not a broadcast group call, may include the Permission to Request the Floor field set to '1'; and

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications;

..

5. shall set the general state to 'G: Floor Taken' state; and

..."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.6]

"Upon receiving a Floor Release message the floor control arbitration logic in the floor control server:

1. shall request the media distributor in the MCPTT server to stop forwarding RTP media packets received from the participant that sent the Floor Release message;

...

4. if mutli-talker floor control is not supported shall enter the 'G: Floor Idle' state as specified in the subclause 6.3.4.3.2; and

..."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.7]

"If the group is not configured to support multi-talker feature control, on receipt of a floor request message with effective priority indicating pre-emptive priority, and if the effective priority of the floor participant with permission to send media is not the pre-emptive priority, or if the group is configured for audio cut-in, the floor control arbitration logic in the floor control server:

1. based on local policy, select one of the following options:

a. revoke the current speaker; or

...

2. if revoking current speaker is selected:

...

c. shall include a Reject Cause field with the <Reject Cause> value set to #4 (Media Burst pre-empted) in the Floor Revoke message sent in subclause 6.3.4.5.2;

d. shall enter the 'G: pending Floor Revoke' state as specified in the subclause 6.3.4.5.2;

e. shall insert the floor participant into the active floor request queue to the position in front of all queued requests, if not inserted yet or update the position of the floor participant in the active floor request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Floor Queue Position Info message to the requesting floor participant, if negotiated support of queueing of floor requests as specified in clause 14. The Floor Queue Position Info message:

i. include the queue position and floor priority in the Queue Info field; and

ii. if the Floor Request message included a Track Info field, shall include the received Track Info field; and

..."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.12]

"Upon receiving an implicit floor request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the floor control arbitration logic in the floor control server:

...

3. *shall set the Reject Cause field in the Floor Revoke message to #4 (Media Burst pre-empted);*
4. *shall enter the 'G: pending Floor Revoke' state as specified in the subclause 6.3.4.5.2;*
5. *shall insert the floor participant into the active floor request queue to the position in front of all queued requests, if not inserted yet or update the position of the floor participant in the active floor request queue to the position in front of all other queued requests, if already inserted; and*
6. *shall send a Floor Queue Position Info message to the requesting floor participant, if negotiated support of queueing floor requests as specified in clause 14. The Floor Queue Position Request message:*
    a. *shall include the queue position and floor priority in the Queue Info field; and*
    b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.5.2]

"When entering this state the floor control arbitration logic in the floor control server:

1. *shall send the Floor Revoke message to the permitted floor participant. The Floor Revoke message:*
    a. *shall include the reason for sending the Floor Revoke message in the <Reject Cause> value in the Reject Cause field; and*
    b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, group is configured as multi-talker group, or a temporary group session, shall include the Floor Indicator field with appropriate indications;*
2. *shall start timer T3 (Stop talking grace) for which a Floor Revoke message has been sent; and*
3. *shall set the general state to 'G: pending Floor Revoke'."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.5.4]

"Upon receiving a Floor Release message for which a Floor Revoke message has been sent, the floor control arbitration logic in the floor control server:

1. *shall request the media distributor in the MCPTT server to stop forwarding RTP media packets;*

…

6. *if the active floor request queue is not empty the floor control server shall enter the 'G: Idle' state as specified in the subclause 6.3.4.3.2;*
7. *if the active floor request queue is empty and the list of currently granted talkers is empty the floor control server shall enter the 'G: Idle' state as specified in the subclause 6.3.4.3.2; and*
8. *if the active floor request queue is empty and the list of currently granted talkers is not empty the floor control server shall enter the 'G: Floor Taken' state as specified in subclause 6.3.4.4.2.*

..."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.3.3]

"When a Floor Taken message is received from the floor control server arbitration logic, the floor control interface towards the MCPTT client in the floor control server:

1. *shall forward the Floor Taken message to the associated floor participant;*

2. *may set the first bit in the subtype of the Floor Taken message to '1' (Acknowledgment is required) as described in subclause 8.2.2, and*

NOTE: It is an implementation option to handle the receipt of the Floor Ack message and what action to take if the Floor Ack message is not received.

3. *shall enter the 'U: not permitted and Floor Taken' state as specified in the subclause 6.3.5.4.2."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.3.4]

"Upon receiving a Floor Request message from the associated floor participant, the floor control interface towards the MCPTT client in the floor control server:

1. *if the session is not a broadcast group call or if the session is a broadcast group call and the associated floor participant is the initiator of the broadcast group call, shall forward the Floor Request message to the floor control server arbitration logic;*

...

3. *shall remain in the 'U: not permitted and Floor Idle' state."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.3.5]

"When a Floor Granted message is received from the floor control arbitration logic in the MCPTT server, the floor control interface towards the MCPTT client in the floor control server:

1. *shall forward the Floor Granted messages to the associated floor participant;*

2. *may set the first bit in the subtype of the Floor Granted message to '1' (Acknowledgment is required) as described in subclause 8.2.2; and*

NOTE: It is an implementation option to handle the receipt of the Floor Ack message and what action to take if the Floor Ack message is not received.

3. *shall enter the state 'U: permitted' as specified in subclause 6.3.5.5.2."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.3.10]

"When receiving a Floor Idle message from the floor control server arbitration logic in the MCPTT server, the floor control interface towards the MCPTT client in the floor control server:

1. *shall forward the Floor Idle message to the associated floor participant;*

2. *may set the first bit in the subtype of the Floor Idle message to '1' (Acknowledgment is required) as described in subclause 8.2.2; and*

NOTE: It is an implementation option to handle the receipt of the Floor Ack message and what action to take if the Floor Ack message is not received.

3. *shall remain in the 'U: not permitted and Floor Idle' state."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.4.4]

"…

If the group is not configured for multi-talker floor control, upon receiving a Floor Request message, without a Floor Indicator field or with the Floor Indicator field included where the D-bit (Emergency call) and the E-bit (Imminent peril call) are set to '0', from the associated floor participant, and if the MCPTT client did not negotiate queueing of floor requests or did not include a priority in the "mc_priority" fmtp attribute as specified in clause 14, the floor control interface towards the MCPTT client in the floor control server:

1. *shall send a Floor Deny message to the associated floor participant. The Floor Deny message:*

   a. *shall include in the Reject Cause field the <Reject Cause> value cause #1 (Another MCPTT client has permission);*

b. may include in the Reject Cause field an additional text string explaining the reason for rejecting the floor request in the <Reject Phrase> value;

c. if the Floor Request included a Track Info field, shall include the received Track Info field; and

d. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications;

2. may set the first bit in the subtype of the Floor Deny message to '1' (Acknowledgment is required) as described in subclause 8.2.2; and

NOTE 1: It is an implementation option to handle the receipt of the Floor Ack message and what action to take if the Floor Ack message is not received.

3. shall remain in the 'U: not permitted and Floor Taken' state.

...

Upon receiving a Floor Request message from the associated floor participant and if the MCPTT client negotiated support of queueing of floor requests or included a floor priority in the "mc_priority" or both as described in specified in clause 14 and according to local policy, the floor control interface towards the MCPTT client in the floor control server:

...

1. shall determine the effective priority level as described in subclause 4.1.1.4 by using the following parameters:

a. the floor priority shall be:

i. the lower of the floor priority included in Floor Request message and the negotiated maximum floor priority that the MCPTT client is permitted to request, if the MCPTT client negotiated floor priority "mc_priority" and floor priority is included in the Floor Request message;

ii. the receive only floor priority, if the MCPTT client negotiated floor priority in the "mc_priority" fmtp attribute and if the negotiated maximum floor priority that the MCPTT client is permitted to request is "receive only";

iii. the default priority, if the MCPTT client negotiated floor priority in the "mc_priority" fmtp attribute, if the negotiated maximum floor priority that the MCPTT client is permitted to request is not receive only and if the floor priority is not included in the Floor Request message; and

iv. the default priority, if the MCPTT client did not negotiate floor priority in the "mc_priority" fmtp attribute; and

b. the type of the call shall be

i. if the Floor Indicator field is included in the message and the D-bit (Emergency call bit) is set to '1', determined to be an emergency call;

ii. if the Floor Indicator field is included in the message and the E-bit (Imminent peril call) is set to '1', determined to be an imminent peril call; and

iii. if the Floor Indicator field is not included in the message or the Floor Indicator field is included and neither the D-bit (Emergency call bit) nor the E-bit (Imminent peril call) is set to '1', determined to be a normal call;

...

5. if the effective priority level is pre-emptive and there are no other pre-emptive requests in the active floor request queue and the effective priority level of the current MCPTT client with permission to send a media is not the pre-emptive priority, the floor control interface towards the MCPTT client in the floor control server:

a. shall forward the Floor Request message to the floor control server arbitration logic indicating that a Floor Request message with pre-emptive priority is received; and

b. shall remain in the 'U: not permitted and Floor Taken' state

NOTE 4: The Floor control server arbitration logic initiates revoking the permission to send media towards the current MCPTT client with the permission to send media as specified in the subclause 6.3.4.4.7;

...

7. if the MCPTT client did not negotiate "queueing" and the effective priority level is not pre-emptive, the floor control interface towards the MCPTT client in the floor control server:

a. shall send a Floor Deny message to the associated floor participant. The Floor Deny message:

i. shall include in the Reject Cause field the <Reject Cause> value cause #1 (Another MCPTT client has permission);

ii. may include in the Reject Cause field an additional text string explaining the reason for rejecting the floor request in the <Reject Phrase> value;

iii. if the Floor Request included a Track Info field, shall include the received Track Info field; and

iv. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications; and

b. shall remain in the 'U: not permitted and Floor Taken' state; and

8. if the MCPTT client negotiated support of queueing of floor requests as specified in clause 14 and the effective priority level is not pre-emptive, the floor control interface towards the MCPTT client in the floor control server:

a. shall insert the MCPTT client into the active floor request queue, if not inserted yet, or update the position of the MCPTT client in the active floor request queue, if already inserted, to the position immediately following all queued requests at the same effective priority level;

b. the floor control server shall send a Floor Queue Position Info message to the floor participant. The Floor Queue Position Info message:

i. shall include the queue position and floor priority in the Queue Info field;

ii. if the Floor Request included a Track Info field, shall include the received Track Info field; and

iii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications;

c. shall remain in the 'U: not permitted and Floor Taken' state; and

d. may set the first bit in the subtype of the Floor Queue Position message to '1' (Acknowledgment is required) as described in subclause 8.2.2.

NOTE 5: It is an implementation option to handle the receipt of the Floor Ack message and what action to take if the Floor Ack message is not received."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.4.7]

"Upon receiving a Floor Queue Position Request message from the associated floor participant, the floor control interface towards the MCPTT client in the floor control server:

1. shall send the Floor Queue Position Info message. The Floor Queue Position Info message:

a. shall include the queue position and floor priority in the Queue Info field;

b. if a Track Info field is included in the Floor Queue Position Info message, shall include the received Track Info field;

c. may include the first bit in the subtype of the Floor Queue Position Info message set to '1' (Acknowledgment is required) as described in subclause 8.2.2; and

NOTE: It is an implementation option to handle the receipt of the Floor Ack message and what action to take if the Floor Ack message is not received.

d. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications; and

3. shall remain in the 'U: not permitted and Floor Taken' state."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.4.8]

"When an ongoing session is upgraded to an emergency group call and when the application and signalling plane indicates that a subsequent SDP offer included the "mc_implicit_request" fmtp attribute as specified in clause 14, the floor control interface towards the MCPTT client in the floor control server:

1. shall indicate to the floor control server arbitration logic that an implicit floor request is received due to an upgrade to an emergency group call; and

2. shall remain in the 'U: not permitted and Floor Taken' state."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.5.3]

"Upon receiving a Floor Release message from the associated floor participant, the floor control interface towards the MCPTT client in the floor control server:

1. if the first bit in the subtype of the Floor Release message is set to '1' (Acknowledgment is required) as described in subclause 8.2.2, shall send a Floor Ack message. The Floor Ack message:

a. shall include the Message Type field set to '4' (Floor Release); and

b. shall include the Source field set to '2' (the controlling MCPTT function is the source);

...

4. if no indication is stored:

a. shall forward the Floor Release message to the general floor control operation state machine of the floor control arbitration logic in the MCPTT server with the first bit in the subtype of the Floor Release message set to '0' (Acknowledgment is not required), if not already set; and

b. shall remain in the 'U: permitted' state."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.5.5]

"When receiving the Floor Revoke message from the floor control server arbitration logic in the MCPTT server, the floor control interface towards the MCPTT client in the floor control server:

1. shall forward the Floor Revoke message to the floor participant;

2. if the Floor Revoke message includes the Track Info field, shall store the Track Info field; and

3. shall enter the state 'U pending Floor Revoke' as specified in the subclause 6.3.5.6.2."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.6.2]

"When entering this state the floor control interface towards the MCPTT client in the floor control server:

1. shall start timer T8 (Floor Revoke); and

2. shall enter the state 'U: pending Floor Revoke'."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.5.6.5]

"Upon receiving a Floor Release message from the associated floor participant, the floor control interface towards the MCPTT client in the floor control server:

1. if the first bit in the subtype of the Floor Release message is set to '1' (Acknowledgment is required) as described in subclause 8.2.2, shall send a Floor Ack message. The Floor Ack message:

a. shall include the Message Type field set to '4' (Floor Release); and

b. shall include the Source field set to '2' (the controlling MCPTT function is the source);

...

3. *if the G-bit in the Floor Indicator is set to '0':*

   a. *shall forward the Floor Release message to the floor control server arbitration logic; and*

   b. *shall remain in the state 'U: pending Floor Revoke'."*

5.2.1.1.3 Test description

5.2.1.1.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the participating MCPTT functions.
- SS-UE1 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.1.3.1-1.
- SS-UE2 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.1.3.1-1.

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the controlling MCPTT function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2, and to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are registered and authorized to use the MCPTT system.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.

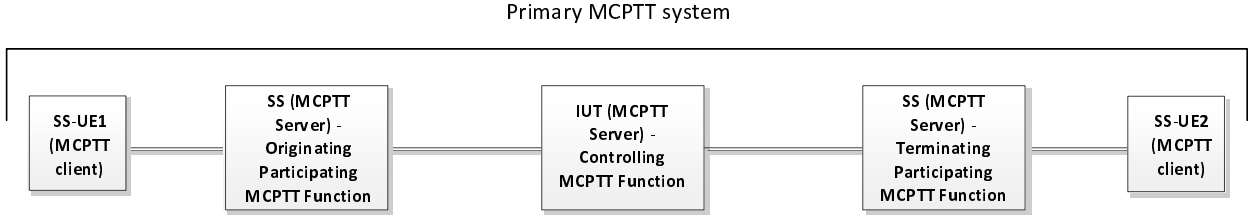


Figure 5.2.1.1.3.1-1: Functions of the testing components

5.2.1.1.3.2 Test procedure sequence

Table 5.2.1.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) to initiate an on-demand pre-arranged group call with automatic commencement mode and implicit floor control	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P
3	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP INVITE	1	P
4	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 183 (Session Progress) message sent reliably.	<--	SIP 183 (Session Progress)		-
5	Check: Does the IUT (MCPTT Server) send a SIP PRACK message to the SS (MCPTT Server) providing the terminating participating MCPTT function in response to the SIP 183 (Session Progress)?	-->	SIP PRACK	1	P
6	The SS (MCPTT Server) responds to the SIP PRACK with a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
7	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 8-9 and steps 10-11 might be performed in parallel.	-	-	-	-
8	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	1	P
9	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
10	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	1	P
11	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message.	<--	SIP ACK		-
12	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor.	<--	Floor Release		-
13	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P
14	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	2	P
15	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
16	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
17	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	2	P
18	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
19	The SS (MCPTT Server) providing the terminating participating MCPTT function sends a Floor Request message from SS-UE2 with a higher priority than SS-UE1 (MCPTT client).	<--	Floor Request		-
20	Check: Does the IUT (MCPTT Server) send a Floor Revoke message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Revoke	2	P
21	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1.	<--	Floor Release		-
22	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Granted	2	P
23	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the originating participating MCPTT function informing that the floor is taken by SS-UE2 (MCPTT client)?	-->	Floor Taken	2	P
24	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1 with a lower priority than SS-UE2 (MCPTT client) and the Floor Indicator F-bit set to 0.	<--	Floor Request		-
25	Check: Does the IUT (MCPTT Server) send a Floor Deny message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Deny	2	P
26	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1 with a lower priority than SS-UE2 (MCPTT client) and the Floor Indicator F-bit set to 1.	<--	Floor Request		-
27	Check: Does the IUT (MCPTT Server) send a Queue Position Info message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Queue Position Info	2	P
28	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Queue Position Request message from SS-UE1.	<--	Floor Queue Position Request		-
29	Check: Does the IUT (MCPTT Server) send a Queue Position Info message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Queue Position Info	2	P
30	The SS (MCPTT Server) providing the terminating participating MCPTT function sends a Floor Release message from SS-UE2 with an acknowledgement required to release the floor	<--	Floor Release		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
31	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P
32	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	2	P
33	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
34	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
35	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P
36	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	2	P
37	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	2	P
38	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) to upgrade the call to an emergency group call with implicit floor request.	<--	SIP INVITE		-
39	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	3	P
40	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP INVITE	3	P
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 41-43 and steps 44-46 might be performed in parallel.	-	-	-	-
41	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
42	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	3	P
43	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	4	P
44	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	3	P
45	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
46	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	4	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
47	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
48	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	4	P
49	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	4	P
50	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	4	P
51	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
52	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	4	P
53	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	4	P
54	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
55	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	4	P
56	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	4	P
57	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	4	P
58	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) to downgrade the call to a normal group call without implicit floor request.	<--	SIP INVITE		-
59	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	5	P
60	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP INVITE	5	P
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 61-62 and steps 63-64 might be performed in parallel.	-	-	-	-
61	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
62	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	5	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
63	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	5	P
64	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
65	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	6	P
66	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	6	P
67	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
68	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	6	P
69	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	6	P
70	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
71	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	6	P
72	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	6	P
73	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	6	P
74	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) to upgrade the call to an imminent-peril group call with implicit floor request.	<--	SIP INVITE		-
75	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	7	P
76	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP INVITE	7	P
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 77-79 and steps 80-82 might be performed in parallel.	-	-	-	-
77	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
78	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	7	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
79	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	8	P
80	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	7	P
81	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
82	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	8	P
83	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
84	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	8	P
85	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	8	P
86	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	8	P
87	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
88	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	8	P
89	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	8	P
90	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
91	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	8	P
92	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	8	P
93	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	8	P
94	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) to downgrade the call to a normal group call without implicit floor request.	<--	SIP INVITE		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
95	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	9	P
96	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP INVITE	9	P
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 97-98 and steps 99-100 might be performed in parallel.	-	-	-	-
97	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
98	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	9	P
99	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	9	P
100	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
101	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	10	P
102	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	10	P
103	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
104	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	10	P
105	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	10	P
106	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
107	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	10	P
108	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	10	P
109	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	10	P
110	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP BYE request from SS-UE1	<--	SIP BYE		-
111	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) response to SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	11	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
112	Check: Does the IUT (MCPTT Server) send a SIP BYE message to the SS (MCPTT Server) providing the terminating participating MCPTT function to end the call?	-->	SIP BYE	11	P
113	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-

5.2.1.1.3.3 Specific message contents

Table 5.2.1.1.3.3-1: SIP INVITE from the SS (MCPTT-Server) (Step 1, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.1.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.1.3.3-1B			

Table 5.2.1.1.3.3-1A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions INITIAL_SDP_OFFER, IMPLICIT_GRANT_REQUESTED, IMPLICIT_FLOOR_GRANTED

Table 5.2.1.1.3.3-1B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.1.1.3.3-2: SIP INVITE from the IUT (MCPTT-Server) (Step 3, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing terminating participating function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing terminating participating function.		
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			
Session-Expires				
generic-param	any allowed value			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Max-Forwards				
value	any allowed value			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.1.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.1.3.3-2B			

5.2.1.1.3.3-2A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the controlling MCPTT Server			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the controlling MCPTT Server			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the controlling MCPTT Server			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the controlling MCPTT Server			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.1.3.3-2B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A				

Table 5.2.1.1.3.3-3: SIP 183 (Session Progress) from the SS (MCPTT-Server) (Step 4, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.1-1, conditions MCPTT, 100rel

Table 5.2.1.1.3.3-4: SIP PRACK from the IUT (MCPTT-Server) (Step 5, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.10.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Request-URI	same URI as the SS has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server), providing terminating participating function.		
Max-Forwards				
value	any allowed value			

Table 5.2.1.1.3.3-5: SIP 200 (OK) to PRACK from the SS (MCPTT-Server) (Step 6, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Content-Type	not present			
Content-Length	not present			

Table 5.2.1.1.3.3-6: SIP 200 (OK) from the SS (MCPTT Server) (Step 7, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	3600			
refresher	"uas"			
Message-body				
SDP message	SDP message as described in Table 5.2.1.1.3.3-6A			

Table 5.2.1.1.3.3-6A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.1.3.3-7: SIP 200 (OK) from the IUT (MCPTT Server) (Step 10, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	any allowed value			
refresher	"uac"			
Message-body				
SDP message	SDP message as described in Table 5.2.1.1.3.3-7A			

Table 5.2.1.1.3.3-7A: SDP in SIP 200 (OK) from the IUT (MCPTT Server) (Table 5.2.1.1.3.3-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_FLOOR_GRANTED, IMPLICIT_GRANT_REQUESTED				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the MCPTT Server			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the MCPTT Server			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the MCPTT Server (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the MCPTT Server			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the MCPTT Server being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			

Table 5.2.1.1.3.3-8: SIP INVITE from the SS (MCPTT-Server) (Step 38, Table 5.2.1.1.3.2-1)

Derivation Path: Table 5.2.1.1.3.3-1 conditions MCPTT, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.1.3.3-8A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.1.3.3-8B			

Table 5.2.1.1.3.3-8A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-8)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_OFFER, IMPLICIT_GRANT_REQUESTED

Table 5.2.1.1.3.3-8B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-8)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A				

Table 5.2.1.1.3.3-9: SIP INVITE from the IUT (MCPTT-Server) (Step 40, Table 5.2.1.1.3.2-1)

Derivation Path: Table 5.2.1.1.3.3-2, conditions MCPTT, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 5.2.1.1.3.3-9A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.1.3.3-9B			

Table 5.2.1.1.3.3-9A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-9)

Derivation Path: Table 5.2.1.1.3.3-2A, condition SDP_OFFER instead of INITIAL_SDP_OFFER

Table 5.2.1.1.3.3-9B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-9)

Derivation Path: Table 5.2.1.1.3.3-2B, conditions EMERGENCY-CALL

Table 5.2.1.1.3.3-10: SIP 200 (OK) from the SS (MCPTT-Server) (Steps 41, 61, 77 and 97, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.1.3.3-10A			

Table 5.2.1.1.3.3-10A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-10)

Derivation Path: Table 5.2.1.1.3.3-6A

Table 5.2.1.1.3.3-11: SIP 200 (OK) from the IUT (MCPTT-Server) (Steps 44 and 80, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.1.3.3-11A			

Table 5.2.1.1.3.3-11A: SDP in SIP 200 (OK) from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_GRANT_REQUESTED

Table 5.2.1.1.3.3-12: SIP 200 (OK) from the IUT (MCPTT-Server) (Steps 63 and 99, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.1.3.3-12A			

Table 5.2.1.1.3.3-12A: SDP in SIP 200 (OK) from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER

Table 5.2.1.1.3.3-13: SIP INVITE from the SS (MCPTT-Server) (Step 58, Table 5.2.1.1.3.2-1)

Derivation Path: Table 5.2.1.1.3.3-1 conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.1.3.3-13A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.1.3.3-13B			

Table 5.2.1.1.3.3-13A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-13)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_OFFER

Table 5.2.1.1.3.3-13B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-13)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
emergency-ind	Encrypted (see note) <emergency-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A				

Table 5.2.1.1.3.3-14: SIP INVITE from the IUT (MCPTT-Server) (Step 60, Table 5.2.1.1.3.2-1)

Derivation Path: Table 5.2.1.1.3.3-2, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 5.2.1.1.3.3-14A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.1.3.3-14B			

Table 5.2.1.1.3.3-14A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-14)

Derivation Path: Table 5.2.1.1.3.3-9A

Table 5.2.1.1.3.3-14B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-14)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
emergency-ind	Encrypted (see note) <emergency-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A				

Table 5.2.1.1.3.3-15: SIP INVITE from the SS (MCPTT-Server) (Step 74, Table 5.2.1.1.3.2-1)

Derivation Path: Table 5.2.1.1.3.3-1, conditions MCPTT, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.1.3.3-15A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.1.3.3-15B			

Table 5.2.1.1.3.3-15A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-15)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_OFFER, IMPLICIT_GRANT_REQUESTED,

Table 5.2.1.1.3.3-15B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-15)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A				

Table 5.2.1.1.3.3-16: SIP INVITE from the IUT (MCPTT-Server) (Step 76, Table 5.2.1.1.3.2-1)

Derivation Path: Table 5.2.1.1.3.3-2, conditions MCPTT, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 5.2.1.1.3.3-16A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.1.3.3-16B			

Table 5.2.1.1.3.3-16A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-16)

Derivation Path: Table 5.2.1.1.3.3-9A

Table 5.2.1.1.3.3-16B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-16)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, INVITE_REFER, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A				

Table 5.2.1.1.3.3-17: SIP INVITE from the SS (MCPTT-Server) (Step 94, Table 5.2.1.1.3.2-1)

Derivation Path: Table 5.2.1.1.3.3-1, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.1.3.3-17A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.1.3.3-17B			

Table 5.2.1.1.3.3-17A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-17)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_OFFER

Table 5.2.1.1.3.3-17B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.1.3.3-17)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
imminentperil-ind	Encrypted (see note) <imminentperil-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.1.1.3.3-18: SIP INVITE from the IUT (MCPTT-Server) (Step 96, Table 5.2.1.1.3.2-1)

Derivation Path: Table 5.2.1.1.3.3-2, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 5.2.1.1.3.3-18A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.1.3.3-18B			

Table 5.2.1.1.3.3-18A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-18)

Derivation Path: Table 5.2.1.1.3.3-9A

Table 5.2.1.1.3.3-18B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.1.3.3-18)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
imminentperil-ind	Encrypted (see note) <imminentperil-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.1.3.3-19: Floor Taken (Steps 9, 18, 33, 69 and 105, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.1.1.3.3-20: Floor Release (Steps 12, 30, 34, 70 and 106, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.1.1.3.3-21: Floor Ack (Steps 13, 31, 35, 48, 55, 71, 84, 91 and 107, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 5.2.1.1.3.3-22: Floor Idle
(Steps 14, 15, 36, 37, 65, 66, 72, 73, 101, 102, 108 and 109, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.1.1.3.3-23: Floor Request (Steps 16, 26, 67 and 103, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 5.2.1.1.3.3-24: Floor Request (Step 24, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	
Floor Indicator			
Floor Indicator	"1000000000000000"	bit A=1 (Normal call) bit F=0 (Queueing not supported)	

Table 5.2.1.1.3.3-25: Floor Granted (Steps 17, 32, 68 and 104, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

Table 5.2.1.1.3.3-26: Floor Request (Step 19, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"2"	The highest priority for the user	

Table 5.2.1.1.3.3-27: Floor Revoke (Step 20, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.8-1

Table 5.2.1.1.3.3-28: Floor Release (Step 21, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1

Table 5.2.1.1.3.3-29: Floor Granted (Step 22, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"2"	The highest priority for the user	

Table 5.2.1.1.3.3-30: Floor Taken (Step 23, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1

Table 5.2.1.1.3.3-31: Floor Deny (Step 25, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.4-1

Table 5.2.1.1.3.3-32: Floor Queue Position Info (Steps 27 and 29, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.10-1

Table 5.2.1.1.3.3-33: Floor Queue Position Request (Step 28, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.9-1

Table 5.2.1.1.3.3-34: Floor Taken (Steps 43, 53, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, condition EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.1.1.3.3-35: Floor Idle (Steps 49, 50, 56, 57, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, condition EMERGENCY-CALL

Table 5.2.1.1.3.3-36: Floor Request (Step 51, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1 condition EMERGENCY-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 5.2.1.1.3.3-37: Floor Granted (Step 46, 52, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition EMERGENCY-CALL

Table 5.2.1.1.3.3-38: Floor Release (Steps 47, 54, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition EMERGENCY-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.1.1.3.3-39: Floor Taken (Steps 79, 89, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1 condition IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.1.1.3.3-40: Floor Idle (Steps 85, 86, 92, 93, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1 condition IMMPERIL-CALL

Table 5.2.1.1.3.3-41: Floor Request (Step 87, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1 condition IMMPERIL-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 5.2.1.1.3.3-42: Floor Granted (Step 82, 88, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition IMMPERIL-CALL

Table 5.2.1.1.3.3-43: Floor Release (Steps 83, 90, Table 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition IMMPERIL-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.1.1.3.3-44: SIP BYE from the SS (MCPTT-Server) (Step 110, 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
From				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Local URI of the dialog (from the SS' point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Local tag of the dialog (from the SS' point of view)		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the SS' point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the SS' point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	70	The recommended initial value is 70 in IETF RFC 3261 [17].		

Table 5.2.1.1.3.3-45: SIP BYE from the IUT (MCPTT-Server) (Step 112, 5.2.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261[17]	
value	any allowed value	Non-zero value		

5.2.1.2 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Automatic Commencement Mode / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / Originating Participating server

5.2.1.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE1 (MCPTT client) initiates an on-demand pre-arranged group call with automatic
commencement mode and implicit floor control }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) and interacts with SS-UE1 (MCPTT
client) to set up the on-demand pre-arranged group call by sending SIP messages }
           }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
Automatic Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted, Floor Request, Floor Release, Floor Revoke, Floor Deny, Floor Queue Position Info, Floor
Queue Position Request ) by forwarding messages to the SS (MCPTT Server) and to the SS-UE1 (MCPTT
client) }
           }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
Automatic Commencement Mode }
ensure that {
  when { the SS-UE1 (MCPTT client) upgrades the call to an emergency group call with implicit floor
request }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) and interacts with SS-UE1 (MCPTT
client) to set up an emergency group call by sending SIP messages }
           }
```

(4)

```
with { IUT (MCPTT Server) having established an MCPTT emergency group call with implicit floor
request }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted, Floor Request, Floor Release ) by forwarding messages via the SS (MCPTT Server) and to the
SS-UE1 (MCPTT client) }
            }
```

(5)

```
with { IUT (MCPTT Server) having established an MCPTT emergency group call with implicit floor
request }
ensure that {
  when { the SS-UE1 (MCPTT client) downgrades the call to a normal group call without implicit floor
request }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) and interacts with SS-UE1 (MCPTT
client) to set up the normal group call by sending SIP messages }
            }
```

(6)

```
with { IUT (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted, Floor Request, Floor Release ) by sending messages via the SS (MCPTT Server) and to the SS-
UE1 (MCPTT client) }
            }
```

(7)

```
with { IUT (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) upgrades the call to imminent-peril group call with implicit
floor request }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) and interacts with SS-UE1 (MCPTT
client) to set up the imminent-peril group call by sending SIP messages }
            }
```

(8)

```
with { IUT (MCPTT Server) having established an MCPTT imminent-peril group call with implicit floor
request }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted, Floor Request, Floor Release ) by sending messages via the SS (MCPTT Server) and to the SS-
UE1 (MCPTT client) }
            }
```

(9)

```
with { IUT (MCPTT Server) having established an MCPTT imminent-peril group call with implicit floor
request }
ensure that {
  when { the SS-UE1 (MCPTT client) downgrades the call to a normal group call without implicit floor
request }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) and interacts with SS-UE1 (MCPTT
client) to set up the normal group call by sending SIP messages }
            }
```

(10)

```
with { IUT (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted, Floor Request, Floor Release ) by sending messages via the SS (MCPTT Server) and to the SS-
UE1 (MCPTT client) }
            }
```

(11)

```
with { IUT (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the pre-arranged group call }
    then { IUT (MCPTT Server) forwards a SIP BYE message to SS (MCPTT Server) and responds to SS-UE1
(MCPTT client) by forwarding a SIP 200 (OK) message }
            }
```

5.2.1.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.1.1, 6.3.2.1.3, 6.3.2.1.1.1, 6.3.2.1.5.2, 6.3.2.1.2.1, 10.1.1.3.1.3, 6.3.2.1.9, 10.1.1.3.3.1 and 6.3.2.1.6, and ETSI TS 124 380 [5] (3GPP TS 24.380) clause 6.4.2. Unless otherwise stated these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

...

Upon receipt of a "SIP INVITE request for originating participating MCPTT function" containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "prearranged", the participating MCPTT function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The participating MCPTT function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [24]. Otherwise, continue with the rest of the steps;

NOTE 1: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCPTT function can according to local policy choose to accept the request.

2) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

NOTE 2: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

...

8) shall determine the public service identity of the controlling MCPTT function associated with the group identity in the SIP INVITE request. If the participating MCPTT function is unable to identify the controlling MCPTT function for the on demand prearranged group call, it shall reject the SIP INVITE request with a SIP 404 (Not Found) response with the warning text "142 unable to determine the controlling function" in a Warning header field as specified in clause 4.4, and shall not continue with any of the remaining steps;

NOTE 6: The public service identity can identify the controlling MCPTT function in the primary MCPTT system or a partner MCPTT system.

NOTE 7: How the participating MCPTT server discovers the public service identity of the controlling MCPTT function associated with the group identity is out of scope of the current release.

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the group identity which was present in the incoming SIP INVITE request;

11) shall not copy the following header fields from the incoming SIP INVITE request to the outgoing SIP INVITE request, if they were present in the incoming SIP INVITE request:

a) Answer-Mode header field as specified in IETF RFC 5373 [18]; and

b) Priv-Answer-Mode header field as specified in IETF RFC 5373 [18];

12) shall set the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request to the MCPTT ID of the calling user;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCPTT client as specified in clause 6.3.2.1.1.1;

...

14) if the received SIP INVITE request contains an application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause F.3 and if not already copied, shall copy the contents of the application/vnd.3gpp.mcptt-location-info+xml MIME body received in the SIP INVITE request into an application/vnd.3gpp.mcptt-location-info+xml MIME body included in the outgoing SIP request;

15) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCPTT client;

NOTE 9: The participating MCPTT function will leave verification of the Resource-Priority header field to the controlling MCPTT function.

…

Upon receipt of a SIP 2xx response in response to the above SIP INVITE request, the participating MCPTT function:

1) void;

2) shall generate a SIP 200 (OK) response as in clause 6.3.2.1.5.2 with the clarification that if an <MKFC-GKTPs> element was contained in the received SIP 2xx response it is not included in the generated SIP 200 (OK) response;

NOTE 10: If an <MKFC-GKTPs> element is received in a SIP 2xx response, the participating MCPTT function essentially ignores it and does not forward it, resulting in unicast media plane transmission being used for the originating client.

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) shall include an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response;

6A) shall include the answer state into the P-Answer-State header field of the outgoing SIP 200 (OK) response, if received in the P-Answer-State header field of the incoming SIP 200 (OK) response;

7) if the procedures of clause 9.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 9.2.2.2.13;

8) shall send the SIP 200 (OK) response to the MCPTT client according to 3GPP TS 24.229 [4];

9) shall interact with Media Plane as specified in 3GPP TS 24.380 [5]; and

10) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [7].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.3]

"This clause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [4], on receipt of an incoming SIP INVITE request, the participating MCPTT function:

1) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [6] if included in the incoming SIP INVITE request;

2) should include the Session-Expires header field according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

3) shall include the option tag "timer" in the Supported header field;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

5) shall include the g.3gpp.mcptt media feature tag into the Contact header field of the outgoing SIP INVITE request;

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;

...

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request; and

..."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCPTT function:

1) shall contain only one SDP media-level section for MCPTT speech as contained in the received SDP offer; and

2) shall contain one SDP media-level section for media plane control messages, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [4], the participating MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCPTT function, if required;

NOTE 1: Requirements can exist for the participating MCPTT function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the participating MCPTT function; and

NOTE 2: If the participating MCPTT function and the controlling MCPTT function or the participating MCPTT function and the non-controlling MCPTT function are in the same MCPTT server, and the participating MCPTT function does not have a dedicated IP address or a dedicated port number for media plane control channel or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.5.2]

"This clause is referenced from other procedures.

When sending SIP 200 (OK) responses, the participating MCPTT function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [4] and:

1) shall include the option tag "timer" in a Require header field;

2) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7], "UAS Behavior". If the "refresher" parameter is not included in the received request, the "refresher" parameter in the Session-Expires header field shall be set to "uac";

3) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and

c) the isfocus media feature tag;

4) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];

5) shall include the option tag "norefersub" in a Supported header field according to rules and procedures of IETF RFC 4488 [22];

6) may include a Resource-Share header field in accordance with clause 5.7.1.20.2 in 3GPP TS 24.229 [4]; and

7) if the incoming SIP 200 (OK) response contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP 200 (OK) response."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [4], the participating MCPTT function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCPTT function, for the accepted media stream in the received SDP offer, if required; and

NOTE 1: Requirements can exist for the participating MCPTT function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCPTT function, for the accepted media plane control channel, if present in the received SDP offer.

NOTE 2: If the participating MCPTT function and the controlling MCPTT function or the participating MCPTT function and the non-controlling MCPTT function are in the same MCPTT server, and the participating MCPTT function does not have a dedicated IP address or a dedicated port number for media plane control channel or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.1.3]

"This clause covers both on-demand session and pre-established sessions.

Upon receipt of a SIP re-INVITE request for an MCPTT session identifying an on-demand prearranged MCPTT group session, the participating MCPTT function:

…

3) shall generate an outgoing SIP re-INVITE request as specified in clause 6.3.2.1.9;

4) shall include the MCPTT ID of the originating user in <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request;

5) shall, if the SIP re-INVITE request was received within an on-demand session, include in the SIP re-INVITE request an SDP offer based on the SDP offer in the received SIP re-INVITE request as specified in clause 6.3.2.1.1;

...

7) if the received SIP re-INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field; and

NOTE 3: The controlling MCPTT function will determine the validity of the Resource-Priority header field.

8) shall forward the SIP re-INVITE request according to 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response to the above SIP re-INVITE request in step 7) the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

3) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

4) shall copy the contents received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

5) shall send the SIP 200 (OK) response towards the MCPTT client according to 3GPP TS 24.229 [4]; and

6) shall interact with the media plane as specified in 3GPP TS 24.380 [5].

..."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.9]

"This clause is referenced from other procedures.

When generating a SIP re-INVITE request according to 3GPP TS 24.229 [4] on receipt of an incoming SIP re-INVITE request, the participating MCPTT function:

1) if the incoming SIP re-INVITE request contained a MIME resource-lists body with the MCPTT ID of the invited MCPTT user, shall copy the MIME resource-lists body, according to rules and procedures of IETF RFC 5366 [20];

2) if the incoming SIP re-INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body; and

3) if the incoming SIP re-INVITE request contained an application/vnd.3gpp.mcptt-location-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-location-info+xml MIME body."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.1.3.3.1]

"Upon receiving from the MCPTT client a SIP BYE request the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.6]

"Upon receiving a SIP BYE request from the MCPTT client, the participating MCPTT function:

1) shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5];

2) shall generate a SIP BYE request as specified in 3GPP TS 24.229 [4];

3) shall set the Request-URI to the MCPTT session identity as included in the received SIP BYE request;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

5) if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and

6) shall send the SIP BYE request toward the controlling MCPTT function, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the terminating MCPTT function shall forward a SIP 200 (OK) response to the MCPTT client and shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the controlling MCPTT function."

5.2.1.2.3 Test description

5.2.1.2.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the controlling MCPTT function.
- SS-UE1 (MCPTT client)

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the originating participating MCPTT function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2, and to the SS-UE1 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- The SS-UE1 (MCPTT client) is affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.

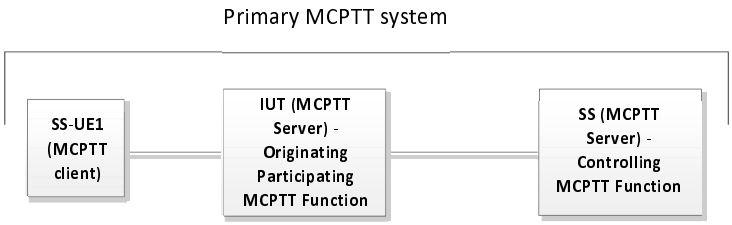


Figure 5.2.1.2.3.1-1: Functions of the testing components

5.2.1.2.3.2 Test procedure sequence

Table 5.2.1.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client) initiates an on-demand pre-arranged group call with automatic commencement mode and implicit floor control.	<--	SIP INVITE		-
2	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS-UE1 (MCPTT client)?	-->	SIP 100 Trying	1	P
3	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE	1	P
4	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 5-7 might be performed in parallel.	-	-	-	-
5	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	1	P
6	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS-UE1 (MCPTT client)?	-->	SIP 200 (OK)	1	P
7	The SS-UE1 (MCPTT client) responds with a SIP ACK message	<--	SIP ACK		-
8	The SS-UE1 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
9	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	2	P
10	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
11	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE1 (MCPTT client)?	-->	Floor Ack	2	P
12	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
13	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	2	P
14	The SS-UE1 (MCPTT client) sends a Floor Request message	<--	Floor Request		-
15	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	2	P
16	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
17	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE1 (MCPTT client)?	-->	Floor Granted	2	P
18	The SS (MCPTT Server) sends a Floor Revoke message with no acknowledgment required.	<--	Floor Revoke		-
19	Check: Does the IUT (MCPTT Server) forward the Floor Revoke message to the SS-UE1 (MCPTT client)?	-->	Floor Revoke	2	P
20	The SS-UE1 (MCPTT client) sends a Floor Release message	<--	Floor Release		-
21	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	2	P
22	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE1 (MCPTT client) that the floor is taken by SS-UE2 (MCPTT client)	<--	Floor Taken		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
23	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
24	The SS-UE1 (MCPTT client) sends a Floor Request message with a lower priority than SS-UE2 (MCPTT client) and the Floor Indicator F-bit set to 0.	<--	Floor Request		-
25	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT server)?	-->	Floor Request	2	P
26	The SS (MCPTT Server) sends a Floor Deny message to the IUT (MCPTT Server)	<--	Floor Deny		-
27	Check: Does the IUT (MCPTT Server) forward the Floor Deny message to the SS-UE1 (MCPTT client)?	-->	Floor Deny	2	P
28	The SS-UE1 (MCPTT client) sends a Floor Request message with a lower priority than SS-UE2 (MCPTT client) and the Floor Indicator F-bit set to 1.	<--	Floor Request		-
29	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT server)?	-->	Floor Request	2	P
30	The SS (MCPTT Server) sends a Floor Queue Position Info message to the IUT (MCPTT Server)	<--	Floor Queue Position Info		-
31	Check: Does the IUT (MCPTT Server) forward the Floor Queue Position Info message to the SS-UE1 (MCPTT client)?	-->	Floor Queue Position Info	2	P
32	The SS-UE1 (MCPTT client) sends a Floor Queue Position Request	<--	Floor Queue Position Request		-
33	Check: Does the IUT (MCPTT Server) forward the Floor Queue Position Request message to the SS (MCPTT Server)?	-->	Floor Queue Position Request	2	P
34	The SS (MCPTT Server) sends a Floor Queue Position Info message to the IUT (MCPTT Server)	<--	Floor Queue Position Info		-
35	Check: Does the IUT (MCPTT Server) forward the Floor Queue Position Info message to the SS-UE1 (MCPTT client)?	-->	Floor Queue Position Info	2	P
36	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
37	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE1 (MCPTT client)?	-->	Floor Granted	2	P
38	The SS-UE1 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
39	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	2	P
40	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
41	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE1 (MCPTT client)?	-->	Floor Ack	2	P
42	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
43	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	2	P
44	The SS-UE1 (MCPTT Client) sends a SIP INVITE requesting the modification of a call to upgrade the call to an emergency group call with implicit floor request.	<--	SIP INVITE		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
45	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS-UE1 (MCPTT client)?	-->	SIP 100 Trying	3	P
46	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE		P
47	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 48-50 might be performed in parallel.	-	-	-	-
48	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	3	P
49	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS-UE1 (MCPTT client)?	-->	SIP 200 (OK)	3	P
50	The SS-UE1 (MCPTT client) responds with a SIP ACK message	<--	SIP ACK		-
50a1	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
50a2	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE1 (MCPTT client)?	-->	Floor Granted	4	P
51	The SS-UE1 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
52	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	4	P
53	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
54	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE1 (MCPTT client)?	-->	Floor Ack	4	P
55	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
56	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	4	P
57	The SS-UE1 (MCPTT client) sends a Floor Request message	<--	Floor Request		-
58	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	4	P
59	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
60	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE1 (MCPTT client)?	-->	Floor Granted	4	P
61	The SS-UE1 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
62	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	4	P
63	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
64	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE1 (MCPTT client)?	-->	Floor Ack	4	P
65	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
66	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	4	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
67	The SS-UE1 (MCPTT Client) sends a SIP INVITE requesting the modification of a call to downgrade the call to a normal group call without implicit floor request.	<--	SIP INVITE		-
68	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS-UE1 (MCPTT client)?	-->	SIP 100 Trying	5	P
69	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE	5	P
70	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 71-73 might be performed in parallel.	-	-	-	-
71	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	5	P
72	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS-UE1 (MCPTT client)?	-->	SIP 200 (OK)	5	P
73	The SS-UE1 (MCPTT client) responds with a SIP ACK message	<--	SIP ACK		-
74	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
75	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	6	P
76	The SS-UE1 (MCPTT client) sends a Floor Request message	<--	Floor Request		-
77	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	6	P
78	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
79	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE1 (MCPTT client)?	-->	Floor Granted	6	P
80	The SS-UE1 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
81	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	6	P
82	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
83	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE1 (MCPTT client)?	-->	Floor Ack	6	P
84	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
85	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	6	P
86	The SS-UE1 (MCPTT Client) sends a SIP INVITE requesting the modification of a call to upgrade the call to an imminent-peril group call with implicit floor request.	<--	SIP INVITE		-
87	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS-UE1 (MCPTT client)?	-->	SIP 100 Trying	7	P
88	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE	7	P
89	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 90-92 might be performed in parallel.	-	-	-	-
90	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	7	P
91	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS-UE1 (MCPTT client)?	-->	SIP 200 (OK)	7	P
92	The SS-UE1 (MCPTT client) responds with a SIP ACK message	<--	SIP ACK		-
92a1	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
92a2	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE1 (MCPTT client)?	-->	Floor Granted	8	P
93	The SS-UE1 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
94	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	8	P
95	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
96	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE1 (MCPTT client)?	-->	Floor Ack	8	P
97	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
98	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	8	P
99	The SS-UE1 (MCPTT client) sends a Floor Request message	<--	Floor Request		-
100	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	8	P
101	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
102	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE1 (MCPTT client)?	-->	Floor Granted	8	P
103	The SS-UE1 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
104	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	8	P
105	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
106	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE1 (MCPTT client)?	-->	Floor Ack	8	P
107	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
108	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	8	P
109	The SS-UE1 (MCPTT Client) sends a SIP INVITE requesting the modification of a call to downgrade the call to a normal group call without implicit floor request.	<--	SIP INVITE		-
110	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS-UE1 (MCPTT client)?	-->	SIP 100 Trying	9	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
111	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE	9	P
112	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
-	EXCEPTION: Depending on the MCPTT Server implementation, steps 113-115 might be performed in parallel.	-	-	-	-
113	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	9	P
114	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS-UE1 (MCPTT client)?	-->	SIP 200 (OK)	9	P
115	The SS-UE1 (MCPTT client) responds with a SIP ACK message	<--	SIP ACK		-
116	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
117	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	10	P
118	The SS-UE1 (MCPTT client) sends a Floor Request message	<--	Floor Request		-
119	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	10	P
120	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
121	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE1 (MCPTT client)?	-->	Floor Granted	10	P
122	The SS-UE1 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
123	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	10	P
124	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
125	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE1 (MCPTT client)?	-->	Floor Ack	10	P
126	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
127	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	10	P
128	The SS-UE1 (MCPTT client) sends a SIP BYE request	<--	SIP BYE		-
129	Check: Does the IUT (MCPTT Server) forward the SIP BYE message to the SS (MCPTT Server) to end the call?	-->	SIP BYE	11	P
130	SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
131	Check: Does the IUT (MCPTT Server) forward the SIP 200 (OK) response to SS-UE1 (MCPTT client)	-->	SIP 200 (OK)	11	P

5.2.1.2.3.3 Specific message contents

Table 5.2.1.2.3.3-1: SIP INVITE from the SS-UE1 (Step 1, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [25]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [25].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.2.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.2.3.3-1B			

Table 5.2.1.2.3.3-1A: SDP in SIP INVITE from the SS-UE1 (Table 5.2.1.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions INITIAL_SDP_OFFER, IMPLICIT_GRANT_REQUESTED

Table 5.2.1.2.3.3-1B: MCPTT-Info in SIP INVITE from the SS-UE1 (Table 5.2.1.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER

Table 5.2.1.2.3.3-2: SIP INVITE from the IUT (MCPTT-Server) (Step 3, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server which is simulated by the SS, providing the controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the IUT (MCPTT-Server)			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server, providing originating participating function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server which is simulated by the SS, providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server, providing originating participating function.		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.2.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.2.3.3-2B			

Table 5.2.1.2.3.3-2A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions INITIAL_SDP_OFFER, IMPLICIT_GRANT_REQUESTED				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the IUT (MCPTT Server)			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (MCPTT Server)			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT Server)			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT Server)			

Table 5.2.1.2.3.3-2B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.1.2.3.3-3: SIP 200 (OK) from the SS (MCPTT Server) (Step 4, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.2.3.3-3A			

Table 5.2.1.2.3.3-3A: SDP in SIP 200 (OK) from the SS (MCPTT Server) (Table 5.2.1.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_FLOOR_GRANTED, IMPLICIT_GRANT_REQUESTED

Table 5.2.1.2.3.3-4: SIP 200 (OK) from the IUT (MCPTT-Server) (Step 6, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.2.3.3-4A			

Table 5.2.1.2.3.3-4A: SDP in SIP 200 (OK) from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_FLOOR_GRANTED, IMPLICIT_GRANT_REQUESTED				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the IUT (MCPTT Server)			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (MCPTT Server)			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the IUT (MCPTT Server) (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the IUT (MCPTT Server)			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the MCPTT Server being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT Server)			

Table 5.2.1.2.3.3-5: SIP INVITE from the SS-UE1 (Step 44, Table 5.2.1.2.3.2-1)

Derivation Path: Table 5.2.1.2.3.3-1, conditions MCPTT, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.2.3.3-5A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.2.3.3-5B			

Table 5.2.1.2.3.3-5A: SDP in SIP INVITE from the SS-UE1 (Table 5.2.1.2.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions SDP_OFFER, IMPLICIT_GRANT_REQUESTED

Table 5.2.1.2.3.3-5B: MCPTT-Info in SIP INVITE from the SS-UE1 (Table 5.2.1.2.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, EMERGENCY-CALL

Table 5.2.1.2.3.3-6: SIP INVITE from the IUT (MCPTT-Server) (Step 46, Table 5.2.1.2.3.2-1)

Derivation Path: Table 5.2.1.2.3.3-2 conditions MCPTT, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.2.3.3-6A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.2.3.3-6B			

Table 5.2.1.2.3.3-6A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-6)

Derivation Path: Table 5.2.1.2.3.3-2A, condition SDP_OFFER instead of INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin	Same o=line as in the previous SDP message sent by the IUT (MCPTT-Server) except that sess-version is incremented by one			

Table 5.2.1.2.3.3-6B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.1.2.3.3-7: SIP 200 (OK) from the SS (MCPTT Server) (Steps 47 and 89, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.2.3.3-7A			

Table 5.2.1.2.3.3-7A: SDP in SIP 200 (OK) from the SS (MCPTT Server) (Table 5.2.1.2.3.3-7)

Derivation Table 5.2.1.2.3.3-3A

Table 5.2.1.2.3.3-8: SIP 200 (OK) from the IUT (MCPTT-Server) (Steps 49 and 91, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.2.3.3-8A			

Table 5.2.1.2.3.3-8A: SDP in SIP 200 (OK) from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-8)

Derivation Path: Table 5.2.1.2.3.3-4A				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin	Same o=line as in the previous SDP message sent by the IUT (MCPTT-Server) except that sess-version is incremented by one			

Table 5.2.1.2.3.3-9: SIP INVITE from the SS-UE1 (Step 67, Table 5.2.1.2.3.2-1)

Derivation Path: Table 5.2.1.2.3.3-1 conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.2.3.3-9A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.2.3.3-9B			

Table 5.2.1.2.3.3-9A: SDP in SIP INVITE from the SS-UE1 (Table 5.2.1.2.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions SDP_OFFER

Table 5.2.1.2.3.3-9B: MCPTT-Info in SIP INVITE from the SS-UE1 (Table 5.2.1.2.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
emergency-ind	Encrypted <emergency-ind> with mcpttBoolean set to false	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1		

Table 5.2.1.2.3.3-10: SIP INVITE from the IUT (MCPTT-Server) (Step 69, Table 5.2.1.2.3.2-1)

Derivation Path: Table 5.2.1.2.3.3-2 conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.2.3.3-10A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.2.3.3-10B			

Table 5.2.1.2.3.3-10A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the IUT (MCPTT Server)			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (MCPTT Server)			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT Server)			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT Server)			

Table 5.2.1.2.3.3-10B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
emergency-ind	Encrypted (see note) <emergency-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.1.2.3.3-11: SIP INVITE from the SS-UE1 (Step 86, Table 5.2.1.2.3.2-1)

Derivation Path: Table 5.2.1.2.3.3-1 conditions MCPTT, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.2.3.3-11A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.2.3.3-11B			

Table 5.2.1.2.3.3-11A: SDP in SIP INVITE from the SS-UE1 (Table 5.2.1.2.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions SDP_OFFER, IMPLICIT_GRANT_REQUESTED

Table 5.2.1.2.3.3-11B: MCPTT-Info in SIP INVITE from the SS-UE1 (Table 5.2.1.2.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, IMMPERIL-CALL

Table 5.2.1.2.3.3-12: SIP INVITE from the IUT (MCPTT-Server) (Step 88, Table 5.2.1.2.3.2-1)

Derivation Path: Table 5.2.1.2.3.3-2 conditions MCPTT, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.2.3.3-12A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.2.3.3-12B			

Table 5.2.1.2.3.3-12A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-12)

Derivation Path: Table 5.2.1.2.3.3-6A

Table 5.2.1.2.3.3-12B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.1.2.3.3-13: SIP INVITE from the SS-UE1 (Steps 109, Table 5.2.1.2.3.2-1)

Derivation Path: Table 5.2.1.2.3.3-9				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.2.3.3-13A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.2.3.3-13B			

Table 5.2.1.2.3.3-13A: SDP in SIP INVITE from the SS-UE1 (Table 5.2.1.2.3.3-13)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions SDP_OFFER

Table 5.2.1.2.3.3-13B: MCPTT-Info in SIP INVITE from the SS-UE1 (Table 5.2.1.2.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
imminentperil-ind	Encrypted <imminentperil-ind> with mcpttBoolean set to false	Encryption according to note 2 in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1		

Table 5.2.1.2.3.3-14: SIP INVITE from the IUT (MCPTT-Server) (Step 111, Table 5.2.1.2.3.2-1)

Derivation Path: Table 5.2.1.2.3.3-2 conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.2.1.2.3.3-14A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.2.3.3-14B			

Table 5.2.1.2.3.3-14A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-14)

Derivation Path: Table 5.2.1.2.3.3-10A

Table 5.2.1.2.3.3-14B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-14)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
imminentperil-ind	Encrypted (see note) <imminentperil-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.1.2.3.3-15: Floor Release (Steps 8, 9, 38, 39, 80, 81, 122 and 123, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.1.2.3.3-16: Floor Ack (Steps 10, 11, 40, 41, 82, 83, 124 and 125, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 5.2.1.2.3.3-17: Floor Release (Steps 20 and 21, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1

Table 5.2.1.2.3.3-18: Floor Request (Steps 14, 15, 28, 29, 76, 77, 118 and 119, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 5.2.1.2.3.3-19: Floor Request (Steps 24 and 25, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	
Floor Indicator			
Floor Indicator	"1000000000000000"	bit A=1 (Normal call) bit F=0 (Queueing not supported)	

Table 5.2.1.2.3.3-20: Floor Taken (Steps 22 and 23, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1

Table 5.2.1.2.3.3-21: Floor Granted (Steps 16, 17, 36, 37, 78, 79, 120 and 121, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

Table 5.2.1.2.3.3-22: Floor Deny (Steps 26 and 27, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.4-1

Table 5.2.1.2.3.3-23: Floor Queue Position Info (Steps 30, 31, 34 and 35, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.10-1

Table 5.2.1.2.3.3-24: Floor Queue Position Request (Steps 32 and 33, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.9-1

Table 5.2.1.2.3.3-25: Floor Idle
(Steps 12, 13, 42, 43, 74, 75, 84, 85, 116, 117, 126 and 127, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.1.2.3.3-26: Floor Revoke (Steps 12 and 13, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.8-1

Table 5.2.1.2.3.3-27: Floor Release (Steps 51, 52, 61 and 62, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition ACK, EMERGENCY-CALL

Table 5.2.1.2.3.3-28: Floor Ack (Steps 53, 54, 63 and 64, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition DOWNLINK, EMERGENCY-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 5.2.1.2.3.3-29: Floor Request (Steps 57 and 58, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1 condition EMERGENCY-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 5.2.1.2.3.3-30: Floor Granted (Steps 50a1, 50a2, 59 and 60, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition EMERGENCY-CALL

Table 5.2.1.2.3.3-31: Floor Idle (Steps 55, 56, 65 and 66 Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.9-1 condition EMERGENCY-CALL

Table 5.2.1.2.3.3-32: Floor Release (Steps 93, 94, 103 and 104, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition ACK, IMMPERIL-CALL

Table 5.2.1.2.3.3-33: Floor Ack (Steps 95, 96, 105 and 106, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition DOWNLINK, IMMPERIL-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 5.2.1.2.3.3-34: Floor Request (Steps 99 and 100, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1 condition IMMPERIL-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 5.2.1.2.3.3-35: Floor Granted (Steps 92a1, 92a2, 101 and 102, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition IMMPERIL-CALL

Table 5.2.1.2.3.3-36: Floor Idle (Steps 97, 98, 107 and 108 Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.9-1 condition IMMPERIL-CALL

Table 5.2.1.2.3.3-37: SIP BYE from the SS-UE1 (Step 128, 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT (MCPTT Server)		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the UE's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the UE's point of view)		

Table 5.2.1.2.3.3-38: SIP BYE from the IUT (MCPTT-Server) (Step 129, 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261[17]	
value	any allowed value	Non-zero value		

Table 5.2.1.2.3.3-39: SIP 200 (OK) from the IUT (MCPTT-Server) (Steps 72 and 114, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.2.3.3-39A			

Table 5.2.1.2.3.3-39A: SDP in SIP 200 (OK) from the IUT (MCPTT-Server) (Table 5.2.1.2.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER

Table 5.2.1.2.3.3-40: SIP 200 (OK) from the SS (MCPTT-Server) (Steps 70, 112, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.2.3.3-40A			

Table 5.2.1.2.3.3-40A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 5.2.1.2.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER

5.2.1.3 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Automatic Commencement Mode / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / Terminating Participating server

5.2.1.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS (MCPTT Server) initiates an on-demand pre-arranged group call }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the call by sending SIP
messages and interacts with SS-UE2 (MCPTT client) to set up the call by sending SIP messages }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted, Floor Request, Floor Release ) by sending messages to the SS (MCPTT Server) and to the SS-
UE2 (MCPTT client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call }
ensure that {
  when { the SS (MCPTT Server) upgrades the call to an emergency group call }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the emergency call by
sending SIP messages and interacts with SS-UE2 (MCPTT client) to set up the emergency call by
sending SIP messages }
            }
```

(4)

```
with { IUT (MCPTT Server) having established an MCPTT emergency group call }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Idle ) by sending messages
to the SS-UE2 (MCPTT client) }
            }
```

(5)

```
with { IUT (MCPTT Server) having established an MCPTT emergency group call }
ensure that {
  when { the SS (MCPTT Server) downgrades the call to a normal group call }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the normal group call by
sending SIP messages and interacts with SS-UE2 (MCPTT client) to set up the normal group call by
sending SIP messages }
            }
```

(6)

```
with { IUT (MCPTT Server) having established an MCPTT group call }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Idle ) by sending messages
to the SS-UE2 (MCPTT client) }
            }
```

(7)

```
with { IUT (MCPTT Server) having established an MCPTT group call }
ensure that {
  when { the SS (MCPTT Server) upgrades the call to imminent-peril group call }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the imminent-peril group
call by sending SIP messages and interacts with SS-UE2 (MCPTT client) to set up the imminent-peril
group call by sending SIP messages }
            }
```

(8)

```
with { IUT (MCPTT Server) having established an MCPTT imminent-peril group call }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Idle ) by sending messages
to the SS-UE2 (MCPTT client) }
            }
```

(9)

```
with { IUT (MCPTT Server) having established an MCPTT imminent-peril group call }
ensure that {
  when { the SS (MCPTT Server) downgrades the call to a normal group call }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the normal group call by
sending SIP messages and interacts with SS-UE2 (MCPTT client) to set up the normal group call by
sending SIP messages }
            }
```

(10)

```
with { IUT (MCPTT Server) having established an MCPTT group call }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Idle ) by sending messages
to the SS-UE2 (MCPTT client) }
            }
```

(11)

```
with { IUT (MCPTT Server) having established an MCPTT group call }
ensure that {
  when { the SS (MCPTT Server) ends the pre-arranged group call }
    then { IUT (MCPTT Server) sends a SIP BYE message to SS-UE2 (MCPTT client) and forwards a SIP
200 (OK) message to the SS (MCPTT Server) }
            }
```

5.2.1.3.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 10.1.1.4.1.1, 6.3.3.1.2, 10.1.1.3.2, 6.3.2.2.5.1, 6.3.2.2.5.2, 6.3.2.2.4.1, 6.3.2.2.3, 6.3.2.2.9, 6.3.2.2.1, 6.3.2.1.1.1, 6.3.2.2.4.2, 6.3.2.2.2.1, 6.3.2.1.2.1, 10.1.1.3.6, 10.1.1.3.4.1, 6.3.2.2.8.1, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 6.4.2. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.1.1]

"This clause describes the procedures for inviting an MCPTT user to an MCPTT session. The procedure is initiated by the controlling MCPTT function as the result of an action in clause 10.1.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the terminating participating MCPTT function associated to the MCPTT user to be invited.;

NOTE 1: How the controlling MCPTT function finds the address of the terminating MCPTT participating function is out of the scope of the current release.

NOTE 2: If the terminating MCPTT user is part of a partner MCPTT system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCPTT function.

3) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCPTT function;

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcptt-request-uri> element set to the MCPTT ID of the terminating user; and

b) the <mcptt-calling-group-id> element set to the group identity;

NOTE 3: The <mcptt-calling-user-id> is already included in the MIME body as a result of calling clause 6.3.3.1.2 in step 1).

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

6) if the in-progress emergency state of the group is set to a value of "true" the controlling MCPTT function:

a) shall include a Resource-Priority header field populated with the values for an MCPTT emergency group call as specified in clause 6.3.3.1.19;

b) if the received SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "true":

i) shall include in the outgoing SIP INVITE request in the application/vnd.3gpp.mcptt-info+xml MIME body an <emergency-ind> element set to a value of "true"; and

ii) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCPTT user and MCPTT group are authorised for the initiation of MCPTT emergency alerts as determined by the procedures of clause 6.3.3.1.13.1, shall populate the application/vnd.3gpp.mcptt-info+xml MIME body and the application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause 6.3.3.1.12. Otherwise, shall set the <alert-ind> element to a value of "false"; and

c) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCPTT function:

a) shall include a Resource-Priority header field populated with the values for an MCPTT imminent peril group call as specified in clause 6.3.3.1.19; and

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true";

8) void

9) shall send the SIP INVITE request towards the terminating network in accordance with 3GPP TS 24.229 [4].

Upon receiving a SIP 183 (Session Progress) response containing a Require header field with the option tag "100rel" and containing a P-Answer-State header field with the value "Unconfirmed" in response to the SIP INVITE request the controlling MCPTT function:

1) shall send a SIP PRACK request towards the MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

..."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.2]

"This clause is referenced from other procedures.

The controlling MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4].

The controlling MCPTT function:

1) shall include in the Contact header field an MCPTT session identity for the MCPTT session with the g.3gpp.mcptt media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" according to IETF RFC 3840 [16];

2) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [9] in the SIP INVITE request;

4) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

5) shall include a Referred-By header field with the public user identity of the inviting MCPTT client;

6) should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7]. The refresher parameter shall be omitted;

7) shall include the Supported header field set to "timer";

...

11) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing INVITE request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

This clause covers both on-demand session and pre-established sessions.

Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

...

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [18] if the "SIP INVITE request for terminating participating MCPTT function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; and

..."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.5.1]

"When receiving a "SIP INVITE request for terminating participating MCPTT function" that requires automatic commencement mode:

1) if:

a) the invited MCPTT client has one or more pre-established sessions without an associated MCPTT session;

b) the media-level section for the offered MCPTT speech media stream is the same as the media-level section for MCPTT speech media stream in an existing pre-established session; and

c) the media-level section of the offered media-floor control entity is the same as the media-level section for media-floor control entity in an existing pre-established session;

then the participating MCPTT function shall perform the actions specified in clause 6.3.2.2.5.3; or

2) otherwise the participating MCPTT function shall perform the actions specified in clause 6.3.2.2.5.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.5.2]

"When receiving a "SIP INVITE request for terminating participating MCPTT function" for an on-demand session that requires automatic commencement mode the participating MCPTT function:

1) if:

a) the incoming SIP INVITE request contained a Priv-Answer-Mode header field set to the value of "Auto";

b) no Answer-Mode header field or Priv-Answer-Mode header field were received in the incoming SIP INVITE request and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as defined in clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; or

c) the incoming SIP INVITE request contained an Answer-Mode header field set to "Auto" and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as defined in clause 7.3.3 or clause 7.3.4 is set to "auto-answer";

then:

a) shall generate a SIP 183 (Session Progress) response to the "SIP INVITE request for terminating participating MCPTT function" as specified in clause 6.3.2.2.4.1;

b) if the received SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body with the <ambient-listening-type> element set to a value of "remote-init" shall include in the SIP 183 (Session Progress) response an alert-info header field set to a value of "<file:///dev/null>" according to IETF RFC 3261 [24] and as updated by IETF RFC 7462 [77]; and

NOTE: The SIP 183 (session Progress) response can be sent reliably or unreliably depending on the content of the received SIP INVITE request. Regardless of if the SIP 183 (Session Progress) response is sent reliably or unreliably, SDP is not included in the SIP 183 (Session Progress) response.

c) shall set the P-Answer-State header field to "Unconfirmed" in the SIP 183 (Session Progress) response;

2) shall copy the public user identity contained in the Request-URI of the incoming "SIP INVITE request for terminating participating MCPTT function" to the P-Asserted-Identity header field of the SIP 183 (Session Progress) response;

3) shall generate a SIP INVITE request as specified in clause 6.3.2.2.3;

4) shall set the Request-URI to the public user identity associated to the MCPTT ID of the MCPTT user to be invited;

5) shall perform the procedures specified in clause 6.3.2.2.9 to include any MIME bodies in the received SIP INVITE request, into the outgoing SIP INVITE request;

6) shall copy the contents of the P-Asserted-Identity header field of the incoming "SIP INVITE request for terminating participating MCPTT function" to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

7) if the Priv-Answer-Mode header field is present in the incoming SIP INVITE request with a value of "Auto", shall include a Priv-Answer-Mode header field with the value "Auto" in the outgoing SIP INVITE request. Otherwise, if the Answer-Mode header field is present in the incoming SIP INVITE request, the participating MCPTT function shall include an Answer-Mode header field with the value "Auto" in the outgoing SIP INVITE request;

8) if no Answer-Mode header field or Priv-Answer-Mode header field were received in the incoming SIP INVITE request and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as defined in clause 7.3.3 or clause 7.3.4 is set to "auto-answer", shall set the Answer-Mode header field to "Auto" in the outgoing SIP INVITE request;

9) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for terminating participating MCPTT function" as specified in clause 6.3.2.2.1;

10) if the received SIP INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field; and

11) shall send the SIP INVITE request towards the MCPTT client according to 3GPP TS 24.229 [4].

If the SIP 183 (Session Progress) response was sent reliably, then upon receiving a SIP PRACK request, the participating MCPTT function shall generate a SIP 200 (OK) response to the SIP PRACK request and forward the SIP 200 (OK) response, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the above SIP INVITE request sent to the MCPTT client, the participating MCPTT function:

1) if the SIP 183 (Session Progress) was sent unreliably, shall send the SIP 200 (OK) response immediately; and

2) if the SIP 183 (Session Progress) was sent reliably and,

a) if the SIP PRACK request to the SIP 183 (Session Progress) response has been received by the participating MCPTT function and the SIP 200 (OK) response to the SIP PRACK request has been sent, shall send the SIP 200 (OK) response immediately;

b) if the SIP PRACK request to the SIP 183 (Session Progress) response has not yet been received, then upon receipt of the SIP PRACK request, the participating MCPTT function shall generate a SIP 200 (OK) response to the SIP PRACK request and forward the SIP 200 (OK) response, according to 3GPP TS 24.229 [4], before sending the SIP 200 (OK) response to the "SIP INVITE request for terminating participating MCPTT function".

When the participating MCPTT function sends the SIP 200 (OK) response to the "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

1) shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1;

3) shall copy the P-Asserted-Identity header field from the incoming SIP 200 (OK) response to the outgoing SIP 200 (OK) response;

3A) if the SIP 183 (Session Progress) was sent with the P-Answer-State header field set to "Unconfirmed", shall set the P-Answer-State header field to "Confirmed" in the outgoing SIP 200 (OK) response;

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

5) shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [4].

The participating MCPTT function shall forward any other SIP response that does not contain SDP along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.4.1]

"This clause is referenced from other procedures.

When sending a SIP provisional response other than the SIP 100 (Trying) response to the SIP INVITE request, the participating MCPTT function shall generate a SIP provisional response according to 3GPP TS 24.229 [4] and:

1) *shall include the following in the Contact header field:*

   a) *the g.3gpp.mcptt media feature tag; and*

   b) *the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";*

2) *if the outgoing SIP provisional response is to be sent in response to the receipt of a SIP provisional response and the response contains an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP provisional response; and*

3) *if the incoming SIP INVITE request included the Supported header field with the value "100rel" and according to local policy, may include the Require header field with the value "100rel"."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.3]

"The participating MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4] and:

1) *shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [6] if included in the incoming SIP INVITE request;*

2) *should include the Session-Expires header field according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";*

3) *shall include the option tag "timer" in the Supported header field;*

4) *shall include the following in the Contact header field:*

   a) *the g.3gpp.mcptt media feature tag;*

   b) *the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";*

   c) *the isfocus media feature tag;*

   d) *an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the incoming SIP INVITE request; and*

   e) *any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;*

5) *shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];*

6) *shall include the option tag "norefersub" in a Supported header field according to rules and procedures of IETF RFC 4488 [22];*

7) *may include a Resource-Share header field in accordance with clause 5.7.1.20.3 in 3GPP TS 24.229 [4]; and*

8) *if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP INVITE request."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.9]

"This clause is referenced from other procedures.

If the incoming SIP request contains an application/vnd.3gpp.mcptt-info+xml MIME body, the participating MCPTT function:

1) if not already copied:

a) shall copy the contents of the application/vnd.3gpp.mcptt-info+xml MIME body received in the SIP request into an application/vnd.3gpp.mcptt-info+xml MIME body included in the outgoing SIP request; and

b) if an <MKFC-GKTP> element was included in the application/vnd.3gpp.mcptt-info+xml MIME body received in the SIP request, the <MKFC-GKTP> element shall not be copied into the application/vnd.3gpp.mcptt-info+xml MIME body included in the outgoing SIP request.

If the received SIP request contains an application/vnd.3gpp.location-info+xml MIME body as specified in Annex F.3:

1) if not already copied, shall copy the contents of the application/vnd.3gpp.mcptt-location-info+xml MIME body received in the SIP request into an application/vnd.3gpp.mcptt-location-info+xml MIME body included in the outgoing SIP request.

If the received SIP request contains an application/resource-lists+xml MIME body:

1) if not already copied, shall copy the contents of the application/resource-lists+xml MIME body received in the SIP request into an application/resource-lists+xml MIME body included in the outgoing SIP request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.1]

"The participating MCPTT function shall follow the procedures in clause 6.3.2.1.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCPTT function:

1) shall contain only one SDP media-level section for MCPTT speech as contained in the received SDP offer; and

2) shall contain one SDP media-level section for media plane control messages, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [4], the participating MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCPTT function, if required;

NOTE 1: Requirements can exist for the participating MCPTT function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the participating MCPTT function; and

NOTE 2: If the participating MCPTT function and the controlling MCPTT function or the participating MCPTT function and the non-controlling MCPTT function are in the same MCPTT server, and the participating MCPTT function does not have a dedicated IP address or a dedicated port number for media plane control channel or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.4.2]

"This clause is referenced from other procedures.

When sending SIP 200 (OK) responses, the participating MCPTT function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [4] and:

1) shall include the option tag "timer" in a Require header field;

2) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";

3) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and

c) an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCPTT function;

4) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23]; and

5) if the incoming SIP response contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP 200 (OK) response."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.2.1]

"The participating MCPTT function shall follow the procedures in clause 6.3.2.1.2.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [4], the participating MCPTT function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCPTT function, for the accepted media stream in the received SDP offer, if required; and

NOTE 1: Requirements can exist for the participating MCPTT function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCPTT function, for the accepted media plane control channel, if present in the received SDP offer.

NOTE 2: If the participating MCPTT function and the controlling MCPTT function or the participating MCPTT function and the non-controlling MCPTT function are in the same MCPTT server, and the participating MCPTT function does not have a dedicated IP address or a dedicated port number for media plane control channel or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.3.6]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a SIP re-INVITE request for a terminating MCPTT client of a MCPTT group containing an emergency indication or imminent peril indication, the participating MCPTT function:

1) *shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may perform further actions outside the scope of this specification to act upon an included Resource-Priority header field as specified in 3GPP TS 24.229 [4];*

2) *shall generate an outgoing SIP re-INVITE request as specified in clause 6.3.2.2.10;*

3) *shall include in the SIP re-INVITE request an SDP offer based on the SDP offer in the received SIP re-INVITE request as specified in clause 6.3.2.2.1; and*

4) *shall send the SIP re-INVITE request towards the MCPTT client according to 3GPP TS 24.229 [4].*

Upon receiving a SIP 200 (OK) response to the above SIP re-INVITE request sent to the MCPTT client, the participating MCPTT function:

1) *shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;*

2) *shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1;*

3) *shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and*

4) *shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [4]."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.10]

"This clause is referenced from other procedures.

The participating MCPTT function shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [4] and:

1) *shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];*

2) *may include a Resource-Share header field in accordance with clause 5.7.1.20.3 in 3GPP TS 24.229 [4];*

3) *shall perform the procedures specified in clause 6.3.2.2.9 to copy any MIME bodies in the received SIP (re-)INVITE request to the outgoing SIP re-INVITE request; and*

4) *if the received SIP (re-)INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), 10.1.1.3.4.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), 6.3.2.2.8.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function:

1) *shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5] for releasing media plane resource associated with the SIP session with the controlling MCPTT function;*

2) *shall generate a SIP BYE request according to 3GPP TS 24.229 [4];*

3) *shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;*

4) *if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and*

5) *shall send the SIP BYE request to the MCPTT client according to 3GPP TS 24.229 [4].*

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCPTT function:

1) shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCPTT function according to 3GPP TS 24.229 [4]; and

2) shall interact with the media plane as specified in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT client."

5.2.1.3.3 Test description

5.2.1.3.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the controlling MCPTT function.
- SS-UE2 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.3.3.1-1.

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the terminating participating MCPTT function
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.3, and to the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- The SS-UE2 (MCPTT client) is affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.

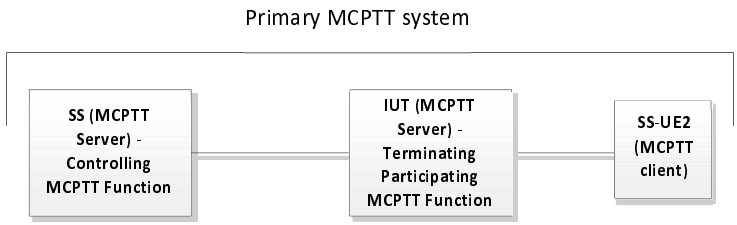


Figure 5.2.1.3.3.1-1: Functions of the testing components

5.2.1.3.3.2 Test procedure sequence

Table 5.2.1.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) initiates an on-demand pre-arranged group call.	<--	SIP INVITE		-
2	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place when one or the other is the case.	-	-	-	-
3a1	Check: Does the IUT (MCPTT Server) respond to the SIP INVITE message with a SIP 183 (Session Progress) message sent reliably?	<--	SIP 183 (Session Progress)	1	P
3b1	Check: Does the IUT (MCPTT Server) respond to the SIP INVITE message with a SIP 183 (Session Progress) message sent unreliably?	<--	SIP 183 (Session Progress)	1	P
-	EXCEPTION: Depending on MCPTT Server implementation, steps 4a1-5 will take place in parallel if the SIP 183 (Session Progress) is sent reliably.	-	-	-	-
4a1	The SS (MCPTT Server) sends a SIP PRACK message to the IUT (MCPTT Server)	-->	SIP PRACK		-
4a2	Check: Does the IUT (MCPTT Server) respond to the SIP PRACK with a SIP 200 (OK) message?	<--	SIP 200 OK	1	P
5	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS-UE2 (MCPTT client)?	-->	SIP INVITE	1	P
6	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
7	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS-UE2 (MCPTT client)?	-->	SIP ACK	1	P
8	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server)?	-->	SIP 200 (OK)	1	P
9	The SS (MCPTT Server) responds with a SIP ACK message	<--	SIP ACK		-
10	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
11	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	2	P
12	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
13	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	2	P
14	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
15	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	2	P
16	The SS-UE2 (MCPTT client) sends a Floor Request message with a higher priority than SS-UE1 (MCPTT client).	<--	Floor Request		-
17	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT server)?	-->	Floor Request	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
18	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
19	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE2 (MCPTT client)?	-->	Floor Granted	2	P
20	The SS-UE2 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
21	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	2	P
22	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
23	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE2 (MCPTT client)?	-->	Floor Ack	2	P
24	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
25	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	2	P
26	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
27	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	2	P
28	The SS (MCPTT Server) sends a SIP INVITE requesting the modification of a call to upgrade the call to an emergency group call.	<--	SIP INVITE		-
29	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	3	P
30	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS-UE2 (MCPTT client)?	-->	SIP INVITE	3	P
31	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
32	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS-UE2 (MCPTT client)?	-->	SIP ACK	3	P
33	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server)?	-->	SIP 200 (OK)	3	P
34	The SS (MCPTT Server) responds with a SIP ACK message	<--	SIP ACK		-
35	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
36	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	4	P
37	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
38	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	4	P
39	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
40	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	4	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
41	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
42	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	4	P
43	The SS (MCPTT Server) sends a SIP INVITE requesting the modification of a call to downgrade the call to a normal group call.	<--	SIP INVITE		-
44	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	5	P
45	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS-UE2 (MCPTT client)?	-->	SIP INVITE	5	P
46	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
47	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS-UE2 (MCPTT client)?	-->	SIP ACK	5	P
48	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server)?	-->	SIP 200 (OK)	5	P
49	The SS (MCPTT Server) responds with a SIP ACK message	<--	SIP ACK		-
50	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
51	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	6	P
52	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
53	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	6	P
54	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
55	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	6	P
56	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
57	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	6	P
58	The SS (MCPTT Server) sends a SIP INVITE requesting the modification of a call to upgrade the call to an imminent-peril group call.	<--	SIP INVITE		-
59	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	7	P
60	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS-UE2 (MCPTT client)?	-->	SIP INVITE	7	P
61	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
62	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS-UE2 (MCPTT client)?	-->	SIP ACK	7	P
63	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server)?	-->	SIP 200 (OK)	7	P
64	The SS (MCPTT Server) responds with a SIP ACK message	<--	SIP ACK		-
65	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
66	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	8	P
67	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
68	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	8	P
69	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
70	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	8	P
71	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
72	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	8	P
73	The SS (MCPTT Server) sends a SIP INVITE requesting the modification of a call to downgrade the call to a normal group call.	<--	SIP INVITE		-
74	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	9	P
75	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS-UE2 (MCPTT client)?	-->	SIP INVITE	9	P
76	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
77	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS-UE2 (MCPTT client)?	-->	SIP ACK	9	P
78	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server)?	-->	SIP 200 (OK)	9	P
79	The SS (MCPTT Server) responds with a SIP ACK message	<--	SIP ACK		-
80	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
81	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	10	P
82	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
83	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	10	P
84	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
85	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	10	P
86	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
87	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	10	P
88	The SS (MCPTT Server) sends a SIP BYE request	<--	SIP BYE		-
89	Check: Does the IUT (MCPTT Server) forward the SIP BYE message to the SS-UE2 (MCPTT client) to end the call?	-->	SIP BYE	11	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
90	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
91	Check: Does the IUT (MCPTT Server) forward the SIP 200 (OK) response to SS (MCPTT Server)	-->	SIP 200 (OK)	11	P

5.2.1.3.3.3 Specific message contents

Table 5.2.1.3.3.3-1: SIP INVITE from the SS (MCPTT Server) (Step 1, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_A	The URI of the IUT (MCPTT Server), providing the terminating participating function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server which is simulated by the SS, providing controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the IUT (MCPTT Server), providing the terminating participating function.		
Max-Forwards				
value	"70"	The recommended initial value is 70 in IETF RFC 3261 [17].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.3.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.3.3.3-1B			

Table 5.2.1.3.3.3-1A: SDP in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.3.3.3-1B: MCPTT-Info in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.3.3.3-2: SIP 183 (Session Progress) from the IUT (MCPTT-Server) (Step 3a1, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.1-1, conditions MCPTT, 100rel

Table 5.2.1.3.3.3-3: SIP 183 (Session Progress) from the IUT (MCPTT-Server) (Step 3b1, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.1-1, conditions MCPTT

Table 5.2.1.3.3.3-4: SIP PRACK from the SS (MCPTT Server) (Step 4a1, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.10.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Request-URI	same URI as the MCPTT-Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT (MCPTT-Server)		
Max-Forwards				
value	"70"	The recommended initial value is 70 in IETF RFC 3261 [17].		

Table 5.2.1.3.3.3-5: SIP 200 OK to PRACK from the IUT (MCPTT-Server) (Step 4a2, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Content-Type	not present			
Content-Length	not present			

Table 5.2.1.3.3.3-6: SIP INVITE from the IUT (MCPTT-Server) (Step 5, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the IUT (MCPTT-Server)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	MCPTT Server's port number		
via-branch	Value assigned by the IUT (MCPTT-Server)			
Record-Route	not present			
From				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server, providing terminating participating function.		
To			IETF RFC 3261 [17] IETF RFC 5031 [19]	
addr-spec				
user-info and host	px_MCX_SIP_PublicUserId_B	Default public user ID (IMPU) as stored in the UICC		
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
Max-Forwards				
value	"70"	The recommended initial value is 70 in IETF RFC 3261 [17].		
Message-body				
MIME body part		**SDP message**		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME-part-body	SDP message as described in Table 5.2.1.3.3.3-6A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.3.3.3-6B			

Table 5.2.1.3.3.3-6A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the IUT (MCPTT-Server)			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (MCPTT-Server)			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT-Server)			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT-Server)			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.3.3.3-6B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.3.3.3-7: SIP 200 (OK) from the SS-UE2 (Step 6, Table 5.2.1.3.3.2-1)

Derivation Path: Annex A, Table A.3.2.4-7

Table 5.2.1.3.3.3-8: SIP 200 (OK) from the IUT (MCPTT-Server) (Step 8, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.3.3.3-8A			

Table 5.2.1.3.3.3-8A: SDP in SIP 200 (OK) from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-8)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the IUT (MCPTT-Server)			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (MCPTT-Server)			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the IUT (MCPTT-Server) (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the IUT (MCPTT-Server)			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the IUT (MCPTT-Server) being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT-Server)			

Table 5.2.1.3.3.3-9: SIP INVITE from the SS (MCPTT Server) (Step 28, Table 5.2.1.3.3.2-1)

Derivation Path: Table 5.2.1.3.3.3-1, conditions MCPTT, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IUT (MCPTT-Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT (MCPTT-Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 5.2.1.3.3.3-9A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.3.3.3-9B			

Table 5.2.1.3.3.3-9A: SDP in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.3.3.3-9)

Derivation Path: Table 5.2.1.3.3.3-1A, condition SDP_OFFER instead of INITIAL_SDP_OFFER

Table 5.2.1.3.3.3-9B: MCPTT-Info in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.3.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, INVITE_REFER, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.3.3.3-10: SIP INVITE from the IUT (MCPTT-Server) (Step 30, Table 5.2.1.3.3.2-1)

Derivation Path: Table 5.2.1.3.3.3-6, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.3.3.3-10A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.3.3.3-10B			

Table 5.2.1.3.3.3-10A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-10)

Derivation Path: Table 5.2.1.3.3.3-6A, condition SDP_OFFER instead of INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin	Same o=line as in the previous SDP message sent by the MCPTT-Server except that sess-version is incremented by one			

Table 5.2.1.3.3.3-10B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.3.3.3-11: SIP 200 (OK) from the SS-UE2 (Steps 31, 46, 61 and 76, Table 5.2.1.3.3.2-1)

Derivation Path: Annex A, Table A.3.2.4-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
SDP message	SDP message as described in Table 5.2.1.3.3.3-11A			

Table 5.2.1.3.3.3-11A: SDP in SIP 200 (OK) (Table 5.2.1.3.3.2-11)

Derivation Path: Annex A, Table A.3.2.4-8				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin	Same o=line as in the previous SDP message sent by the SS-UE2 except that sess-version is incremented by one			

Table 5.2.1.3.3.3-12: SIP 200 (OK) from the IUT (MCPTT-Server) (Steps 33, 48, 63 and 78, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.3.3.3-12A			

Table 5.2.1.3.3.3-12A: SDP in SIP 200 (OK) from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.2.1.3.3.3-8A				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin	Same o=line as in the previous SDP message sent by the IUT (MCPTT-Server) except that sess-version is incremented by one			

Table 5.2.1.3.3.3-13: SIP INVITE from the SS (MCPTT Server) (Step 43, Table 5.2.1.3.3.2-1)

Derivation Path: Table 5.2.1.3.3.3-1, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IUT (MCPTT-Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT (MCPTT-Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 5.2.1.3.3.3-13A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.3.3.3-13B			

Table 5.2.1.3.3.3-13A: SDP in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.3.3.3-13)

Derivation Path: Table 5.2.1.3.3.3-9A

Table 5.2.1.3.3.3-13B: MCPTT-Info in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.3.3.3-13)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
emergency-ind	Encrypted (see note) <emergency-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.3.3.3-14: SIP INVITE from the IUT (MCPTT-Server) (Step 45, Table 5.2.1.3.3.2-1)

Derivation Path: Table 5.2.1.3.3.3-6, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.3.3.3-14A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.3.3.3-14B			

Table 5.2.1.3.3.3-14A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-14)

Derivation Path: Table 5.2.1.3.3.3-10A

Table 5.2.1.3.3.3-14B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-14)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
emergency-ind	Encrypted (see note) <emergency-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.3.3.3-15: SIP INVITE from the SS (MCPTT Server) (Step 58, Table 5.2.1.3.3.2-1)

Derivation Path: Table 5.2.1.3.3.3-1, conditions MCPTT, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IUT (MCPTT-Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT (MCPTT-Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 5.2.1.3.3.3-15A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.3.3.3-15B			

Table 5.2.1.3.3.3-15A: SDP in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.3.3.3-15)

Derivation Path: Table 5.2.1.3.3.3-9A

Table 5.2.1.3.3.3-15B: MCPTT-Info in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.3.3.3-15)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, INVITE_REFER, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.3.3.3-16: SIP INVITE from the IUT (MCPTT-Server) (Step 60, Table 5.2.1.3.3.2-1)

Derivation Path: Table 5.2.1.3.3.3-6, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.3.3.3-16A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.3.3.3-16B			

Table 5.2.1.3.3.3-16A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-16)

Derivation Path: Table 5.2.1.3.3.3-10A

Table 5.2.1.3.3.3-16B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-16)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.3.3.3-16a: SIP INVITE from the SS (MCPTT Server) (Step 73, Table 5.2.1.3.3.2-1)

Derivation Path: Table 5.2.1.3.3.3-1, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IUT (MCPTT-Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT (MCPTT-Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 5.2.1.3.3.3-16aA			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 5.2.1.3.3.3-16aB			

Table 5.2.1.3.3.3-16aA: SDP in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.3.3.3-16a)

Derivation Path: Table 5.2.1.3.3.3-9A

Table 5.2.1.3.3.3-16aB: MCPTT-Info in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.3.3.3-16a)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
imminentperil-ind	Encrypted (see note) <imminentperil-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.3.3.3-16b: SIP INVITE from the IUT (MCPTT-Server) (Step 75, Table 5.2.1.3.3.2-1)

Derivation Path: Table 5.2.1.3.3.3-6, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.3.3.3-16bA			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.3.3.3-16bB			

Table 5.2.1.3.3.3-16bA: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-16b)

Derivation Path: Table 5.2.1.3.3.3-10A

Table 5.2.1.3.3.3-16bB: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.3.3.3-16b)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
imminentperil-ind	Encrypted (see note) <imminentperil-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.3.3.3-17: Floor Taken (Steps 10, 11, 14, 15, 24, 25, 50, 51, 54, 55, 80, 81, 84 and 85, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.1.3.3.3-18: Floor Idle (Steps 12, 13, 26, 27, 52, 53, 56, 57, 82, 83, 86 and 87, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.1.3.3.3-19: Floor Request (Steps 16 and 17, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"2"	The highest priority for the user	

Table 5.2.1.3.3.3-20: Floor Granted (Steps 18 and 19, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"2"	The highest priority for the user	

Table 5.2.1.3.3.3-20: Floor Release (Steps 20 and 21, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.1.3.3.3-21: Floor Ack (Steps 22 and 23, Table 5.2.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 5.2.1.3.3.3-22: Floor Taken (Steps 35, 36, 39 and 40, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1 condition EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.1.3.3.3-23: Floor Idle (Steps 37, 38, 41 and 42, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1 condition EMERGENCY-CALL

Table 5.2.1.3.3.3-24: Floor Taken (Steps 65, 66, 69 and 70, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1 condition IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.1.3.3.3-25: Floor Idle (Steps 67, 68, 71 and 72, Table 5.2.1.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1 condition IMMPERIL-CALL

Table 5.2.1.3.3.3-26: SIP BYE from the SS (MCPTT Server) (Step 88, 5.2.1.3.3.2-1)

Derivation Path: Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IUT (MCPTT-Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT (MCPTT-Server)		
To				
addr-spec	Same URI of the IUT (MCPTT-Server) as used earlier in the dialog	Remote URI of the dialog (from the UE's point of view)		
tag	Same tag of the IUT (MCPTT-Server) as used earlier in the dialog	Remote tag of the dialog ID (from the UE's point of view)		

Table 5.2.1.3.3.3-27: SIP BYE from the IUT (MCPTT-Server) (Step 89, 5.2.1.3.3.2-1)

Derivation Path: Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From				
addr-spec	Same URI of the IUT (MCPTT-Server) as used earlier in the dialog	Local URI of the dialog (from the IUT (MCPTT-Server)'s point of view)		
tag	Same tag of the IUT (MCPTT-Server) as used earlier in the dialog	Local tag of the dialog (from the IUT (MCPTT-Server)'s point of view)		
Max-Forwards			**IETF** RFC 3261 [**17**]	
value	any allowed value	Non-zero value		

5.2.1.4 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Temporary Group / Controlling server / Trusted mode

5.2.1.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE1 (MCPTT client) initiates an on-demand pre-arranged group call to a temporary
group with implicit floor control }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the calls by sending SIP
messages }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call to a
temporary group with implicit floor control }
ensure that {
  when { the SS-UE1 (MCPTT client), SS-UE2 (MCPTT client), SS-UE3 (MCPTT client) and SS-UE4 (MCPTT
client) engage in communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted ) by sending messages via the SS (MCPTT Server) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call to a
temporary group with implicit floor control }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the pre-arranged group call }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) by sending a SIP 200 (OK) message
to the client ending the call and sends a SIP BYE message towards the other participants }
            }
```

5.2.1.4.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 10.1.1.4.1.1, 10.1.1.4.1.2, 10.1.1.4.2, 6.3.3.1.2, 6.3.3.1.1, 6.3.5.2, 6.3.5.5, 6.3.3.2.3.2, 6.3.3.2.1, 10.1.1.4.3, 6.3.8.1, 6.3.3.1.5, 4.8, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 6.3.2.2, 6.3.3, 14.3.4. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.1.1]

"This clause describes the procedures for inviting an MCPTT user to an MCPTT session. The procedure is initiated by the controlling MCPTT function as the result of an action in clause 10.1.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

…

NOTE 1: How the controlling MCPTT function finds the address of the terminating MCPTT participating function is out of the scope of the current release.

…

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcptt-request-uri> element set to the MCPTT ID of the terminating user; and

b) the <mcptt-calling-group-id> element set to the group identity;

NOTE 3: The <mcptt-calling-user-id> is already included in the MIME body as a result of calling clause 6.3.3.1.2 in step 1).

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

…

Upon receiving a SIP 183 (Session Progress) response containing a Require header field with the option tag "100rel" and containing a P-Answer-State header field with the value "Unconfirmed" in response to the SIP INVITE request the controlling MCPTT function:

1) shall send a SIP PRACK request towards the MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

2) shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as

…

NOTE 5: The procedures executed by the controlling MCPTT function prior to sending a response to the inviting MCPTT client are specified in clause 10.1.1.4.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.1.2]

"The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the non-controlling MCPTT function serving the group identity of the MCPTT group owned by the partner MCPTT system;

3) shall set the P-Asserted-Identity to the public service identity of the controlling MCPTT function;

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcptt-request-uri> element set to the group identity of the MCPTT group hosted by the non-controlling MCPTT function in the partner MCPTT system; and

b) the <mcptt-calling-group-id> element set to the group identity of the group served by the controlling MCPTT function;

…

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

8) shall send the SIP INVITE request towards the partner MCPTT system in accordance with 3GPP TS 24.229 [4].

Upon receiving SIP 403 (Forbidden) response for the SIP INVITE request, if according to local policy and if:

1) the response contains a Warning header field with the MCPTT warning code "128"; and

2) the response contains a P-Refused-URI-List header field and an application/resource-lists+xml MIME body as specified in IETF RFC 5318 [36];

NOTE 1: The application/resource-lists+xml MIME body contains MCPTT IDs identifying MCPTT users in a partner MCPTT system that needs to be invited to the prearranged group call in case of group regrouping using interrogating method as specified in 3GPP TS 23.379 [3] clause 10.6.2.4.2.

then the controlling MCPTT function:

1) shall check if the number of members of the MCPTT group exceeds the value contained in the <on-network-max-participant-count> element of the group document as specified in 3GPP TS 24.481 [31]. If exceeded, the controlling MCPTT function shall invite only <on-network-max-participant-count> members from the application/resource-lists+xml MIME body; and

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the prearranged group session It is operator policy that determines which participants in the application/resource-lists+xml MIME body are invited to the group call.

2) shall invite MCPTT users as specified in this clause using the list of MCPTT IDs in URI-List.

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 3: The procedures executed by the controlling MCPTT function prior to sending a response to the inviting MCPTT client are specified in clause 10.1.1.4.2.

2) if at least one of the invited MCPTT clients has subscribed to the conference package, shall subscribe to the conference event package in the non-controlling MCPTT function as specified in clause 10.1.3.4.3; and

3) if the 200 (OK) response includes the <floor-state> element set to "floor-taken", shall wait for a SIP INFO request containing a floor request from the non-controlling MCPTT function.

Upon receiving a SIP INFO request containing a floor request where:

1) the Request-URI contains an MCPTT session ID identifying an ongoing temporary group session; and

2) the application/vnd.3gpp.mcptt-info+xml MIME body contains the <mcptt-calling-group-id> element with the MCPTT group ID of a MCPTT group invited to the temporary group session;

then the controlling MCPTT function:

1) shall send a SIP 200 (OK) response to the SIP INFO request to the non-controlling MCPTT function as specified in 3GPP TS 24.229 [4]; and

2) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.2]

"In the procedures in this clause:

1) MCPTT ID in an incoming SIP INVITE request refers to the MCPTT ID of the originating user from the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

3) MCPTT ID in an outgoing SIP INVITE request refers to the MCPTT ID of the called user in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the outgoing SIP INVITE request;

4) indication of required group members in a SIP 183 (Session Progress) response refers to the <required> element of the application/vnd.3gpp.mcptt-info+xml MIME body set to "true" in a SIP 183 (Session Progress) sent by the non-controlling MCPTT function of an MCPTT group;

5) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

6) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a "SIP INVITE request for controlling MCPTT function of an MCPTT group", the controlling MCPTT function:

…

NOTE 1: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating an MCPTT emergency group call as determined by clause 6.3.3.1.13.2, or for originating an MCPTT imminent peril group call as determined by clause 6.3.3.1.13.5, the controlling MCPTT function can according to local policy choose to accept the request.

…

4) if received SIP INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

5) if the group identity is associated with a group document maintained by the GMS:

NOTE 2: How the MCPTT server determines that a group identity represents a group for which a group document is stored in the GMS is an implementation detail.

a) shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2;

b) if the group referred to by the group identity has been regrouped, shall:

i) stop processing the SIP INVITE request;

ii) shall reject the SIP INVITE request with a SIP 403 (Forbidden) response with the warning text set to "148 group is regrouped" as specified in clause 4.4 "Warning header field";

iii) if the group referred to by the group identity has been regrouped based on a preconfigured group, shall send a copy of the notifying SIP MESSAGE that was generated and sent per clause 16.2.4.1 to the participating function for the MCPTT ID of the incoming SIP INVITE request; and

iv) skip the rest of the steps;

c) if the result of the initial processing in clause 6.3.5.2 was:

i) that authorization of the MCPTT ID is required at a non-controlling MCPTT function of an MCPTT group is required, perform the actions in clause 6.3.3.1.13.7 and do not continue with the rest of the steps in this clause; and

ii) that a SIP 3xx, 4xx, 5xx or 6xx response to the "SIP INVITE request for controlling MCPTT function of an MCPTT group" has been sent, do not continue with the rest of the steps in this clause;

6) if the group identity is associated with a user or group regroup based on a preconfigured group:

a) shall retrieve the stored information for the group identity;

b) if there is no stored information for the group identity, the controlling MCPTT function:

i) shall return a SIP 404 (Not Found) response with the warning text set to "163 the group identity indicated in the request does not exist" as specified in clause 4.4 "Warning header field" and shall not continue with the rest of the steps;

NOTE 3: The user or group regroup can have been removed very recently and the client has sent the group call request prior to receiving the removal notification.

7) shall perform the actions as described in clause 6.3.3.2.2;

…

14) if the MCPTT group call is not ongoing then:

…

c) if the received SIP INVITE request contains an authorised request for an MCPTT emergency group call as determined by clause 6.3.3.1.13.2 or MCPTT imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCPTT user is eligible to be implicitly affiliated with the MCPTT group as determined in step 14) a) iv) above, shall perform the implicit affiliation as specified in clause 9.2.2.3.7;

d) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [4];

e) shall create a prearranged group session and allocate an MCPTT session identity for the prearranged group call, and shall handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

f) if the group identity in the "SIP INVITE request for controlling MCPTT function of an MCPTT group" is a TGI:

i) shall for each of the constituent MCPTT groups homed on the primary MCPTT system:

A) if the controlling MCPTT function does not own the MCPTT group identified by the MCPTT group ID, then generate a SIP INVITE request towards the MCPTT server that owns the MCPTT group identity by following the procedures in clause 10.1.1.4.1.2; and

NOTE 4: The MCPTT server that the SIP INVITE request is sent to acts as a non-controlling MCPTT function;

B) if the controlling MCPTT function owns the MCPTT group identified by the MCPTT group ID then:

I) determine the members to invite to the prearranged MCPTT group call as specified in clause 6.3.5.5;

II) invite each group member determined in step A) above, to the group session, as specified in clause 10.1.1.4.1.1; and

III) interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3; and

ii) shall for each of the constituent MCPTT groups homed on the partner MCPTT system generate a SIP INVITE request for the MCPTT group identity homed on the partner MCPTT system as specified in clause 10.1.1.4.1.2; and

g) if the group identity in the SIP INVITE request for controlling MCPTT function of an MCPTT group is an MCPTT group ID:

i) shall determine the members to invite to the prearranged MCPTT group call as specified in clause 6.3.5.5. If:

A) the number of affiliated members of the MCPTT group is lower than the value contained in the <on-network-minimum-number-of-affiliated-members> element of the group document as specified in 3GPP TS 24.481 [31]; or

B) the group document contains any <on-network-affiliation-to-group-required> group member as specified in 3GPP TS 24.481 [31] that is not affiliated;

then the controlling MCPTT function shall send a SIP 480 (Temporarily Unavailable) response to the MCPTT client that originated the group session with the warning text set to "112 group call abandoned due to required group members not part of the group session" in a Warning header field as specified in clause 4.4 and skip the rest of the steps below;

ii) if necessary, shall start timer TNG1 (acknowledged call setup timer) according to the conditions stated in clause 6.3.3.3;

iii) if the received SIP INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element set to a value of "true":

A) shall cache the information that the MCPTT user has initiated an MCPTT emergency call;

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCPTT emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that the MCPTT user has initiated an MCPTT emergency alert; and

C) if the in-progress emergency state of the group is set to a value of "false":

I) shall set the value of the in-progress emergency state of the group to "true"; and

II) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iv) if the in-progress emergency state of the group is set to a value of "false" and if the received SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCPTT function shall:

A) shall cache the information that the MCPTT user has initiated an MCPTT imminent peril call; and

B) if the in-progress imminent peril state of the group is set to a value of "false", shall set the in-progress imminent peril state of the group to a value of "true";

v) shall invite each group member determined in step 13)g)i) above, to the group session, as specified in clause 10.1.1.4.1.1; and

vi) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3; and

15) if the MCPTT group call is ongoing then:

…

e) if the received SIP INVITE request contains an authorised request for an MCPTT emergency group call as determined by clause 6.3.3.1.13.2 or MCPTT imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCPTT user is eligible to be implicitly affiliated with the MCPTT group as determined in step 15) a) iv) above, shall perform the implicit affiliation as specified in clause 9.2.2.3.7;

f) if the received SIP INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element set to a value of "true":

i) shall cache the information that the MCPTT user has initiated an MCPTT emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCPTT emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that the MCPTT user has initiated an MCPTT emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16; and

C) shall generate SIP re-INVITE requests for the MCPTT emergency group call to the other call participants of the MCPTT group as specified in clause 6.3.3.1.6;

iv) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCPTT user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4]; and

v) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5];

g) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCPTT function:

i) shall cache the information that the MCPTT user has initiated an MCPTT imminent peril call; and

ii) if the in-progress imminent peril state of the group is set to a value of "false":

A) shall set the in-progress imminent peril state of the group to a value of "true";

B) shall generate SIP re-INVITE requests for the MCPTT imminent peril group call to the other call participants of the MCPTT group as specified in clause 6.3.3.1.15; and

C) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

iii) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCPTT user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4];

h) shall generate a SIP 200 (OK) response as specified in the clause 6.3.3.2.3.2;

i) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

j) shall include in the SIP 200 (OK) response with the warning text set to "123 MCPTT session already exists" as specified in clause 4.4;

k) if the received SIP re-INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCPTT emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

l) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 6: In this case, the request was for an imminent peril call but a higher priority MCPTT emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

m) shall interact with media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 7: Resulting media plane processing is completed before the next step is performed.

n) shall send the SIP 200 (OK) response towards the inviting MCPTT client or inviting non-controlling MCPTT function according to 3GPP TS 24.229 [4];

o) shall generate a notification to the MCPTT clients, which have subscribed to the conference event package that the inviting MCPTT User has joined in the MCPTT group session, as specified in clause 6.3.3.4;

NOTE 8: As a group document can potentially have a large content, the controlling MCPTT function can notify using content-indirection as defined in IETF RFC 4483 [32].

p) shall send a SIP NOTIFY request to each MCPTT client according to 3GPP TS 24.229 [4];

q) Upon receiving a SIP ACK to the above SIP 200 (OK) response and the SIP 200 (OK) response contained a Warning header field as specified in clause 4.4 with the warning text containing the mcptt-warn-code set to "149", shall follow the procedures in clause 6.3.3.1.18; and

r) shall not continue with the rest of the clause.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 10.1.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [34], the timer TNG1 (acknowledged call setup timer) is not running, the controlling MCPTT function supports media buffering and the SIP final response is not yet sent to the inviting MCPTT client:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4,4 in the SIP 200 (OK) response, if the prearranged MCPTT group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [31];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) shall include a P-Answer-State header field with the value "Unconfirmed";

5) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCPTT emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) if the received SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

7) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 9: Resulting user plane processing is completed before the next step is performed.

8) shall send the SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4];

9) shall generate a notification to the MCPTT clients, which have subscribed to the conference event package that the inviting MCPTT User has joined in the MCPTT group session, as specified in clause 6.3.3.4; and

NOTE 10: As a group document can potentially have a large content, the controlling MCPTT function can notify using content-indirection as defined in IETF RFC 4483 [32].

10) shall send a SIP NOTIFY request to each MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 183 (Session Progress) response for a SIP INVITE request as specified in clause 10.1.1.4.1.2 containing an indication of required group members, the timer TNG1 (acknowledged call setup timer) is running and all SIP 200 (OK) responses have been received to all SIP INVITE requests sent to MCPTT clients specified in clause 10.1.1.4.1.1, then the controlling MCPTT function shall wait until the SIP 200 (OK) response has been received to the SIP INVITE request specified in clause 10.1.1.4.1.2 before generating a SIP 200 (OK) response to the "SIP INVITE request for controlling MCPTT function of an MCPTT group".

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 10.1.1.4.1 that was sent to an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [31]; and

1) if the MCPTT ID in the SIP 200 (OK) response matches to the MCPTT ID in the corresponding SIP INVITE request;

2) there are no outstanding SIP 200 (OK) responses to SIP INVITE requests which were sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [31]; and

3) there is no outstanding SIP 200 (OK) response to a SIP INVITE request sent in clause 10.1.1.4.1.2 where the SIP 183 (Session Progress) response contained an indication of required group members;

the controlling MCPTT function:

1) shall stop timer TNG1 (acknowledged call setup timer) as described in clause 6.3.3.3;

2) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

3) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCPTT group has been exceeded the <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [31];

4) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 11: Resulting media plane processing is completed before the next step is performed.

6) shall send a SIP 200 (OK) response to the inviting MCPTT client according to 3GPP TS 24.229 [4];

7) shall generate a notification to the MCPTT clients, which have subscribed to the conference event package that the inviting MCPTT user has joined in the MCPTT group session, as specified in clause 6.3.3.4; and

NOTE 12: As a group document can potentially have a large content, the controlling MCPTT function can notify using content-indirection as defined in IETF RFC 4483 [32].

8) shall send the SIP NOTIFY request to the MCPTT clients according to 3GPP TS 24.229 [4].

Upon:

1) receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 10.1.1.4.1;

2) the timer TNG1 (acknowledged call setup timer) is not running;

3) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document;

4) the controlling MCPTT function supports media buffering; and

5) the SIP final response has not yet been sent to the inviting MCPTT client;

the controlling MCPTT function according to local policy:

1) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCPTT group has exceeded the <max-participant-count> members as specified in 3GPP TS 24.481 [31];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCPTT emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

5) if the received SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 13: Resulting media plane processing is completed before the next step is performed.

7) shall send a SIP 200 (OK) response to the inviting MCPTT client according to 3GPP TS 24.229 [4];

8) shall generate a notification to the MCPTT clients, which have subscribed to the conference event package that the inviting MCPTT user has joined in the MCPTT group session, as specified in clause 6.3.3.4; and

NOTE 14: As a group document can potentially have a large content, the controlling MCPTT function can notify using content-indirection as defined in IETF RFC 4483 [32].

9) shall send the SIP NOTIFY request to the MCPTT clients according to 3GPP TS 24.229 [4].

Upon expiry of timer TNG1 (acknowledged call setup timer), if there are outstanding SIP 200 (OK) responses to SIP INVITE requests sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [31], the controlling MCPTT function shall follow the procedures specified in clause 6.3.3.3.

If timer TNG1 (acknowledged call setup timer) is running and a final SIP 4xx, 5xx or 6xx response is received from an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [31], the controlling MCPTT function shall follow the relevant procedures specified in clause 6.3.3.3.

If:

1) timer TNG1 (acknowledged call setup timer) is not running;

2) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document; and

3) a final SIP 4xx, 5xx or 6xx response is received from an invited MCPTT client;

then the controlling MCPTT function shall perform one of the following based on policy:

1) send the SIP final response towards the inviting MCPTT client, according to 3GPP TS 24.229 [4], if a SIP final response was received from all the other invited MCPTT clients and the SIP 200 (OK) response is not yet sent; or

2) remove the invited MCPTT client from the MCPTT Session as specified in clause 6.3.3.1.5, if a SIP final response other than 2xx or 3xx was received from all the invited MCPTT clients and the SIP 200 (OK) response is already sent. The controlling MCPTT function may invite an additional member of the prearranged MCPTT group as specified in clause 10.1.1.4.1 that has not already been invited, if the prearranged MCPTT group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [31], and all members have not yet been invited.

If the group identity in the "SIP INVITE request for controlling MCPTT function of an MCPTT group" is a TGI and constituent MCPTT groups were invited as specified in clause 10.1.1.4.1.2 and,

1) if all non-controlling MCPTT functions hosting the constituent MCPTT groups have responded with a SIP 2xx, SIP 3xx, SIP 4xx, SIP 5xx or SIP 6xx responses to the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group"; and

2) if all expected SIP INFO requests containing a floor request are received;

then the controlling MCPTT function shall indicate to the media plane that all final responses are received.

NOTE 15: If the SIP 200 (OK) response to the SIP INVITE request for non-controlling MCPTT function of an MCPTT group included the application/vnd.3gpp.mcptt-info+xml MIME body with the <floor-state> element set to "floor-taken", the controlling MCPTT function expects that the non-controlling MCPTT functions sends a SIP INFO request containing a floor request.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCPTT client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.2]

"The controlling MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4].

The controlling MCPTT function:

1) shall include in the Contact header field an MCPTT session identity for the MCPTT session with the g.3gpp.mcptt media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" according to IETF RFC 3840 [16];

2) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [9] in the SIP INVITE request;

4) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

5) shall include a Referred-By header field with the public user identity of the inviting MCPTT client;

6) should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7]. The refresher parameter shall be omitted;

7) shall include the Supported header field set to "timer";

8) if the received SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body containing an <ambient-listening-type> element and:

a) if the Priv-Answer-Mode header field specified in IETF RFC 5373 [18] was included in the received SIP INVITE request with a value of "Auto" or if no Priv-Answer-Mode header field was received in the received SIP INVITE request; or

b) a Priv-Answer-Mode header field was received containing a value other than "Auto";

shall include a Priv-Answer-Mode header field set to a value of "Auto" in the outgoing SIP INVITE request;

9) if the received SIP INVITE request contained a Priv-Answer-Mode header field and did not contain an application/vnd.3gpp.mcptt-info+xml MIME body containing an <ambient-listening-type> element, shall include an unmodified Priv-Answer-Mode header field;

10) if the received SIP INVITE request did not contain an application/vnd.3gpp.mcptt-info+xml MIME body containing an <ambient-listening-type> element and did not contain a Priv-Answer-Mode header field, shall include an unmodified Answer-Mode header field if present in the incoming SIP INVITE request; and

11) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing INVITE request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the controlling MCPTT function:

1) when initiating a new MCPTT session the SDP offer;

a) shall contain only one SDP media-level section for MCPTT speech media stream as contained in the received SDP offer; and

b) shall contain one SDP media-level section for media- plane control messages, if present in the received SDP offer; and

2) when adding a new MCPTT user to an existing MCPTT Session, the SDP offer shall contain the media stream currently used in the MCPTT session.

When composing the SDP offer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the controlling MCPTT function;

2) for the MCPTT speech media stream, shall include all media-level attributes from the received SDP offer;

3) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the controlling MCPTT function; and

4) for the offered media plane control channel, shall include the offered media planecontrol channel 'fmtp' attributes as specified in 3GPP TS 24.380 [5] clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.2]

"NOTE 1: In this clause, "MCPTT server" can refer to either the controlling MCPTT function of an MCPTT group or the non-controlling MCPTT function of an MCPTT group.

Upon receipt of a SIP INVITE request:

1) if the MCPTT server is not yet subscribed to the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request, the MCPTT server shall subscribe to the "xcap-diff" event-package for the group document of this group identity as specified in 3GPP TS 24.481 [31];

NOTE 2: The group identity in the <mcptt-request-uri> element is either an MCPTT group ID or a temporary MCPTT group identity (TGI).

…

4) upon receipt of a notification from the group management server containing the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request, or if the group document is already cached:

a) if the MCPTT server is a non-controlling function of an MCPTT group, then the MCPTT server shall exit this clause; and

b) if the MCPTT server is a controlling function of an MCPTT group, then the MCPTT server shall determine if the group document is for a TGI or an MCPTT group ID as follows:

i) if the group document includes an <on-network-temporary> element, then the group document is associated with a TGI;

ii) if the group document does not include an <on-network-temporary> element or an <on-network-regrouped> element, then the group document is associated with an MCPTT ID that has not been regrouped; and

iii) if the group document does not include an <on-network-temporary> element but includes an <on-network-regrouped> element, then the group document is associated with an MCPTT ID that has been regrouped;

5) if the SIP INVITE request is a "SIP INVITE request for controlling function of an MCPTT group" and the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP-INVITE request is an MCPTT group ID that has not been re-grouped, the MCPTT server shall access the following elements from the group document as specified in 3GPP TS 24.481 [31]:

a) if the <on-network-disabled> element is present in the group document, shall send a SIP 403 (Forbidden) response with the warning text set to "115 group is disabled" in a Warning header field as specified in clause 4.4 and shall not continue with the rest of the steps;

b) if the <list> element of the <list-service> element does not contain an entry matching the MCPTT ID of the user in the SIP INVITE request, shall send a SIP 403 (Forbidden) response with the warning text set to "116 user is not part of the MCPTT group" in a Warning header field as specified in clause 4.4 and shall not continue with the rest of the steps;

c) if the <on-network-invite-members> element is set to "true" and if the SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element containing a value not set to "prearranged", shall return a SIP 404 (Not Found) response with the warning text set to "117 the group identity indicated in the request is a prearranged group" as specified in clause 4.4 "Warning header field" and shall not continue with the rest of the steps; and

d) if the <on-network-invite-members> element is set to "false" and if the SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element containing a value not set to "chat" shall return a SIP 404 (Not Found) response with the warning text set to "118 the group identity indicated in the request is a chat group" as specified in clause 4.4 "Warning header field" and shall not continue with the rest of the steps;

6) if the SIP INVITE request is a "SIP INVITE request for controlling function of an MCPTT group" and the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request is associated with an MCPTT group ID that has been regrouped, the MCPTT server:

a) shall obtain the TGI associated with the regrouped group, by accessing the "temporary-MCPTT-group-ID" attribute of the <regrouped> element of the group document associated with the MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request;

b) if not hosting the TGI, shall:

i) stop processing the SIP INVITE request; and

ii) return a SIP 302 (Moved Temporarily) response with:

A) a Contact header field set to the PSI of the MCPTT server hosting the TGI; and

B) an application/vnd.3gpp.mcptt-info MIME body with a <mcptt-request-uri> element set to the TGI; and

c) if hosting the TGI, and the call to the temporary group is in progress, shall:

i) associate the MCPTT ID of the calling user with the temporary group call;

ii) interact with the media plane as specified in 3GPP TS 24.380 [5]; and

iii) exit this clause;

d) if hosting the TGI and the call to the temporary group is not in progress, shall subscribe to the "xcap-diff" event-package for the group document of the TGI as specified in 3GPP TS 24.481 [31], if not already subscribed;

…

g) upon receipt of a notification containing the group document for the TGI, or if the group document is already cached, shall obtain the MCPTT IDs of the constituent groups by accessing the <constituent-MCPTT-group-ID> element(s) of the group document for the TGI; and

h) if:

i) the <associated-group-id> element with an MCPTT ID is included in the application/vnd.3gpp.mcptt-info MIME body;

ii) the MCPTT ID is present in one of the instances of the <constituent-MCPTT-group-ID> element in the group document; and

iii) the group is not homed by the MCPTT server;

shall exit this procedure and authorize the MCPTT user at a non-controlling MCPTT function of a MCPTT group;

NOTE 4: The non-controlling function of an MCPTT group can be located in the primary MCPTT system or a partner MCPTT system.

7) for the MCPTT ID of each constituent group, shall follow the actions below in step 8); and

8) if the SIP INVITE request is a "SIP INVITE request for controlling function of an MCPTT group" and the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request retrieved is associated with a TGI and if:

a) the application/vnd.3gpp.mcptt-info+xml includes an <associated-group-id> element, determine if the constituent MCPTT group identified by the <associated-group-id> element is homed at the MCPTT server and if that is the case:

i) shall subscribe to the "xcap-diff" event-package for the group document for the constituent MCPTT group ID as specified in 3GPP TS 24.481 [31], if not already subscribed;

…

b) the application/vnd.3gpp.mcptt-info+xml does not include an <associated-group-id> element, for each MCPTT ID contained in each instance of the <constituent-MCPTT-group-ID> element of the group document for the TGI:

i) shall determine if the group identity is homed on the controlling MCPTT function of an MCPTT group or homed on a non-controlling MCPTT function of an MCPTT group;

NOTE 5: The non-controlling function of an MCPTT group can be located in the primary MCPTT system or a partner MCPTT system.

ii) for each constituent MCPTT group ID that is homed on the controlling MCPTT function of an MCPTT group shall subscribe to the "xcap-diff" event-package for the group document for the constituent MCPTT group ID as specified in 3GPP TS 24.481 [31], if not already subscribed;

NOTE 6: As soon as an error occurs when subscribing for a group document of a constituent MCPTT group ID, the controlling MCPTT function of an MCPTT group stops subscribing to any further group documents of constituent MCPTT group IDs.

…

c) when all group document(s) for all constituent groups homed at the MCPTT server have been retrieved and if the MCPTT ID of the user identified in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body:

i) is a member of one the retrieved constituent MCPTT groups, received, shall exit this procedure; and

ii) is not a member of any of the retrieved constituent group documents, shall determine that the MCPTT ID of the MCPTT user needs to be authorised by a non-controlling MCPTT function of an MCPTT group and exit this procedure."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.5]

"The MCPTT server shall only invite affiliated group members to a group session.

If the group is not a regroup based on a preconfigured group, the MCPTT server determines the affiliated members from the entries contained in the <list> element of the group document by following the procedures specified in clause 6.3.6.

If the group is a regroup based on a preconfigured group, the MCPTT server determines the affiliated members from the list of users that was stored during successful processing of the creation of the regroup per clause 16 by following the procedures specified in clause 6.3.6.

NOTE 1: The term "affiliated group members" used above also includes those members that are implicitly affiliated by the controlling MCPTT function.

If the number of members of the MCPTT group exceeds the value contained in the <on-network-max-participant-count> element the MCPTT server shall invite only <on-network-max-participant-count> members from the list, but shall prioritise inviting those group members to the group session that have an <entry> element in the <list> element with a <on-network-required> element present.

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the group session. The <on-network-required> element is used to determine which group members need to acknowledge the group call before audio transmission can proceed.

NOTE 3: Other requirements for how the controlling MCPTT function selects which of the <on-network-max-participant-count> members to invite is outside the scope of this specification.

NOTE 4: It is assumed that validation checks are performed at the group management server to ensure that the <on-network-max-participant-count> cannot be less than the number of <on-network-required> users."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.3.2]

"When sending a SIP 200 (OK) response to the initial SIP INVITE request, the controlling MCPTT function:

1) shall generate the SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [4];

2) shall include the Session-Expires header field and start supervising the SIP session according to rules and procedures of IETF RFC 4028 [7], "UAS Behavior". The "refresher" parameter in the Session-Expires header field shall be set to "uac";

3) shall include the option tag "timer" in a Require header field;

4) shall include a P-Asserted-Identity header field with the public service identity of the controlling MCPTT function;

5) shall include a SIP URI for the MCPTT session identity in the Contact header field identifying the MCPTT session at the controlling MCPTT function;

6) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and

c) the isfocus media feature tag;

7) shall include Warning header field(s) received in incoming responses to the SIP INVITE request;

8) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];

9) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [22];

10) shall include the "explicitsub" and "nosub" option tags in a Supported header field according to IETF RFC 7614 [35]; and

11) void

12) shall interact with the media plane as specified in 3GPP TS 24.380 [5]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) for the accepted media stream in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCPTT function; and

2) for the accepted media channel control channel, if present in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCPTT function, for the accepted media plane control channel, if present in the received SDP offer; and

b) shall include 'fmtp' attributes as specified in 3GPP TS 24.380 clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.3]

"Upon receiving a SIP BYE request the controlling MCPTT function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCPTT function:

1) shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing the media plane resource associated with the SIP session towards the MCPTT client;

…

2) shall generate a SIP 200 (OK) response and send the SIP response towards the MCPTT client according to 3GPP TS 24.229 [4];

3) shall check the MCPTT session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCPTT session needs to be released for each participant of the MCPTT session;

4) if release of the MCPTT session is required:

a) shall perform the procedures as specified in the clause 6.3.3.1.5 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and

...

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT participant."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.8.1]

"If:

1) the call is a pre-arranged group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 24.380 [5] expired;

2) there are only one or no participants in the MCPTT session;

3) the call is a pre-arranged group call and if it is according to local policy, the initiator of the group call leaves the MCPTT session;

4) the minimum number of affiliated MCPTT group members is not present;

5) timer TNG3 (group call timer) expires; or

6) the call is a broadcast group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 24.380 [5] expired;

the controlling MCPTT function shall release the MCPTT session for the group call."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.5]

"When a participant needs to be removed from the MCPTT session, the controlling MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5] for the MCPTT session release;

2) shall generate a SIP BYE request according to 3GPP TS 24.229 [4]; and

3) shall send the SIP BYE request to the MCPTT clients according to 3GPP TS 24.229 [4].

If timer TNG3 (group call timer) has not expired, then when the last MCPTT client is removed from the MCPTT session, the controlling MCPTT function shall stop timer TNG3 (group call timer).

When the MCPTT group session needs to be released, the controlling MCPTT function shall send SIP BYE requests as described in this clause, to all participants of the group session.

Upon receiving a SIP 200 (OK) response to a SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT clients."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 4.8]

"In certain deployments, for example, in the case that the MCPTT operator uses the underlying SIP core infrastructure from the carrier operator, the MCPTT operator can prevent certain sensitive application data from being visible in the clear to the SIP layer. The following data are classed as sensitive application data:

- *MCPTT ID;*
- *MCPTT group ID;*
- *user location information;*
- *emergency, alert and imminent-peril indicators;*
- *access token (containing the MCPTT ID);*
- *MCPTT client ID; and*
- *functional alias.*

The above data is transported as XML content in SIP messages. in XML elements or XML attributes.

Data is transported in attributes in the following circumstances in the procedures in the present document:

…

- *an MCPTT ID in application/resource-lists+xml document included in an SIP INVITE request setting up a group call to a temporary group involving a non-controlling function that works in "Trusted Mode" according to IETF RFC 5366 [20], whereby the participants are returned to the controlling function in a MIME body of a SIP 403 (Forbidden) with the P-Refused-URI-List header field according to IETF RFC 5318 [36];*

3GPP TS 33.180 [78] describes a method to provide confidentiality protection of sensitive application data in elements by using XML encryption (i.e. xmlenc) and in attributes by using an attribute confidentiality protection scheme described in clause 6.6.2.3 of the present document. Integrity protection can also be provided by using XML signatures (i.e. xmlsig).

Protection of the data relies on a shared XML protection key (XPK) used to encrypt and sign data:

- *between the MCPTT client and the MCPTT server, the XPK is a client-server key (CSK); and*

- *between MCPTT servers and between MCPTT domains, the XPK is a signalling protection key (SPK).*

The CSK (XPK) and a key-id CSK-ID (XPK-ID) are generated from keying material provided by the key management server. Identity based public key encryption based on MIKEY-SAKKE is used to transport the CSK between SIP end-points. The encrypted CSK is transported from the MCPTT client to the MCPTT server when the MCPTT client performs service authorisation as described in clause 7 and is also used during service authorisation to protect the access token.

The SPK (XPK) and a key-id SPK-ID (XPK-ID) are directly provisioned in the MCPTT servers.

Configuration in the MCPTT client and MCPTT server is used to determine whether one or both of confidentiality protection and integrity protection are required.

The content can be re-encrypted and signed again using the SPK between MCPTT servers."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.2.2]

"When an MCPTT call is established a new instance of the floor control server state machine for 'general floor control operation' is created.

For each MCPTT client added to the MCPTT call, a new instance of the floor control server state machine for 'basic floor control operation towards the floor participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the floor control server could queue the implicit floor control request for the media-floor control entity.

The initial SIP INVITE request or SIP REFER request to establish an MCPTT chat group call or to rejoin an ongoing MCPTT call is not handled as an implicit floor control request message by the floor control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCPTT client due to implicit floor control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCPTT function (see 3GPP TS 24.379 [2]) the floor control server optionally can give an early indication to send RTP media packets, to the inviting MCPTT client.

If an early indication to send RTP media packets is given to the inviting MCPTT client, the floor participant is granted the permission to send media and the MCPTT server buffers RTP media packets received from the MCPTT client at least until the first invited MCPTT client accepts the invitation or until the RTP media packet buffer maximum limit is exceeded.

If the MCPTT server does not support or does not allow media buffering then an early indication to send RTP media packets is not given to the inviting MCPTT client, the floor participant is granted the permission to send media when the first invited MCPTT client accepts the media.

Before the floor control server sends the first floor control message in the MCPTT call, the floor control server has to assign itself a SSRC identifier to be included in media floor control messages and quality feedback messages if the MCPTT server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The floor participant and the floor control server can negotiate the maximum priority level that the floor participant is permitted to request. The floor control server can pre-empt the current sender based on the negotiated maximum priority level that the floor participant is permitted to request and the priority level included in the Floor Request message.

NOTE: The maximum priority level that a floor participant can use is negotiated as specified in subclause 14.3.3 and is based on group configuration data retrieved by the controlling MCPTT function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCPTT function from the configuration management server as described in 3GPP TS 24.484 [13].

For groups configured for audio cut-in floor control, pre-empting of the current sender is not according to priority. Each new floor request results in the floor being revoked from the current talker and being granted to the new requesting talker.

The floor participant and the floor control server can negotiate queueing of floor requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the floor control server queues the floor control request if a Floor Request message is received when another floor participant has the floor and the priority of the current speaker is the same or higher. Queueing is not permitted for groups configured for audio cut-in floor control."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.3]

"When an MCPTT client leaves an MCPTT call and the MCPTT call remains ongoing with the other MCPTT clients, the floor control server follows a two-step procedure.

Step 1 The MCPTT server stops sending floor control messages and RTP media packets to the MCPTT client leaving the MCPTT call and, the MCPTT server discards floor control messages and RTP media packets received from the MCPTT client leaving the MCPTT call.

Step 2 When the application and signalling plane has determined that the MCPTT call with this floor participant has been released, the corresponding instance of the floor control server state machine for 'basic floor control operation towards the floor participant' is released.

When an MCPTT call is released, the floor control server follows a two-step procedure.

Step 1 The MCPTT server stops sending floor control messages and RTP media packets to all floor participants in the MCPTT call.

Step 2 When the application and signalling plane has determined that the MCPTT call has been released, the corresponding instance of the floor control server state machine for 'general floor control operation' are also terminated, along with any 'basic floor control operation towards the floor participant' state machines for the floor participants of this call.

The floor control server state machine for general floor control operation initiates the MCPTT call release depending on the release policy specified in 3GPP TS 24.379 [2]."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 14.3.4]

"If the mc_granted" fmtp attribute is included in an SDP offer, the controlling MCPTT function:

1. if the MCPTT call is not a temporary group session, may use the SIP 200 (OK) response to indicate that the implicit floor request is granted; and

2. if the MCPTT call is a temporary group session, shall not indicate that the implicit floor request is granted.

NOTE: A MCPTT group call is a temporary group session when the <on-network-temporary> element is present in the <list-service> element as specified in 3GPP TS 24.481 [12].

If the controlling MCPTT function grants the implicit request and decide to use the SIPP 200 (OK) response to grant the implicit request, the controlling MCPTT function shall include the "mc_granted" fmtp attribute in the SDP answer.

If the controlling MCPTT function decides not to grant the implicit request, the controlling MCPTT function shall not include the "mc_granted" fmtp attribute in the SDP answer."

5.2.1.4.3 Test description

5.2.1.4.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the originating participating MCPTT function.
  - SS (MCPTT Server) provides the terminating participating MCPTT function.
  - SS (MCPTT Server) provides the non-controlling MCPTT function.

SS-UE1 (MCPTT client)

- Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.4.3.1-1.

SS-UE2 (MCPTT client)

- Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.4.3.1-1.

SS-UE3 (MCPTT client)

- Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.4.3.1-1.

SS-UE4 (MCPTT client)

- Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.4.3.1-1.

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the controlling MCPTT function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2 and A.4.6.
- The SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client), the SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are registered and authorized to use the MCPTT system.
- Group A and Group B have been regrouped in Group T.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.
- The SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are affiliated with Group B and are authorized to initiate group calls.
- Group B is controlled by the non-controlling MCPTT function.
- There are no ongoing calls in any group.
- There is a trusted mutual aid relationship between the partner MCPTT system and the primary MCPTT system

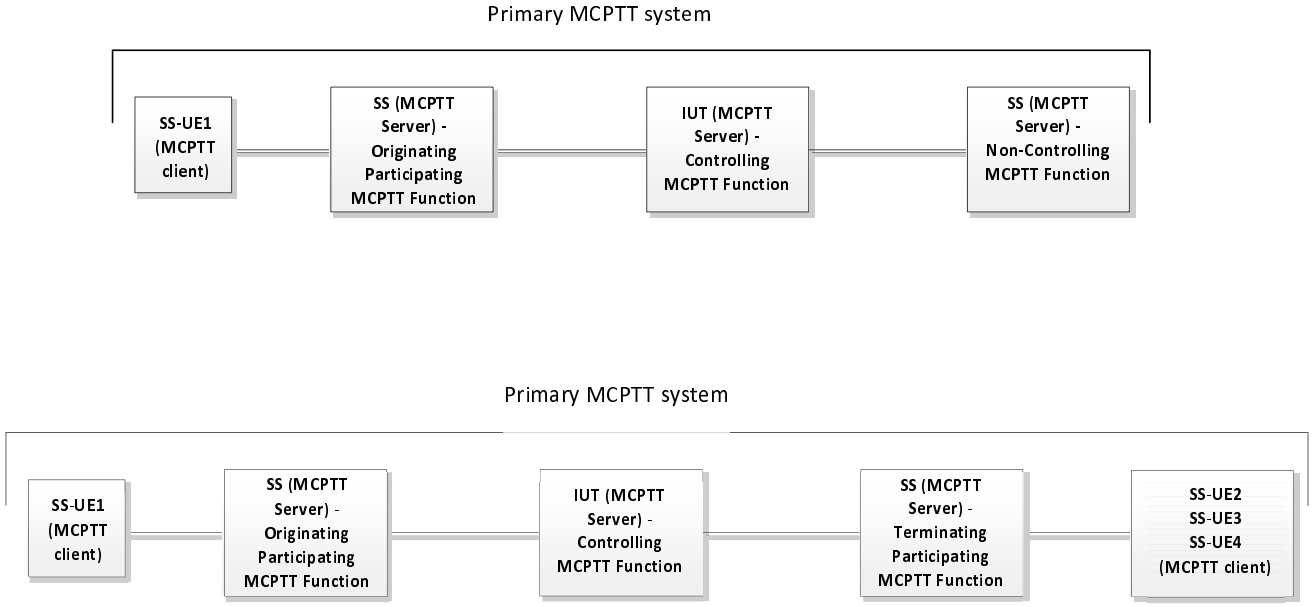


Figure 5.2.1.4.3.1-1: Functions of the testing components

5.2.1.4.3.2 Test procedure sequence

Table 5.2.1.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) (member of constituent Group A) to initiate an on-demand pre-arranged group call to a temporary group (Group T) with implicit floor control	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P
3	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the non-controlling MCPTT function to initiate an on-demand pre-arranged group call in constituent Group B?	-->	SIP INVITE	1	P
4	The SS (MCPTT Server) providing the non-controlling MCPTT function responds with a SIP 403 (Forbidden) message with a P-Refused-URI-List header and a list of two users to be invited to the call	<--	SIP 403 (Forbidden)		-
	- EXCEPTION: The IUT (MCPTT Server) sends three SIP INVITE requests to invite the other participants to the call. The following sequence of steps identified with "lower case letters" can happen in parallel.				
5a	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function to invite SS-UE2 to the call?	-->	SIP INVITE	1	P
6a	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 183 (Session Progress) message sent reliably.	<--	SIP 183 (Session Progress)		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
7a	Check: Does the IUT (MCPTT Server) send a SIP PRACK message to the SS (MCPTT Server) providing the terminating participating MCPTT function in response to the SIP 183 (Session Progress)?	-->	SIP PRACK	1	P
8a	The SS (MCPTT Server) responds to the SIP PRACK with a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
9a	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
10a	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	1	P
11a	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating participating MCPTT function informing SS-UE2 that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
5b	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function to invite SS-UE3 to the call?	-->	SIP INVITE	1	P
6b	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 183 (Session Progress) message sent reliably.	<--	SIP 183 (Session Progress)		-
7b	Check: Does the IUT (MCPTT Server) send a SIP PRACK message to the SS (MCPTT Server) providing the terminating participating MCPTT function in response to the SIP 183 (Session Progress)?	-->	SIP PRACK	1	P
8b	The SS (MCPTT Server) responds to the SIP PRACK with a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
9b	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
10b	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	1	P
11b	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating participating MCPTT function informing SS-UE3 that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
5c	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function to invite SS-UE4 to the call?	-->	SIP INVITE	1	P
6c	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 183 (Session Progress) message sent reliably.	<--	SIP 183 (Session Progress)		-
7c	Check: Does the IUT (MCPTT Server) send a SIP PRACK message to the SS (MCPTT Server) providing the terminating participating MCPTT function in response to the SIP 183 (Session Progress)?	-->	SIP PRACK	1	P
8c	The SS (MCPTT Server) responds to the SIP PRACK with a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
9c	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
10c	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
11c	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating participating MCPTT function informing SS-UE4 that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
	- EXCEPTION: Step 12 and 13 below can happen just after steps 9a, 9b or 9c above and in parallel to other sequences of steps identified by "lower case letters".				
12	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	1	P
13	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
12	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
13	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P
14	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
15	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
16	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
17	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
18	The SS (MCPTT Server) providing the participating MCPTT function sends a Floor Request message from SS-UE3	<--	Floor Request		-
19	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Granted	2	P
20	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the participating MCPTT function informing SS-UE1 that the floor is taken by SS-UE3 (MCPTT client)?	-->	Floor Taken	2	P
21	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the participating MCPTT function informing SS-UE2 that the floor is taken by SS-UE3 (MCPTT client)?	-->	Floor Taken	2	P
22	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the participating MCPTT function informing SS-UE4 that the floor is taken by SS-UE3 (MCPTT client)?	-->	Floor Taken	2	P
23	The SS (MCPTT Server) providing the participating MCPTT function sends a Floor Release message from SS-UE3 with an acknowledgement required to release the floor	<--	Floor Release		-
24	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
25	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
26	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
27	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
28	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
29	The SS (MCPTT Server) providing the participating MCPTT function sends a SIP BYE request from SS-UE1	<--	SIP BYE		-
30	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) response to SS (MCPTT Server) providing the participating MCPTT function?	-->	SIP 200 (OK)	3	P
	- EXCEPTION: Depending on local policy, the controlling MCPTT Server can decide to end the call if the initiator leaves the call. Steps 31A and 32A ("upper case A letter") below represent the call finalization process when this local policy applies. "Lower case letter" notation represent messages that can be sent in parallel, since the MCPTT server needs to send SIP BYE messages to the rest of participants in the call.				
31Aa	Check: Does the IUT (MCPTT Server) send a SIP BYE message for SS-UE2 to the SS (MCPTT Server) providing the participating MCPTT function to end the call?	-->	SIP BYE	3	P
32Aa	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
31Ab	Check: Does the IUT (MCPTT Server) send a SIP BYE message for SS-UE3 to the SS (MCPTT Server) providing the participating MCPTT function to end the call?	-->	SIP BYE	3	P
32Ab	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
31Ac	Check: Does the IUT (MCPTT Server) send a SIP BYE message for SS-UE4 to the SS (MCPTT Server) providing the participating MCPTT function to end the call?	-->	SIP BYE	3	P
32Ac	The SS (MCPTT Server) providing the participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
	- EXCEPTION: If according to local policy, the call continues even if the initiator leaves, the SS will need to close the call until a single participant is left. Steps 31B - 36B ("upper case B letter") below represent the call finalization process when this local policy applies.				
31B	The SS (MCPTT Server) providing the participating MCPTT function sends a SIP BYE request from SS-UE2	<--	SIP BYE		-
32B	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) response to SS (MCPTT Server) providing the participating MCPTT function?	-->	SIP 200 (OK)	3	P
33B	The SS (MCPTT Server) providing the participating MCPTT function sends a SIP BYE request from SS-UE3	<--	SIP BYE		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
34B	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) response to SS (MCPTT Server) providing the participating MCPTT function?	-->	SIP 200 (OK)	3	P
35B	Check: Does the IUT (MCPTT Server) send a SIP BYE message for SS-UE4 to the SS (MCPTT Server) providing the participating MCPTT function to end the call?	-->	SIP BYE	3	P
36B	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-

5.2.1.4.3.3 Specific message contents

Table 5.2.1.4.3.3-1: SIP INVITE from the SS (MCPTT-Server) (Step 1, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.4.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.4.3.3-1B			

Table 5.2.1.4.3.3-1A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.4.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions INITIAL_SDP_OFFER, IMPLICIT_GRANT_REQUESTED, IMPLICIT_FLOOR_GRANTED

Table 5.2.1.4.3.3-1B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.4.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_T_ID			
associated-group-id	Encrypted (see note) <associated-group-id> with mcpttURI set to px_MCPTT_Group_A_ID			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.1.4.3.3-2: SIP INVITE from the IUT (MCPTT-Server) (Step 3, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server which is simulated by the SS, providing non-controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server which is simulated by the SS, providing non-controlling function.		
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	any allowed value			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Max-Forwards				
value	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.4.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.4.3.3-2B			

5.2.1.4.3.3-2A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.4.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the controlling MCPTT Server			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the controlling MCPTT Server			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the controlling MCPTT Server			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the controlling MCPTT Server			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_priority	"10"			
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.4.3.3-2B: MCPTT-Info in SIP INVITE from IUT (MCPTT Server) (Table 5.2.1.4.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_B_ID	The URI of the group		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.4.3.3-3: SIP 403 (Forbidden) from the SS (MCPTT-Server) (Step 4, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.19.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Warning	not present			
Content-Length				
value	any value	length of message-body		
Content-Type				
value	"multipart/mixed"			
Message-body				
MIME body part		**Resource list**		
MIME-part-headers				
Content-Type	"application/resource-lists+xml"			
Content-ID	any value	Unique URL identifying the Resource-lists XML MIME body; used as reference in the signature MIME body and P-Refused-URI-List header		
MIME-part-body	As described in Table 5.2.1.4.3.3-3A			
MIME body part		**Signature**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-signed+xml"			
MIME-part-body	Signatures for XML MIME bodies as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.13.1-1			

Table 5.2.1.4.3.3-3A: Resource-lists from the SS (MCPTT-Server) (Table 5.2.1.4.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_C	The MCPTT ID of one affiliated member of Group B		
display-name	Not present			
entry[2]				
uri attribute	px_MCPTT_ID_User_D	The MCPTT ID of another affiliated member of Group B		
display-name	Not present			

Table 5.2.1.4.3.3-4: SIP INVITE from the IUT (MCPTT-Server) (Step 5a, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing terminating participating function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing terminating participating function.		
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			
Session-Expires				
generic-param	any allowed value			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Max-Forwards				
value	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.4.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.4.3.3-4A			

Table 5.2.1.4.3.3-4A: MCPTT-Info in SIP INVITE from IUT (MCPTT Server) (Table 5.2.1.4.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.4.3.3-5: SIP 183 (Session Progress) from the SS (MCPTT-Server) (Steps 6a, 6b and 6c, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.1-1, conditions MCPTT, 100rel

Table 5.2.1.4.3.3-6: SIP PRACK from the IUT (MCPTT-Server) (Steps 7a, 7b and 7c, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.10.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Request-URI	same URI as the SS has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server), providing terminating participating function.		
Max-Forwards				
value	any allowed value			

Table 5.2.1.4.3.3-7: SIP 200 (OK) to PRACK from the SS (MCPTT-Server) (Steps 8a, 8b and 8c, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Content-Type	not present			
Content-Length	not present			

Table 5.2.1.4.3.3-8: SIP 200 (OK) from the SS (MCPTT Server) (Steps 9a, 9b and 9c, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	3600			
refresher	"uas"			
Message-body				
SDP message	SDP message as described in Table 5.2.1.4.3.3-8A			

Table 5.2.1.4.3.3-8A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 5.2.1.4.3.3-8)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.4.3.3-9: SIP INVITE from the IUT (MCPTT-Server) (Step 5b, Table 5.2.1.4.3.2-1)

Derivation Path: Table 5.2.1.4.3.3-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.4.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.4.3.3-9A			

Table 5.2.1.4.3.3-9A: MCPTT-Info in SIP INVITE from IUT (MCPTT Server) (Table 5.2.1.4.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_C	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.4.3.3-10: SIP INVITE from the IUT (MCPTT-Server) (Step 5c, Table 5.2.1.4.3.2-1)

Derivation Path: Table 5.2.1.4.3.3-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.4.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.4.3.3-10A			

Table 5.2.1.4.3.3-10A: MCPTT-Info in SIP INVITE from IUT (MCPTT Server) (Table 5.2.1.4.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_D	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.4.3.3-11: SIP 200 (OK) from the IUT (MCPTT Server) (Step 12, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	any allowed value			
refresher	"uac"			
Message-body				
SDP message	SDP message as described in Table 5.2.1.4.3.3-11A			

Table 5.2.1.4.3.3-11A: SDP in SIP 200 (OK) from the IUT (MCPTT Server) (Table 5.2.1.4.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_FLOOR_GRANTED, IMPLICIT_GRANT_REQUESTED				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the MCPTT Server			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the MCPTT Server			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the MCPTT Server (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the MCPTT Server			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the MCPTT Server being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			

Table 5.2.1.4.3.3-12: SIP BYE from the SS (MCPTT-Server) (Steps 29, 31B and 33B, 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
From				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Local URI of the dialog (from the SS' point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Local tag of the dialog (from the SS' point of view)		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the SS' point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the SS' point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	70	The recommended initial value is 70 in IETF RFC 3261 [17].		

Table 5.2.1.4.3.3-13: SIP BYE from the IUT (MCPTT-Server) (Steps 31Aa, 31Ab, 31Ac and 35B, 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

Table 5.2.1.4.3.3-14: Floor Taken (Steps 11a, 11b and 11c, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.1.4.3.3-15: Floor Taken (Steps 20, 21 and 22, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_C			

Table 5.2.1.4.3.3-16: Floor Release (Steps 12 and 23, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.1.4.3.3-17: Floor Ack (Steps 13 and 24, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 5.2.1.4.3.3-18: Floor Idle (Steps 14, 15, 16, 17, 25, 26, 27 and 28, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.1.4.3.3-19: Floor Request (Step 18, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1

Table 5.2.1.4.3.3-20: Floor Granted (Step 19, Table 5.2.1.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

5.2.1.5 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Temporary Group / Non-Controlling server / Trusted mode

5.2.1.5.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server) }
ensure that {
  when { the SS (MCPTT Server) initiates an on-demand pre-arranged group call }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) by sending SIP 403 (Forbidden)
message }
            }
```

5.2.1.5.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 10.1.1.4.2, 10.1.1.4.1.2, 6.3.3.1.2, 6.3.3.1.1, 10.1.1.5.2.1, 10.1.1.5.2.2, 10.1.1.5.5, 6.3.5.2, 6.3.5.4 and 6.3.5.5. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.2]

"In the procedures in this clause:

1) MCPTT ID in an incoming SIP INVITE request refers to the MCPTT ID of the originating user from the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

...

Upon receipt of a "SIP INVITE request for controlling MCPTT function of an MCPTT group", the controlling MCPTT function:

...

14) if the MCPTT group call is not ongoing then:

...

f) if the group identity in the "SIP INVITE request for controlling MCPTT function of an MCPTT group" is a TGI:

i) shall for each of the constituent MCPTT groups homed on the primary MCPTT system:

A) if the controlling MCPTT function does not own the MCPTT group identified by the MCPTT group ID, then generate a SIP INVITE request towards the MCPTT server that owns the MCPTT group identity by following the procedures in clause 10.1.1.4.1.2; and

NOTE 4: The MCPTT server that the SIP INVITE request is sent to acts as a non-controlling MCPTT function;

...

ii) shall for each of the constituent MCPTT groups homed on the partner MCPTT system generate a SIP INVITE request for the MCPTT group identity homed on the partner MCPTT system as specified in clause 10.1.1.4.1.2;"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.1.2]

"The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the non-controlling MCPTT function serving the group identity of the MCPTT group owned by the partner MCPTT system;

3) shall set the P-Asserted-Identity to the public service identity of the controlling MCPTT function;

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcptt-request-uri> element set to the group identity of the MCPTT group hosted by the non-controlling MCPTT function in the partner MCPTT system; and

b) the <mcptt-calling-group-id> element set to the group identity of the group served by the controlling MCPTT function;

5) shall include the Recv-Info header field set to g.3gpp.mcptt-floor-request;

6) void

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

8) shall send the SIP INVITE request towards the partner MCPTT system in accordance with 3GPP TS 24.229 [4].

Upon receiving SIP 403 (Forbidden) response for the SIP INVITE request, if according to local policy and if:

1) the response contains a Warning header field with the MCPTT warning code "128"; and

2) the response contains a P-Refused-URI-List header field and an application/resource-lists+xml MIME body as specified in IETF RFC 5318 [36];

NOTE 1: The application/resource-lists+xml MIME body contains MCPTT IDs identifying MCPTT users in a partner MCPTT system that needs to be invited to the prearranged group call in case of group regrouping using interrogating method as specified in 3GPP TS 23.379 [3] clause 10.6.2.4.2.

then the controlling MCPTT function:

1) shall check if the number of members of the MCPTT group exceeds the value contained in the <on-network-max-participant-count> element of the group document as specified in 3GPP TS 24.481 [31]. If exceeded, the controlling MCPTT function shall invite only <on-network-max-participant-count> members from the application/resource-lists+xml MIME body; and

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the prearranged group session It is operator policy that determines which participants in the application/resource-lists+xml MIME body are invited to the group call.

2) shall invite MCPTT users as specified in this clause using the list of MCPTT IDs in URI-List.

..."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.2]

"This clause is referenced from other procedures.

The controlling MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4].

The controlling MCPTT function:

1) shall include in the Contact header field an MCPTT session identity for the MCPTT session with the g.3gpp.mcptt media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" according to IETF RFC 3840 [16];

2) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [9] in the SIP INVITE request;

4) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

5) shall include a Referred-By header field with the public user identity of the inviting MCPTT client;

6) should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7]. The refresher parameter shall be omitted;

7) shall include the Supported header field set to "timer";

...

11) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing INVITE request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the controlling MCPTT function:

1) when initiating a new MCPTT session the SDP offer;

a) shall contain only one SDP media-level section for MCPTT speech media stream as contained in the received SDP offer; and

b) shall contain one SDP media-level section for media- plane control messages, if present in the received SDP offer; and

...

When composing the SDP offer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the controlling MCPTT function;

2) for the MCPTT speech media stream, shall include all media-level attributes from the received SDP offer;

3) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the controlling MCPTT function; and

4) for the offered media plane control channel, shall include the offered media planecontrol channel 'fmtp' attributes as specified in 3GPP TS 24.380 [5] clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.5.2.1]

"When receiving the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" the MCPTT server can be acting as a controller MCPTT function in an ongoing prearranged group call or, if an prearranged group call is not ongoing, be initiated as an non-controlling MCPTT function and invite MCPTT users.

If a prearranged group call is not ongoing the MCPTT server shall perform the actions specified in clause 10.1.1.5.2.2.

...

When operating in the non-controlling mode and a SIP BYE request is received from the controlling MCPTT function, the non-controlling MCPTT function shall change from operating in the non-controlling mode to operating in the controlling mode as specified in clause 10.1.1.5.2.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.5.2.2]

"Upon receipt of a "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" and if a prearranged group call is not ongoing, the non-controlling MCPTT function of an MCPTT group:

NOTE 1: The Contact header field of the SIP INVITE request contains the isfocus media feature tag.

..

5) if a trusted mutual aid relationship exists between the partner MCPTT system and the primary MCPTT system and the procedure in 3GPP TS 23.379 [3] clause 10.6.2.4.2 is supported:

a) shall generate a SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [4];

b) shall retrieve the group members of the prearranged group identified by the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request, as specified in clause 6.3.5.2;

c) if the retrieval of group members was successful shall include a P-Refused-URI-List header field populated with affiliated members of the prearranged group in accordance with the IETF RFC 5318 [36]:

..

e) shall send the SIP 403 (Forbidden) response towards the controlling MCPTT function as specified in 3GPP TS 24.229 [4]; and

f) shall not process the remaining steps;

6) shall retrieve the group document from the group management server for the MCPTT group ID contained in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2 and continue with the rest of the steps if the checks in clause 6.3.5.2 succeed;

7) shall cache the content of the SIP INVITE request, if received in the Contact header field and if the specific feature tags are supported;

8) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [4];

9) determine the members to invite to the prearranged MCPTT group call as specified in clause 6.3.5.5;

10) if the group document retrieved from the group management server contains <on-network-required> group members as specified in 3GPP TS 24.481 [31], shall send a SIP 183 (Session Progress) response to the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" as specified in clause 6.3.4.2.2.1 and shall populate the response with an application/vnd.3gpp.mcptt-info+xml MIME body containing the <required> element set to "true".

11) if the group document retrieved from the group management server does not contain any <on-network-required> group members as specified in 3GPP TS 24.481 [31], may, according to local policy, send a SIP 183 (Session Progress) response to the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" as specified in clause 6.3.4.2.2.1;

12) shall invite each group member determined in step 9) above, to the group session, as specified in clause 10.1.1.5.1; and

13) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

Unless a SIP response has been sent to the controlling MCPTT function as specified in step 10 or 11 above, the non-controlling MCPTT function of an MCPTT group shall wait for the first SIP provisional response or first SIP 200 (OK) response from one of the invited MCPTT clients, before sending a response to the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group".

…

If the group document contains <on-network-required> group member(s) as specified in 3GPP TS 24.481 [31], then the non-controlling MCPTT function of an MCPTT group shall wait until all SIP 200 (OK) responses to SIP INVITE requests have been received from the <on-network-required> MCPTT clients before sending a SIP 200 (OK) response back to the controlling MCPTT function, as specified above.

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.5.5]

"Upon receiving a "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" when a prearranged group session is not ongoing, the non-controlling MCPTT-function shall:

NOTE 1: The difference between a "SIP INVITE request for controlling MCPTT function of an MCPTT group" and a "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" is that the latter SIP INVITE request contains the isfocus media feature tag in the Contact header field.

…

4) shall retrieve the group document from the group management server for the MCPTT group ID contained in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2 and continue with the rest of the steps if the checks in clause 6.3.5.2 succeed;

NOTE 2: If the checks are not successful, the SIP response to the "SIP INVITE request for controlling MCPTT function of an MCPTT group" is already sent in the clause 6.3.5.2.

5) shall cache the content of the SIP INVITE request;

6) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [4];

7) shall authorize the MCPTT user in the <mcptt-calling-user-id> element in the application/vnd.3gpp.mcptt-info+xml MIME body of the "SIP INVITE request for controlling MCPTT function of an MCPTT group" as specified in clause 6.3.5.4, if the MCPTT user is unauthorized to initiated a pre-arranged group session the non-controlling MCPTT function shall send a SIP 403 (Forbidden) response with the warning text set to "119 user is not authorised to initiate the group call" in a Warning header field as specified in clause 4.4.

8) shall generate a SIP INVITE request to the controlling MCPTT function as specified in clause 6.3.4.1.4; and

9) shall send the SIP INVITE request to the controlling MCPTT function as specified in 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response to the SIP INVITE request sent to the controlling MCPTT function as specified above, the non-controlling MCPTT function:

1) shall send the SIP ACK request to the controlling MCPTT function as specified in 3GPP TS 24.229 [4];

2) shall generate a SIP 200 (OK) to the "SIP INVITE request for controlling MCPTT function of an MCPTT group" as specified in 3GPP TS 24.229 populated as follows:

a) shall include an SDP answer as specified in clause 6.3.4.2.1 based on the SDP answer in the SIP 200 (OK) response;

b) shall include the public service identifier of the non-controlling MCPTT function in the P-Asserted-Identity header field;

c) shall include the warning text set to "148 group is regrouped" in a Warning header field as specified in clause 4.4; and

d) shall send the SIP 200 (OK) request according to 3GPP TS 24.229;

NOTE 3: As long as the MCPTT group is regrouped the floor control messages in the media plane includes a grouped regrouped indication as specified in 3GPP TS 24.380 [5].

3) shall start acting as a non-controlling MCPTT function and interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.5;

4) shall determine the members to invite to the prearranged MCPTT group call as specified in clause 6.3.5.5; and

5) shall invite each group member determined in step 4) immediately above, to the group session, as specified in clause 10.1.1.5.1.

Upon receipt of other final SIP responses with the exception of the SIP 2xx response to the INVITE request sent to the controlling MCPTT function as specified above, the non-controlling MCPTT function:

1) shall send the SIP ACK response to the controlling MCPTT function as specified in 3GPP TS 24.229 [4]; and

2) shall start acting as a controlling MCPTT function as specified in clause 10.1.1.4 and invite members as specified in clause 6.3.4.1.2.

NOTE 4: Regardless if the controlling MCPTT function accepts or rejects the SIP INVITE request sent above the prearranged group session continues to be initiated with only the members of the group homed on the non-controlling MCPTT function of the group being invited to the group call.

The non-controlling MCPTT function shall handle SIP responses (other than the SIP 2xx response) to the SIP INVITE requests sent to invited members as specified in 3GPP TS 24.229.

Upon receipt of a SIP 2xx response to SIP INVITE requests sent to invited members, the non-controlling MCPTT function:

1) shall send the SIP ACK request as specified in 3GPP TS 24.229 [4]; and

2) shall interact with the media plane as specified in 3GPP TS 24.380 [5]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.2]

"NOTE 1: In this clause, "MCPTT server" can refer to either the controlling MCPTT function of an MCPTT group or the non-controlling MCPTT function of an MCPTT group.

Upon receipt of a SIP INVITE request:

…

4) upon receipt of a notification from the group management server containing the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request, or if the group document is already cached:

a) if the MCPTT server is a non-controlling function of an MCPTT group, then the MCPTT server shall exit this clause; and

…

NOTE 4: The non-controlling function of an MCPTT group can be located in the primary MCPTT system or a partner MCPTT system.

…

NOTE 5: The non-controlling function of an MCPTT group can be located in the primary MCPTT system or a partner MCPTT system.

ii) for each constituent MCPTT group ID that is homed on the controlling MCPTT function of an MCPTT group shall subscribe to the "xcap-diff" event-package for the group document for the constituent MCPTT group ID as specified in 3GPP TS 24.481 [31], if not already subscribed;

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.4]

"The following conditions shall be met for a controlling MCPTT function or non-controlling MCPTT function of an MCPTT group to initiate a group session for the requesting MCPTT user:

1) if the <on-network-regrouped> element in the <list-service> element is present in the group document and if the MCPTT group ID indicated in the incoming INVITE request is the same as the MCPTT group ID in the "temporary-MCPTT-group-ID" attribute of the <on-network-regrouped> element; or

2) if the <on-network-regrouped> element in the <list-service> element of the MCPTT group document is not present in the group document;

and:

1) an <entry> element exists in the <list> element of the group document for the MCPTT user;

2) if the group is a user regroup based on a preconfigured group and the MCPTT ID contained in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body in the SIP INVITE request is included in the list of users that was stored during successful processing of the creation of the user regroup per clause 16;

3) a <rule> exists in the group document with:

a) the <is-list-member> element of the <conditions> element present and with the <allow-initiate-conference> element of the corresponding <actions> element set to "true"; or

b) the <identity> element of the <conditions> element containing an entry matching the MCPTT ID in the SIP INVITE request, with the <allow-initiate-conference> element of the <actions> element is set to "true"; and

4) if the <supported-services> element is present, it contains:

a) a <service> element containing an "enabler" attribute which is set to the MCPTT ICSI; and

b) if a <group-media> element is present, an entry set to "MCPTT speech".

then the MCPTT user shall be authorised to initiate the group session."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.5]

"The MCPTT server shall only invite affiliated group members to a group session.

If the group is not a regroup based on a preconfigured group, the MCPTT server determines the affiliated members from the entries contained in the <list> element of the group document by following the procedures specified in clause 6.3.6.

If the group is a regroup based on a preconfigured group, the MCPTT server determines the affiliated members from the list of users that was stored during successful processing of the creation of the regroup per clause 16 by following the procedures specified in clause 6.3.6.

NOTE 1: The term "affiliated group members" used above also includes those members that are implicitly affiliated by the controlling MCPTT function.

If the number of members of the MCPTT group exceeds the value contained in the <on-network-max-participant-count> element the MCPTT server shall invite only <on-network-max-participant-count> members from the list, but shall prioritise inviting those group members to the group session that have an <entry> element in the <list> element with a <on-network-required> element present.

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the group session. The <on-network-required> element is used to determine which group members need to acknowledge the group call before audio transmission can proceed.

NOTE 3: Other requirements for how the controlling MCPTT function selects which of the <on-network-max-participant-count> members to invite is outside the scope of this specification.

NOTE 4: It is assumed that validation checks are performed at the group management server to ensure that the <on-network-max-participant-count> cannot be less than the number of <on-network-required> users."

5.2.1.5.3 Test description

5.2.1.5.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the controlling MCPTT function.

  SS-UE1 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.5.3.1-1.

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the non-controlling MCPTT function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.º

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Figure A.4.4.

- The SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client), the SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are registered and authorized to use the MCPTT system.
- Group A and Group B have been regrouped in Group T
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.
- The SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are affiliated with Group B and are authorized to initiate group calls.
- Group B is controlled by the non-controlling MCPTT function.
- There are no ongoing calls in any of the groups.
- There is a trusted mutual aid relationship between the partner MCPTT system and the primary MCPTT system

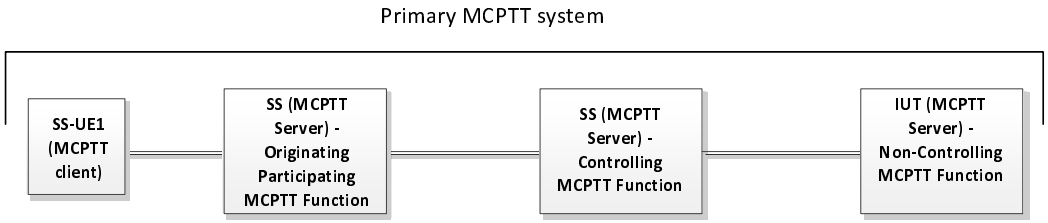


Figure 5.2.1.5.3.1-1: Functions of the testing components

5.2.1.5.3.2 Test procedure sequence

Table 5.2.1.5.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) providing the controlling MCPTT function sends a SIP INVITE to the IUT (MCPTT Server) to initiate an on-demand pre-arranged group call in constituent Group B.	<--	SIP INVITE		-
2	Check: Does the IUT (MCPTT Server) respond with a SIP 403 (Forbidden) message with a P-Refused-URI-List header and a list of two users to be invited to the call?	-->	SIP 403 (Forbidden)		P

5.2.1.5.3.3 Specific message contents

Table 5.2.1.5.3.3-1: SIP INVITE from the SS (MCPTT-Server) (Step 1, Table 5.2.1.5.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_C	The public service identity of the MCPTT server under test.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The public service identity of the MCPTT Server which is simulated by the SS, providing controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_C			
Max-Forwards				
value	"70"	The recommended initial value is 70 IETF RFC 3261 [17].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.5.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.5.3.3-1B			

Table 5.2.1.5.3.3-1A: SDP in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.5.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"10"			
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.5.3.3-1B: MCPTT-Info in SIP INVITE from SS (MCPTT Server) (Table 5.2.1.5.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_B_ID	The URI of the group		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.5.3.3-2: SIP 403 (Forbidden) from the IUT (MCPTT-Server) (Step 2, Table 5.2.1.5.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.19.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Warning	not present			
Content-Length				
value	any value	length of message-body		
Content-Type				
value	"multipart/mixed"			
Message-body				
MIME body part		**Resource list**		
MIME-part-headers				
Content-Type	"application/resource-lists+xml"			
Content-ID	any value	Unique URL identifying the Resource-lists XML MIME body; used as reference in the signature MIME body and P-Refused-URI-List header		
MIME-part-body	As described in Table 5.2.1.5.3.3-2A			
MIME body part		**Signature**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcptt-signed+xml"			
MIME-part-body	Signatures for XML MIME bodies as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.13.1-1			

Table 5.2.1.5.3.3-2A: Resource-lists from the IUT (MCPTT-Server) (Table 5.2.1.5.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
list				
entry[1]				
uri attribute	px_MCPTT_ID_User_C	The MCPTT ID of one affiliated member of Group B		
display-name	Not present			
entry[2]				
uri attribute	px_MCPTT_ID_User_D	The MCPTT ID of another affiliated member of Group B		
display-name	Not present			

5.2.1.6 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Temporary Group / Participating server / Trusted and Untrusted modes

5.2.1.6.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE1 (MCPTT client) initiates an on-demand pre-arranged group call to a temporary
group with automatic commencement mode and implicit floor control }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) and SS-UE2 (MCPTT client) and SS-UE3
(MCPTT client) and SS-UE4 (MCPTT client) to set up the call by sending SIP
messages }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call to a
temporary group with automatic commencement mode and implicit floor control }
ensure that {
  when { the SS-UE1 (MCPTT client), SS-UE2 (MCPTT client), SS-UE3 (MCPTT client) and SS-UE4 (MCPTT
client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted, Floor Release, Floor Request ) by sending messages to the SS (MCPTT Server), SS-UE2 (MCPTT
client), SS-UE3 (MCPTT client) and SS-UE4 (MCPTT client) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call to a
temporary group with automatic commencement mode and implicit floor control }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the pre-arranged group call }
    then { IUT (MCPTT Server) forwards a SIP BYE message to the SS (MCPTT Server) and forwards a SIP
200 OK message to the SS-UE1 (MCPTT client)and forwards a SIP BYE message to the other participants
and forwards a SIP 200 OK messages from other participants to the SS (MCPTT Server) }
            }
```

5.2.1.6.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 10.1.1.3.1.1, 6.3.2.1.3, 6.3.2.1.1.1, 6.3.2.1.2.1, 10.1.1.3.2, 6.3.2.2.3, 6.3.2.2.1, 6.3.2.2.2, 6.3.2.1.6, 6.3.2.2.8.1 and 4.8, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 6.4.2 and 6.4.4. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCPTT function" containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "prearranged", the participating MCPTT function:

2) shall determine the MCPTT ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request;

NOTE 2: The MCPTT ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

…

8) shall determine the public service identity of the controlling MCPTT function associated with the group identity in the SIP INVITE request. If the participating MCPTT function is unable to identify the controlling MCPTT function for the on demand prearranged group call, it shall reject the SIP INVITE request with a SIP 404 (Not Found) response with the warning text "142 unable to determine the controlling function" in a Warning header field as specified in clause 4.4, and shall not continue with any of the remaining steps;

NOTE 6: The public service identity can identify the controlling MCPTT function in the primary MCPTT system or a partner MCPTT system.

NOTE 7: How the participating MCPTT server discovers the public service identity of the controlling MCPTT function associated with the group identity is out of scope of the current release.

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the group identity which was present in the incoming SIP INVITE request;

11) shall not copy the following header fields from the incoming SIP INVITE request to the outgoing SIP INVITE request, if they were present in the incoming SIP INVITE request:

a) Answer-Mode header field as specified in IETF RFC 5373 [18]; and

b) Priv-Answer-Mode header field as specified in IETF RFC 5373 [18];

12) shall set the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request to the MCPTT ID of the calling user;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCPTT client as specified in clause 6.3.2.1.1.1;

13a) if the received SIP request contains a <functional-alias-URI> element of the application/vnd.3gpp.mcptt-info+xml MIME body, then check if the status of the functional alias is activated for the MCPTT ID. If the functional alias status is activated, then set the <functional-alias-URI> element of the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request to the received value, if the status is unequal activated then do not include a <functional-alias-URI> element;

NOTE 8: The participating MCPTT server learns the functional alias state for an MCPTT ID from procedures specified in clause 9A.2.2.2.7.

14) if the received SIP INVITE request contains an application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause F.3 and if not already copied, shall copy the contents of the application/vnd.3gpp.mcptt-location-info+xml MIME body received in the SIP INVITE request into an application/vnd.3gpp.mcptt-location-info+xml MIME body included in the outgoing SIP request;

15) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [4] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCPTT client;

NOTE 9: The participating MCPTT function will leave verification of the Resource-Priority header field to the controlling MCPTT function.

16) if, according to clause 6.4, the SIP INVITE request is regarded as being received with an implicit request to grant the floor to the originating MCPTT client:

if:

a) the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-location-info+xml MIME body with a <Report> element included in the <location-info> root element; and

b) the <allow-location-info-when-talking> element of the <ruleset> element of the MCPTT user profile document identified by the MCPTT ID of the calling MCPTT user (see the MCPTT user profile document in 3GPP TS 24.484 [50]) is set to a value of "true";

then shall copy the application/vnd.3gpp.mcptt-location-info+xml MIME body from the received SIP INVITE request into the outgoing SIP INVITE request;

otherwise:

if:

a) the participating MCPTT function has the location of the originating MCPTT client available; and

b) the <allow-location-info-when-talking> element of the <ruleset> element of the MCPTT user profile document identified by the MCPTT ID of the calling MCPTT user (see the MCPTT user profile document in 3GPP TS 24.484 [50]) is set to a value of "true";

then shall include an application/vnd.3gpp.mcptt-location-info+xml MIME body with a <Report> element included in the <location-info> root element; and

17) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [4].

…

Upon receipt of a SIP 2xx response in response to the above SIP INVITE request, the participating MCPTT function:

1) void;

2) shall generate a SIP 200 (OK) response as in clause 6.3.2.1.5.2 with the clarification that if an <MKFC-GKTPs> element was contained in the received SIP 2xx response it is not included in the generated SIP 200 (OK) response;

NOTE 10: If an <MKFC-GKTPs> element is received in a SIP 2xx response, the participating MCPTT function essentially ignores it and does not forward it, resulting in unicast media plane transmission being used for the originating client.

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) shall include an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response;

6A) shall include the answer state into the P-Answer-State header field of the outgoing SIP 200 (OK) response, if received in the P-Answer-State header field of the incoming SIP 200 (OK) response;

7) if the procedures of clause 9.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 9.2.2.2.13;

8) shall send the SIP 200 (OK) response to the MCPTT client according to 3GPP TS 24.229 [4];

9) shall interact with Media Plane as specified in 3GPP TS 24.380 [5]; and

10) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [7].

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.3]

"This clause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [4], on receipt of an incoming SIP INVITE request, the participating MCPTT function:

1) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [6] if included in the incoming SIP INVITE request;

2) should include the Session-Expires header field according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

3) shall include the option tag "timer" in the Supported header field;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

5) shall include the g.3gpp.mcptt media feature tag into the Contact header field of the outgoing SIP INVITE request;

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;

7) if the incoming SIP INVITE request contained a MIME resource-lists body, shall copy the MIME resource-lists body, according to rules and procedures of IETF RFC 5366 [20];

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request; and

9) if, according to clause 6.4, the SIP INVITE request is regarded as being not received with an implicit floor request, then:

a) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-location-info+xml MIME body with a <Report> element included in the <location-info> root element; and

b) if the <allow-location-info-when-talking> element of the <ruleset> element of the MCPTT user profile document identified by the MCPTT ID of the calling MCPTT user (see the MCPTT user profile document in 3GPP TS 24.484 [50]) is set to a value of "true";

shall copy the application/vnd.3gpp.mcptt-location-info+xml MIME body from the SIP INVITE request into the outgoing SIP INVITE."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCPTT function:

1) shall contain only one SDP media-level section for MCPTT speech as contained in the received SDP offer; and

2) shall contain one SDP media-level section for media plane control messages, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [4], the participating MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCPTT function, if required;

NOTE 1: Requirements can exist for the participating MCPTT function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the participating MCPTT function; and

NOTE 2: If the participating MCPTT function and the controlling MCPTT function or the participating MCPTT function and the non-controlling MCPTT function are in the same MCPTT server, and the participating MCPTT function does not have a dedicated IP address or a dedicated port number for media plane control channel or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [4], the participating MCPTT function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCPTT function, for the accepted media stream in the received SDP offer, if required; and

NOTE 1: Requirements can exist for the participating MCPTT function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCPTT function, for the accepted media plane control channel, if present in the received SDP offer.

NOTE 2: If the participating MCPTT function and the controlling MCPTT function or the participating MCPTT function and the non-controlling MCPTT function are in the same MCPTT server, and the participating MCPTT function does not have a dedicated IP address or a dedicated port number for media plane control channel or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

This clause covers both on-demand session and pre-established sessions.

Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the participating MCPTT function:

…

4) shall use the MCPTT ID present in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCPTT ID and public user identity;

5) if the binding between the MCPTT ID and public user identity does not exist, then the participating MCPTT function shall reject the SIP INVITE request with a SIP 404 (Not Found) response. Otherwise, continue with the rest of the steps;

6) void;

NOTE 2: If an <MKFC-GKTPs> element is received in a SIP INVITE request, the participating MCPTT function essentially ignores it and does not forward it, resulting in unicast media plane transmission being used for the terminating client.

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [18] if the "SIP INVITE request for terminating participating MCPTT function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; and

8) shall perform the manual commencement procedures specified in clause 6.3.2.2.6.1 and according to IETF RFC 5373 [18] if the "SIP INVITE request for terminating participating MCPTT function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCPTT client as per clause 7.3.3 or clause 7.3.4 is set to "manual-answer"."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.3]

"The participating MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4] and:

1) *shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [6] if included in the incoming SIP INVITE request;*

2) *should include the Session-Expires header field according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";*

3) *shall include the option tag "timer" in the Supported header field;*

4) *shall include the following in the Contact header field:*

   a) *the g.3gpp.mcptt media feature tag;*

   b) *the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";*

   c) *the isfocus media feature tag;*

   d) *an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the incoming SIP INVITE request; and*

   e) *any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;*

5) *shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];*

6) *shall include the option tag "norefersub" in a Supported header field according to rules and procedures of IETF RFC 4488 [22];*

7) *may include a Resource-Share header field in accordance with clause 5.7.1.20.3 in 3GPP TS 24.229 [4]; and*

8) *if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing SIP INVITE request."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.1]

"The participating MCPTT function shall follow the procedures in clause 6.3.2.1.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.2]

"The participating MCPTT function shall follow the procedures in clause 6.3.2.1.2.1."

[ETSI TS 123 379 [29] (3GPP TS 23.379), clause 10.1.1.3.3.1]

"Upon receiving from the MCPTT client a SIP BYE request the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.1.6]

"Upon receiving a SIP BYE request from the MCPTT client, the participating MCPTT function:

1) *shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5];*

2) *shall generate a SIP BYE request as specified in 3GPP TS 24.229 [4];*

3) *shall set the Request-URI to the MCPTT session identity as included in the received SIP BYE request;*

4) *shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;*

5) *if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and*

6) shall send the SIP BYE request toward the controlling MCPTT function, according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the terminating MCPTT function shall forward a SIP 200 (OK) response to the MCPTT client and shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the controlling MCPTT function."

[ETSI TS 123 379 [29] (3GPP TS 23.379), clause 10.1.1.3.4.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.2.2.8.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the participating MCPTT function:

1) shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.380 [5] for releasing media plane resource associated with the SIP session with the controlling MCPTT function;

2) shall generate a SIP BYE request according to 3GPP TS 24.229 [4];

3) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

4) if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and

5) shall send the SIP BYE request to the MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCPTT function:

1) shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCPTT function according to 3GPP TS 24.229 [4]; and

2) shall interact with the media plane as specified in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT client."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 4.8]

"In certain deployments, for example, in the case that the MCPTT operator uses the underlying SIP core infrastructure from the carrier operator, the MCPTT operator can prevent certain sensitive application data from being visible in the clear to the SIP layer. The following data are classed as sensitive application data:

- *MCPTT ID;*
- *MCPTT group ID;*
- *user location information;*
- *emergency, alert and imminent-peril indicators;*
- *access token (containing the MCPTT ID);*
- *MCPTT client ID; and*
- *functional alias.*

The above data is transported as XML content in SIP messages. in XML elements or XML attributes.

Data is transported in attributes in the following circumstances in the procedures in the present document:

- *an MCPTT ID, an MCPTT Group ID, and an MCPTT client ID in an XML document published in SIP PUBLISH request for affiliation according to IETF RFC 3856 [51];*
- *an MCPTT ID or an MCPTT Group ID in XML document notified in a SIP NOTIFY request for affiliation according to IETF RFC 3856 [51];*

- *an MCPTT ID and functional alias in an XML document published in SIP PUBLISH request for functional alias management according to IETF RFC 3856 [51];*

- *an MCPTT ID and functional alias in an XML document notified in a SIP NOTIFY request for functional alias management according to IETF RFC 3856 [51];*

- *an MCPTT ID in application/resource-lists+xml document included in an SIP INVITE request setting up a private call according to IETF RFC 5366 [20];*

- *an MCPTT ID in application/resource-lists+xml document included in an SIP INVITE request setting up a group call to a temporary group involving a non-controlling function that works in "Trusted Mode" according to IETF RFC 5366 [20], whereby the participants are returned to the controlling function in a MIME body of a SIP 403 (Forbidden) with the P-Refused-URI-List header field according to IETF RFC 5318 [36];*

- *an MCPTT ID in XML document provided in SIP NOTIFY request of a conference event package according to IETF RFC 4575 [30]; and*

- *an MCPTT ID or MCPTT Group ID in application/resource-lists+xml document according to IETF RFC 5366 [20], included in a SIP REFER request when using a pre-established session (the application/resource-lists+xml MIME body is pointed to by a Cid-URL as specified in IETF RFC 2392 [62] contained in the Refer-To header field of the SIP REFER request);*

- *an MCPTT ID in XML document provided in SIP INFO request according to IETF RFC 6086 [64], after receiving the SIP ACK for a SIP 200 (OK) with the Warning header field set with the Warning text "111 group call proceeded without all required group members".*

3GPP TS 33.180 [78] describes a method to provide confidentiality protection of sensitive application data in elements by using XML encryption (i.e. xmlenc) and in attributes by using an attribute confidentiality protection scheme described in clause 6.6.2.3 of the present document. Integrity protection can also be provided by using XML signatures (i.e. xmlsig).

Protection of the data relies on a shared XML protection key (XPK) used to encrypt and sign data:

- *between the MCPTT client and the MCPTT server, the XPK is a client-server key (CSK); and*

- *between MCPTT servers and between MCPTT domains, the XPK is a signalling protection key (SPK).*

The CSK (XPK) and a key-id CSK-ID (XPK-ID) are generated from keying material provided by the key management server. Identity based public key encryption based on MIKEY-SAKKE is used to transport the CSK between SIP end-points. The encrypted CSK is transported from the MCPTT client to the MCPTT server when the MCPTT client performs service authorisation as described in clause 7 and is also used during service authorisation to protect the access token.

The SPK (XPK) and a key-id SPK-ID (XPK-ID) are directly provisioned in the MCPTT servers.

Configuration in the MCPTT client and MCPTT server is used to determine whether one or both of confidentiality protection and integrity protection are required."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.4.2]

"Upon receiving a floor control message the participating MCPTT function:

1. *shall immediately forward the floor control message to the floor control server if the message is received from the floor participant;*

2. *if an MBMS subchannel is not used for a conversation in the session the floor control message is associated with, shall immediately forward the floor control message to the floor participant if the message is received from the floor control server; and*

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCPTT function:

1. *shall stop sending floor control messages towards the floor participant and the floor control server; and*
2. *shall stop sending RTP media packets towards the MCPTT client and towards the controlling MCPTT function.*

When the participating MCPTT function receives an indication from the application and signalling plane that the session is released, the participating MCPTT function:

1. *in case of a pre-established session, shall perform the actions in subclause 9.3.2; and*
2. *in case of an on-demand session, shall release the media resources associated with the session."*

5.2.1.6.3 Test description

5.2.1.6.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the controlling MCPTT function.
  - SS (MCPTT Server) provides the non-controlling MCPTT function.
- SS-UE1 (MCPTT client), SS-UE2 (MCPTT client), SS-UE3 (MCPTT client) and SS-UE4 (MCPTT client)

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the originating participating MCPTT function.
  - IUT (MCPTT Server) provides the terminating participating MCPTT function.
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2 and 4.3 and to the SS-UE1 (MCPTT client), SS-UE2 (MCPTT client), SS-UE3 (MCPTT client) and SS-UE4 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client), the SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are registered and authorized to use the MCPTT system.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.
- Group A and Group B have been regrouped in Group T
- Group A is controlled by the controlling MCPTT function.
- The SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are affiliated with Group B and are authorized to initiate group calls.
- Group B is controlled by the non-controlling MCPTT function.

- There are no ongoing calls in Group A and Group B.

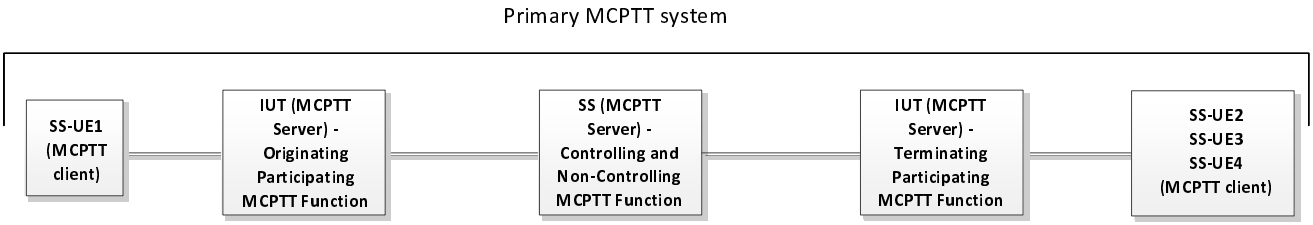


Figure 5.2.1.6.3.1-1: Functions of the testing components, trusted mode

5.2.1.6.3.2 Test procedure sequence

Table 5.2.1.6.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT client), which is a member of the constituent Group A, initiates an on-demand pre-arranged group call to a temporary group (Group T) with automatic commencement mode and implicit floor control.	<--	SIP INVITE		-
2	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS-UE1 (MCPTT client)?	-->	SIP 100 Trying	1	P
3	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the controlling MCPTT function?	-->	SIP INVITE	1	P
	- EXCEPTION: The SS (MCPTT Server) sends three SIP INVITE requests to invite the other participants to the call. The following sequence of steps identified with first "lower case letters" can happen in parallel.				
4a	The SS (MCPTT Server) sends a SIP INVITE to the IUT (MCPTT Server) to invite SS-UE2 to the call?	<--	SIP INVITE		-
5aa1	Check: Does the IUT (MCPTT Server) respond to the SIP INVITE message with a SIP 183 (Session Progress) message sent reliably?	-->	SIP 183 (Session Progress)	1	P
5ab1	Check: Does the IUT (MCPTT Server) respond to the SIP INVITE message with a SIP 183 (Session Progress) message sent unreliably?	-->	SIP 183 (Session Progress)	1	P
-	EXCEPTION: Depending on MCPTT Server implementation, steps 6aa1-6aa2 will take place in parallel if the SIP 183 (Session Progress) is sent reliably (second "lower case letters")..	-	-	-	-
6aa1	The SS (MCPTT Server) sends a SIP PRACK message to the IUT (MCPTT Server)	<--	SIP PRACK		-
6aa2	Check: Does the IUT (MCPTT Server) respond to the SIP PRACK with a SIP 200 (OK) message?	-->	SIP 200 OK	1	P
7a	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS-UE2 (MCPTT client)?	-->	SIP INVITE	1	P
8a	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
9a	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS-UE2 (MCPTT client)?	-->	SIP ACK	1	P
10a	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server)?	-->	SIP 200 (OK)	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
11a	The SS (MCPTT Server) responds with a SIP ACK message	<--	SIP ACK		-
12a	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
13a	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	2	P
4b	The SS (MCPTT Server) sends a SIP INVITE to the IUT (MCPTT Server) to invite SS-UE3 to the call?	<--	SIP INVITE		-
5ba1	Check: Does the IUT (MCPTT Server) respond to the SIP INVITE message with a SIP 183 (Session Progress) message sent reliably?	-->	SIP 183 (Session Progress)	1	P
5bb1	Check: Does the IUT (MCPTT Server) respond to the SIP INVITE message with a SIP 183 (Session Progress) message sent unreliably?	-->	SIP 183 (Session Progress)	1	P
-	EXCEPTION: Depending on MCPTT Server implementation, steps 6ba1-6ba2 will take place in parallel if the SIP 183 (Session Progress) is sent reliably (second "lower case letters")..	-	-	-	-
6ba1	The SS (MCPTT Server) sends a SIP PRACK message to the IUT (MCPTT Server)	<--	SIP PRACK		-
6ba2	Check: Does the IUT (MCPTT Server) respond to the SIP PRACK with a SIP 200 (OK) message?	-->	SIP 200 OK	1	P
7b	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS-UE3 (MCPTT client)?	-->	SIP INVITE	1	P
8b	The SS-UE3 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
9b	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS-UE3 (MCPTT client)?	-->	SIP ACK	1	P
10b	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server)?	-->	SIP 200 (OK)	1	P
11b	The SS (MCPTT Server) responds with a SIP ACK message	<--	SIP ACK		-
12b	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE3 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
13b	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE3 (MCPTT client)?	-->	Floor Taken	2	P
4c	The SS (MCPTT Server) sends a SIP INVITE to the IUT (MCPTT Server) to invite SS-UE4 to the call?	<--	SIP INVITE		-
5ca1	Check: Does the IUT (MCPTT Server) respond to the SIP INVITE message with a SIP 183 (Session Progress) message sent reliably?	-->	SIP 183 (Session Progress)	1	P
5cb1	Check: Does the IUT (MCPTT Server) respond to the SIP INVITE message with a SIP 183 (Session Progress) message sent unreliably?	-->	SIP 183 (Session Progress)	1	P
-	EXCEPTION: Depending on MCPTT Server implementation, steps 6ca1-6ca2 will take place in parallel if the SIP 183 (Session Progress) is sent reliably (second "lower case letters")..	-	-	-	-
6ca1	The SS (MCPTT Server) sends a SIP PRACK message to the IUT (MCPTT Server)	<--	SIP PRACK		-
6ca2	Check: Does the IUT (MCPTT Server) respond to the SIP PRACK with a SIP 200 (OK) message?	-->	SIP 200 OK	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
7c	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS-UE4 (MCPTT client)?	-->	SIP INVITE	1	P
8c	The SS-UE4 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
9c	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS-UE4 (MCPTT client)?	-->	SIP ACK	1	P
10c	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server)?	-->	SIP 200 (OK)	1	P
11c	The SS (MCPTT Server) responds with a SIP ACK message	<--	SIP ACK		-
12c	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE4 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken		-
13c	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE4 (MCPTT client)?	-->	Floor Taken	2	P
14	The SS (MCPTT server) responds with a SIP 200 (OK) message to the IUT (MCPTT Server)	<--	SIP 200 (OK)		-
15	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	1	P
16	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS-UE1 (MCPTT Client)?	-->	SIP 200 (OK)	1	P
17	The SS-UE1 (MCPTT Client) responds with a SIP ACK message	<--	SIP ACK		-
18	The SS-UE1 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
19	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	2	P
20	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
21	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE1 (MCPTT client)?	-->	Floor Ack	2	P
22	The SS (MCPTT server) sends a Floor Idle message for SS-UE1	<--	Floor Idle		-
23	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	2	P
24	The SS (MCPTT server) sends a Floor Idle message for SS-UE1	<--	Floor Idle		-
25	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	2	P
25	The SS (MCPTT server) sends a Floor Idle message for SS-UE2	<--	Floor Idle		-
26	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	2	P
27	The SS (MCPTT server) sends a Floor Idle message for SS-UE3	<--	Floor Idle		-
28	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE3 (MCPTT client)?	-->	Floor Idle	2	P
29	The SS (MCPTT server) sends a Floor Idle message for SS-UE4	<--	Floor Idle		-
30	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE4 (MCPTT client)?	-->	Floor Idle	2	P
31	The SS-UE3 (MCPTT client) sends a Floor Request message	<--	Floor Request		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
32	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	2	P
33	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
34	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE3 (MCPTT client)?	-->	Floor Granted	2	P
35	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE1 (MCPTT client) that the floor is taken by SS-UE3 (MCPTT client)	<--	Floor Taken		-
36	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
37	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE3 (MCPTT client)	<--	Floor Taken		-
38	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	2	P
39	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE4 (MCPTT client) that the floor is taken by SS-UE3 (MCPTT client)	<--	Floor Taken		-
40	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE4 (MCPTT client)?	-->	Floor Taken	2	P
41	The SS-UE3 (MCPTT client) sends a Floor Release message with an acknowledgement required to release the floor	<--	Floor Release		-
42	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	2	P
43	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
44	Check: Does the IUT (MCPTT Server) forward the Floor Ack message to the SS-UE3 (MCPTT client)?	-->	Floor Ack	2	P
45	The SS (MCPTT server) sends a Floor Idle message for SS-UE1	<--	Floor Idle		-
46	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	2	P
47	The SS (MCPTT server) sends a Floor Idle message for SS-UE2	<--	Floor Idle		-
48	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	2	P
49	The SS (MCPTT server) sends a Floor Idle message for SS-UE3	<--	Floor Idle		-
50	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE3 (MCPTT client)?	-->	Floor Idle	2	P
51	The SS (MCPTT server) sends a Floor Idle message for SS-UE4	<--	Floor Idle		-
52	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE4 (MCPTT client)?	-->	Floor Idle	2	P
53	The SS-UE1 (MCPTT client) sends a SIP BYE request	<--	SIP BYE		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
54	Check: Does the IUT (MCPTT Server) forward the SIP BYE message to the SS (MCPTT Server) to end the call?	-->	SIP BYE	3	P
55	SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
56	Check: Does the IUT (MCPTT Server) forward the SIP 200 (OK) response to SS-UE1 (MCPTT client)	-->	SIP 200 (OK)	3	P
57	The SS (MCPTT Server) sends a SIP BYE request for SS-UE2	<--	SIP BYE		-
58	Check: Does the IUT (MCPTT Server) forward the SIP BYE message to the SS-UE2 (MCPTT client) to end the call?	-->	SIP BYE	3	P
59	The SS-UE2 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
60	Check: Does the IUT (MCPTT Server) forward the SIP 200 (OK) response to SS (MCPTT Server)	-->	SIP 200 (OK)	3	P
61	The SS (MCPTT Server) sends a SIP BYE request for SS-UE3	<--	SIP BYE		-
62	Check: Does the IUT (MCPTT Server) forward the SIP BYE message to the SS-UE3 (MCPTT client) to end the call?	-->	SIP BYE	3	P
63	The SS-UE3 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
64	Check: Does the IUT (MCPTT Server) forward the SIP 200 (OK) response to SS (MCPTT Server)	-->	SIP 200 (OK)	3	P
65	The SS (MCPTT Server) sends a SIP BYE request for SS-UE4	<--	SIP BYE		-
66	Check: Does the IUT (MCPTT Server) forward the SIP BYE message to the SS-UE4 (MCPTT client) to end the call?	-->	SIP BYE	3	P
67	The SS-UE4 (MCPTT client) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
68	Check: Does the IUT (MCPTT Server) forward the SIP 200 (OK) response to SS (MCPTT Server)	-->	SIP 200 (OK)	3	P

5.2.1.6.3.3 Specific message contents

Table 5.2.1.6.3.3-1: SIP INVITE from the SS-UE1 (Step 1, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [25]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [25].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.6.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.6.3.3-1B			

Table 5.2.1.6.3.3-1A: SDP in SIP INVITE from the SS-UE1 (Table 5.2.1.6.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions INITIAL_SDP_OFFER, IMPLICIT_GRANT_REQUESTED

Table 5.2.1.6.3.3-1B: MCPTT-Info in SIP INVITE from the SS-UE1 (Table 5.2.1.6.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
associated-group-id	px_MCPTT_Group_A_ID	The URI of the constituent group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.6.3.3-2: SIP INVITE from the IUT (MCPTT-Server) (Step 3, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server which is simulated by the SS, providing the controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the IUT (MCPTT-Server)			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server, providing originating participating function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server which is simulated by the SS, providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server, providing originating participating function.		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.6.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.6.3.3-2B			

Table 5.2.1.6.3.3-2A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.6.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions INITIAL_SDP_OFFER, IMPLICIT_GRANT_REQUESTED				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the IUT (MCPTT Server)			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (MCPTT Server)			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT Server)			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT Server)			

Table 5.2.1.6.3.3-2B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.6.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
associated-group-id	px_MCPTT_Group_A_ID	The URI of the constituent group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.1.6.3.3-3: SIP 200 (OK) from the SS (MCPTT Server) (Step 14, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.6.3.3-3A			

Table 5.2.1.6.3.3-3A: SDP in SIP 200 (OK) from the SS (MCPTT Server) (Table 5.2.1.6.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_FLOOR_GRANTED, IMPLICIT_GRANT_REQUESTED

Table 5.2.1.6.3.3-4: SIP 200 (OK) from the IUT (MCPTT-Server) (Step 16, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.6.3.3-4A			

Table 5.2.1.6.3.3-4A: SDP in SIP 200 (OK) from the IUT (MCPTT-Server) (Table 5.2.1.6.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_FLOOR_GRANTED, IMPLICIT_GRANT_REQUESTED				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the IUT (MCPTT Server)			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (MCPTT Server)			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the IUT (MCPTT Server) (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the IUT (MCPTT Server)			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the MCPTT Server being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT Server)			

Table 5.2.1.6.3.3-5: SIP INVITE from the SS (MCPTT Server) (Step 4a, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_A	The URI of the IUT (MCPTT Server), providing the terminating participating function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server which is simulated by the SS, providing controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the IUT (MCPTT Server), providing the terminating participating function.		
Max-Forwards				
value	"70"	The recommended initial value is 70 IETF RFC 3261 [17].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.6.3.3-5A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.6.3.3-5B			

Table 5.2.1.6.3.3-5A: SDP in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.6.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.6.3.3-5B: MCPTT-Info in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.6.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.6.3.3-6: SIP 183 (Session Progress) from the IUT (MCPTT-Server) (Steps 5aa1, 5ba1 and 5ca1, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.1-1, conditions MCPTT, 100rel

Table 5.2.1.6.3.3-7: SIP 183 (Session Progress) from the IUT (MCPTT-Server) (Steps 5ab1, 5bb1 and 5cb1, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.1-1, conditions MCPTT

Table 5.2.1.6.3.3-8: SIP PRACK from the SS (MCPTT Server) (Steps 6aa1, 6ba1 and 6ca1, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.10.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Request-URI	same URI as the MCPTT-Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT (MCPTT-Server)		
Max-Forwards				
value	"70"	The recommended initial value is 70 in IETF RFC 3261 [17].		

Table 5.2.1.6.3.3-9: SIP 200 OK to PRACK from the IUT (MCPTT-Server) (Steps 6aa2, 6ba2 and 6ca2, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Content-Type	not present			
Content-Length	not present			

Table 5.2.1.6.3.3-10: SIP INVITE from the IUT (MCPTT-Server) (Step 7a, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the IUT (MCPTT-Server)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	MCPTT Server's port number		
via-branch	Value assigned by the IUT (MCPTT-Server)			
Record-Route	not present			
From				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server, providing terminating participating function.		
To			IETF RFC 3261 [17] IETF RFC 5031 [19]	
addr-spec				
user-info and host	px_MCX_SIP_PublicUserId_B	Default public user ID (IMPU) as stored in the UICC		
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
Max-Forwards				
value	"70"	The recommended initial value is 70 IETF RFC 3261 [17]		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.6.3.3-10A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.6.3.3-10B			

Table 5.2.1.6.3.3-10A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.6.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the IUT (MCPTT-Server)			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (MCPTT-Server)			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT-Server)			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT-Server)			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.6.3.3-10B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.6.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.6.3.3-11: SIP 200 (OK) from the SS-UE2, SS-UE3 and SS-UE4 (Steps 8a, 8b and 8c, Table 5.2.1.6.3.2-1)

Derivation Path: Annex A, Table A.3.2.4-7

Table 5.2.1.6.3.3-12: SIP 200 (OK) from the IUT (MCPTT-Server) (Steps 10a, 10b and 10c, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.1.6.3.3-12A			

Table 5.2.1.6.3.3-12A: SDP in SIP 200 (OK) from the IUT (MCPTT-Server) (Table 5.2.1.6.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the IUT (MCPTT-Server)			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (MCPTT-Server)			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the IUT (MCPTT-Server) (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the IUT (MCPTT-Server)			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the IUT (MCPTT-Server) being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (MCPTT-Server)			

Table 5.2.1.6.3.3-13: SIP INVITE from the SS (MCPTT Server) (Step 4b, Table 5.2.1.6.3.2-1)

Derivation Path: Table 5.2.1.6.3.3-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server which is simulated by the SS, providing controlling function.		TRUSTED_MODE
addr-spec	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server which is simulated by the SS, providing non-controlling function.		UNTRUSTED_MODE
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.6.3.3-5A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.6.3.3-13A			

Table 5.2.1.6.3.3-13A: MCPTT-Info in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.6.3.3-13)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_C	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.6.3.3-14: SIP INVITE from the IUT (MCPTT-Server) (Step 7b, Table 5.2.1.6.3.2-1)

Derivation Path: Table 5.2.1.6.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
To			IETF RFC 3261 [17] IETF RFC 5031 [19]	
addr-spec				
user-info and host	px_MCX_SIP_PublicUserId_C	Default public user ID (IMPU) as stored in the UICC		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.6.3.3-10A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.6.3.3-13A			

Table 5.2.1.6.3.3-15: SIP INVITE from the SS (MCPTT Server) (Step 4c, Table 5.2.1.6.3.2-1)

Derivation Path: Table 5.2.1.6.3.3-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server which is simulated by the SS, providing controlling function.		TRUSTED_MODE
addr-spec	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server which is simulated by the SS, providing non-controlling function.		UNTRUSTED_MODE
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.6.3.3-5A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.6.3.3-15A			

Table 5.2.1.6.3.3-15A: MCPTT-Info in SIP INVITE from the SS (MCPTT Server) (Table 5.2.1.6.3.3-15)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_D	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.6.3.3-16: SIP INVITE from the IUT (MCPTT-Server) (Step 7c, Table 5.2.1.6.3.2-1)

Derivation Path: Table 5.2.1.6.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
To			IETF RFC 3261 [17] IETF RFC 5031 [19]	
addr-spec				
user-info and host	px_MCX_SIP_PublicUserId_D	Default public user ID (IMPU) as stored in the UICC		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.6.3.3-10A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.6.3.3-15A			

Table 5.2.1.6.3.3-17: SIP BYE from the SS-UE1 (Step 53, 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT (MCPTT Server)		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the UE's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the UE's point of view)		

Table 5.2.1.6.3.3-18: SIP BYE from the IUT (MCPTT-Server) (Step 54, 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

Table 5.2.1.6.3.3-19: SIP BYE from the SS (MCPTT Server) (Steps 57, 61 and 65, 5.2.1.6.3.2-1)

Derivation Path: Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IUT (MCPTT-Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT (MCPTT-Server)		
To				
addr-spec	Same URI of the IUT (MCPTT-Server) as used earlier in the dialog	Remote URI of the dialog (from the UE's point of view)		
tag	Same tag of the IUT (MCPTT-Server) as used earlier in the dialog	Remote tag of the dialog ID (from the UE's point of view)		

Table 5.2.1.6.3.3-20: SIP BYE from the IUT (MCPTT-Server) (Steps 58, 62 and 66, 5.2.1.6.3.2-1)

Derivation Path: Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From				
addr-spec	Same URI of the IUT (MCPTT-Server) as used earlier in the dialog	Local URI of the dialog (from the IUT (MCPTT-Server)'s point of view)		
tag	Same tag of the IUT (MCPTT-Server) as used earlier in the dialog	Local tag of the dialog (from the IUT (MCPTT-Server)'s point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

Table 5.2.1.6.3.3-21: Floor Taken (Steps 12a, 13a, 12b, 13b, 12c and 13c, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.1.6.3.3-22: Floor Taken (Steps 35- 40, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_C			

Table 5.2.1.6.3.3-23: Floor Release (Steps 18, 19, 41 and 42, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.1.6.3.3-24: Floor Ack (Steps 20, 21, 43 and 44, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 5.2.1.6.3.3-25: Floor Idle (Steps 22-30 and 45-52, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.1.6.3.3-26: Floor Request (Steps 31 and 32, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1

Table 5.2.1.6.3.3-27: Floor Granted (Steps 33 and 34, Table 5.2.1.6.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

5.2.1.7 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Temporary Group / Controlling server / Untrusted mode

5.2.1.7.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE1 (MCPTT client) initiates an on-demand pre-arranged group call to a temporary
group with implicit floor control }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the call by sending SIP
messages }
           }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call to a
temporary group with implicit floor control }
ensure that {
  when { the SS-UE1 (MCPTT client), SS-UE2 (MCPTT client), SS-UE3 (MCPTT client) and SS-UE4 (MCPTT
client) engage in communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted ) by sending messages via the SS (MCPTT Server) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call to a
temporary group with implicit floor control }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the pre-arranged group call }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) by sending a SIP 200 (OK) message
to the client ending the call and sends a SIP BYE message to the other participants }
            }
```

5.2.1.7.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: clauses 10.1.1.4.1.1, 10.1.1.4.1.2, 10.1.1.4.2, 6.3.3.1.2, 6.3.3.1.1, 6.3.5.2, 6.3.5.5, 6.3.3.2.3.2, 6.3.3.2.1, 10.1.1.4.3, 6.3.8.1, 6.3.3.1.5, 4.8, 5.3.2, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 6.3.2.2, 6.3.3, 14.3.4. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.1.1]

"This clause describes the procedures for inviting an MCPTT user to an MCPTT session. The procedure is initiated by the controlling MCPTT function as the result of an action in clause 10.1.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

…

NOTE 1: How the controlling MCPTT function finds the address of the terminating MCPTT participating function is out of the scope of the current release.

…

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcptt-request-uri> element set to the MCPTT ID of the terminating user; and

b) the <mcptt-calling-group-id> element set to the group identity;

NOTE 3: The <mcptt-calling-user-id> is already included in the MIME body as a result of calling clause 6.3.3.1.2 in step 1).

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

…

Upon receiving a SIP 183 (Session Progress) response containing a Require header field with the option tag "100rel" and containing a P-Answer-State header field with the value "Unconfirmed" in response to the SIP INVITE request the controlling MCPTT function:

1) shall send a SIP PRACK request towards the MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

2) shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as

…

NOTE 5: The procedures executed by the controlling MCPTT function prior to sending a response to the inviting MCPTT client are specified in clause 10.1.1.4.2."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.1.2]

"The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the non-controlling MCPTT function serving the group identity of the MCPTT group owned by the partner MCPTT system;

3) shall set the P-Asserted-Identity to the public service identity of the controlling MCPTT function;

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcptt-request-uri> element set to the group identity of the MCPTT group hosted by the non-controlling MCPTT function in the partner MCPTT system; and

b) the <mcptt-calling-group-id> element set to the group identity of the group served by the controlling MCPTT function;

…

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

8) shall send the SIP INVITE request towards the partner MCPTT system in accordance with 3GPP TS 24.229 [4].

Upon receiving SIP 403 (Forbidden) response for the SIP INVITE request, if according to local policy and if:

1) the response contains a Warning header field with the MCPTT warning code "128"; and

2) the response contains a P-Refused-URI-List header field and an application/resource-lists+xml MIME body as specified in IETF RFC 5318 [36];

NOTE 1: The application/resource-lists+xml MIME body contains MCPTT IDs identifying MCPTT users in a partner MCPTT system that needs to be invited to the prearranged group call in case of group regrouping using interrogating method as specified in 3GPP TS 23.379 [3] clause 10.6.2.4.2.

then the controlling MCPTT function:

1) shall check if the number of members of the MCPTT group exceeds the value contained in the <on-network-max-participant-count> element of the group document as specified in 3GPP TS 24.481 [31]. If exceeded, the controlling MCPTT function shall invite only <on-network-max-participant-count> members from the application/resource-lists+xml MIME body; and

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the prearranged group session It is operator policy that determines which participants in the application/resource-lists+xml MIME body are invited to the group call.

2) shall invite MCPTT users as specified in this clause using the list of MCPTT IDs in URI-List.

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 3: The procedures executed by the controlling MCPTT function prior to sending a response to the inviting MCPTT client are specified in clause 10.1.1.4.2.

2) if at least one of the invited MCPTT clients has subscribed to the conference package, shall subscribe to the conference event package in the non-controlling MCPTT function as specified in clause 10.1.3.4.3; and

3) if the 200 (OK) response includes the <floor-state> element set to "floor-taken", shall wait for a SIP INFO request containing a floor request from the non-controlling MCPTT function.

Upon receiving a SIP INFO request containing a floor request where:

1) the Request-URI contains an MCPTT session ID identifying an ongoing temporary group session; and

2) the application/vnd.3gpp.mcptt-info+xml MIME body contains the <mcptt-calling-group-id> element with the MCPTT group ID of a MCPTT group invited to the temporary group session;

then the controlling MCPTT function:

1) shall send a SIP 200 (OK) response to the SIP INFO request to the non-controlling MCPTT function as specified in 3GPP TS 24.229 [4]; and

2) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.2]

"In the procedures in this clause:

1) MCPTT ID in an incoming SIP INVITE request refers to the MCPTT ID of the originating user from the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

3) MCPTT ID in an outgoing SIP INVITE request refers to the MCPTT ID of the called user in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the outgoing SIP INVITE request;

4) indication of required group members in a SIP 183 (Session Progress) response refers to the <required> element of the application/vnd.3gpp.mcptt-info+xml MIME body set to "true" in a SIP 183 (Session Progress) sent by the non-controlling MCPTT function of an MCPTT group;

5) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

6) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a "SIP INVITE request for controlling MCPTT function of an MCPTT group", the controlling MCPTT function:

…

NOTE 1: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating an MCPTT emergency group call as determined by clause 6.3.3.1.13.2, or for originating an MCPTT imminent peril group call as determined by clause 6.3.3.1.13.5, the controlling MCPTT function can according to local policy choose to accept the request.

…

4) if received SIP INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

5) if the group identity is associated with a group document maintained by the GMS:

NOTE 2: How the MCPTT server determines that a group identity represents a group for which a group document is stored in the GMS is an implementation detail.

a) shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2;

b) if the group referred to by the group identity has been regrouped, shall:

i) stop processing the SIP INVITE request;

ii) shall reject the SIP INVITE request with a SIP 403 (Forbidden) response with the warning text set to "148 group is regrouped" as specified in clause 4.4 "Warning header field";

iii) if the group referred to by the group identity has been regrouped based on a preconfigured group, shall send a copy of the notifying SIP MESSAGE that was generated and sent per clause 16.2.4.1 to the participating function for the MCPTT ID of the incoming SIP INVITE request; and

iv) skip the rest of the steps;

c) if the result of the initial processing in clause 6.3.5.2 was:

i) that authorization of the MCPTT ID is required at a non-controlling MCPTT function of an MCPTT group is required, perform the actions in clause 6.3.3.1.13.7 and do not continue with the rest of the steps in this clause; and

ii) that a SIP 3xx, 4xx, 5xx or 6xx response to the "SIP INVITE request for controlling MCPTT function of an MCPTT group" has been sent, do not continue with the rest of the steps in this clause;

6) if the group identity is associated with a user or group regroup based on a preconfigured group:

a) shall retrieve the stored information for the group identity;

b) if there is no stored information for the group identity, the controlling MCPTT function:

i) shall return a SIP 404 (Not Found) response with the warning text set to "163 the group identity indicated in the request does not exist" as specified in clause 4.4 "Warning header field" and shall not continue with the rest of the steps;

NOTE 3: The user or group regroup can have been removed very recently and the client has sent the group call request prior to receiving the removal notification.

7) shall perform the actions as described in clause 6.3.3.2.2;

…

14) if the MCPTT group call is not ongoing then:

…

c) if the received SIP INVITE request contains an authorised request for an MCPTT emergency group call as determined by clause 6.3.3.1.13.2 or MCPTT imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCPTT user is eligible to be implicitly affiliated with the MCPTT group as determined in step 14) a) iv) above, shall perform the implicit affiliation as specified in clause 9.2.2.3.7;

d) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [4];

e) shall create a prearranged group session and allocate an MCPTT session identity for the prearranged group call, and shall handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

f) if the group identity in the "SIP INVITE request for controlling MCPTT function of an MCPTT group" is a TGI:

i) shall for each of the constituent MCPTT groups homed on the primary MCPTT system:

A) if the controlling MCPTT function does not own the MCPTT group identified by the MCPTT group ID, then generate a SIP INVITE request towards the MCPTT server that owns the MCPTT group identity by following the procedures in clause 10.1.1.4.1.2; and

NOTE 4: The MCPTT server that the SIP INVITE request is sent to acts as a non-controlling MCPTT function;

B) if the controlling MCPTT function owns the MCPTT group identified by the MCPTT group ID then:

I) determine the members to invite to the prearranged MCPTT group call as specified in clause 6.3.5.5;

II) invite each group member determined in step A) above, to the group session, as specified in clause 10.1.1.4.1.1; and

III) interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3; and

ii) shall for each of the constituent MCPTT groups homed on the partner MCPTT system generate a SIP INVITE request for the MCPTT group identity homed on the partner MCPTT system as specified in clause 10.1.1.4.1.2; and

g) if the group identity in the SIP INVITE request for controlling MCPTT function of an MCPTT group is an MCPTT group ID:

i) shall determine the members to invite to the prearranged MCPTT group call as specified in clause 6.3.5.5. If:

A) the number of affiliated members of the MCPTT group is lower than the value contained in the <on-network-minimum-number-of-affiliated-members> element of the group document as specified in 3GPP TS 24.481 [31]; or

B) the group document contains any <on-network-affiliation-to-group-required> group member as specified in 3GPP TS 24.481 [31] that is not affiliated;

then the controlling MCPTT function shall send a SIP 480 (Temporarily Unavailable) response to the MCPTT client that originated the group session with the warning text set to "112 group call abandoned due to required group members not part of the group session" in a Warning header field as specified in clause 4.4 and skip the rest of the steps below;

ii) if necessary, shall start timer TNG1 (acknowledged call setup timer) according to the conditions stated in clause 6.3.3.3;

iii) if the received SIP INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element set to a value of "true":

A) shall cache the information that the MCPTT user has initiated an MCPTT emergency call;

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCPTT emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that the MCPTT user has initiated an MCPTT emergency alert; and

C) if the in-progress emergency state of the group is set to a value of "false":

I) shall set the value of the in-progress emergency state of the group to "true"; and

II) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iv) if the in-progress emergency state of the group is set to a value of "false" and if the received SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCPTT function shall:

A) shall cache the information that the MCPTT user has initiated an MCPTT imminent peril call; and

B) if the in-progress imminent peril state of the group is set to a value of "false", shall set the in-progress imminent peril state of the group to a value of "true";

v) shall invite each group member determined in step 13)g)i) above, to the group session, as specified in clause 10.1.1.4.1.1; and

vi) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3; and

15) if the MCPTT group call is ongoing then:

…

e) if the received SIP INVITE request contains an authorised request for an MCPTT emergency group call as determined by clause 6.3.3.1.13.2 or MCPTT imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCPTT user is eligible to be implicitly affiliated with the MCPTT group as determined in step 15) a) iv) above, shall perform the implicit affiliation as specified in clause 9.2.2.3.7;

f) if the received SIP INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element set to a value of "true":

i) shall cache the information that the MCPTT user has initiated an MCPTT emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCPTT emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that the MCPTT user has initiated an MCPTT emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16; and

C) shall generate SIP re-INVITE requests for the MCPTT emergency group call to the other call participants of the MCPTT group as specified in clause 6.3.3.1.6;

iv) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCPTT user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4]; and

v) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5];

g) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCPTT function:

i) shall cache the information that the MCPTT user has initiated an MCPTT imminent peril call; and

ii) if the in-progress imminent peril state of the group is set to a value of "false":

A) shall set the in-progress imminent peril state of the group to a value of "true";

B) shall generate SIP re-INVITE requests for the MCPTT imminent peril group call to the other call participants of the MCPTT group as specified in clause 6.3.3.1.15; and

C) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

iii) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCPTT user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4];

h) shall generate a SIP 200 (OK) response as specified in the clause 6.3.3.2.3.2;

i) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

j) shall include in the SIP 200 (OK) response with the warning text set to "123 MCPTT session already exists" as specified in clause 4.4;

k) if the received SIP re-INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCPTT emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

l) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 6: In this case, the request was for an imminent peril call but a higher priority MCPTT emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

m) shall interact with media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 7: Resulting media plane processing is completed before the next step is performed.

n) shall send the SIP 200 (OK) response towards the inviting MCPTT client or inviting non-controlling MCPTT function according to 3GPP TS 24.229 [4];

o) shall generate a notification to the MCPTT clients, which have subscribed to the conference event package that the inviting MCPTT User has joined in the MCPTT group session, as specified in clause 6.3.3.4;

NOTE 8: As a group document can potentially have a large content, the controlling MCPTT function can notify using content-indirection as defined in IETF RFC 4483 [32].

p) shall send a SIP NOTIFY request to each MCPTT client according to 3GPP TS 24.229 [4];

q) Upon receiving a SIP ACK to the above SIP 200 (OK) response and the SIP 200 (OK) response contained a Warning header field as specified in clause 4.4 with the warning text containing the mcptt-warn-code set to "149", shall follow the procedures in clause 6.3.3.1.18; and

r) shall not continue with the rest of the clause.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 10.1.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [34], the timer TNG1 (acknowledged call setup timer) is not running, the controlling MCPTT function supports media buffering and the SIP final response is not yet sent to the inviting MCPTT client:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4,4 in the SIP 200 (OK) response, if the prearranged MCPTT group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [31];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) shall include a P-Answer-State header field with the value "Unconfirmed";

5) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCPTT emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) if the received SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

7) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 9: Resulting user plane processing is completed before the next step is performed.

8) shall send the SIP 200 (OK) response towards the inviting MCPTT client according to 3GPP TS 24.229 [4];

9) shall generate a notification to the MCPTT clients, which have subscribed to the conference event package that the inviting MCPTT User has joined in the MCPTT group session, as specified in clause 6.3.3.4; and

NOTE 10: As a group document can potentially have a large content, the controlling MCPTT function can notify using content-indirection as defined in IETF RFC 4483 [32].

10) shall send a SIP NOTIFY request to each MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP 183 (Session Progress) response for a SIP INVITE request as specified in clause 10.1.1.4.1.2 containing an indication of required group members, the timer TNG1 (acknowledged call setup timer) is running and all SIP 200 (OK) responses have been received to all SIP INVITE requests sent to MCPTT clients specified in clause 10.1.1.4.1.1, then the controlling MCPTT function shall wait until the SIP 200 (OK) response has been received to the SIP INVITE request specified in clause 10.1.1.4.1.2 before generating a SIP 200 (OK) response to the "SIP INVITE request for controlling MCPTT function of an MCPTT group".

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 10.1.1.4.1 that was sent to an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [31]; and

1) if the MCPTT ID in the SIP 200 (OK) response matches to the MCPTT ID in the corresponding SIP INVITE request;

2) there are no outstanding SIP 200 (OK) responses to SIP INVITE requests which were sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [31]; and

3) there is no outstanding SIP 200 (OK) response to a SIP INVITE request sent in clause 10.1.1.4.1.2 where the SIP 183 (Session Progress) response contained an indication of required group members;

the controlling MCPTT function:

1) shall stop timer TNG1 (acknowledged call setup timer) as described in clause 6.3.3.3;

2) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

3) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCPTT group has been exceeded the <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [31];

4) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

NOTE 11: Resulting media plane processing is completed before the next step is performed.

6) shall send a SIP 200 (OK) response to the inviting MCPTT client according to 3GPP TS 24.229 [4];

7) shall generate a notification to the MCPTT clients, which have subscribed to the conference event package that the inviting MCPTT user has joined in the MCPTT group session, as specified in clause 6.3.3.4; and

NOTE 12: As a group document can potentially have a large content, the controlling MCPTT function can notify using content-indirection as defined in IETF RFC 4483 [32].

8) shall send the SIP NOTIFY request to the MCPTT clients according to 3GPP TS 24.229 [4].

Upon:

1) *receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 10.1.1.4.1;*

2) *the timer TNG1 (acknowledged call setup timer) is not running;*

3) *the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document;*

4) *the controlling MCPTT function supports media buffering; and*

5) *the SIP final response has not yet been sent to the inviting MCPTT client;*

the controlling MCPTT function according to local policy:

1) *shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.2;*

2) *shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCPTT group has exceeded the <max-participant-count> members as specified in 3GPP TS 24.481 [31];*

3) *shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;*

4) *if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCPTT emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;*

5) *if the received SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;*

6) *shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;*

NOTE 13: Resulting media plane processing is completed before the next step is performed.

7) *shall send a SIP 200 (OK) response to the inviting MCPTT client according to 3GPP TS 24.229 [4];*

8) *shall generate a notification to the MCPTT clients, which have subscribed to the conference event package that the inviting MCPTT user has joined in the MCPTT group session, as specified in clause 6.3.3.4; and*

NOTE 14: As a group document can potentially have a large content, the controlling MCPTT function can notify using content-indirection as defined in IETF RFC 4483 [32].

9) *shall send the SIP NOTIFY request to the MCPTT clients according to 3GPP TS 24.229 [4].*

Upon expiry of timer TNG1 (acknowledged call setup timer), if there are outstanding SIP 200 (OK) responses to SIP INVITE requests sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [31], the controlling MCPTT function shall follow the procedures specified in clause 6.3.3.3.

If timer TNG1 (acknowledged call setup timer) is running and a final SIP 4xx, 5xx or 6xx response is received from an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [31], the controlling MCPTT function shall follow the relevant procedures specified in clause 6.3.3.3.

If:

1) *timer TNG1 (acknowledged call setup timer) is not running;*

2) *the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document; and*

3) *a final SIP 4xx, 5xx or 6xx response is received from an invited MCPTT client;*

then the controlling MCPTT function shall perform one of the following based on policy:

1) *send the SIP final response towards the inviting MCPTT client, according to 3GPP TS 24.229 [4], if a SIP final response was received from all the other invited MCPTT clients and the SIP 200 (OK) response is not yet sent; or*

2) *remove the invited MCPTT client from the MCPTT Session as specified in clause 6.3.3.1.5, if a SIP final response other than 2xx or 3xx was received from all the invited MCPTT clients and the SIP 200 (OK) response is already sent. The controlling MCPTT function may invite an additional member of the prearranged MCPTT group as specified in clause 10.1.1.4.1 that has not already been invited, if the prearranged MCPTT group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [31], and all members have not yet been invited.*

If the group identity in the "SIP INVITE request for controlling MCPTT function of an MCPTT group" is a TGI and constituent MCPTT groups were invited as specified in clause 10.1.1.4.1.2 and,

1) *if all non-controlling MCPTT functions hosting the constituent MCPTT groups have responded with a SIP 2xx, SIP 3xx, SIP 4xx, SIP 5xx or SIP 6xx responses to the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group"; and*

2) *if all expected SIP INFO requests containing a floor request are received;*

then the controlling MCPTT function shall indicate to the media plane that all final responses are received.

NOTE 15: If the SIP 200 (OK) response to the SIP INVITE request for non-controlling MCPTT function of an MCPTT group included the application/vnd.3gpp.mcptt-info+xml MIME body with the <floor-state> element set to "floor-taken", the controlling MCPTT function expects that the non-controlling MCPTT functions sends a SIP INFO request containing a floor request.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCPTT client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.2]

"The controlling MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4].

The controlling MCPTT function:

1) *shall include in the Contact header field an MCPTT session identity for the MCPTT session with the g.3gpp.mcptt media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" according to IETF RFC 3840 [16];*

2) *shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];*

3) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [9] in the SIP INVITE request;*

4) *shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];*

5) *shall include a Referred-By header field with the public user identity of the inviting MCPTT client;*

6) *should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7]. The refresher parameter shall be omitted;*

7) *shall include the Supported header field set to "timer";*

8) if the received SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body containing an <ambient-listening-type> element and:

a) if the Priv-Answer-Mode header field specified in IETF RFC 5373 [18] was included in the received SIP INVITE request with a value of "Auto" or if no Priv-Answer-Mode header field was received in the received SIP INVITE request; or

b) a Priv-Answer-Mode header field was received containing a value other than "Auto";

shall include a Priv-Answer-Mode header field set to a value of "Auto" in the outgoing SIP INVITE request;

9) if the received SIP INVITE request contained a Priv-Answer-Mode header field and did not contain an application/vnd.3gpp.mcptt-info+xml MIME body containing an <ambient-listening-type> element, shall include an unmodified Priv-Answer-Mode header field;

10) if the received SIP INVITE request did not contain an application/vnd.3gpp.mcptt-info+xml MIME body containing an <ambient-listening-type> element and did not contain a Priv-Answer-Mode header field, shall include an unmodified Answer-Mode header field if present in the incoming SIP INVITE request; and

11) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing INVITE request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the controlling MCPTT function:

1) when initiating a new MCPTT session the SDP offer;

a) shall contain only one SDP media-level section for MCPTT speech media stream as contained in the received SDP offer; and

b) shall contain one SDP media-level section for media- plane control messages, if present in the received SDP offer; and

2) when adding a new MCPTT user to an existing MCPTT Session, the SDP offer shall contain the media stream currently used in the MCPTT session.

When composing the SDP offer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the controlling MCPTT function;

2) for the MCPTT speech media stream, shall include all media-level attributes from the received SDP offer;

3) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the controlling MCPTT function; and

4) for the offered media plane control channel, shall include the offered media planecontrol channel 'fmtp' attributes as specified in 3GPP TS 24.380 [5] clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.2]

"NOTE 1: In this clause, "MCPTT server" can refer to either the controlling MCPTT function of an MCPTT group or the non-controlling MCPTT function of an MCPTT group.

Upon receipt of a SIP INVITE request:

1) if the MCPTT server is not yet subscribed to the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request, the MCPTT server shall subscribe to the "xcap-diff" event-package for the group document of this group identity as specified in 3GPP TS 24.481 [31];

NOTE 2: The group identity in the <mcptt-request-uri> element is either an MCPTT group ID or a temporary MCPTT group identity (TGI).

…

4) upon receipt of a notification from the group management server containing the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request, or if the group document is already cached:

a) if the MCPTT server is a non-controlling function of an MCPTT group, then the MCPTT server shall exit this clause; and

b) if the MCPTT server is a controlling function of an MCPTT group, then the MCPTT server shall determine if the group document is for a TGI or an MCPTT group ID as follows:

i) if the group document includes an <on-network-temporary> element, then the group document is associated with a TGI;

ii) if the group document does not include an <on-network-temporary> element or an <on-network-regrouped> element, then the group document is associated with an MCPTT ID that has not been regrouped; and

iii) if the group document does not include an <on-network-temporary> element but includes an <on-network-regrouped> element, then the group document is associated with an MCPTT ID that has been regrouped;

5) if the SIP INVITE request is a "SIP INVITE request for controlling function of an MCPTT group" and the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP-INVITE request is an MCPTT group ID that has not been re-grouped, the MCPTT server shall access the following elements from the group document as specified in 3GPP TS 24.481 [31]:

a) if the <on-network-disabled> element is present in the group document, shall send a SIP 403 (Forbidden) response with the warning text set to "115 group is disabled" in a Warning header field as specified in clause 4.4 and shall not continue with the rest of the steps;

b) if the <list> element of the <list-service> element does not contain an entry matching the MCPTT ID of the user in the SIP INVITE request, shall send a SIP 403 (Forbidden) response with the warning text set to "116 user is not part of the MCPTT group" in a Warning header field as specified in clause 4.4 and shall not continue with the rest of the steps;

c) if the <on-network-invite-members> element is set to "true" and if the SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element containing a value not set to "prearranged", shall return a SIP 404 (Not Found) response with the warning text set to "117 the group identity indicated in the request is a prearranged group" as specified in clause 4.4 "Warning header field" and shall not continue with the rest of the steps; and

d) if the <on-network-invite-members> element is set to "false" and if the SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element containing a value not set to "chat" shall return a SIP 404 (Not Found) response with the warning text set to "118 the group identity indicated in the request is a chat group" as specified in clause 4.4 "Warning header field" and shall not continue with the rest of the steps;

6) if the SIP INVITE request is a "SIP INVITE request for controlling function of an MCPTT group" and the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request is associated with an MCPTT group ID that has been regrouped, the MCPTT server:

a) shall obtain the TGI associated with the regrouped group, by accessing the "temporary-MCPTT-group-ID" attribute of the <regrouped> element of the group document associated with the MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request;

b) if not hosting the TGI, shall:

i) stop processing the SIP INVITE request; and

ii) return a SIP 302 (Moved Temporarily) response with:

A) a Contact header field set to the PSI of the MCPTT server hosting the TGI; and

B) an application/vnd.3gpp.mcptt-info MIME body with a <mcptt-request-uri> element set to the TGI; and

c) if hosting the TGI, and the call to the temporary group is in progress, shall:

i) associate the MCPTT ID of the calling user with the temporary group call;

ii) interact with the media plane as specified in 3GPP TS 24.380 [5]; and

iii) exit this clause;

d) if hosting the TGI and the call to the temporary group is not in progress, shall subscribe to the "xcap-diff" event-package for the group document of the TGI as specified in 3GPP TS 24.481 [31], if not already subscribed;

…

g) upon receipt of a notification containing the group document for the TGI, or if the group document is already cached, shall obtain the MCPTT IDs of the constituent groups by accessing the <constituent-MCPTT-group-ID> element(s) of the group document for the TGI; and

h) if:

i) the <associated-group-id> element with an MCPTT ID is included in the application/vnd.3gpp.mcptt-info MIME body;

ii) the MCPTT ID is present in one of the instances of the <constituent-MCPTT-group-ID> element in the group document; and

iii) the group is not homed by the MCPTT server;

shall exit this procedure and authorize the MCPTT user at a non-controlling MCPTT function of a MCPTT group;

NOTE 4: The non-controlling function of an MCPTT group can be located in the primary MCPTT system or a partner MCPTT system.

7) for the MCPTT ID of each constituent group, shall follow the actions below in step 8); and

8) if the SIP INVITE request is a "SIP INVITE request for controlling function of an MCPTT group" and the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request retrieved is associated with a TGI and if:

a) the application/vnd.3gpp.mcptt-info+xml includes an <associated-group-id> element, determine if the constituent MCPTT group identified by the <associated-group-id> element is homed at the MCPTT server and if that is the case:

i) shall subscribe to the "xcap-diff" event-package for the group document for the constituent MCPTT group ID as specified in 3GPP TS 24.481 [31], if not already subscribed;

…

b) the application/vnd.3gpp.mcptt-info+xml does not include an <associated-group-id> element, for each MCPTT ID contained in each instance of the <constituent-MCPTT-group-ID> element of the group document for the TGI:

i) shall determine if the group identity is homed on the controlling MCPTT function of an MCPTT group or homed on a non-controlling MCPTT function of an MCPTT group;

NOTE 5: The non-controlling function of an MCPTT group can be located in the primary MCPTT system or a partner MCPTT system.

ii) for each constituent MCPTT group ID that is homed on the controlling MCPTT function of an MCPTT group shall subscribe to the "xcap-diff" event-package for the group document for the constituent MCPTT group ID as specified in 3GPP TS 24.481 [31], if not already subscribed;

NOTE 6: As soon as an error occurs when subscribing for a group document of a constituent MCPTT group ID, the controlling MCPTT function of an MCPTT group stops subscribing to any further group documents of constituent MCPTT group IDs.

…

c) when all group document(s) for all constituent groups homed at the MCPTT server have been retrieved and if the MCPTT ID of the user identified in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body:

i) is a member of one the retrieved constituent MCPTT groups, received, shall exit this procedure; and

ii) is not a member of any of the retrieved constituent group documents, shall determine that the MCPTT ID of the MCPTT user needs to be authorised by a non-controlling MCPTT function of an MCPTT group and exit this procedure."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.5]

"The MCPTT server shall only invite affiliated group members to a group session.

If the group is not a regroup based on a preconfigured group, the MCPTT server determines the affiliated members from the entries contained in the <list> element of the group document by following the procedures specified in clause 6.3.6.

If the group is a regroup based on a preconfigured group, the MCPTT server determines the affiliated members from the list of users that was stored during successful processing of the creation of the regroup per clause 16 by following the procedures specified in clause 6.3.6.

NOTE 1: The term "affiliated group members" used above also includes those members that are implicitly affiliated by the controlling MCPTT function.

If the number of members of the MCPTT group exceeds the value contained in the <on-network-max-participant-count> element the MCPTT server shall invite only <on-network-max-participant-count> members from the list, but shall prioritise inviting those group members to the group session that have an <entry> element in the <list> element with a <on-network-required> element present.

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the group session. The <on-network-required> element is used to determine which group members need to acknowledge the group call before audio transmission can proceed.

NOTE 3: Other requirements for how the controlling MCPTT function selects which of the <on-network-max-participant-count> members to invite is outside the scope of this specification.

NOTE 4: It is assumed that validation checks are performed at the group management server to ensure that the <on-network-max-participant-count> cannot be less than the number of <on-network-required> users."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.3.2]

"When sending a SIP 200 (OK) response to the initial SIP INVITE request, the controlling MCPTT function:

1) shall generate the SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [4];

2) shall include the Session-Expires header field and start supervising the SIP session according to rules and procedures of IETF RFC 4028 [7], "UAS Behavior". The "refresher" parameter in the Session-Expires header field shall be set to "uac";

3) shall include the option tag "timer" in a Require header field;

4) shall include a P-Asserted-Identity header field with the public service identity of the controlling MCPTT function;

5) shall include a SIP URI for the MCPTT session identity in the Contact header field identifying the MCPTT session at the controlling MCPTT function;

6) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt"; and

c) the isfocus media feature tag;

7) shall include Warning header field(s) received in incoming responses to the SIP INVITE request;

8) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23];

9) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [22];

10) shall include the "explicitsub" and "nosub" option tags in a Supported header field according to IETF RFC 7614 [35]; and

11) void

12) shall interact with the media plane as specified in 3GPP TS 24.380 [5]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) for the accepted media stream in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCPTT function; and

2) for the accepted media channel control channel, if present in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCPTT function, for the accepted media plane control channel, if present in the received SDP offer; and

b) shall include 'fmtp' attributes as specified in 3GPP TS 24.380 clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.3]

"Upon receiving a SIP BYE request the controlling MCPTT function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCPTT function:

1) shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing the media plane resource associated with the SIP session towards the MCPTT client;

…

2) shall generate a SIP 200 (OK) response and send the SIP response towards the MCPTT client according to 3GPP TS 24.229 [4];

3) shall check the MCPTT session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCPTT session needs to be released for each participant of the MCPTT session;

4) if release of the MCPTT session is required:

a) shall perform the procedures as specified in the clause 6.3.3.1.5 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and

...

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT participant."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.8.1]

"If:

1) *the call is a pre-arranged group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 24.380 [5] expired;*

2) *there are only one or no participants in the MCPTT session;*

3) *the call is a pre-arranged group call and if it is according to local policy, the initiator of the group call leaves the MCPTT session;*

4) *the minimum number of affiliated MCPTT group members is not present;*

5) *timer TNG3 (group call timer) expires; or*

6) *the call is a broadcast group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 24.380 [5] expired;*

the controlling MCPTT function shall release the MCPTT session for the group call."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.5]

"When a participant needs to be removed from the MCPTT session, the controlling MCPTT function:

1) *shall interact with the media plane as specified in 3GPP TS 24.380 [5] for the MCPTT session release;*

2) *shall generate a SIP BYE request according to 3GPP TS 24.229 [4]; and*

3) *shall send the SIP BYE request to the MCPTT clients according to 3GPP TS 24.229 [4].*

If timer TNG3 (group call timer) has not expired, then when the last MCPTT client is removed from the MCPTT session, the controlling MCPTT function shall stop timer TNG3 (group call timer).

When the MCPTT group session needs to be released, the controlling MCPTT function shall send SIP BYE requests as described in this clause, to all participants of the group session.

Upon receiving a SIP 200 (OK) response to a SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT clients."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 4.8]

"In certain deployments, for example, in the case that the MCPTT operator uses the underlying SIP core infrastructure from the carrier operator, the MCPTT operator can prevent certain sensitive application data from being visible in the clear to the SIP layer. The following data are classed as sensitive application data:

- *MCPTT ID;*
- *MCPTT group ID;*
- *user location information;*
- *emergency, alert and imminent-peril indicators;*
- *access token (containing the MCPTT ID);*
- *MCPTT client ID; and*
- *functional alias.*

The above data is transported as XML content in SIP messages. in XML elements or XML attributes.

Data is transported in attributes in the following circumstances in the procedures in the present document:

…

- *an MCPTT ID in application/resource-lists+xml document included in an SIP INVITE request setting up a group call to a temporary group involving a non-controlling function that works in "Trusted Mode" according to IETF RFC 5366 [20], whereby the participants are returned to the controlling function in a MIME body of a SIP 403 (Forbidden) with the P-Refused-URI-List header field according to IETF RFC 5318 [36];*

3GPP TS 33.180 [78] describes a method to provide confidentiality protection of sensitive application data in elements by using XML encryption (i.e. xmlenc) and in attributes by using an attribute confidentiality protection scheme described in clause 6.6.2.3 of the present document. Integrity protection can also be provided by using XML signatures (i.e. xmlsig).

Protection of the data relies on a shared XML protection key (XPK) used to encrypt and sign data:

- *between the MCPTT client and the MCPTT server, the XPK is a client-server key (CSK); and*

- *between MCPTT servers and between MCPTT domains, the XPK is a signalling protection key (SPK).*

The CSK (XPK) and a key-id CSK-ID (XPK-ID) are generated from keying material provided by the key management server. Identity based public key encryption based on MIKEY-SAKKE is used to transport the CSK between SIP end-points. The encrypted CSK is transported from the MCPTT client to the MCPTT server when the MCPTT client performs service authorisation as described in clause 7 and is also used during service authorisation to protect the access token.

The SPK (XPK) and a key-id SPK-ID (XPK-ID) are directly provisioned in the MCPTT servers.

Configuration in the MCPTT client and MCPTT server is used to determine whether one or both of confidentiality protection and integrity protection are required.

The content can be re-encrypted and signed again using the SPK between MCPTT servers."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 5.3.2]

"…

Figure 5.3.2-5 shows the roles of the MCPTT server in a mutual aid relationship between a primary MCPTT system and a partner MCPTT with the use of a non-controlling MCPTT function of an MCPTT group within the partner MCPTT system. This can occur due to group re-grouping where the MCPTT server(s) of one or more of the constituent groups are homed on the partner system. If the primary MCPTT system and partner MCPTT system operate in a trusted mutual aid relationship, then the non-controlling MCPTT function of an MCPTT group can provide the identities of the users of the group to the controlling MCPTT function. If the primary MCPTT system and partner MCPTT system operate in an untrusted mutual aid relationship, then the non-controlling MCPTT function of an MCPTT group invites the users of the group on behalf of the controlling MCPTT function.

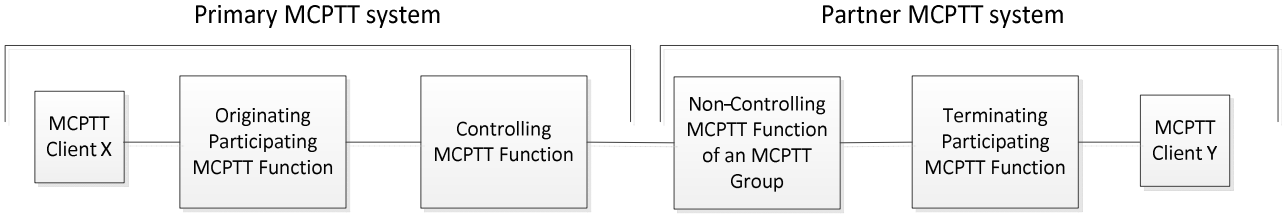


Figure 5.3.2-5: Mutual aid relationship between the primary MCPTT system and a partner MCPTT system involving the use of a non-controlling MCPTT function of an MCPTT group in the partner MCPTT system"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.2.2]

"When an MCPTT call is established a new instance of the floor control server state machine for 'general floor control operation' is created.

For each MCPTT client added to the MCPTT call, a new instance of the floor control server state machine for 'basic floor control operation towards the floor participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the floor control server could queue the implicit floor control request for the media-floor control entity.

The initial SIP INVITE request or SIP REFER request to establish an MCPTT chat group call or to rejoin an ongoing MCPTT call is not handled as an implicit floor control request message by the floor control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCPTT client due to implicit floor control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCPTT function (see 3GPP TS 24.379 [2]) the floor control server optionally can give an early indication to send RTP media packets, to the inviting MCPTT client.

If an early indication to send RTP media packets is given to the inviting MCPTT client, the floor participant is granted the permission to send media and the MCPTT server buffers RTP media packets received from the MCPTT client at least until the first invited MCPTT client accepts the invitation or until the RTP media packet buffer maximum limit is exceeded.

If the MCPTT server does not support or does not allow media buffering then an early indication to send RTP media packets is not given to the inviting MCPTT client, the floor participant is granted the permission to send media when the first invited MCPTT client accepts the media.

Before the floor control server sends the first floor control message in the MCPTT call, the floor control server has to assign itself a SSRC identifier to be included in media floor control messages and quality feedback messages if the MCPTT server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The floor participant and the floor control server can negotiate the maximum priority level that the floor participant is permitted to request. The floor control server can pre-empt the current sender based on the negotiated maximum priority level that the floor participant is permitted to request and the priority level included in the Floor Request message.

NOTE: The maximum priority level that a floor participant can use is negotiated as specified in subclause 14.3.3 and is based on group configuration data retrieved by the controlling MCPTT function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCPTT function from the configuration management server as described in 3GPP TS 24.484 [13].

For groups configured for audio cut-in floor control, pre-empting of the current sender is not according to priority. Each new floor request results in the floor being revoked from the current talker and being granted to the new requesting talker.

The floor participant and the floor control server can negotiate queueing of floor requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the floor control server queues the floor control request if a Floor Request message is received when another floor participant has the floor and the priority of the current speaker is the same or higher. Queueing is not permitted for groups configured for audio cut-in floor control."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.3]

"When an MCPTT client leaves an MCPTT call and the MCPTT call remains ongoing with the other MCPTT clients, the floor control server follows a two-step procedure.

Step 1 The MCPTT server stops sending floor control messages and RTP media packets to the MCPTT client leaving the MCPTT call and, the MCPTT server discards floor control messages and RTP media packets received from the MCPTT client leaving the MCPTT call.

Step 2 When the application and signalling plane has determined that the MCPTT call with this floor participant has been released, the corresponding instance of the floor control server state machine for 'basic floor control operation towards the floor participant' is released.

When an MCPTT call is released, the floor control server follows a two-step procedure.

Step 1 The MCPTT server stops sending floor control messages and RTP media packets to all floor participants in the MCPTT call.

Step 2 When the application and signalling plane has determined that the MCPTT call has been released, the corresponding instance of the floor control server state machine for 'general floor control operation' are also terminated, along with any 'basic floor control operation towards the floor participant' state machines for the floor participants of this call.

The floor control server state machine for general floor control operation initiates the MCPTT call release depending on the release policy specified in 3GPP TS 24.379 [2]."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 14.3.4]

"If the mc_granted" fmtp attribute is included in an SDP offer, the controlling MCPTT function:

1. if the MCPTT call is not a temporary group session, may use the SIP 200 (OK) response to indicate that the implicit floor request is granted; and

2. if the MCPTT call is a temporary group session, shall not indicate that the implicit floor request is granted.

NOTE: A MCPTT group call is a temporary group session when the <on-network-temporary> element is present in the <list-service> element as specified in 3GPP TS 24.481 [12].

If the controlling MCPTT function grants the implicit request and decide to use the SIPP 200 (OK) response to grant the implicit request, the controlling MCPTT function shall include the "mc_granted" fmtp attribute in the SDP answer.

If the controlling MCPTT function decides not to grant the implicit request, the controlling MCPTT function shall not include the "mc_granted" fmtp attribute in the SDP answer."

5.2.1.7.3 Test description

5.2.1.7.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the originating participating MCPTT function.
  - SS (MCPTT Server) provides the terminating participating MCPTT function.
  - SS (MCPTT Server) provides the non-controlling MCPTT function.

  SS-UE1 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.7.3.1-1.

  SS-UE2 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.7.3.1-1.

  SS-UE3 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.7.3.1-1.

  SS-UE4 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.7.3.1-1.

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the controlling MCPTT function.

- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2 and A.4.6.
- The SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client), the SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are registered and authorized to use the MCPTT system.
- Group A and Group B have been regrouped in Group T.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.
- The SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are affiliated with Group B and are authorized to initiate group calls.
- Group B is controlled by the non-controlling MCPTT function.
- There are no ongoing calls in any group.
- There is not a trusted mutual aid relationship between the partner MCPTT system and the primary MCPTT system

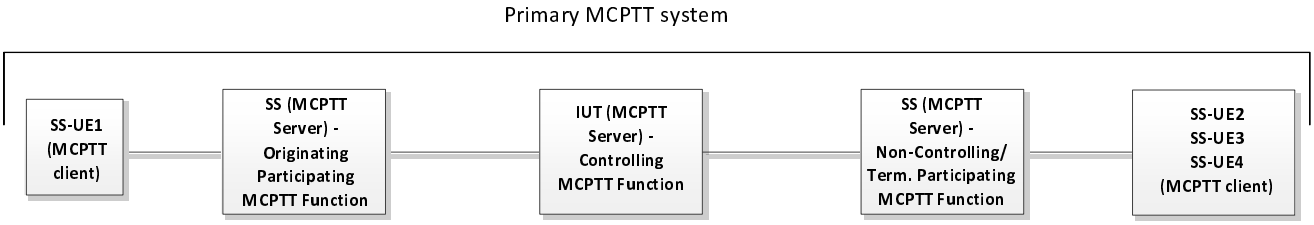


Figure 5.2.1.7.3.1-1: Functions of the testing components

5.2.1.7.3.2 Test procedure sequence

Table 5.2.1.7.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) (member of constituent Group A) to initiate an on-demand pre-arranged group call to a temporary group (Group T) with implicit floor control	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
3	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the non-controlling MCPTT function to initiate an on-demand pre-arranged group call in constituent Group B?	-->	SIP INVITE	1	P
4	The SS (MCPTT Server) providing the non-controlling MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
5	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	SIP ACK	1	P
6	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the non-controlling MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
7	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function to invite SS-UE2 to the call?	-->	SIP INVITE	1	P
8	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 183 (Session Progress) message sent reliably.	<--	SIP 183 (Session Progress)		-
9	Check: Does the IUT (MCPTT Server) send a SIP PRACK message to the SS (MCPTT Server) providing the terminating participating MCPTT function in response to the SIP 183 (Session Progress)?	-->	SIP PRACK	1	P
10	The SS (MCPTT Server) responds to the SIP PRACK with a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
11	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
12	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	1	P
13	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating participating MCPTT function informing SS-UE2 that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
14	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	1	P
15	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
16	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
17	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P
18	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
19	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	Floor Idle	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
20	The SS (MCPTT Server) providing the non-controlling MCPTT function sends a Floor Request message from SS-UE3 including a Track-Info field with the participant reference	<--	Floor Request		-
21	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the non-controlling MCPTT function including the Track-Info field received in the Floor Request?	-->	Floor Granted	2	P
22	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the participating MCPTT function informing SS-UE1 that the floor is taken by SS-UE3 (MCPTT client)?	-->	Floor Taken	2	P
23	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the participating MCPTT function informing SS-UE2 that the floor is taken by SS-UE3 (MCPTT client)?	-->	Floor Taken	2	P
24	The SS (MCPTT Server) providing the non-controlling MCPTT function sends a Floor Release message from SS-UE3 with an acknowledgement required to release the floor and including a Track-Info field with the participant reference	<--	Floor Release		-
25	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P
26	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
27	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	Floor Idle	2	P
28	The SS (MCPTT Server) providing the participating MCPTT function sends a SIP BYE request from SS-UE1	<--	SIP BYE		-
29	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) response to SS (MCPTT Server) providing the participating MCPTT function?	-->	SIP 200 (OK)	3	P
	- EXCEPTION: Depending on local policy, the controlling MCPTT Server can decide to end the call if the initiator leaves the call. Steps 27A and 28A ("upper case A letter") below represent the call finalization process when this local policy applies. "Lower case letter" notation represent messages that can be sent in parallel, since the MCPTT server needs to send SIP BYE messages to the rest of participants in the call.				
30Aa	Check: Does the IUT (MCPTT Server) send a SIP BYE message for SS-UE2 to the SS (MCPTT Server) providing the participating MCPTT function to end the call?	-->	SIP BYE	3	P
31Aa	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
30Ab	Check: Does the IUT (MCPTT Server) send a SIP BYE message to the SS (MCPTT Server) providing the non-controlling MCPTT function to end the call?	-->	SIP BYE	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
31Ab	The SS (MCPTT Server) providing the non-controlling MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
	- EXCEPTION: If according to local policy, the call continues even if the initiator leaves, the SS will need to close the call until a single participant is left. Steps 30B - 33B ("upper case B letter") below represent the call finalization process when this local policy applies.				
30B	The SS (MCPTT Server) providing the non-controlling MCPTT function sends a SIP BYE request	<--	SIP BYE		-
31B	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) response to SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	SIP 200 (OK)	3	P
32B	Check: Does the IUT (MCPTT Server) send a SIP BYE message for SS-UE2 to the SS (MCPTT Server) providing the participating MCPTT function to end the call?	-->	SIP BYE	3	P
33B	The SS (MCPTT Server) providing the participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-

5.2.1.7.3.3 Specific message contents

Table 5.2.1.7.3.3-1: SIP INVITE from the SS (MCPTT-Server) (Step 1, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
feature-param	"isfocus" not present			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.7.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.7.3.3-1B			

Table 5.2.1.7.3.3-1A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.7.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions INITIAL_SDP_OFFER, IMPLICIT_GRANT_REQUESTED, IMPLICIT_FLOOR_GRANTED

Table 5.2.1.7.3.3-1B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.7.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_T_ID			
associated-group-id	Encrypted (see note) <associated-group-id> with mcpttURI set to px_MCPTT_Group_A_ID			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.1.7.3.3-2: SIP INVITE from the IUT (MCPTT-Server) (Step 3, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server which is simulated by the SS, providing non-controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server which is simulated by the SS, providing non-controlling function.		
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			
Session-Expires				
generic-param	any allowed value			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Max-Forwards				
value	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.7.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.7.3.3-2B			

5.2.1.7.3.3-2A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.7.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the controlling MCPTT Server			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the controlling MCPTT Server			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the controlling MCPTT Server			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the controlling MCPTT Server			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"10"			
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.7.3.3-2B: MCPTT-Info in SIP INVITE from IUT (MCPTT Server) (Table 5.2.1.7.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_B_ID	The URI of the group		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.7.3.3-3: SIP 200 (OK) from the SS (MCPTT Server) (Step 4, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	3600			
refresher	"uas"			
Message-body				
SDP message	SDP message as described in Table 5.2.1.7.3.3-3A			

Table 5.2.1.7.3.3-3A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 5.2.1.7.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"10"		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.7.3.3-4: SIP INVITE from the IUT (MCPTT-Server) (Step 7, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing terminating participating function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing terminating participating function.		
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			
Session-Expires				
generic-param	any allowed value			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Max-Forwards				
value	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.7.3.3-4A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.7.3.3-4B			

5.2.1.7.3.3-4A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.7.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the controlling MCPTT Server			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the controlling MCPTT Server			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the controlling MCPTT Server			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the controlling MCPTT Server			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (**ETSI TS 136 579-1 [1]** (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.7.3.3-4B: MCPTT-Info in SIP INVITE from IUT (MCPTT Server) (Table 5.2.1.7.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.7.3.3-5: SIP 183 (Session Progress) from the SS (MCPTT-Server) (Step 8, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.1-1, conditions MCPTT, 100rel

Table 5.2.1.7.3.3-6: SIP PRACK from the IUT (MCPTT-Server) (Step 9, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.10.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Request-URI	same URI as the SS has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server), providing terminating participating function.		
Max-Forwards				
value	any allowed value			

Table 5.2.1.7.3.3-7: SIP 200 (OK) to PRACK from the SS (MCPTT-Server) (Step 10, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Content-Type	not present			
Content-Length	not present			

Table 5.2.1.7.3.3-8: SIP 200 (OK) from the SS (MCPTT Server) (Step 11, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	3600			
refresher	"uas"			
Message-body				
SDP message	SDP message as described in Table 5.2.1.7.3.3-8A			

Table 5.2.1.7.3.3-8A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 5.2.1.7.3.3-8)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.7.3.3-9: SIP 200 (OK) from the IUT (MCPTT Server) (Step 14, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	any allowed value			
refresher	"uac"			
Message-body				
SDP message	SDP message as described in Table 5.2.1.7.3.3-9A			

Table 5.2.1.7.3.3-9A: SDP in SIP 200 (OK) from the IUT (MCPTT Server) (Table 5.2.1.7.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_FLOOR_GRANTED, IMPLICIT_GRANT_REQUESTED				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the MCPTT Server			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the MCPTT Server			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the MCPTT Server (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the MCPTT Server			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the MCPTT Server being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			

Table 5.2.1.7.3.3-10: SIP BYE from the SS (MCPTT-Server) (Steps 28 and 30B, 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
From				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Local URI of the dialog (from the SS' point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Local tag of the dialog (from the SS' point of view)		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the SS' point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the SS' point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	70	The recommended initial value is 70 in IETF RFC 3261 [17].		

Table 5.2.1.7.3.3-11: SIP BYE from the IUT (MCPTT-Server) (Steps 30Aa, 30Ab and 32B, 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

Table 5.2.1.7.3.3-12: Floor Taken (Steps 6 and 13, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.1.7.3.3-13: Floor Taken (Steps 22 and 23, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_C			

Table 5.2.1.7.3.3-14: Floor Release (Step 16, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.1.7.3.3-15: Floor Release (Step 24, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	
Track Info			
Queueing Capability	"1"	The client supports queueing	
Participant Type Value	px_MCX_User_C_ParticipantType	ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1	
Floor Participant Reference	0x00000001	The selected value is randomly chosen - 32 bit value	

Table 5.2.1.7.3.3-16: Floor Ack (Step 13, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 5.2.1.7.3.3-17: Floor Ack (Step 25, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	
Track Info			
Queueing Capability	"1"	The client supports queueing	
Participant Type Value	px_MCX_User_C_ParticipantType	ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1	
Floor Participant Reference	0x00000001	The selected value is randomly chosen - 32 bit value	

Table 5.2.1.7.3.3-18: Floor Idle (Steps 18, 19, 26 and 27, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.1.7.3.3-19: Floor Request (Step 20, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Track Info			
Queueing Capability	"1"	The client supports queueing	
Participant Type Value	px_MCX_User_C_ParticipantType	ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1	
Floor Participant Reference	0x00000001	The selected value is randomly chosen - 32 bit value	

Table 5.2.1.7.3.3-20: Floor Granted (Step 21, Table 5.2.1.7.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Track Info	The same value as received in Floor Request		
Queueing Capability	"1"		
Participant Type Value	px_MCX_User_C_ParticipantType		
Floor Participant Reference	0x00000001		

5.2.1.8 MCPTT Server - MCPTT Server / Pre-arranged Group Call / Temporary Group / Non-Controlling server / Untrusted mode

5.2.1.8.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS (MCPTT Server) initiates an on-demand pre-arranged group call in regrouped group }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the call by sending SIP
messages }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call in regrouped
group }
ensure that {
  when { the SS-UE1 (MCPTT client), SS-UE3 (MCPTT client) and SS-UE4 (MCPTT client) engage in
communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted, Floor Request, Floor Release ) by sending messages via the SS (MCPTT Server) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call in regrouped
group }
ensure that {
  when { the SS-UE3 (MCPTT client) ends the pre-arranged group call }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) by sending a SIP 200 (OK) message
to the client ending the call and sends a SIP BYE message to the other participant }
            }
```

5.2.1.8.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 10.1.1.4.2, 10.1.1.4.1.2, 6.3.3.1.2, 6.3.3.1.1, 10.1.1.5.2.1, 10.1.1.5.2.2, 10.1.1.5.5, 6.3.5.2, 6.3.5.4 and 6.3.5.5, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 6.5.2.2, 6.5.4.2, 6.5.4.3, 6.5.4.5, 6.5.4.6, 6.5.4.8, 6.5.4.9, 6.5.4.14, 6.5.4.15, 6.5.3 and 14.3.4. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.2]

"In the procedures in this clause:

1) MCPTT ID in an incoming SIP INVITE request refers to the MCPTT ID of the originating user from the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

...

Upon receipt of a "SIP INVITE request for controlling MCPTT function of an MCPTT group", the controlling MCPTT function:

...

14) if the MCPTT group call is not ongoing then:

...

f) if the group identity in the "SIP INVITE request for controlling MCPTT function of an MCPTT group" is a TGI:

i) shall for each of the constituent MCPTT groups homed on the primary MCPTT system:

A) if the controlling MCPTT function does not own the MCPTT group identified by the MCPTT group ID, then generate a SIP INVITE request towards the MCPTT server that owns the MCPTT group identity by following the procedures in clause 10.1.1.4.1.2; and

NOTE 4: The MCPTT server that the SIP INVITE request is sent to acts as a non-controlling MCPTT function;

...

ii) shall for each of the constituent MCPTT groups homed on the partner MCPTT system generate a SIP INVITE request for the MCPTT group identity homed on the partner MCPTT system as specified in clause 10.1.1.4.1.2;"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.4.1.2]

"The controlling MCPTT function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the non-controlling MCPTT function serving the group identity of the MCPTT group owned by the partner MCPTT system;

3) shall set the P-Asserted-Identity to the public service identity of the controlling MCPTT function;

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcptt-request-uri> element set to the group identity of the MCPTT group hosted by the non-controlling MCPTT function in the partner MCPTT system; and

b) the <mcptt-calling-group-id> element set to the group identity of the group served by the controlling MCPTT function;

5) shall include the Recv-Info header field set to g.3gpp.mcptt-floor-request;

6) void

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

8) shall send the SIP INVITE request towards the partner MCPTT system in accordance with 3GPP TS 24.229 [4].

Upon receiving SIP 403 (Forbidden) response for the SIP INVITE request, if according to local policy and if:

1) the response contains a Warning header field with the MCPTT warning code "128"; and

2) the response contains a P-Refused-URI-List header field and an application/resource-lists+xml MIME body as specified in IETF RFC 5318 [36];

NOTE 1: The application/resource-lists+xml MIME body contains MCPTT IDs identifying MCPTT users in a partner MCPTT system that needs to be invited to the prearranged group call in case of group regrouping using interrogating method as specified in 3GPP TS 23.379 [3] clause 10.6.2.4.2.

then the controlling MCPTT function:

1) shall check if the number of members of the MCPTT group exceeds the value contained in the <on-network-max-participant-count> element of the group document as specified in 3GPP TS 24.481 [31]. If exceeded, the controlling MCPTT function shall invite only <on-network-max-participant-count> members from the application/resource-lists+xml MIME body; and

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the prearranged group session It is operator policy that determines which participants in the application/resource-lists+xml MIME body are invited to the group call.

2) shall invite MCPTT users as specified in this clause using the list of MCPTT IDs in URI-List.

..."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.2]

"This clause is referenced from other procedures.

The controlling MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4].

The controlling MCPTT function:

1) shall include in the Contact header field an MCPTT session identity for the MCPTT session with the g.3gpp.mcptt media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" according to IETF RFC 3840 [16];

2) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [9] in the SIP INVITE request;

4) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];

5) shall include a Referred-By header field with the public user identity of the inviting MCPTT client;

6) should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7]. The refresher parameter shall be omitted;

7) shall include the Supported header field set to "timer";

...

11) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing INVITE request."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the controlling MCPTT function:

1) when initiating a new MCPTT session the SDP offer;

a) shall contain only one SDP media-level section for MCPTT speech media stream as contained in the received SDP offer; and

b) shall contain one SDP media-level section for media- plane control messages, if present in the received SDP offer; and

...

When composing the SDP offer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the controlling MCPTT function;

2) for the MCPTT speech media stream, shall include all media-level attributes from the received SDP offer;

3) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the controlling MCPTT function; and

4) for the offered media plane control channel, shall include the offered media planecontrol channel 'fmtp' attributes as specified in 3GPP TS 24.380 [5] clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.5.2.1]

"When receiving the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" the MCPTT server can be acting as a controller MCPTT function in an ongoing prearranged group call or, if an prearranged group call is not ongoing, be initiated as an non-controlling MCPTT function and invite MCPTT users.

If a prearranged group call is not ongoing the MCPTT server shall perform the actions specified in clause 10.1.1.5.2.2.

...

When operating in the non-controlling mode and a SIP BYE request is received from the controlling MCPTT function, the non-controlling MCPTT function shall change from operating in the non-controlling mode to operating in the controlling mode as specified in clause 10.1.1.5.2.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.5.2.2]

"Upon receipt of a "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" and if a prearranged group call is not ongoing, the non-controlling MCPTT function of an MCPTT group:

NOTE 1: The Contact header field of the SIP INVITE request contains the isfocus media feature tag.

..

5) if a trusted mutual aid relationship exists between the partner MCPTT system and the primary MCPTT system and the procedure in 3GPP TS 23.379 [3] clause 10.6.2.4.2 is supported:

a) shall generate a SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [4];

b) shall retrieve the group members of the prearranged group identified by the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request, as specified in clause 6.3.5.2;

c) if the retrieval of group members was successful shall include a P-Refused-URI-List header field populated with affiliated members of the prearranged group in accordance with the IETF RFC 5318 [36]:

..

e) shall send the SIP 403 (Forbidden) response towards the controlling MCPTT function as specified in 3GPP TS 24.229 [4]; and

f) shall not process the remaining steps;

6) shall retrieve the group document from the group management server for the MCPTT group ID contained in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2 and continue with the rest of the steps if the checks in clause 6.3.5.2 succeed;

7) shall cache the content of the SIP INVITE request, if received in the Contact header field and if the specific feature tags are supported;

8) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [4];

9) determine the members to invite to the prearranged MCPTT group call as specified in clause 6.3.5.5;

10) if the group document retrieved from the group management server contains <on-network-required> group members as specified in 3GPP TS 24.481 [31], shall send a SIP 183 (Session Progress) response to the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" as specified in clause 6.3.4.2.2.1 and shall populate the response with an application/vnd.3gpp.mcptt-info+xml MIME body containing the <required> element set to "true".

11) if the group document retrieved from the group management server does not contain any <on-network-required> group members as specified in 3GPP TS 24.481 [31], may, according to local policy, send a SIP 183 (Session Progress) response to the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" as specified in clause 6.3.4.2.2.1;

12) shall invite each group member determined in step 9) above, to the group session, as specified in clause 10.1.1.5.1; and

13) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3;

Unless a SIP response has been sent to the controlling MCPTT function as specified in step 10 or 11 above, the non-controlling MCPTT function of an MCPTT group shall wait for the first SIP provisional response or first SIP 200 (OK) response from one of the invited MCPTT clients, before sending a response to the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group".

…

If the group document contains <on-network-required> group member(s) as specified in 3GPP TS 24.481 [31], then the non-controlling MCPTT function of an MCPTT group shall wait until all SIP 200 (OK) responses to SIP INVITE requests have been received from the <on-network-required> MCPTT clients before sending a SIP 200 (OK) response back to the controlling MCPTT function, as specified above.

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.1.5.5]

"Upon receiving a "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" when a prearranged group session is not ongoing, the non-controlling MCPTT-function shall:

NOTE 1: The difference between a "SIP INVITE request for controlling MCPTT function of an MCPTT group" and a "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" is that the latter SIP INVITE request contains the isfocus media feature tag in the Contact header field.

…

4) shall retrieve the group document from the group management server for the MCPTT group ID contained in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2 and continue with the rest of the steps if the checks in clause 6.3.5.2 succeed;

NOTE 2: If the checks are not successful, the SIP response to the "SIP INVITE request for controlling MCPTT function of an MCPTT group" is already sent in the clause 6.3.5.2.

5) shall cache the content of the SIP INVITE request;

6) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [4];

7) shall authorize the MCPTT user in the <mcptt-calling-user-id> element in the application/vnd.3gpp.mcptt-info+xml MIME body of the "SIP INVITE request for controlling MCPTT function of an MCPTT group" as specified in clause 6.3.5.4, if the MCPTT user is unauthorized to initiated a pre-arranged group session the non-controlling MCPTT function shall send a SIP 403 (Forbidden) response with the warning text set to "119 user is not authorised to initiate the group call" in a Warning header field as specified in clause 4.4.

8) shall generate a SIP INVITE request to the controlling MCPTT function as specified in clause 6.3.4.1.4; and

9) shall send the SIP INVITE request to the controlling MCPTT function as specified in 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response to the SIP INVITE request sent to the controlling MCPTT function as specified above, the non-controlling MCPTT function:

1) shall send the SIP ACK request to the controlling MCPTT function as specified in 3GPP TS 24.229 [4];

2) shall generate a SIP 200 (OK) to the "SIP INVITE request for controlling MCPTT function of an MCPTT group" as specified in 3GPP TS 24.229 populated as follows:

a) shall include an SDP answer as specified in clause 6.3.4.2.1 based on the SDP answer in the SIP 200 (OK) response;

b) shall include the public service identifier of the non-controlling MCPTT function in the P-Asserted-Identity header field;

c) shall include the warning text set to "148 group is regrouped" in a Warning header field as specified in clause 4.4; and

d) shall send the SIP 200 (OK) request according to 3GPP TS 24.229;

NOTE 3: As long as the MCPTT group is regrouped the floor control messages in the media plane includes a grouped regrouped indication as specified in 3GPP TS 24.380 [5].

3) shall start acting as a non-controlling MCPTT function and interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.5;

4) shall determine the members to invite to the prearranged MCPTT group call as specified in clause 6.3.5.5; and

5) shall invite each group member determined in step 4) immediately above, to the group session, as specified in clause 10.1.1.5.1.

Upon receipt of other final SIP responses with the exception of the SIP 2xx response to the INVITE request sent to the controlling MCPTT function as specified above, the non-controlling MCPTT function:

1) shall send the SIP ACK response to the controlling MCPTT function as specified in 3GPP TS 24.229 [4]; and

2) shall start acting as a controlling MCPTT function as specified in clause 10.1.1.4 and invite members as specified in clause 6.3.4.1.2.

NOTE 4: Regardless if the controlling MCPTT function accepts or rejects the SIP INVITE request sent above the prearranged group session continues to be initiated with only the members of the group homed on the non-controlling MCPTT function of the group being invited to the group call.

The non-controlling MCPTT function shall handle SIP responses (other than the SIP 2xx response) to the SIP INVITE requests sent to invited members as specified in 3GPP TS 24.229.

Upon receipt of a SIP 2xx response to SIP INVITE requests sent to invited members, the non-controlling MCPTT function:

1) shall send the SIP ACK request as specified in 3GPP TS 24.229 [4]; and

2) shall interact with the media plane as specified in 3GPP TS 24.380 [5]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.2]

"NOTE 1: In this clause, "MCPTT server" can refer to either the controlling MCPTT function of an MCPTT group or the non-controlling MCPTT function of an MCPTT group.

Upon receipt of a SIP INVITE request:

…

4) upon receipt of a notification from the group management server containing the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request, or if the group document is already cached:

a) if the MCPTT server is a non-controlling function of an MCPTT group, then the MCPTT server shall exit this clause; and

…

NOTE 4: The non-controlling function of an MCPTT group can be located in the primary MCPTT system or a partner MCPTT system.

…

NOTE 5: The non-controlling function of an MCPTT group can be located in the primary MCPTT system or a partner MCPTT system.

ii) for each constituent MCPTT group ID that is homed on the controlling MCPTT function of an MCPTT group shall subscribe to the "xcap-diff" event-package for the group document for the constituent MCPTT group ID as specified in 3GPP TS 24.481 [31], if not already subscribed;

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.4]

"The following conditions shall be met for a controlling MCPTT function or non-controlling MCPTT function of an MCPTT group to initiate a group session for the requesting MCPTT user:

1) if the <on-network-regrouped> element in the <list-service> element is present in the group document and if the MCPTT group ID indicated in the incoming INVITE request is the same as the MCPTT group ID in the "temporary-MCPTT-group-ID" attribute of the <on-network-regrouped> element; or

2) if the <on-network-regrouped> element in the <list-service> element of the MCPTT group document is not present in the group document;

and:

1) an <entry> element exists in the <list> element of the group document for the MCPTT user;

2) if the group is a user regroup based on a preconfigured group and the MCPTT ID contained in the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body in the SIP INVITE request is included in the list of users that was stored during successful processing of the creation of the user regroup per clause 16;

3) a <rule> exists in the group document with:

a) the <is-list-member> element of the <conditions> element present and with the <allow-initiate-conference> element of the corresponding <actions> element set to "true"; or

b) the <identity> element of the <conditions> element containing an entry matching the MCPTT ID in the SIP INVITE request, with the <allow-initiate-conference> element of the <actions> element is set to "true"; and

4) if the <supported-services> element is present, it contains:

a) a <service> element containing an "enabler" attribute which is set to the MCPTT ICSI; and

b) if a <group-media> element is present, an entry set to "MCPTT speech".

then the MCPTT user shall be authorised to initiate the group session."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.5]

"The MCPTT server shall only invite affiliated group members to a group session.

If the group is not a regroup based on a preconfigured group, the MCPTT server determines the affiliated members from the entries contained in the <list> element of the group document by following the procedures specified in clause 6.3.6.

If the group is a regroup based on a preconfigured group, the MCPTT server determines the affiliated members from the list of users that was stored during successful processing of the creation of the regroup per clause 16 by following the procedures specified in clause 6.3.6.

NOTE 1: The term "affiliated group members" used above also includes those members that are implicitly affiliated by the controlling MCPTT function.

If the number of members of the MCPTT group exceeds the value contained in the <on-network-max-participant-count> element the MCPTT server shall invite only <on-network-max-participant-count> members from the list, but shall prioritise inviting those group members to the group session that have an <entry> element in the <list> element with a <on-network-required> element present.

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the group session. The <on-network-required> element is used to determine which group members need to acknowledge the group call before audio transmission can proceed.

NOTE 3: Other requirements for how the controlling MCPTT function selects which of the <on-network-max-participant-count> members to invite is outside the scope of this specification.

NOTE 4: It is assumed that validation checks are performed at the group management server to ensure that the <on-network-max-participant-count> cannot be less than the number of <on-network-required> users."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.2.2]

"When receiving an indication from the application and signalling plane that a group session is initiated, the floor control server interface:

1. *shall initiate and store a message sequence number value with the value to be used in the Message Sequence Number field in the Floor Idle and Floor Taken messages;*
2. *shall for each MCPTT client in the MCPTT group controlled by the non-controlling MCPTT function that are participating in the session:*
    a. *generate a random temporary identifier between '0' and '4294967295';*
    b. *store an association between the generated temporary identifier and the floor participant interface;*
    c. *store information about capabilities negotiated in the "mc_queueing" and "mc_priority" fmtp attributes as described in clause 14;*
    d. *store information whether the MCPTT client requested privacy or not; and*
    e. *initiate an instance of the 'floor participant state transition' state machine as specified in subclause 6.5.5; and*
3. *shall perform the actions in the subclause 6.5.4.*

When receiving an indication from the application and signalling plane that an MCPTT client has accepted an invitation to the session, the floor participant interface shall perform the actions in subclause 6.5.5."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.2]

"Upon receiving a Floor Request message from one floor participant interface, the floor control server interface:

1. *shall forward the Floor Request message to the controlling MCPTT function. The Floor Request message:*
    a. *shall include all fields included by the floor participant;*

b. if a Track Info field is included, shall include the temporary identifier at the end of the <Floor Participant Reference> value item; and

c. if a Track Info field is not included, shall include a Track Info field populated as follows:

i. shall include the "mc_queueing" fmtp attribute value negotiated as specified in clause 14 in the <Queueing Capability> value;

ii. shall include a <Participant Type> value based on the <participant-type> element specified in 3GPP TS 24.481 [12], if value in the <participant-type> element is available, otherwise set the <Participant Type> value to "unknown"; and

iii. shall include the temporary identifier as the first <Floor Participant Reference> value; and

2. if the value of the <Queueing Capability> in the Track Info is '1' (the floor participant in the MCPTT client supports queueing), shall store the outgoing Floor Request message in the passive floor request queue."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.3]

"Upon receiving a Floor Release message from one floor participant interface, the floor control server interface:

NOTE: A Floor Release message can be received from the permitted floor participant and from any participant that is queued in the floor control server.

1. shall forward a Floor Release message to the controlling MCPTT function. The Floor Release message:

a. shall include all fields included by the floor participant in the Floor Release message;

b. if a Track Info field is included, shall include the temporary identifier at the end of the <Floor Participant Reference> value item; and

c. if a Track Info field is not included, shall include a Track Info field as follows:

i. shall include the "mc_queueing" fmtp attribute value negotiated as specified in clause 14 in the <Queueing Capability> value; and

ii. shall include the temporary identifier as the first <Floor Participant Reference> value; and

2. if a Floor Request message received from this floor participant is in the passive floor request queue, shall remove the floor request from the passive floor request queue."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.5]

"Upon receiving a Floor Ack message from one floor participant interface the floor control server interface:

1. shall send the Floor Ack message towards the controlling MCPTT function. The Floor Ack message:

a. shall include all fields included by the floor participant in the Floor Ack message;

b. if Track Info field is included, shall include the temporary identifier at the end of the <Floor Participant Reference> value item; and

c. if a Track Info field is not included, shall include a Track Info field with temporary identifier as the first <Floor Participant Reference>."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.6]

"Upon receiving a Floor Granted message sent from the controlling MCPTT function, the floor control server interface:

1. shall send the Floor Granted to the floor participant interface identified by the <Participant Reference> value at the end of the Track Info field. The Floor Granted message:

a. shall include the fields as received with the following exceptions:

i. if the Track Info field only contains one <Participant Reference> value, shall remove the Track Info field from the outgoing Floor Granted message; and

ii. if the Track Info field contains more than one <Participant Reference> value, shall remove the last <Participant Reference> value from the Track Info field from the outgoing Floor Granted message; and

b if the Floor Indicator field is included in the Floor Granted message and the G-bit is set to '1' (Dual floor), shall store the SSRC of granted floor participant and associate the stored value with dual floor;

2. if:

a. a SSRC of granted floor participant and associate the stored value with dual floor is not stored, shall send a Floor Taken message populated as specified in step d. below to all participant interfaces with the exception of the floor participant interface to which the Floor Granted message is sent;

b. a SSRC of granted floor participant associate with dual floor is stored and if the Floor Indicator field is not included in the Floor Granted message or if the Floor Indicator field is included in the Floor Granted message and the G-bit is set to '0' (Not dual floor), shall send a Floor Taken message populated as specified in step d. below to all participant interfaces with the exception of:

i. the floor participant interface to which the Floor Granted message is sent; and

ii. the floor participants only listening to the overriding floor participant;

c. the Floor Indicator field is included in the Floor Granted message and the G-bit is set to '1' (Dual floor):

i. shall send a Floor Taken message populated as specified in step d. below to floor participants that will listen to the RTP media from the overriding MCPTT client according to local policy;

NOTE 1: The MCPTT client overridden by the overriding MCPTT client is still sending voice (overridden). The list of floor participants that receive the overriding, overridden, or both transmissions is based on configuration.

d. The Floor Taken message:

i. shall include the granted MCPTT user's MCPTT ID in the Granted Party's Identity field and may include the functional alias of the granted MCPTT user in the Functional Alias field, if privacy is not requested by the granted floor participant when the MCPTT client was invited to the session;

NOTE 2: The privacy request was stored for each invited MCPTT client when the MCPTT client accepted the invitation as specified in subclause 6.5.2.

ii. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1;

iii. shall include the Permission to Request Floor field to '0', if the group call is a broadcast group call;

iv. may include the Permission to Request the Floor field set to '1', if the group call is not a broadcast group call; and

v. shall set the first bit in the subtype of the Floor Taken message to '0' (acknowledgement is not required); and

NOTE 3: A Floor Taken message sent to all participants does not require acknowledgement.

e. if the Floor Indicator field was included in the Floor Granted message, shall include the received Floor Indicator field; and

3. if the Floor Request message received from the floor participant is in the passive floor request queue, shall remove the floor request from the passive floor request queue; and

4 if the Floor Indicator field is included in the Floor Granted message and the G-bit is set to '1' (Dual floor), shall send a Floor Idle message to those floor participants that will only listen to RTP media from the overriding MCPTT client. The Floor Idle message:

a. shall include the Floor Indicator field as received in the Floor Granted message with the G-bit set to '0' (Not dual floor); and

b. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.8]

"Upon receiving a Floor Idle message sent from the controlling MCPTT function, the floor control server interface:

NOTE 1: The Floor Idle message can be either destined to floor participants in all MCPTT clients or is sent to the floor participant in a specific MCPTT client. In the latter case the Floor Idle message contains the Track Info field.

1. if the Floor Idle message contains a Track Info field;

a. shall use the <Participant Reference> value at the end of the Track Info field to identify the floor participant interface;

b. if:

i. the Track Info field only contains one <Participant Reference> value:

A. shall remove the Track Info field from the outgoing Floor Idle message;

B. shall increase the stored message sequence number value with 1; and

C. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and

ii. if the Track Info field contains more than one <Participant Reference> value, shall remove the last <Participant Reference> value from the Track Info field; and

c. shall send the Floor Idle message to the floor participant interface;

2. if the Floor Idle message does not contain a Track Info field;

a. shall set the first bit in the subtype of the Floor Idle message to '0' (acknowledgement is not required);

NOTE 2: A Floor Idle message sent to all participants does not require acknowledgement.

b if:

i. the Floor Indicator field is not included in the Floor Idle message or if the Floor Indicator field is included and the G-bit is set to '0' (Not dual floor), shall send the Floor Idle message to all floor participant interfaces. The Floor Idle message:

A. shall include received fields; and

B. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and

ii. the Floor Indicator field is included in the Floor Idle message and the G-bit is set to '1' (Dual floor),

A. shall send the Floor Idle message to floor participants listening to the overriding MCPTT client according to local policy. The Floor Idle message:

- *shall include received fields; and*
- *shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and*

B. shall discard the stored SSRC of granted floor participant associated with a dual floor;

c. shall send a Floor Ack message towards the controlling MCPTT function if the first bit in the subtype of the received Floor Idle message is set to '1' (acknowledgement is required) as specified in subclause 8.2.2. The Floor Ack message:

i. shall include the Source field set to '3' (the non-controlling MCPTT function is the source); and

ii. shall include the Message Type field set to '5' (Floor Idle); and

3. shall empty the passive floor request queue."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.9]

"Upon receiving a Floor Taken message sent from the controlling MCPTT function, the floor control server interface:

NOTE 1: The Floor Taken message can be either destined to floor participants in all MCPTT clients or is sent to the floor participant in a specific MCPTT client. In the latter case the Floor Taken message contains the Track Info field.

1. if the Floor Taken message contains a Track Info field;

a. shall use the <Participant Reference> value at the end of the Track Info field to identify the floor participant interface;

b. if the Track Info field only contains one <Participant Reference> value:

A. shall remove the Track Info field from the outgoing Floor Taken message; and

B. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1;

c. if the Track Info field contains more than one <Participant Reference> value, shall remove the last <Participant Reference> value from the Track Info field; and

d. shall send the Floor Taken message to the floor participant interface;

2. if the Floor Taken message does not contain a Track Info field:

a. shall set the first bit in the subtype of the Floor Taken message to '0' (acknowledgement is not required);

NOTE 2: A Floor Taken message sent to all participants does not require acknowledgement.

b. if the Floor Indicator field is included in the Floor Taken message and the G-bit is set to '1' (Dual floor), shall store the SSRC of granted floor participant and associate the stored SSRC with dual floor;

c. if there is no stored SSRC of granted floor participant associated with a dual floor, shall send the Floor Taken message to all floor participant interfaces. The Floor Taken message:

i. shall include all received fields with the exception of the Message Sequence Number field;

ii. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1;

d. if there is a stored SSRC of granted floor participant associated with a dual floor:

i. if the Floor Indicator field is not included in the Floor Taken message or if the Floor Indicator field is included in the Floor Taken message and the G-bit is set to '0' (Not dual floor), shall send the Floor Taken message to all floor participant interfaces with the exception of the floor participants only listening to RTP media from the overriding floor participant. The Floor Idle message:

A. shall include all received fields; and

B. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and

ii. if the Floor Indicator field is included in the Floor Taken message and the G-bit is set to '1' (Dual floor):

A. shall send the Floor Taken message to all floor participants listening to the RTP media from the overriding MCPTT client according to local policy. The Floor Taken message:

- shall include all received fields; and

- *shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and*

B. *shall send a Floor Idle message to those floor participants that will only listen to RTP media from the overriding MCPTT client according to local policy. The Floor Idle message:*

- *shall include all received fields;*

- *shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and*

- *shall update the received Floor Indicator field and set the G-bit to '0' and;*

e. shall send a Floor Ack message towards the controlling MCPTT function if the first bit in the subtype of the received Floor Taken message is set to '1' (acknowledgement is required) as specified in subclause 8.2.2. The Floor Ack message:

i. shall include the Source field set to '3' (the non-controlling MCPTT function is the source); and

ii. shall include the Message Type field set to '2' (Floor Taken);"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.14]

"Upon receiving an MCPTT call release step 1 request from the application and signalling plane e.g. when the session is going to be released, the floor control interface:

1. shall ignore floor control message from the floor control server;

2. shall request the media distributor to stop distributing RTP media packets to the network media interface of the MCPTT clients; and

3. shall ignore any floor control messages received from the floor participant interfaces."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.15]

"Upon receiving an MCPTT call release step 2 request from the application and signalling plane, the floor control server interface:

1. shall release all resources associated with this session."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.3]

"When an MCPTT client leaves an MCPTT call and the MCPTT call remains ongoing with the other MCPTT clients, the non-controlling MCPTT function of an MCPTT group follows a two-step procedure:

Step 1 The floor participant interface stops sending floor control messages and RTP media packets to the MCPTT client leaving the MCPTT call and the floor participant interface discards floor control messages and RTP media packets received from the MCPTT client leaving the MCPTT call; and

Step 2 When the application and signalling plane has determined that the session with this floor participant has been released, the corresponding instance of the 'floor participant state transition' state machine is released.

When an MCPTT call is released, the floor control server interface follows a two-step procedure:

Step 1 The floor control server interface stops sending floor control messages and RTP media packets to MCPTT clients in the MCPTT call; and

Step 2 When the application and signalling plane has determined that the MCPTT call has been released, resources in the floor control server interface are released, along with all 'floor participant state transition' state machines.

The non-controlling MCPTT function of an MCPTT group can initiate an MCPTT call release depending on the release policy specified in 3GPP TS 24.379 [2]."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 14.3.4]

"If the mc_granted" fmtp attribute is included in an SDP offer, the controlling MCPTT function:

1. *if the MCPTT call is not a temporary group session, may use the SIP 200 (OK) response to indicate that the implicit floor request is granted; and*

2. *if the MCPTT call is a temporary group session, shall not indicate that the implicit floor request is granted.*

NOTE: A MCPTT group call is a temporary group session when the <on-network-temporary> element is present in the <list-service> element as specified in 3GPP TS 24.481 [12].

If the controlling MCPTT function grants the implicit request and decide to use the SIPP 200 (OK) response to grant the implicit request, the controlling MCPTT function shall include the "mc_granted" fmtp attribute in the SDP answer.

If the controlling MCPTT function decides not to grant the implicit request, the controlling MCPTT function shall not include the "mc_granted" fmtp attribute in the SDP answer."

5.2.1.8.3 Test description

5.2.1.8.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the participating MCPTT functions.
  - SS (MCPTT Server) provides the controlling MCPTT function.

  SS-UE1 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.8.3.1-1.

  SS-UE2 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.8.3.1-1.

  SS-UE3 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.8.3.1-1.

  SS-UE4 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.1.8.3.1-1.

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the non-controlling MCPTT function
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figures A.4.2 and A.4.5.
- The SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client), the SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are registered and authorized to use the MCPTT system.

- Group A and Group B have been regrouped in Group T.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.
- The SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are affiliated with Group B and are authorized to initiate group calls.
- Group B is controlled by the non-controlling MCPTT function.
- There are no ongoing calls in Group A and Group B.
- There is not a trusted mutual aid relationship between the partner MCPTT system and the primary MCPTT system

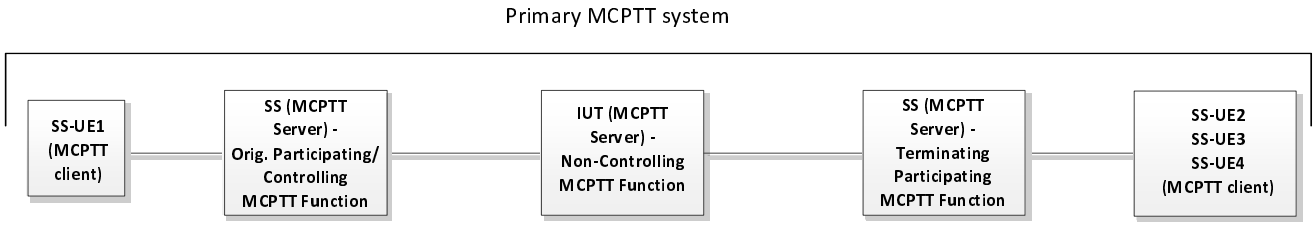


Figure 5.2.1.8.3.1-1: Functions of the testing components

5.2.1.8.3.2 Test procedure sequence

Table 5.2.1.8.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) providing the controlling MCPTT function sends a SIP INVITE request to initiate an on-demand pre-arranged group call in regrouped Group B.	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P
3	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function to invite SS-UE3 to the call?	-->	SIP INVITE	1	P
4	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function to invite SS-UE4 to the call?	-->	SIP INVITE	1	P
5	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 183 (Session Progress) message sent reliably.	<--	SIP 183 (Session Progress)	-	-
6	Check: Does the IUT (MCPTT Server) send a SIP PRACK message to the SS (MCPTT Server) providing the terminating participating MCPTT function in response to the SIP 183 (Session Progress)?	-->	SIP PRACK	1	P
7	The SS (MCPTT Server) responds to the SIP PRACK with a SIP 200 (OK) message.	<--	SIP 200 (OK)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
8	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message from SS-UE3	<--	SIP 200 (OK)	-	-
9	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	1	P
10	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the controlling MCPTT function?	-->	SIP 200 (OK)	1	P
11	The SS (MCPTT Server) providing the controlling MCPTT function responds with a SIP ACK message	<--	SIP ACK	-	-
12	The SS (MCPTT Server) providing the controlling MCPTT function sends a Floor Taken to the IUT (MCPTT Server) providing the non-controlling function informing that the floor is taken by SS-UE1.	<--	Floor Taken	-	-
13	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating MCPTT function informing SS-UE3 that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
14	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 183 (Session Progress) message sent reliably.	<--	SIP 183 (Session Progress)		-
15	Check: Does the IUT (MCPTT Server) send a SIP PRACK message to the SS (MCPTT Server) providing the terminating participating MCPTT function in response to the SIP 183 (Session Progress)?	-->	SIP PRACK	1	P
16	The SS (MCPTT Server) responds to the SIP PRACK with a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
17	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message from SS-UE4	<--	SIP 200 (OK)		-
18	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	1	P
19	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating MCPTT function informing SS-UE4 that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
20	The SS (MCPTT Server) providing the controlling MCPTT function sends a Floor Idle message to the IUT (MCPTT Server)	<--	Floor Idle		-
21	Check: Does the IUT (MCPTT Server) send a Floor Idle message for SS-UE3 to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	2	P
22	Check: Does the IUT (MCPTT Server) send a Floor Idle message for SS-UE4 to the SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	Floor Idle	2	P
23	The SS (MCPTT Server) providing the participating MCPTT function sends a Floor Request message from SS-UE3	<--	Floor Request		-
24	Check: Does the IUT (MCPTT Server) send a Floor Request message from SS-UE3 including a Track-Info field with the participant reference?	-->	Floor Request	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
25	The SS (MCPTT Server) providing the controlling MCPTT function sends a Floor Granted message for SS-UE3 including the Track-Info field sent in the Floor Request.	<--	Floor Granted		-
26	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the participating MCPTT function for SS-UE3 without the Track-Info field?	-->	Floor Granted	2	P
27	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating MCPTT function informing SS-UE4 that the floor is taken by SS-UE3 (MCPTT client)?	-->	Floor Taken	2	P
28	The SS (MCPTT Server) providing the participating MCPTT function sends a Floor Release with an acknowledgement required message from SS-UE3 to release the floor.	<--	Floor Release		-
29	Check: Does the IUT (MCPTT Server) send a Floor Release message from SS-UE3 with an acknowledgement required and including a Track-Info field with the participant reference?	-->	Floor Release	2	P
30	The SS (MCPTT Server) providing the controlling MCPTT function sends a Floor Ack for the Floor Release send by the IUT (MCPTT Server).	<--	Floor Ack		-
31	Check: Does the IUT (MCPTT Server) send a Floor Ack message to the SS (MCPTT Server) providing the participating MCPTT function for the Floor Release?	-->	Floor Ack	2	P
32	The SS (MCPTT Server) providing the controlling MCPTT function sends a Floor Idle message to the IUT (MCPTT Server)-	<--	Floor Idle		-
33	Check: Does the IUT (MCPTT Server) send a Floor Idle message for SS-UE3 to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	2	P
34	Check: Does the IUT (MCPTT Server) send a Floor Idle message for SS-UE4 to the SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	Floor Idle	2	P
35	The SS (MCPTT Server) providing the participating MCPTT function sends a SIP BYE request from SS-UE3 to the IUT (MCPTT Server)	<--	SIP BYE		-
36	Check: Does the IUT (MCPTT Server) providing the non-controlling MCPTT function respond with a SIP 200 (OK) message?	-->	SIP 200 (OK)	3	P
37	The SS (MCPTT Server) providing the controlling MCPTT function sends a SIP BYE request to the IUT (MCPTT Server)	<--	SIP BYE		-
38	Check: Does the IUT (MCPTT Server) providing the non-controlling MCPTT function respond with a SIP 200 (OK) message?	-->	SIP 200 (OK)	3	P
39	Check: Does the IUT (MCPTT Server) providing the non-controlling MCPTT function send a SIP BYE request for SS-UE4 to the SS (MCPTT Server) providing the participating MCPTT function?	-->	SIP BYE	3	P
40	The SS (MCPTT Server) providing the participating MCPTT function sends a SIP 200 (OK) response.	<--	SIP 200 (OK)		-

5.2.1.8.3.3 Specific message contents

Table 5.2.1.8.3.3-1: SIP INVITE from the SS (MCPTT-Server) (Step 1, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server which is simulated by the SS, providing non-controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server which is simulated by the SS, providing non-controlling function.		
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			
Session-Expires				
generic-param	any allowed value			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Max-Forwards				
value	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.8.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.8.3.3-1B			

5.2.1.8.3.3-1A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.1.8.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"10"			
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.8.3.3-1B: MCPTT-Info in SIP INVITE from SS (MCPTT Server) (Table 5.2.1.8.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_B_ID	The URI of the group		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.8.3.3-2: SIP 200 (OK) from the IUT (MCPTT Server) (Step 10, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	any allowed value			
refresher	"uac"			
Message-body				
SDP message	SDP message as described in Table 5.2.1.8.3.3-2A			

Table 5.2.1.8.3.3-2A: SDP in SIP 200 (OK) from the IUT (MCPTT Server) (Table 5.2.1.8.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the MCPTT Server			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the MCPTT Server			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the MCPTT Server (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the MCPTT Server			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the MCPTT Server being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			

Table 5.2.1.8.3.3-3: SIP INVITE from the IUT (MCPTT-Server) (Step 3, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing terminating participating function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server providing the non-controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing terminating participating function.		
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			
Session-Expires				
generic-param	any allowed value			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Max-Forwards				
value	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.8.3.3-3A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.8.3.3-3B			

5.2.1.8.3.3-3A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.1.8.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the non-controlling MCPTT Server			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the non-controlling MCPTT Server			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the non-controlling MCPTT Server			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the non-controlling MCPTT Server			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"10"		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.1.8.3.3-3B: MCPTT-Info in SIP INVITE from IUT (MCPTT Server) (Table 5.2.1.8.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_C	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.8.3.3-4: SIP INVITE from the IUT (MCPTT-Server) (Step 4, Table 5.2.1.8.3.2-1)

Derivation Path: Table 5.2.1.8.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.1.8.3.3-3A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.1.8.3.3-4A			

Table 5.2.1.8.3.3-4A: MCPTT-Info in SIP INVITE from IUT (MCPTT Server) (Table 5.2.1.8.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_D	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
mcptt-calling-group-id	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.1.8.3.3-5: SIP 183 (Session Progress) from the SS (MCPTT-Server) (Steps 5 and 14, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.1-1, conditions MCPTT, 100rel

Table 5.2.1.8.3.3-6: SIP PRACK from the IUT (MCPTT-Server) (Steps 6 and 15, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.10.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Request-URI	same URI as the SS has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server), providing terminating participating function.		
Max-Forwards				
value	any allowed value			

Table 5.2.1.8.3.3-7: SIP 200 (OK) to PRACK from the SS (MCPTT-Server) (Steps 7 and 16, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Content-Type	not present			
Content-Length	not present			

Table 5.2.1.8.3.3-8: SIP 200 (OK) from the SS (MCPTT Server) (Steps 8 and 17, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	3600			
refresher	"uas"			
Message-body				
SDP message	SDP message as described in Table 5.2.1.8.3.3-8A			

Table 5.2.1.8.3.3-8A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 5.2.1.8.3.3-8)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER

Table 5.2.1.8.3.3-9: SIP BYE from the SS (MCPTT-Server) (Steps 33 and 37, 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
From				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Local URI of the dialog (from the SS' point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Local tag of the dialog (from the SS' point of view)		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the SS' point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the SS' point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	70	The recommended initial value is 70 in IETF RFC 3261 [17].		

Table 5.2.1.8.3.3-10: SIP BYE from the IUT (MCPTT-Server) (Step 39, 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server) providing the terminating participating function.		
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

Table 5.2.1.13.3.3-11: Floor Taken (Steps 12, 13 and 19, Table 5.2.1.13.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.1.13.3.3-12: Floor Taken (Step 27, Table 5.2.1.13.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_C			

Table 5.2.1.13.3.3-13: Floor Release (Step 28, Table 5.2.1.13.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.1.13.3.3-14: Floor Release (Step 29, Table 5.2.1.13.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	
Track Info			
Queueing Capability	"1"	The client supports queueing	
Participant Type Value	px_MCX_User_C_ParticipantType	ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1	
Floor Participant Reference	Any allowed value	The selected value is randomly chosen - 32 bit value	

Table 5.2.1.13.3.3-15: Floor Ack (Step 30, Table 5.2.1.13.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	
Track Info	The same value as received in Floor Release		

Table 5.2.1.13.3.3-16: Floor Ack (Step 31, Table 5.2.1.13.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 5.2.1.13.3.3-15: Floor Idle (Steps 20-22 and 32-34, Table 5.2.1.13.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.1.8.3.3-16: Floor Request (Step 23, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1

Table 5.2.1.8.3.3-17: Floor Granted (Step 26, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

Table 5.2.1.8.3.3-18: Floor Request (Step 24, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Track Info			
Queueing Capability	"1"	The client supports queueing	
Participant Type Value	px_MCX_User_C_ParticipantType	ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1	
Floor Participant Reference	Any allowed value	The selected value is randomly chosen - 32 bit value	

Table 5.2.1.8.3.3-19: Floor Granted (Step 25, Table 5.2.1.8.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Track Info	The same value as received in Floor Request		

5.2.2 Chat Group Call

5.2.2.1 MCPTT Server - MCPTT Server / Chat Group Call / Automatic Commencement Mode / Controlling server

5.2.2.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE1 (MCPTT client) initiates a chat group call with automatic commencement mode }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the call by sending SIP
messages }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT chat group call with automatic commencement
mode }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Idle, Floor Granted) by
sending messages via the SS (MCPTT Server) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT chat group call with automatic commencement
mode }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the chat group call with automatic commencement mode }
    then { IUT (MCPTT Server) responds to SS-UE1 (MCPTT client) by sending a SIP 200 (OK) message
and sends a SIP BYE message to SS (MCPTT Server) }
            }
```

5.2.2.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 10.1.2.4.1.1,6.3.2.1.1.1, 6.3.3.1.2, 10.1.2.4.1.2, 10.1.2.4.3.2, 6.3.3.1.5, 6.3.3.2.4, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 6.3.4.3.2, 6.3.4.3.3, 6.3.4.4.2, 6.3.4.4.6 Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.4.1.1]

"In the procedures in this clause:

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

...

10) shall create a chat group session and allocate an MCPTT session identity for the chat group session if the MCPTT chat group session identity does not already exist, and may handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

11) if the chat group session is ongoing and the <on-network-max-participant-count> as specified in 3GPP TS 24.481 [31] is already reached:

a) if, according to local policy, the user identified by the MCPTT ID in the SIP INVITE request is deemed to have a higher priority than an existing user in the chat group session, may remove a participant from the session by following clause 10.1.1.4.4.3, and skip the next step

...

15) shall accept the SIP request and generate a SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [4];

16) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.3.3.2.1 unless the procedures of clause 6.3.3.1.8 were performed in step 13)a) or step 14)a) above;"

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.1.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the controlling MCPTT function:

1) when initiating a new MCPTT session the SDP offer;

a) shall contain only one SDP media-level section for MCPTT speech media stream as contained in the received SDP offer; and

b) shall contain one SDP media-level section for media- plane control messages, if present in the received SDP offer; and

2) when adding a new MCPTT user to an existing MCPTT Session, the SDP offer shall contain the media stream currently used in the MCPTT session.

When composing the SDP offer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the controlling MCPTT function;

2) for the MCPTT speech media stream, shall include all media-level attributes from the received SDP offer;

3) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the controlling MCPTT function; and

4) for the offered media plane control channel, shall include the offered media planecontrol channel 'fmtp' attributes as specified in 3GPP TS 24.380 [5] clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.1.2]

"This clause is referenced from other procedures.

The controlling MCPTT function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [4].

The controlling MCPTT function:

1) *shall include in the Contact header field an MCPTT session identity for the MCPTT session with the g.3gpp.mcptt media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" according to IETF RFC 3840 [16];*

2) *shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [6];*

3) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [9] in the SIP INVITE request;*

4) *shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [6];*

5) *shall include a Referred-By header field with the public user identity of the inviting MCPTT client;*

6) *should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7]. The refresher parameter shall be omitted;*

7) *shall include the Supported header field set to "timer";*

8) *if the received SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body containing an <ambient-listening-type> element and:*

   a) *if the Priv-Answer-Mode header field specified in IETF RFC 5373 [18] was included in the received SIP INVITE request with a value of "Auto" or if no Priv-Answer-Mode header field was received in the received SIP INVITE request; or*

   b) *a Priv-Answer-Mode header field was received containing a value other than "Auto";*

   shall include a Priv-Answer-Mode header field set to a value of "Auto" in the outgoing SIP INVITE request;

9) *if the received SIP INVITE request contained a Priv-Answer-Mode header field and did not contain an application/vnd.3gpp.mcptt-info+xml MIME body containing an <ambient-listening-type> element, shall include an unmodified Priv-Answer-Mode header field;*

10) *if the received SIP INVITE request did not contain an application/vnd.3gpp.mcptt-info+xml MIME body containing an <ambient-listening-type> element and did not contain a Priv-Answer-Mode header field, shall include an unmodified Answer-Mode header field if present in the incoming SIP INVITE request; and*

11) *if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the application/vnd.3gpp.mcptt-info+xml MIME body to the outgoing INVITE request."*

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.4.1.2]

"In the procedures in this clause:

1) *emergency indication in an incoming SIP re-INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and*

2) *imminent peril indication in an incoming SIP re-INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.*

Upon receipt of a SIP re-INVITE request for an MCPTT session identity identifying a chat MCPTT group session, the controlling MCPTT function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP re-INVITE request with a SIP 500 (Server Internal Error) response. The controlling MCPTT function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [24] and skip the rest of the steps;

NOTE 1: if the SIP re-INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating an MCPTT emergency group call as determined by clause 6.3.3.1.13.2, or for originating an MCPTT imminent peril group call as determined by clause 6.3.3.1.13.5, the controlling MCPTT function can according to local policy choose to accept the request.

2) if the received SIP re-INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

3) if the SIP re-INVITE request contains an unauthorised request for an MCPTT emergency call as determined by clause 6.3.3.1.13.2:

a) shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [4] and skip the rest of the steps;

4) if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "true" and is an authorised request to initiate an MCPTT emergency group call as determined by clause 6.3.3.1.13.2, the controlling MCPTT function shall:

a) validate that the SIP re-INVITE request includes a Resource-Priority header field is populated correctly for an MCPTT emergency group call as specified in clause 6.3.3.1.19, and if not:

i) shall perform the actions specified in clause 6.3.3.1.8; and

ii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8 shall proceed with the rest of the steps.

NOTE 2: Verify that the Resource-Priority header is included and properly populated for both ongoing and newly-entered in-progress emergency states of the specified group.

b) if the in-progress emergency state of the group is set to a value of "true" and the MCPTT user is indicating a new emergency indication:

i) shall cache the MCPTT ID of the MCPTT user that has initiated an MCPTT emergency call;

ii) if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "true" and is an authorised request for an MCPTT emergency alert as determined by clause 6.3.3.1.13.1, shall cache the MCPTT ID of the MCPTT user that has initiated an MCPTT emergency alert; and

iii) for each of the other affiliated members of the group, generate a SIP MESSAGE request notification of the MCPTT user's emergency indication as specified in clause 6.3.3.1.11 with the following clarifications:

A) set the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true";

B) if the received SIP re-INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCPTT emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, perform the procedures specified in clause 6.3.3.1.12; and

C) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [4]; and

c) if the in-progress emergency state of the group is set to a value of "false":

i) shall set the value of the in-progress emergency state of the group to "true";

ii) shall cache the MCPTT ID of the MCPTT user that has initiated an MCPTT emergency call;

iii) if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "true" and this is an authorised request for an MCPTT emergency alert as specified in clause 6.3.3.1.13.1, shall cache the MCPTT ID of the MCPTT user that has initiated an MCPTT emergency alert;

iv) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

v) shall generate SIP re-INVITE requests for the MCPTT emergency group call to the other affiliated and joined participants of the chat MCPTT group as specified in clause 6.3.3.1.6. The MCPTT controlling function:

A) for each affiliated and joined member shall send the SIP re-INVITE request towards the MCPTT client as specified in 3GPP TS 24.229 [4]; and

B) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

vi) shall generate SIP INVITE requests for the MCPTT emergency group call to the affiliated but not joined members of the chat MCPTT group as specified in clause 6.3.3.1.7. The controlling MCPTT function:

A) for each affiliated but not joined member shall send the SIP INVITE request towards the MCPTT client as specified in 3GPP TS 24.229 [4]; and

B) Upon receiving a SIP 200 (OK) response to the SIP INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

vii) if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false";

5) if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is an unauthorised request for an MCPTT emergency group call cancellation as determined by clause 6.3.3.1.13.4:

a) shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response;

b) shall include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcptt-info+xml MIME body as specified in annex F.1 with an <emergency-ind> element set to a value of "true";

c) if an <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body is included set to "false" and there is an outstanding MCPTT emergency alert for the MCPTT user, shall include in the application/vnd.3gpp.mcptt-info+xml MIME body and <alert-ind> element set to a value of "true"; and

d) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [4] and skip the rest of the steps;

6) if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is determined to be an authorised request for an MCPTT emergency call cancellation as specified in clause 6.3.3.1.13.4 and the in-progress emergency state of the group to is set to a value of "true" the controlling MCPTT function shall:

a) validate that the SIP re-INVITE request includes a Resource-Priority header field is populated correctly for a normal priority MCPTT group call as specified in clause 6.3.3.1.19, and if not:

i) shall perform the actions specified in clause 6.3.3.1.8; and

ii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8 shall proceed with the rest of the steps;

NOTE 3: Verify that the Resource-Priority header is included and properly populated for an in-progress emergency state cancellation of the specified group.

b) shall set the in-progress emergency group state of the group to a value of "false";

c) shall clear the cache of the MCPTT ID of the MCPTT user identified by the <originated-by> element as having an outstanding MCPTT emergency group call;

d) if an <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body is included and set to "false" and is determined to be an authorised request for an MCPTT emergency alert cancellation as specified in clause 6.3.3.1.13.3 and there is an outstanding MCPTT emergency alert for the MCPTT user shall:

i) if the received SIP re-INVITE request contains an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, clear the cache of the MCPTT ID of the MCPTT user identified by the <originated-by> element as having an outstanding MCPTT emergency alert; and

ii) if the received SIP re-INVITE request does not contain an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, clear the cache of the MCPTT ID of the sender of the SIP re-INVITE request as having an outstanding MCPTT emergency alert;

e) shall generate SIP re-INVITE requests to the other affiliated and joined members of the MCPTT group as specified in clause 6.3.3.1.6. The MCPTT controlling function:

i) for each affiliated and joined member shall send the SIP re-INVITE request towards the MCPTT client as specified in 3GPP TS 24.229 [4]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCPTT function shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

NOTE 4: Clause 6.3.3.1.6 will inform the affiliated and joined members of the cancellation of the MCPTT group's in-progress emergency state and the cancellation of the MCPTT emergency alert if applicable.

f) for each of the affiliated but not joined members of the group shall:

i) generate a SIP MESSAGE request notification of the cancellation of the MCPTT user's emergency call as specified in clause 6.3.3.1.11;

ii) set the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "false";

iii) if indicated above in step d), set the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "false"; and

iv) send the SIP MESSAGE request according to 3GPP TS 24.229 [4];

7) if a Resource-Priority header field is included in the SIP re-INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the received SIP re-INVITE request does not contain an authorised request for an MCPTT emergency call as determined in step 4) above and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps; or

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the received SIP re-INVITE request does not contain an authorised request for an MCPTT imminent peril call as determined by the procedures of clause 6.3.3.1.13.5 and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps;

8) if the received SIP re-INVITE request contains an imminent peril indication, shall perform the procedures specified in clause 10.1.2.4.1.3 and skip the rest of the steps;

9) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.3.3.2.1 unless the procedures of clause 6.3.3.1.8 were performed in step 6) a) i) above;

10) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [23];

11) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "true" and if this is an unauthorised request for an MCPTT emergency alert as determined by clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

12) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "false" and if this is an unauthorised request for an MCPTT emergency alert cancellation as determined by clause 6.3.3.1.13.3, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

13) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true", this is an authorised request for an MCPTT imminent peril group call and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCPTT emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

14) shall interact with media plane as specified in 3GPP TS 24.380 [5]; and

15) shall send the SIP 200 (OK) response towards the MCPTT client according to 3GPP TS 24.229 [4].

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCPTT client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.4.3.2]

"When the MCPTT session for group call needs to be released as specified in clause 6.3.8.1, the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.5."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.3.1.5]

"When a participant needs to be removed from the MCPTT session, the controlling MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5] for the MCPTT session release;

2) shall generate a SIP BYE request according to 3GPP TS 24.229 [4]; and

3) shall send the SIP BYE request to the MCPTT clients according to 3GPP TS 24.229 [4].

If timer TNG3 (group call timer) has not expired, then when the last MCPTT client is removed from the MCPTT session, the controlling MCPTT function shall stop timer TNG3 (group call timer).

When the MCPTT group session needs to be released, the controlling MCPTT function shall send SIP BYE requests as described in this clause, to all participants of the group session.

Upon receiving a SIP 200 (OK) response to a SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT clients."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCPTT function:

1) shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing the media plane resource associated with the SIP session towards the MCPTT client;

…

2) shall generate a SIP 200 (OK) response and send the SIP response towards the MCPTT client according to 3GPP TS 24.229 [4];

3) shall check the MCPTT session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCPTT session needs to be released for each participant of the MCPTT session;

4) if release of the MCPTT session is required:

a) shall perform the procedures as specified in the clause 6.3.3.1.5 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and

...

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT participant."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCPTT client negotiated support of queueing floor requests as described in clause 14, and the state machine specified in subclause 6.3.6 does not exist, the floor control arbitration logic in the floor control server:

1. if there is a Track Info field associated with the floor control server state transition diagram for 'general floor control operation' stored, shall remove the Track Info field from the storage;

2. if the active floor request queue is empty the floor control server:

a. shall send Floor Idle message to all floor participants. The Floor Idle message:

i. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications;

b. shall start timer T7 (Floor Idle) and initialise counter C7 (Floor Idle) to 1;

c. shall start timer T4 (Inactivity); and

d. shall set the general state to the 'G: Floor Idle' state; and

3. if the active floor request queue is not empty the floor control server:

a. shall select a queued floor request from the top of the active floor request queue;

b. shall remove that queued floor request from the active floor request queue;

c. if the queued floor request includes a Track Info field, shall store the Track Info field and associate it with the floor control server state transition diagram for 'general floor control operation'; and

d. shall enter the 'G: Floor Taken' state as specified in the subclause 6.3.4.4.2 with respect to that floor participant.

..."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.3.3]

"Upon receiving a floor request message (from a floor participant that is permitted to make a floor request) the floor control arbitration logic in the floor control server:

...

3. if the floor request is granted the floor control server:

...

c. shall store the SSRC of floor participant granted the permission to send media until the floor is released associated to that floor request;

d. if a Track Info field is included in the Floor Request message, shall store the received Track Info field, and

e. shall enter the 'G: Floor Taken' state as specified in the subclause 6.3.4.4.2."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.2]

"When entering this state the floor control arbitration logic in the floor control server:

1. shall send a Floor Granted message to the floor participant to which the floor is granted. The Floor Granted message:

a. shall include the value of timer T2 (Stop talking)in the Duration field;

b. shall include the granted priority in the Floor priority field;

c. if a Track Info field associated with the floor control server state transition diagram for 'general floor control operation' is stored, shall include the stored Track Info field;

...

3. shall send Floor Taken message to all other floor participants. The Floor Taken message:

a. if the floor is currently granted only to one particpant:

i shall include the granted MCPTT user's MCPTT ID in the Granted Party's Identity field, if privacy is not requested;

ii. may include the functional alias of the granted MCPTT user in the Functional Alias field, if privacy is not requested; and

iii. shall include the Location field:

A. if the current location of the talker is not available or is not to be reported according to the MCPTT user profile, then the location type is set to '0' (Not provided); or

B. if the current location of the talker is available and may be reported according to the MCPTT user profile, then the location type and location value are set as specified in table 8.2.3.21-3;

…

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

…

e. if the session is not a broadcast group call, may include the Permission to Request the Floor field set to '1'; and

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Floor Indicator field with appropriate indications;

..

5. shall set the general state to 'G: Floor Taken' state; and

..."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.4.4.6]

"Upon receiving a Floor Release message the floor control arbitration logic in the floor control server:

1. *shall request the media distributor in the MCPTT server to stop forwarding RTP media packets received from the participant that sent the Floor Release message;*

...

4. *if mutli-talker floor control is not supported shall enter the 'G: Floor Idle' state as specified in the subclause 6.3.4.3.2; and*

..."

5.2.2.1.3 Test description

5.2.2.1.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server) provides the participating MCPTT function.
- SS-UE1 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.2.1.3.1-1.
- SS-UE2 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.2.1.3.1-1.

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the controlling MCPTT function
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figures A.4.2.
- The SS (MCPTT Server) is connected to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are registered and authorized to use the MCPTT system.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.

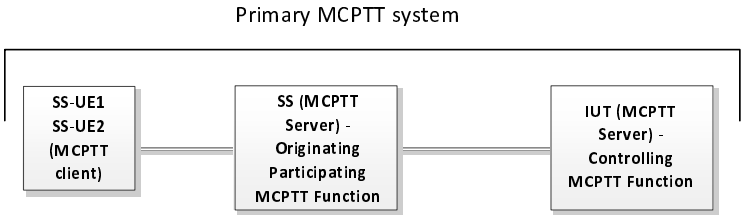


Figure 5.2.2.1.3.1-1: Functions of the testing components

5.2.2.1.3.2 Test procedure sequence

Table 5.2.2.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS(MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) sends a SIP INVITE message initiating a chat group call with automatic commencement mode.	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCPTT Server) send a SIP 100 (Trying) message to the participating server SS (MCPTT Server)?	-->	SIP 100 (Trying)	1	P
3	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) message to SS (MCPTT Server)?	-->	SIP 200 OK	1	P
4	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message.	<--	SIP ACK	-	-
5	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE2 (joining MCPTT client)	<--	SIP INVITE	-	-
6	Check: Does the IUT (MCPTT Server) send a SIP 100 (Trying) message to the SS (MCPTT Server)?	-->	SIP 100 (Trying)	1	P
7	Check: Does the IUT (MCPTT controlling Server) send a SIP 200 (OK) message to SS (MCPTT Server)?	-->	SIP 200 OK	1	P
8	The SS (MCPTT Server) responds with a SIP ACK message.	<--	SIP ACK	-	-
9	The SS (MCPTT Server) sends a Floor Request message with no acknowledgement required to request the floor from SS-UE1 (MCPTT client)	<--	Floor Request	-	-
10	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS-UE1 (MCPTT client) via the participating server SS (MCPTT Server)?	-->	Floor Granted	2	P
11	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS-UE2 (MCPTT client) via the participating server SS (MCPTT Server)?	-->	Floor Taken	2	P
12	The SS (MCPTT Server) sends a Floor Release message with no acknowledgement required to release the floor from SS-UE1 (MCPTT client)	<--	Floor Release	-	-
13	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS-UE2 (MCPTT client) via the participating server SS (MCPTT Server)?	-->	Floor Idle	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
14	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS-UE1 (MCPTT client) via the participating server SS (MCPTT Server)?	-->	Floor Idle	2	P
15	The SS (MCPTT Server) sends a Floor Request message with no acknowledgement required to request the floor from SS-UE2 (MCPTT client)	<--	Floor Request	-	-
16	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS-UE2 (MCPTT client) via the participating server SS (MCPTT Server)?	-->	Floor Granted	2	P
17	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS-UE1 (MCPTT client) via the participating server SS (MCPTT Server)?	-->	Floor Taken	2	P
18	The SS (MCPTT Server) sends a Floor Release message with no acknowledgement required to release the floor from SS-UE2 (MCPTT client)	<--	Floor Release	-	-
19	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS-UE2 (MCPTT client) via the participating server SS (MCPTT Server)?	-->	Floor Idle	2	P
20	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS-UE1 (MCPTT client) via the participating server SS (MCPTT Server)?	-->	Floor Idle	2	P
21	The SS (MCPTT Server) sends a SIP BYE request from SS-UE1 (MCPTT client)	<--	SIP BYE	-	-
22	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK) message to SS-UE1 (MCPTT client) via the participating server SS (MCPTT Server)?	-->	SIP 200 (OK)	3	P
23	Check: Does the IUT (MCPTT Server) send a SIP BYE message to the SS-UE2 (MCPTT client) to end the call via the participating server SS (MCPTT Server)?	-->	SIP BYE	3	P
24	The SS (MCPTT Server) responds with a SIP 200 (OK) message from SS-UE2 (MCPTT client)	<--	SIP 200 (OK)	-	-

5.2.2.1.3.3 Specific message contents

Table 5.2.2.1.3.3-1: SIP INVITE from the SS (MCPTT Server) (Step 1, Table 5.2.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of a second MCPTT Server which is implemented in the DUT used in MCPTT Server testing.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B			
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.1.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.1.3.3-2			

Table 5.2.2.1.3.3-1A: SDP in SIP INVITE (Table 5.2.2.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 condition SDP_OFFER, INITIAL_SDP_OFFER

Table 5.2.2.1.3.3-2: MCPTT-INFO in SIP INVITE (Table 5.2.2.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions CHAT-GROUP-CALL				
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 5.2.2.1.3.3-3: SIP 200 OK from the IUT (MCPTT Server) (Steps 3 and 7 Table 5.2.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	any allowed value			
refresher	"uac"			
Message-body				
SDP message	SDP message as described in Table 5.2.2.1.3.3-3A			

Table 5.2.2.1.3.3-3A: SDP in SIP 200 (OK) from the IUT (MCPTT Server) (Table 5.2.2.1.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 condition SDP_ANSWER				
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the MCPTT Server			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the MCPTT Server			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the MCPTT Server (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the MCPTT Server			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the MCPTT Server being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			

Table 5.2.2.1.3.3-4: SIP INVITE from the SS (MCPTT Server) (Step 5, Table 5.2.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of a second MCPTT Server which is implemented in the DUT used in MCPTT Server testing.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B			
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.1.3.3-4A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.1.3.3-5			

Table 5.2.2.1.3.3-4A: SDP in SIP INVITE from the SS (MCPTT Server) (Table 5.2.2.1.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 condition SDP_OFFER, INITIAL_SDP_OFFER

Table 5.2.2.1.3.3-5: MCPTT-INFO in SIP INVITE from the SS (MCPTT Server) (Table 5.2.2.1.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions CHAT-GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-1A.				

Table 5.2.2.1.3.3-6: Floor Request (Steps 9 and 15 Table 5.2.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1

Table 5.2.2.1.3.3-7: Floor Granted (Steps 10 and 16 Table 5.2.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

Table 5.2.2.1.3.3-8: Floor Taken (Step 11, Table 5.2.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.2.1.3.3-9: Floor Idle (Steps 13, 14, 19 and 20, Table 5.2.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.2.1.3.3-10: Floor Taken (Step 17, Table 5.2.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_B			

Table 5.2.2.1.3.3-11: SIP BYE from the SS (MCPTT-Server) (Step 21, Table 5.2.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
From				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Local URI of the dialog (from the SS' point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Local tag of the dialog (from the SS' point of view)		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the SS' point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the SS' point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	70	The recommended initial value is 70 in IETF RFC 3261 [17].		

Table 5.2.2.1.3.3-12: SIP BYE from the IUT (MCPTT-Server) (Step 23, Table 5.2.2.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

5.2.2.2 MCPTT Server - MCPTT Server / Chat Group Call / Automatic Commencement Mode / Originating Participating server

5.2.2.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE1 (MCPTT client) initiates a chat group call with automatic commencement mode }
    then { IUT (MCPTT Server) interacts with SS-UE1 (MCPTT client) to set up the chat group call by
sending SIP messages and interacts with SS (MCPTT Server) to set up the chat group call by sending
SIP messages }
           }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT chat group call with automatic commencement
mode }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Request, Floor Granted, Floor Taken,
Floor Release, Floor Idle) by forwarding messages to the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT
client) and SS (MCPTT Server) }
           }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT chat group call with automatic commencement
mode }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the chat group call with automatic commencement mode }
    then { IUT (MCPTT Server) forwards a SIP BYE message to SS (MCPTT Server) and responds to the
SS-UE1 (MCPTT client) by sending a SIP 200 (OK) message and forwards a SIP BYE message to the SS-UE2
(MCPTT client) and responds to SS (MCPTT Server) by sending a SIP 200 (OK) message }
           }
```

5.2.2.2.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 6.3.2.1.1.1, 6.3.2.1.3, 10.1.2.3.1.1, ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.4.2. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCPTT function:

1) shall contain only one SDP media-level section for MCPTT speech as contained in the received SDP offer; and

2) shall contain one SDP media-level section for media plane control messages, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [4], the participating MCPTT function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCPTT function, if required;

NOTE 1: Requirements can exist for the participating MCPTT function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media plane control channel, if any, in the received SDP offer with the IP address and port number of the participating MCPTT function; and

NOTE 2: If the participating MCPTT function and the controlling MCPTT function or the participating MCPTT function and the non-controlling MCPTT function are in the same MCPTT server, and the participating MCPTT function does not have a dedicated IP address or a dedicated port number for media plane control channel or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 6.3.2.1.3]

"This clause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [4], on receipt of an incoming SIP INVITE request, the participating MCPTT function:

1) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [6] if included in the incoming SIP INVITE request;

2) should include the Session-Expires header field according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

3) shall include the option tag "timer" in the Supported header field;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

5) shall include the g.3gpp.mcptt media feature tag into the Contact header field of the outgoing SIP INVITE request;

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;

7) if the incoming SIP INVITE request contained a MIME resource-lists body, shall copy the MIME resource-lists body, according to rules and procedures of IETF RFC 5366 [20];

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request; and

9) if, according to clause 6.4, the SIP INVITE request is regarded as being not received with an implicit floor request, then:

a) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcptt-location-info+xml MIME body with a <Report> element included in the <location-info> root element; and

b) if the <allow-location-info-when-talking> element of the <ruleset> element of the MCPTT user profile document identified by the MCPTT ID of the calling MCPTT user (see the MCPTT user profile document in 3GPP TS 24.484 [50]) is set to a value of "true";

shall copy the application/vnd.3gpp.mcptt-location-info+xml MIME body from the SIP INVITE request into the outgoing SIP INVITE."

[ETSI TS 124 379 [4] (3GPP TS 24.379) clause 10.1.2.3.1.1]

"...

Upon receipt of a "SIP INVITE request for originating participating MCPTT function" for a group identity identifying a chat MCPTT group containing an application/vnd.3gpp.mcptt-info+xml MIME body with the <session-type> element set to a value of "chat", the participating MCPTT function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The participating MCPTT function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [24]. Otherwise, continue with the rest of the steps;

...

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the group identity present in the incoming SIP INVITE request;

11) shall include the MCPTT ID of the calling user in <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request;

...

12) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request as specified in clause 6.3.2.1.1.1;

...

Upon receipt of a SIP 2xx response to the above SIP INVITE request, the participating MCPTT function:

1) void;

2) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2 with the clarification that if an <MKFC-GKTPs> element was contained in the received SIP 2xx response it is not included in the generated SIP 200 (OK) response;

NOTE 11: If an <MKFC-GKTPs> element is received in a SIP 2xx response, the participating MCPTT function essentially ignores it and does not forward it, resulting in unicast media plane transmission being used for the originating client.

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) shall include an MCPTT session identity mapped to the MCPTT session identity provided in the Contact header field of the received SIP 200 (OK) response;

7) if the procedures of clause 9.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 9.2.2.2.13;

8) shall send the SIP 200 (OK) response to the MCPTT client according to 3GPP TS 24.229 [4];

9) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

10) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [7]."

[ETSI TS 124 380 [5] (3GPP TS 24.380) clause 6.4.2]

"Upon receiving a floor control message the participating MCPTT function:

1. shall immediately forward the floor control message to the floor control server if the message is received from the floor participant;

..."

5.2.2.2.3 Test description

5.2.2.2.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the controlling MCPTT function
- SS-UE1 (MCPTT client)
- SS-UE2 (MCPTT client)

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the originating participating MCPTT function
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2, and to the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.

  The SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) perform the Procedure for MCPTT Authorization, Configuration and Key Generation as specified in clause 5.1.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.

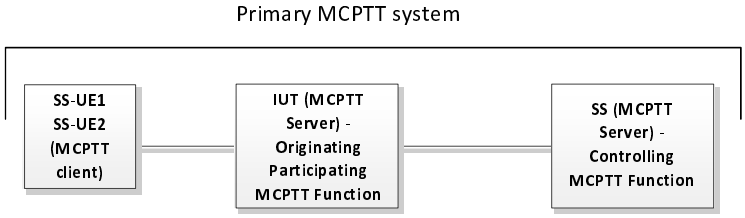


Figure 5.2.2.2.3.1-1: Functions of the testing components

5.2.2.2.3.2 Test procedure sequence

Table 5.2.2.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1(MCPTT client) sends a SIP INVITE message initiating a chat group call with automatic commencement mode.	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS-UE1 (MCPTT client)?	-->	SIP 100 Trying	1	P
3	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE	1	P
4	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
5	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	1	P
6	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS-UE1 (MCPTT client)?	-->	SIP 200 (OK)	1	P
7	The SS-UE1 (MCPTT client) responds with a SIP ACK message	<--	SIP ACK		-
8	The SS-UE2(MCPTT client) sends a SIP INVITE message initiating a chat group call with automatic commencement mode.	<--	SIP INVITE	-	-
9	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS-UE2 (MCPTT client)?	-->	SIP 100 Trying	1	P
10	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE	1	P
11	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)	-	-
12	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	1	P
13	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS-UE2 (MCPTT client)?	-->	SIP 200 (OK)	1	P
14	The SS-UE2 (MCPTT client) responds with a SIP ACK message	<--	SIP ACK	-	-
15	The SS-UE1 (MCPTT client) sends a Floor Request message with no acknowledgement required to request the floor.	<--	Floor Request	-	-
16	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	2	P
17	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required informing the SS-UE1 (MCPTT client) the floor grant.	<--	Floor Granted	-	-
18	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE1 (MCPTT client)?	-->	Floor Granted	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
19	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE2 (MCPTT client) that the floor is taken by SS-UE1 (MCPTT client)	<--	Floor Taken	-	-
20	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE2 (MCPTT client)?	-->	Floor Taken	2	P
21	The SS-UE1 (MCPTT client) sends a Floor Release message	<--	Floor Release	-	-
22	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	2	P
23	The SS (MCPTT server) sends a Floor Idle message informing the SS-UE1 that floor is no longer taken	<--	Floor Idle	-	-
24	The SS (MCPTT server) sends a Floor Idle message informing the SS-UE2 that floor is no longer taken	<--	Floor Idle	-	-
25	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	2	P
26	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	2	P
27	The SS-UE2 (MCPTT client) sends a Floor Request message with no acknowledgement required to request the floor	<--	Floor Request		-
28	Check: Does the IUT (MCPTT Server) forward the Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	2	P
29	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required indicating floor grant to the SS-UE2.	<--	Floor Granted	-	-
30	Check: Does the IUT (MCPTT Server) forward the Floor Granted message to the SS-UE2 (MCPTT client)?	-->	Floor Granted	2	P
31	The SS (MCPTT Server) sends a Floor Taken message to the IUT (MCPTT Server) informing the SS-UE1 (MCPTT client) that the floor is taken by SS-UE2 (MCPTT client)	<--	Floor Taken	-	-
32	Check: Does the IUT (MCPTT Server) forward the Floor Taken message to the SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
33	The SS-UE2 (MCPTT client) sends a Floor Release message	<--	Floor Release	-	-
34	Check: Does the IUT (MCPTT Server) forward the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	2	P
35	The SS (MCPTT server) sends a Floor Idle message informing the SS-UE1 that floor is no longer taken	<--	Floor Idle	-	-
36	The SS (MCPTT server) sends a Floor Idle message informing the SS-UE2 that floor is no longer taken	<--	Floor Idle	-	-
37	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE1 (MCPTT client)?	-->	Floor Idle	2	P
38	Check: Does the IUT (MCPTT Server) forward the Floor Idle message to the SS-UE2 (MCPTT client)?	-->	Floor Idle	2	P
39	The SS-UE1 (MCPTT client) sends a SIP BYE request	<--	SIP BYE		-
40	Check: Does the IUT (MCPTT Server) forward the SIP BYE message to the SS (MCPTT Server) to end the call?	-->	SIP BYE	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
41	SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
42	Check: Does the IUT (MCPTT Server) forward the SIP 200 (OK) response to SS-UE1 (MCPTT client)	-->	SIP 200 (OK)	3	P
43	The SS (MCPTT server) sends a SIP BYE request	<--	SIP BYE		-
44	Check: Does the IUT (MCPTT Server) forward the SIP BYE message to the SS-UE2 (MCPTT Client) to end the call?	-->	SIP BYE	3	P
45	SS-UE2 (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
46	Check: Does the IUT (MCPTT Server) forward the SIP 200 (OK) response to SS (MCPTT Server)	-->	SIP 200 (OK)	3	P

5.2.2.2.3.3 Specific message contents

Table 5.2.2.2.3.3-1: SIP INVITE from the SS-UE1 (MCPTT client) (Step 1, Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The public service identity of the MCPTT server under test		
Session-Expires			IETF RFC 4028 [25]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [25].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.2.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.2.3.3-1B			

Table 5.2.2.2.3.3-1A: SDP in SIP INVITE from the SS-UE1 (MCPTT client) (Table 5.2.2.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions INITIAL_SDP_OFFER

Table 5.2.2.2.3.3-1B: MCPTT-Info in SIP INVITE from the SS-UE1 (MCPTT client) (Table 5.2.2.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions CHAT-GROUP-CALL, INVITE_REFER

Table 5.2.2.2.3.3-2: SIP INVITE from the IUT (MCPTT-Server) (Step 3, Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS		
Session-Expires			IETF RFC 4028 [25]	
delta-seconds	any allowed value			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The public service identity of the MCPTT server under test.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_A			
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			
Supported				
option-tag	"timer"			
P-Asserted-Identity				
addr-spec				
user-info and host	same URI as in the incoming SIP INVITE request			
port	not present			
Contact				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B			
feature-param	"isfocus" not present			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.2.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.2.3.3-2B			

Table 5.2.2.2.3.3-2A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.2.2.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the MCPTT Server			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the MCPTT Server			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			

Table 5.2.2.2.3.3-2B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.2.2.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions CHAT-GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.2.2.3.3-3: SIP 200 (OK) from the SS (MCPTT-Server) (Steps 4 and 11 Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.2.2.3.3-3A			

Table 5.2.2.2.3.3-3A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 5.2.2.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, FIRST_SDP_FROM_SS

Table 5.2.2.2.3.3-4: SIP 200 (OK) from the IUT (MCPTT-Server) (Steps 6 and 13 Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 5.2.2.2.3.3-4A			

Table 5.2.2.2.3.3-4A: SDP in SIP 200 (OK) from the IUT (MCPTT-Server) (Table 5.2.2.2.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions CHAT-GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the MCPTT Server			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the MCPTT Server			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the MCPTT Server (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the MCPTT Server			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the MCPTT Server being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			

Table 5.2.2.2.3.3-5: SIP INVITE from the SS-UE2 (MCPTT client) (Step 8, Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The public service identity of the MCPTT server under test		
Session-Expires			IETF RFC 4028 [25]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [25].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.2.3.3-5A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.2.3.3-5B			

Table 5.2.2.2.3.3-5A: SDP in SIP INVITE from the SS-UE2 (MCPTT client) (Table 5.2.2.2.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions INITIAL_SDP_OFFER

Table 5.2.2.2.3.3-5B: MCPTT-Info in SIP INVITE from the SS-UE2 (MCPTT client) (Table 5.2.2.2.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions CHAT-GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.2.2.3.3-6: SIP INVITE from the IUT (MCPTT-Server) (Step 10, Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS		
Session-Expires			IETF RFC 4028 [25]	
delta-seconds	any allowed value			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The public service identity of the MCPTT server under test.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_A			
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			
Supported				
option-tag	"timer"			
P-Asserted-Identity				
addr-spec				
user-info and host	same URI as in the incoming SIP INVITE request			
port	not present			
Contact				
addr-spec				
user-info and host	tsc_MCPTT_PublicServiceId_B			
feature-param	"isfocus" not present			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.2.3.3-6A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.2.3.3-6B			

Table 5.2.2.2.3.3-6A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.2.2.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the MCPTT Server			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the MCPTT Server			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			

Table 5.2.2.2.3.3-6B: MCPTT-Info in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.2.2.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions CHAT-GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.2.2.3.3-7: Floor Request (Steps 15, 16, 27 and 28 Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1

Table 5.2.2.2.3.3-8: Floor Granted (Steps 17, 18, 29 and 30 Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

Table 5.2.2.2.3.3-9: Floor Taken (Steps 19 and 20 Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.2.2.3.3-10: Floor Idle (Steps 23, 24, 25, 26, 35, 36, 37 and 38 Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.2.2.3.3-11: Floor Taken (Steps 31 and 32 Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_B			

Table 5.2.2.2.3.3-12: Floor Release (Steps 21, 22, 33 and 34, Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1

Table 5.2.2.2.3.3-13: SIP BYE from SS-UE1 (MCPTT client) (Step 39, Table 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1 condition UDP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the UE's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the UE's point of view)		

Table 5.2.2.2.3.3-14: SIP BYE from the IUT (MCPTT-Server) (Step 40, 5.2.2.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

Table 5.2.2.2.3.3-15: SIP BYE from the SS (MCPTT Server) (Step 43, 5.2.2.2.3.2-1)

Derivation Path: Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the UE's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the UE's point of view)		

Table 5.2.2.2.3.3-16: SIP BYE from the IUT (MCPTT-Server) (Step 44, 5.2.2.2.3.2-1)

Derivation Path: Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

5.2.2.3 MCPTT Server - MCPTT Server / Chat Group Call / Temporary Group / Controlling server / Untrusted mode

5.2.2.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE1 (MCPTT client) initiates an on-demand chat group call to a temporary group and
the SS-UE2 (MCPTT client) joins the chat group call in the temporary group and the SS-UE3 (MCPTT
client) joins the chat group call in the temporary group }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the chat group call by
sending SIP messages for all participants }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT chat group call to a temporary group }
ensure that {
  when { the SS-UE1 (MCPTT client), SS-UE2 (MCPTT client)  and SS-UE3 (MCPTT client) engage in
communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Idle, Floor Ack, Floor
Granted) by sending messages via the SS (MCPTT Server) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT chat group call to a temporary group }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the chat group call }
    then { IUT (MCPTT Server) responds to the SS (MCPTT Server) by sending a SIP 200 (OK) message
and sends a SIP BYE message to SS (MCPTT Server) towards other participants  }
            }
```

5.2.2.3.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 10.1.2.4.1.1, 6.3.5.2, 6.3.3.5.1, 6.3.3.1.5, 6.3.3.2.1, 10.1.2.4.2, 6.3.3.2.4, 10.1.2.4.3.2, 10.1.2.4.3.3, 6.3.3.4, 6.3.5.1, 6.3.5.2, 4.8, 5.3.2, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 6.3.2.2, 6.3.3, 14.3.4. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.4.1.1]

"In the procedures in this clause:

1) MCPTT ID in an incoming SIP INVITE request refers to the MCPTT ID of the originating user from the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;

3) MCPTT ID in an outgoing SIP INVITE request refers to the MCPTT ID of the called user in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the outgoing SIP INVITE request;

4) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

5) alert indication in an incoming SIP INVITE request refers to the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a "SIP INVITE request for controlling MCPTT function of an MCPTT group" containing a group identity identifying a chat MCPTT group, the controlling MCPTT function:

…

4) shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2 and continue with the rest of the steps if the checks in clause 6.3.5.2 succeed;

…

10) shall create a chat group session and allocate an MCPTT session identity for the chat group session if the MCPTT chat group session identity does not already exist, and may handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

…

15) shall accept the SIP request and generate a SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [4];

16) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.3.3.2.1 unless the procedures of clause 6.3.3.1.8 were performed in step 13)a) or step 14)a) above;

17) should include the Session-Expires header field and start supervising the SIP session according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

18) shall include the "timer" option tag in a Require header field;

19) shall include the following in a Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";

c) the MCPTT session identity; and

d) the media feature tag isfocus;

20) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [23];

21) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCPTT emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

22) if the received SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCPTT emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

23) shall interact with media plane as specified in 3GPP TS 24.380 [5];

24) shall send the SIP 200 (OK) response to the MCPTT client according to 3GPP TS 24.229 [4]; and

25) if the chat group session was already ongoing and if at least one of the participants has subscribed to the conference event package, shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as specified in clause 10.1.3.4.2.

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.2]

"NOTE 1: In this clause, "MCPTT server" can refer to either the controlling MCPTT function of an MCPTT group or the non-controlling MCPTT function of an MCPTT group.

Upon receipt of a SIP INVITE request:

…

6) if the SIP INVITE request is a "SIP INVITE request for controlling function of an MCPTT group" and the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request is associated with an MCPTT group ID that has been regrouped, the MCPTT server:

a) shall obtain the TGI associated with the regrouped group, by accessing the "temporary-MCPTT-group-ID" attribute of the <regrouped> element of the group document associated with the MCPTT ID in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request;

b) if not hosting the TGI, shall:

i) stop processing the SIP INVITE request; and

ii) return a SIP 302 (Moved Temporarily) response with:

A) a Contact header field set to the PSI of the MCPTT server hosting the TGI; and

B) an application/vnd.3gpp.mcptt-info MIME body with a <mcptt-request-uri> element set to the TGI; and

c) if hosting the TGI, and the call to the temporary group is in progress, shall:

i) associate the MCPTT ID of the calling user with the temporary group call;

ii) interact with the media plane as specified in 3GPP TS 24.380 [5]; and

iii) exit this clause;

d) if hosting the TGI and the call to the temporary group is not in progress, shall subscribe to the "xcap-diff" event-package for the group document of the TGI as specified in 3GPP TS 24.481 [31], if not already subscribed;

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.5.1]

"When the controlling MCPTT function receives a SIP INVITE request to initiate a group session, then after an MCPTT session identity has been allocated for the group session and if the <on-network-maximum-duration> element is present in the group document as specified in 3GPP TS 24.481 [31], the controlling MCPTT function: shall start timer TNG3 (group call timer) with the value obtained from the <on-network-maximum-duration> element of the group document as specified in 3GPP TS 24.481 [31].

If the <on-network-maximum-duration> element is not present in the group document as specified in 3GPP TS 24.481 [31], then the controlling MCPTT function shall not start timer TNG3 (group call timer).

NOTE 1: The configuration of <on-network-maximum-duration> element in 3GPP TS 24.481 [31] is mandated for a pre-arranged group and is optional for a chat group.

When merging two or more active group calls into a temporary group call, the controlling MCPTT function(s) hosting the active group calls shall stop timer TNG3 (group call timer) for each group call, and the controlling MCPTT function hosting the temporary group call shall start timer TNG3 (group call timer) for the temporary group call.

NOTE 2: If the MCPTT server(s) hosting the independent active group calls are different to the MCPTT server that will host the temporary group call, then the MCPTT server(s) hosting the independent active group calls become non-controlling MCPTT function(s) of an MCPTT group, for the temporary group call.

When splitting a temporary group call into independent group calls, the controlling MCPTT function hosting the temporary group call shall stop timer TNG3 (group call timer) and the controlling MCPTT function(s) hosting the independent group calls shall start TNG3 (group call timer) for each group call.

When the last MCPTT client leaves the MCPTT session, the controlling MCPTT function shall stop timer TNG3 (group call timer).

On expiry of timer (group call timer), the controlling MCPTT function shall release the MCPTT session by following the procedures in clause 6.3.3.1.5;"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.1.5]

"When a participant needs to be removed from the MCPTT session, the controlling MCPTT function:

1) shall interact with the media plane as specified in 3GPP TS 24.380 [5] for the MCPTT session release;

2) shall generate a SIP BYE request according to 3GPP TS 24.229 [4]; and

3) shall send the SIP BYE request to the MCPTT clients according to 3GPP TS 24.229 [4].

If timer TNG3 (group call timer) has not expired, then when the last MCPTT client is removed from the MCPTT session, the controlling MCPTT function shall stop timer TNG3 (group call timer).

When the MCPTT group session needs to be released, the controlling MCPTT function shall send SIP BYE requests as described in this clause, to all participants of the group session.

Upon receiving a SIP 200 (OK) response to a SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT clients."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [4], the controlling MCPTT function:

1) for the accepted media stream in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCPTT function; and

2) for the accepted media channel control channel, if present in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCPTT function, for the accepted media plane control channel, if present in the received SDP offer; and

b) shall include 'fmtp' attributes as specified in 3GPP TS 24.380 clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.4.2]

"Upon receiving a SIP BYE request the controlling MCPTT function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCPTT function:

1) shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing the media plane resource associated with the SIP session towards the MCPTT client;

NOTE: The non-controlling MCPTT function is also regarded as a MCPTT client in a temporary MCPTT group session.

2) shall generate a SIP 200 (OK) response and send the SIP response towards the MCPTT client according to 3GPP TS 24.229 [4];

3) shall check the MCPTT session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCPTT session needs to be released for each participant of the MCPTT session;

4) if release of the MCPTT session is required:

a) shall perform the procedures as specified in the clause 6.3.3.1.5 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and

5) if a release of the MCPTT session is not required, shall send a SIP NOTIFY request to all remaining MCPTT clients in the MCPTT session with a subscription to the conference event package as specified in clause 10.1.3.4.2.

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCPTT function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.380 [5] for releasing media plane resources associated with the SIP session with the MCPTT participant."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.4.3.2]

"When the MCPTT session for group call needs to be released as specified in clause 6.3.8.1, the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.5."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.4.3.3]

"When an MCPTT client needs to be removed from the MCPTT session (e.g. due to de-affiliation or admitting a higher priority user), the controlling MCPTT function shall follow the procedures in clause 6.3.3.1.5.

After successful removing the MCPTT client from the MCPTT session, the controlling MCPTT function may generate a notification to the MCPTT clients, which have subscribed to the conference event package that an MCPTT user has been removed from the MCPTT session, as specified in clause 6.3.3.4 and send the SIP NOTIFY request to the MCPTT client according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.3.4]

"The controlling MCPTT function shall generate a SIP NOTIFY request according to 3GPP TS 24.229 [4] with the clarification in this clause.

In the SIP NOTIFY request, the controlling MCPTT function:

1) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCPTT function;

2) shall include an Event header field set to "conference";

3) shall include an Expires header field set to 3600 seconds according to IETF RFC 4575 [30], as default value;

4) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [4]), in a P-Preferred-Service header field according to IETF RFC 6050 [9]; and

5) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <mcptt-calling-group-id> set to the value of the MCPTT group ID;

b) if the target is a MCPTT user, the value of <mcptt-request-uri> element set to the value of MCPTT ID of the targeted MCPTT user; and

c) if the target is the non-controlling MCPTT function, the value of <mcptt-request-uri> element set to the constituent MCPTT group ID.

In the SIP NOTIFY request, the controlling MCPTT function shall include an application/conference-info+xml MIME body according to IETF RFC 4575 [30] with the following limitations:

1) the controlling MCPTT function shall include the MCPTT group ID of the MCPTT group in the "entity" attribute of the <conference-info> element;

2) for each participant in the MCPTT session with the exception of non-controlling MCPTT functions, the controlling MCPTT function shall include a <user> element. The <user> element shall:

NOTE: Non-controlling MCPTT functions will appear as a participant in temporary group sessions.

a) include the "entity" attribute. The "entity" attribute:

i) shall for the MCPTT client, which initiated, joined or rejoined an MCPTT session, include the MCPTT ID of the MCPTT user which originates SIP INVITE request; and

ii) shall for an invited MCPTT client include the MCPTT ID of the invited MCPTT user in case of a prearranged group call or chat group call;

b) shall include a single <endpoint> element. The <endpoint> element:

i) shall include the "entity" attribute; and

ii) shall include the <status> element indicating the status of the MCPTT session according to RFC 4575; and

c) may include <roles> element.

NOTE: The usage of <roles> is only applicable for human consumption."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.1]

"This clause describes how an MCPTT server accesses a group document from a group management server. The MCPTT server which accesses a group document performs the role of a controlling MCPTT function or performs the role of a non-controlling MCPTT function of an MCPTT group when accessing a group document. In such cases, for a group call:

…

The MCPTT server can cache the group document associated with an MCPTT group or temporary group, and can subscribe to be notified of changes to the group document associated with an MCPTT group or temporary group as specified in 3GPP TS 24.481 [31].

NOTE 1: During the group regrouping operation as specified in 3GPP TS 24.481 [31], the controlling MCPTT function is notified of the constituent MCPTT group identities associated with the TGI.

If the group data associated with an MCPTT group ID or TGI cached in the MCPTT server is removed, the MCPTT server re-subscribes for changes in the group information associated with the MCPTT group ID or TGI.

NOTE 2: Re-subscription can occur prior to the receipt of an SIP INVITE request containing an MCPTT group ID or TGI of a group document which is no longer cached on the MCPTT server."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.2]

"…

Upon receipt of a SIP INVITE request:

…

6) if the SIP INVITE request is a "SIP INVITE request for controlling function of an MCPTT group" and the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request is associated with an MCPTT group ID that has been regrouped, the MCPTT server:

…

c) if hosting the TGI, and the call to the temporary group is in progress, shall:

i) associate the MCPTT ID of the calling user with the temporary group call;

ii) interact with the media plane as specified in 3GPP TS 24.380 [5]; and

iii) exit this clause;

d) if hosting the TGI and the call to the temporary group is not in progress, shall subscribe to the "xcap-diff" event-package for the group document of the TGI as specified in 3GPP TS 24.481 [31], if not already subscribed;

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 4.8]

"In certain deployments, for example, in the case that the MCPTT operator uses the underlying SIP core infrastructure from the carrier operator, the MCPTT operator can prevent certain sensitive application data from being visible in the clear to the SIP layer. The following data are classed as sensitive application data:

- *MCPTT ID;*
- *MCPTT group ID;*
- *user location information;*
- *emergency, alert and imminent-peril indicators;*
- *access token (containing the MCPTT ID);*
- *MCPTT client ID; and*
- *functional alias.*

The above data is transported as XML content in SIP messages. in XML elements or XML attributes.

Data is transported in attributes in the following circumstances in the procedures in the present document:

…

- *an MCPTT ID in application/resource-lists+xml document included in an SIP INVITE request setting up a group call to a temporary group involving a non-controlling function that works in "Trusted Mode" according to IETF RFC 5366 [20], whereby the participants are returned to the controlling function in a MIME body of a SIP 403 (Forbidden) with the P-Refused-URI-List header field according to IETF RFC 5318 [36];*

3GPP TS 33.180 [78] describes a method to provide confidentiality protection of sensitive application data in elements by using XML encryption (i.e. xmlenc) and in attributes by using an attribute confidentiality protection scheme described in clause 6.6.2.3 of the present document. Integrity protection can also be provided by using XML signatures (i.e. xmlsig).

Protection of the data relies on a shared XML protection key (XPK) used to encrypt and sign data:

- *between the MCPTT client and the MCPTT server, the XPK is a client-server key (CSK); and*
- *between MCPTT servers and between MCPTT domains, the XPK is a signalling protection key (SPK).*

The CSK (XPK) and a key-id CSK-ID (XPK-ID) are generated from keying material provided by the key management server. Identity based public key encryption based on MIKEY-SAKKE is used to transport the CSK between SIP end-points. The encrypted CSK is transported from the MCPTT client to the MCPTT server when the MCPTT client performs service authorisation as described in clause 7 and is also used during service authorisation to protect the access token.

The SPK (XPK) and a key-id SPK-ID (XPK-ID) are directly provisioned in the MCPTT servers.

Configuration in the MCPTT client and MCPTT server is used to determine whether one or both of confidentiality protection and integrity protection are required.

The content can be re-encrypted and signed again using the SPK between MCPTT servers."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 5.3.2]

"…

Figure 5.3.2-5 shows the roles of the MCPTT server in a mutual aid relationship between a primary MCPTT system and a partner MCPTT with the use of a non-controlling MCPTT function of an MCPTT group within the partner MCPTT system. This can occur due to group re-grouping where the MCPTT server(s) of one or more of the constituent groups are homed on the partner system. If the primary MCPTT system and partner MCPTT system operate in a trusted mutual aid relationship, then the non-controlling MCPTT function of an MCPTT group can provide the identities of the users of the group to the controlling MCPTT function. If the primary MCPTT system and partner MCPTT system operate in an untrusted mutual aid relationship, then the non-controlling MCPTT function of an MCPTT group invites the users of the group on behalf of the controlling MCPTT function.

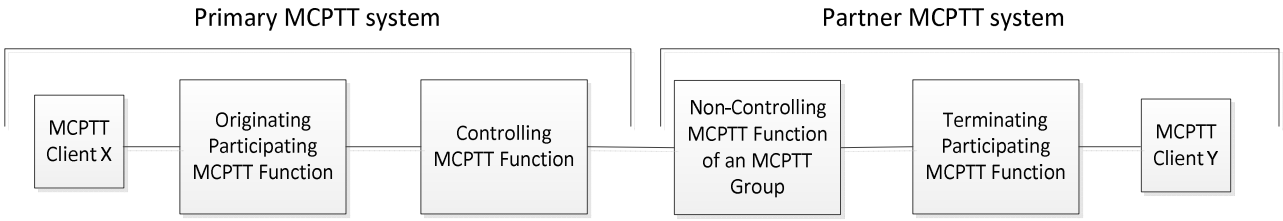


Figure 5.3.2-5: Mutual aid relationship between the primary MCPTT system and a partner MCPTT system involving the use of a non-controlling MCPTT function of an MCPTT group in the partner MCPTT system"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.2.2]

"…

The initial SIP INVITE request or SIP REFER request to establish an MCPTT chat group call or to rejoin an ongoing MCPTT call is not handled as an implicit floor control request message by the floor control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

…"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.3.3]

"When an MCPTT client leaves an MCPTT call and the MCPTT call remains ongoing with the other MCPTT clients, the floor control server follows a two-step procedure.

Step 1 The MCPTT server stops sending floor control messages and RTP media packets to the MCPTT client leaving the MCPTT call and, the MCPTT server discards floor control messages and RTP media packets received from the MCPTT client leaving the MCPTT call.

Step 2 When the application and signalling plane has determined that the MCPTT call with this floor participant has been released, the corresponding instance of the floor control server state machine for 'basic floor control operation towards the floor participant' is released.

When an MCPTT call is released, the floor control server follows a two-step procedure.

Step 1 The MCPTT server stops sending floor control messages and RTP media packets to all floor participants in the MCPTT call.

Step 2 When the application and signalling plane has determined that the MCPTT call has been released, the corresponding instance of the floor control server state machine for 'general floor control operation' are also terminated, along with any 'basic floor control operation towards the floor participant' state machines for the floor participants of this call.

The floor control server state machine for general floor control operation initiates the MCPTT call release depending on the release policy specified in 3GPP TS 24.379 [2]."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 14.3.4]

"If the mc_granted" fmtp attribute is included in an SDP offer, the controlling MCPTT function:

1. *if the MCPTT call is not a temporary group session, may use the SIP 200 (OK) response to indicate that the implicit floor request is granted; and*

2. *if the MCPTT call is a temporary group session, shall not indicate that the implicit floor request is granted.*

NOTE: A MCPTT group call is a temporary group session when the <on-network-temporary> element is present in the <list-service> element as specified in 3GPP TS 24.481 [12].

If the controlling MCPTT function grants the implicit request and decide to use the SIPP 200 (OK) response to grant the implicit request, the controlling MCPTT function shall include the "mc_granted" fmtp attribute in the SDP answer.

If the controlling MCPTT function decides not to grant the implicit request, the controlling MCPTT function shall not include the "mc_granted" fmtp attribute in the SDP answer."

5.2.2.3.3 Test description

5.2.2.3.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the originating participating MCPTT function
  - SS (MCPTT Server) provides the terminating participating MCPTT function
  - SS (MCPTT Server) provides the non-controlling MCPTT function

  SS-UE1 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.2.3.3.1-1.

  SS-UE2 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.2.3.3.1-1.

  SS-UE3 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.2.3.3.1-1.

  SS-UE4 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.2.3.3.1-1.

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the controlling MCPTT function
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2 and A.4.6.
- The SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client), the SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are registered and authorized to use the MCPTT system.
- Group A and Group B have been regrouped in Group T.
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.
- The SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are affiliated with Group B and are authorized to initiate group calls.
- Group B is controlled by the non-controlling MCPTT function.
- There are no ongoing calls in Group A and Group B.
- There is not a trusted mutual aid relationship between the partner MCPTT system and the primary MCPTT system

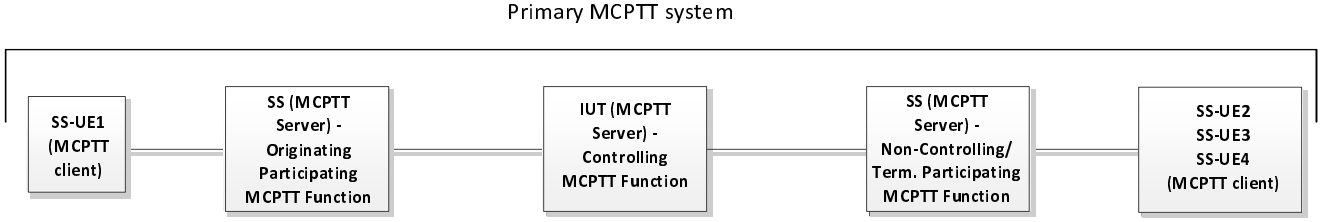


Figure 5.2.2.3.3.1-1: Functions of the testing components

5.2.2.3.3.2 Test procedure sequence

Table 5.2.2.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) (member of constituent Group A) to initiate an on-demand chat group call to a temporary group (Group T)	<--	SIP INVITE	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
2	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P
3	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	1	P
4	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
5	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE2 (MCPTT client) (member of constituent Group A) to join the chat group call in the temporary group (Group T)	<--	SIP INVITE	-	-
6	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P
7	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	1	P
8	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
9	The SS (MCPTT Server) providing the non-controlling MCPTT function sends a SIP INVITE request from constituent Group B to join the chat group call in the temporary group (Group T)	<--	SIP INVITE	-	-
10	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P
11	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) to the SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	SIP 200 (OK)	1	P
12	The SS (MCPTT Server) providing the non-controlling MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
13	The SS (MCPTT Server) providing the participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
14	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the participating MCPTT function for SS-UE1?	-->	Floor Granted	2	P
15	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the participating MCPTT function informing SS-UE2 that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
16	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the non-controlling MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
17	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
18	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
19	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
20	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	Floor Idle	2	P
21	The SS (MCPTT Server) providing the non-controlling MCPTT function sends a Floor Request message from SS-UE3 including a Track-Info field with the participant reference	<--	Floor Request		-
22	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the non-controlling MCPTT function including the Track-Info field received in the Floor Request?	-->	Floor Granted	2	P
23	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the participating MCPTT function informing SS-UE1 that the floor is taken by SS-UE3 (MCPTT client)?	-->	Floor Taken	2	P
24	Check: Does the IUT (MCPTT Server) send a Floor Taken message to the SS (MCPTT Server) providing the participating MCPTT function informing SS-UE2 that the floor is taken by SS-UE3 (MCPTT client)?	-->	Floor Taken	2	P
25	The SS (MCPTT Server) providing the non-controlling MCPTT function sends a Floor Release message from SS-UE3 with an acknowledgement required to release the floor and including a Track-Info field with the participant reference	<--	Floor Release		-
26	Check: Does the IUT (MCPTT Server) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P
27	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the participating MCPTT function?	-->	Floor Idle	2	P
28	Check: Does the IUT (MCPTT Server) send a Floor Idle message to the SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	Floor Idle	2	P
29	The SS (MCPTT Server) providing the participating MCPTT function sends a SIP BYE request from SS-UE1	<--	SIP BYE		-
30	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) response to SS (MCPTT Server) providing the participating MCPTT function?	-->	SIP 200 (OK)	3	P
31	The SS (MCPTT Server) providing the non-controlling MCPTT function sends a SIP BYE request	<--	SIP BYE		-
32	Check: Does the IUT (MCPTT Server) send a SIP 200 (OK) response to SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	SIP 200 (OK)	3	P
33	Check: Does the IUT (MCPTT Server) send a SIP BYE message for SS-UE2 to the SS (MCPTT Server) providing the participating MCPTT function to end the call?	-->	SIP BYE	3	P
34	The SS (MCPTT Server) providing the participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-

5.2.2.3.3.3 Specific message contents

Table 5.2.2.3.3.3-1: SIP INVITE from the SS (MCPTT-Server) (Step 1, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.3.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.3.3.3-1B			

Table 5.2.2.3.3.3-1A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.2.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions INITIAL_SDP_OFFER

Table 5.2.2.3.3.3-1B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.2.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions CHAT-GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_T_ID			
associated-group-id	Encrypted (see note) <associated-group-id> with mcpttURI set to px_MCPTT_Group_A_ID			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.2.3.3.3-2: SIP 200 (OK) from the IUT (MCPTT Server) (Steps 3, 7 and 11, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	any allowed value			
refresher	"uac"			
Message-body				
SDP message	SDP message as described in Table 5.2.2.3.3.3-2A			

Table 5.2.2.3.3.3-2A: SDP in SIP 200 (OK) from the IUT (MCPTT Server) (Table 5.2.2.3.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the MCPTT Server			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the MCPTT Server			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the MCPTT Server (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the MCPTT Server			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the MCPTT Server being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			

Table 5.2.2.3.3.3-3: SIP INVITE from the SS (MCPTT-Server) (Step 5, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.3.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.3.3.3-3A			

Table 5.2.2.3.3.3-3A: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.2.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions CHAT-GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_B			
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_T_ID			
associated-group-id	Encrypted (see note) <associated-group-id> with mcpttURI set to px_MCPTT_Group_A_ID			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.2.3.3.3-4: SIP INVITE from the SS (MCPTT-Server) (Step 9, Table 5.2.2.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Request-URI				
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server which is simulated by the SS, providing non-controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server which is simulated by the SS, providing non-controlling function.		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.3.3.3-4A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.3.3.3-4B			

Table 5.2.2.3.3.3-4A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.2.3.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions INITIAL_SDP_OFFER				
Media description[2]		Media description for media control		
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"10"			
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.2.3.3.3-4B: MCPTT-Info in SIP INVITE from SS (MCPTT Server) (Table 5.2.2.3.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
mcptt-calling-user-id	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_B_ID	The URI of the group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.2.3.3.3-5: SIP BYE from the SS (MCPTT-Server) (Steps 29 and 31, 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
From				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Local URI of the dialog (from the SS' point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Local tag of the dialog (from the SS' point of view)		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the SS' point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the SS' point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	70	The recommended initial value is 70 in IETF RFC 3261 [17].		

Table 5.2.2.3.3.3-6: SIP BYE from the IUT (MCPTT-Server) (Step 33, 5.2.2.3.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

Table 5.2.2.3.3.3-7: Floor Request (Step 13, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1

Table 5.2.2.3.3.3-8: Floor Granted (Step 14, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

Table 5.2.2.3.3.3-9: Floor Taken (Steps 15 and 16, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.2.3.3.3-10: Floor Release (Steps 17, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.2.3.3.3-11: Floor Ack (Step 18, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 5.2.2.3.3.3-12: Floor Idle (Steps 19, 20, 27 and 28, Table 5.2.2.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.2.3.3.3-13: Floor Request (Step 21, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Track Info			
Queueing Capability	"1"	The client supports queueing	
Participant Type Value	px_MCX_User_C_ParticipantType	ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1	
Floor Participant Reference	0x00000001	The selected value is randomly chosen - 32 bit value	

Table 5.2.2.3.3.3-20: Floor Granted (Step 22, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Track Info	The same value as received in Floor Request		
Queueing Capability	"1"		
Participant Type Value	px_MCX_User_C_ParticipantType		
Floor Participant Reference	0x00000001		

Table 5.2.2.3.3.3-21: Floor Taken (Steps 23 and 24, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_C			

Table 5.2.2.3.3.3-22: Floor Release (Step 25, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	
Track Info			
Queueing Capability	"1"	The client supports queueing	
Participant Type Value	px_MCX_User_C_ParticipantType	ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1	
Floor Participant Reference	0x00000001	The selected value is randomly chosen - 32 bit value	

Table 5.2.2.3.3.3-23: Floor Ack (Step 26, Table 5.2.2.3.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	
Track Info			
Queueing Capability	"1"	The client supports queueing	
Participant Type Value	px_MCX_User_C_ParticipantType	ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1	
Floor Participant Reference	0x00000001	The selected value is randomly chosen - 32 bit value	

5.2.2.4 MCPTT Server - MCPTT Server / Chat Group Call / Temporary Group / Non-Controlling server / Untrusted mode

5.2.2.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE3 (MCPTT client) initiates a chat group call to a temporary group and the SS-UE1
(MCPTT client) invited to join the chat group call in the temporary group and the SS-UE4 (MCPTT
client) joins the chat group call in the temporary group }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the chat group call by
sending SIP messages for all participants }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT chat group call to a temporary group }
ensure that {
  when { the SS-UE1 (MCPTT client), SS-UE3 (MCPTT client)  and SS-UE4 (MCPTT client) engage in
communication }
    then { IUT (MCPTT Server) enforces floor control (Floor Taken, Floor Idle, Floor Ack, Floor
Granted, Floor Request, Floor Release) by sending messages via the SS (MCPTT Server) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT chat group call to a temporary group }
ensure that {
  when { the SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) end the chat group call }
    then { IUT (MCPTT Server) providing the non-controlling MCPTT function responds to the SS (MCPTT
Server) by sending a SIP 200 (OK) message and providing the controlling MCPTT function sends a SIP
BYE message to SS (MCPTT Server) }
            }
```

5.2.2.4.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 10.1.2.5.1.1, 10.1.2.5.1.2, 10.1.2.5.1.3, 10.1.2.5.1.4, 10.1.2.5.1.5, 10.1.2.4.1.1, 10.1.2.5.1.8, 6.3.5.1, 6.3.5.2, 6.3.4.2.2.2, 6.3.4.2.1, 6.3.8.1, 4.8, 5.3.2, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 6.5.2.3.1, 6.5.2.2, 6.5.4.2, 6.5.4.3, 6.5.4.5, 6.5.4.6, 6.5.4.8, 6.5.4.9, 6.5.4.14, 6.5.4.15, 6.5.3 and 14.3.4. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.5.1.1]

"When receiving the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" the MCPTT server can be acting as a controller MCPTT function in an ongoing chat group call or, if a chat group call is not ongoing, be initiated as an non-controlling MCPTT function and invite MCPTT users.

If a chat group call is not ongoing the MCPTT server shall perform the actions specified in clause 10.1.2.5.1.2.

If the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" is received when a chat group call is ongoing, the controlling MCPTT function may switch from operating in a controlling MCPTT function mode to operate in a non-controlling MCPTT function mode as specified in clause 10.1.2.5.1.3.

When operating in the non-controlling mode and a SIP BYE request is received from the controlling MCPTT function, the non-controlling MCPTT function shall change from operating in the non-controlling mode to operating in the controlling mode as specified in clause 10.1.2.5.1.4."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.5.1.2]

"Upon receipt of a "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" and if a chat group call is not ongoing, the non-controlling MCPTT function of an MCPTT group:

NOTE 1: The Contact header field of the SIP INVITE request contains the isfocus media feature tag.

…

4) if the partner MCPTT system does not have a mutual aid relationship with the primary MCPTT system identified by the contents of the P-Asserted-Identity, shall reject the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" with a SIP 403 (Forbidden) response, with warning text set to "128 isfocus already assigned" in a Warning header field as specified in clause 4.4, and shall not process the remaining steps;

5) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [4];

6) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.4.2.2.2 before continuing with the rest of the steps;

7) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.4.2.1;

8) shall interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.3.5; and

NOTE 2: Resulting media plane processing is completed before the next step is performed.

9) shall send a SIP 200 (OK) response to the controlling MCPTT function according to 3GPP TS 24.229 [4]."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.5.1.3]

"Upon receipt of a "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" and if a chat group call is already ongoing, the non-controlling MCPTT function of an MCPTT group:

NOTE 1: The Contact header field of the SIP INVITE request contains the isfocus media feature tag.

…

3) if the partner MCPTT system does not have a mutual aid relationship with the primary MCPTT system identified by the contents of the P-Asserted-Identity, shall reject the "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" with a SIP 403 (Forbidden) response, with warning text set to "128 isfocus already assigned" in a Warning header field as specified in clause 4.4, and shall not process the remaining steps;

4) shall cache the content of the SIP INVITE request, if received in the Contact header field and if the specific feature tags are supported;

5) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [4];

6) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.4.2.2.2 before continuing with the rest of the steps;

7) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.4.2.1;

8) shall instruct the media plane to initialise the switch to the non-controlling mode as specified in 3GPP TS 24.380 [5] clause 6.5.2.3;

NOTE 2: Resulting media plane processing is completed before the next step is performed. The media plane indicates the state of the floor and if the state is "floor-taken", information about the current speaker(s).

9) if the media plane provided information about the current speaker(s), cache the information about the current speaker(s); and

10) shall send a SIP 200 (OK) response to the controlling MCPTT function according to 3GPP TS 24.229 [4].

Upon receipt of the SIP ACK request, the non-controlling MCPTT function of an MCPTT group:

1) if information about a current speaker(s) is cached:

a) shall generate a SIP INFO request as specified in clause 6.3.4.1.3; and

b) shall send the SIP INFO request to the controlling MCPTT function as specified in 3GPP TS 24.229 [4];

2) shall instruct the media plane to finalise the switch to the non-controlling mode as specified in 3GPP TS 24.380 [5] clause 6.3.5.3; and

3) if at least one of the MCPTT clients in the chat group session has a subscription to the conference event package, shall subscribe to the conference event package from the controlling MCPTT function as specified in clause 10.1.3.5.3."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.5.1.4]

"Upon receipt of a SIP BYE request, the non-controlling MCPTT function of an MCPTT group:

1) if keeping the chat group call active is according to the release policy in clause 6.3.8.1, shall request media plane to switch to controlling mode as specified in 3GPP TS 24.380 [5] clause 6.3.5;

NOTE 1: Resulting media plane processing is completed before the next step is performed.

2) shall send a SIP 200 (OK) response to the SIP BYE request; and

3) if at least one MCPTT client has subscribed to the conference package, shall send a NOTIFY request to all participants with a subscription to the conference event package as specified in clause 10.1.3.5.2.

NOTE 2: The SIP NOTIFY request will indicate that all participants, with the exception of the MCPTT users belonging to the constituent MCPTT group hosted by the non-controlling MCPTT function, have left the group session."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.5.1.5]

"When acting in the non-controlling connection mode when receiving of a "SIP INVITE request for controlling MCPTT function of an MCPTT group" containing a group identity identifying a constituent chat MCPTT group being part of the temporary group call, the non-controlling MCPTT function shall act as a controlling MCPTT function towards the MCPTT client and shall perform the actions in the clause 10.1.2.4.1.1 with the following clarifications:

1) *the MCPTT session identity in the Contact header field of the SIP 200 (OK) response shall be the MCPTT session identity generated by the non-controlling MCPTT function; and*

2) *the clause 10.1.3.5.2 shall be used when sending the SIP NOTIFY request for subscriptions to the conference event package."*

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.4.1.1]

"In the procedures in this clause:

1) *MCPTT ID in an incoming SIP INVITE request refers to the MCPTT ID of the originating user from the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;*

2) *group identity in an incoming SIP INVITE request refers to the group identity from the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the incoming SIP INVITE request;*

3) *MCPTT ID in an outgoing SIP INVITE request refers to the MCPTT ID of the called user in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the outgoing SIP INVITE request;*

4) *emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and*

5) *alert indication in an incoming SIP INVITE request refers to the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.*

Upon receipt of a "SIP INVITE request for controlling MCPTT function of an MCPTT group" containing a group identity identifying a chat MCPTT group, the controlling MCPTT function:

…

4) *shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2 and continue with the rest of the steps if the checks in clause 6.3.5.2 succeed;*

…

10) *shall create a chat group session and allocate an MCPTT session identity for the chat group session if the MCPTT chat group session identity does not already exist, and may handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;*

…

15) *shall accept the SIP request and generate a SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [4];*

16) *shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [4] with the clarifications specified in clause 6.3.3.2.1 unless the procedures of clause 6.3.3.1.8 were performed in step 13)a) or step 14)a) above;*

17) *should include the Session-Expires header field and start supervising the SIP session according to IETF RFC 4028 [7]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";*

18) *shall include the "timer" option tag in a Require header field;*

19) shall include the following in a Contact header field:

a) the g.3gpp.mcptt media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";

c) the MCPTT session identity; and

d) the media feature tag isfocus;

20) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [23];

21) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCPTT emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

22) if the received SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCPTT emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

23) shall interact with media plane as specified in 3GPP TS 24.380 [5];

24) shall send the SIP 200 (OK) response to the MCPTT client according to 3GPP TS 24.229 [4]; and

25) if the chat group session was already ongoing and if at least one of the participants has subscribed to the conference event package, shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as specified in clause 10.1.3.4.2.

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 10.1.2.5.1.8]

"Upon receiving a "SIP INVITE request for controlling MCPTT function of an MCPTT group" when a chat group session is not ongoing, the non-controlling MCPTT-function shall:

NOTE 1: The difference between a "SIP INVITE request for controlling MCPTT function of an MCPTT group" and a "SIP INVITE request for non-controlling MCPTT function of an MCPTT group" is that the latter SIP INVITE request contains the isfocus media feature tag in the Contact header field.

…

4) shall retrieve the group document from the group management server for the MCPTT group ID contained in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2 and continue with the rest of the steps if the checks in clause 6.3.5.2 succeed;

NOTE 2: If the checks are not successful, the SIP response to the "SIP INVITE request for controlling MCPTT function of an MCPTT group" is already sent in the clause 6.3.5.2.

5) shall cache the content of the SIP INVITE request;

6) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [4];

7) shall authorize the MCPTT user in the <mcptt-calling-user-id> element in the application/vnd.3gpp.mcptt-info+xml MIME body of the "SIP INVITE request for controlling MCPTT function of an MCPTT group" as specified in clause 6.3.5.2, if the MCPTT user is unauthorized to join a chat group session, the non-controlling MCPTT function shall send a SIP 403 (Forbidden) response with the warning text set to "106 user not authorised to join chat group" in a Warning header field as specified in clause 4.4.

8) shall generate a SIP INVITE request to the controlling MCPTT function as specified in clause 6.3.4.1.4; and

9) shall send the SIP INVITE request to the controlling MCPTT function as specified in 3GPP TS 24.229 [4].

Upon receipt of a SIP 2xx response to the SIP INVITE request sent to the controlling MCPTT function as specified above, the non-controlling MCPTT function:

1) shall send the SIP ACK request to the controlling MCPTT function as specified in 3GPP TS 24.229 [4];

2) shall generate a SIP 200 (OK) to the "SIP INVITE request for controlling MCPTT function of an MCPTT group" as specified in 3GPP TS 24.229 populated as follows:

a) shall include an SDP answer as specified in clause 6.3.4.2.1 based on the SDP answer in the SIP 200 (OK) response;

b) shall include the public service identifier of the non-controlling MCPTT function in the P-Asserted-Identity header field; and

c) shall include the warning text set to "148 group is regrouped" in a Warning header field as specified in clause 4.4; and

NOTE 3: As long as the MCPTT group is regrouped the floor control messages in the media plane includes a grouped regrouped indication as specified in 3GPP TS 24.380 [5].

3) shall start acting as a non-controlling MCPTT function and interact with the media plane as specified in 3GPP TS 24.380 [5] clause 6.5.

Upon receipt of other final SIP responses with the exception of the SIP 2xx response to the INVITE request sent to the controlling MCPTT function as specified above, the non-controlling MCPTT function:

1) shall send the SIP ACK response to the controlling MCPTT function as specified in 3GPP TS 24.229 [4]; and

2) perform the actions in the clause 10.1.1.5.2.4.

NOTE 4: Regardless if the controlling MCPTT function accepts or rejects the SIP INVITE request sent above the prearranged group session continues to be initiated with only the members of the group homed on the non-controlling MCPTT function of the group being invited to the group call."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.1]

"This clause describes how an MCPTT server accesses a group document from a group management server. The MCPTT server which accesses a group document performs the role of a controlling MCPTT function or performs the role of a non-controlling MCPTT function of an MCPTT group when accessing a group document. In such cases, for a group call:

…

The MCPTT server can cache the group document associated with an MCPTT group or temporary group, and can subscribe to be notified of changes to the group document associated with an MCPTT group or temporary group as specified in 3GPP TS 24.481 [31].

NOTE 1: During the group regrouping operation as specified in 3GPP TS 24.481 [31], the controlling MCPTT function is notified of the constituent MCPTT group identities associated with the TGI.

If the group data associated with an MCPTT group ID or TGI cached in the MCPTT server is removed, the MCPTT server re-subscribes for changes in the group information associated with the MCPTT group ID or TGI.

NOTE 2: Re-subscription can occur prior to the receipt of an SIP INVITE request containing an MCPTT group ID or TGI of a group document which is no longer cached on the MCPTT server."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.5.2]

"…

Upon receipt of a SIP INVITE request:

…

6) if the SIP INVITE request is a "SIP INVITE request for controlling function of an MCPTT group" and the group document for the group identity in the <mcptt-request-uri> element of the application/vnd.3gpp.mcptt-info MIME body of the SIP INVITE request is associated with an MCPTT group ID that has been regrouped, the MCPTT server:

…

c) if hosting the TGI, and the call to the temporary group is in progress, shall:

i) associate the MCPTT ID of the calling user with the temporary group call;

ii) interact with the media plane as specified in 3GPP TS 24.380 [5]; and

iii) exit this clause;

d) if hosting the TGI and the call to the temporary group is not in progress, shall subscribe to the "xcap-diff" event-package for the group document of the TGI as specified in 3GPP TS 24.481 [31], if not already subscribed;

…"

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.4.2.2.2]

"When sending a SIP 200 (OK) response, the non-controlling MCPTT function of an MCPTT group:

1) shall generate the SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [4];

2) shall include the Session-Expires header field and start supervising the SIP session according to rules and procedures of IETF RFC 4028 [7], "UAS Behavior". The "refresher" parameter in the Session-Expires header field shall be set to "uac";

3) shall include the option tag "timer" in a Require header field;

4) shall include the public service identity of the non-controlling MCPTT function in the P-Asserted-Identity header field;

5) shall include the following in the Contact header field:

a) the g.3gpp.mcptt media feature tag; and

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt";

6) shall include Warning header field(s) received in incoming responses to the SIP INVITE request;

7) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [23]; and

8) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcptt-called-party-id> element set to the constituent MCPTT group ID and the <floor-state> element set to the state of the floor."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.4.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [4], the non-controlling MCPTT function of an MCPTT group:

1) for the accepted media stream in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the non-controlling MCPTT function; and

2) for the accepted media plane control channel, if present in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the non-controlling MCPTT function; and

b) shall include 'fmtp' attributes as specified in 3GPP TS 24.380 [5] clause 14."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 6.3.8.1]

"If:

1) the call is a pre-arranged group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 24.380 [5] expired;

2) there are only one or no participants in the MCPTT session;

3) the call is a pre-arranged group call and if it is according to local policy, the initiator of the group call leaves the MCPTT session;

4) the minimum number of affiliated MCPTT group members is not present;

5) timer TNG3 (group call timer) expires; or

6) the call is a broadcast group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 24.380 [5] expired;

the controlling MCPTT function shall release the MCPTT session for the group call."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 4.8]

"In certain deployments, for example, in the case that the MCPTT operator uses the underlying SIP core infrastructure from the carrier operator, the MCPTT operator can prevent certain sensitive application data from being visible in the clear to the SIP layer. The following data are classed as sensitive application data:

- *MCPTT ID;*
- *MCPTT group ID;*
- *user location information;*
- *emergency, alert and imminent-peril indicators;*
- *access token (containing the MCPTT ID);*
- *MCPTT client ID; and*
- *functional alias.*

The above data is transported as XML content in SIP messages. in XML elements or XML attributes.

Data is transported in attributes in the following circumstances in the procedures in the present document:

…

- *an MCPTT ID in application/resource-lists+xml document included in an SIP INVITE request setting up a group call to a temporary group involving a non-controlling function that works in "Trusted Mode" according to IETF RFC 5366 [20], whereby the participants are returned to the controlling function in a MIME body of a SIP 403 (Forbidden) with the P-Refused-URI-List header field according to IETF RFC 5318 [36];*

3GPP TS 33.180 [78] describes a method to provide confidentiality protection of sensitive application data in elements by using XML encryption (i.e. xmlenc) and in attributes by using an attribute confidentiality protection scheme described in clause 6.6.2.3 of the present document. Integrity protection can also be provided by using XML signatures (i.e. xmlsig).

Protection of the data relies on a shared XML protection key (XPK) used to encrypt and sign data:

- *between the MCPTT client and the MCPTT server, the XPK is a client-server key (CSK); and*
- *between MCPTT servers and between MCPTT domains, the XPK is a signalling protection key (SPK).*

The CSK (XPK) and a key-id CSK-ID (XPK-ID) are generated from keying material provided by the key management server. Identity based public key encryption based on MIKEY-SAKKE is used to transport the CSK between SIP end-points. The encrypted CSK is transported from the MCPTT client to the MCPTT server when the MCPTT client performs service authorisation as described in clause 7 and is also used during service authorisation to protect the access token.

The SPK (XPK) and a key-id SPK-ID (XPK-ID) are directly provisioned in the MCPTT servers.

Configuration in the MCPTT client and MCPTT server is used to determine whether one or both of confidentiality protection and integrity protection are required.

The content can be re-encrypted and signed again using the SPK between MCPTT servers."

[ETSI TS 124 379 [4] (3GPP TS 24.379), clause 5.3.2]

"…

Figure 5.3.2-5 shows the roles of the MCPTT server in a mutual aid relationship between a primary MCPTT system and a partner MCPTT with the use of a non-controlling MCPTT function of an MCPTT group within the partner MCPTT system. This can occur due to group re-grouping where the MCPTT server(s) of one or more of the constituent groups are homed on the partner system. If the primary MCPTT system and partner MCPTT system operate in a trusted mutual aid relationship, then the non-controlling MCPTT function of an MCPTT group can provide the identities of the users of the group to the controlling MCPTT function. If the primary MCPTT system and partner MCPTT system operate in an untrusted mutual aid relationship, then the non-controlling MCPTT function of an MCPTT group invites the users of the group on behalf of the controlling MCPTT function.

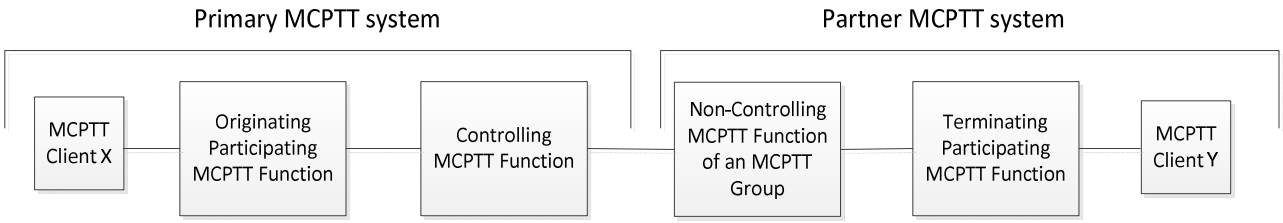


Figure 5.3.2-5: Mutual aid relationship between the primary MCPTT system and a partner MCPTT system involving the use of a non-controlling MCPTT function of an MCPTT group in the partner MCPTT system"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.2.3.1]

"The switching from working in a controlling MCPTT functional mode to a non-controlling MCPTT functional mode is a 2-step procedure.

Step 1 The controlling MCPTT function prepares for acting as a non-controlling MCPTT function. The step 1 procedure is specified in subclause 6.5.2.3.2.

Before continuing with step 2 the application and signalling plane needs to receive a confirmation that the SIP session between the floor control server and the interface to the floor control server is established.

Step 2 The controlling MCPTT functions starts acting as a non-controlling MCPTT function. The step 2 procedure is specified in subclause 6.5.2.3.3."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.2.2]

"When receiving an indication from the application and signalling plane that a group session is initiated, the floor control server interface:

1. *shall initiate and store a message sequence number value with the value to be used in the Message Sequence Number field in the Floor Idle and Floor Taken messages;*

2. *shall for each MCPTT client in the MCPTT group controlled by the non-controlling MCPTT function that are participating in the session:*

    a. *generate a random temporary identifier between '0' and '4294967295';*

    b. *store an association between the generated temporary identifier and the floor participant interface;*

    c. *store information about capabilities negotiated in the "mc_queueing" and "mc_priority" fmtp attributes as described in clause 14;*

    d. *store information whether the MCPTT client requested privacy or not; and*

    e. *initiate an instance of the 'floor participant state transition' state machine as specified in subclause 6.5.5; and*

3. *shall perform the actions in the subclause 6.5.4.*

When receiving an indication from the application and signalling plane that an MCPTT client has accepted an invitation to the session, the floor participant interface shall perform the actions in subclause 6.5.5."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.2]

"Upon receiving a Floor Request message from one floor participant interface, the floor control server interface:

1. *shall forward the Floor Request message to the controlling MCPTT function. The Floor Request message:*

    a. *shall include all fields included by the floor participant;*

    b. *if a Track Info field is included, shall include the temporary identifier at the end of the <Floor Participant Reference> value item; and*

    c. *if a Track Info field is not included, shall include a Track Info field populated as follows:*

        i. *shall include the "mc_queueing" fmtp attribute value negotiated as specified in clause 14 in the <Queueing Capability> value;*

        ii. *shall include a <Participant Type> value based on the <participant-type> element specified in 3GPP TS 24.481 [12], if value in the <participant-type> element is available, otherwise set the <Participant Type> value to "unknown"; and*

        iii. *shall include the temporary identifier as the first <Floor Participant Reference> value; and*

2. *if the value of the <Queueing Capability> in the Track Info is '1' (the floor participant in the MCPTT client supports queueing), shall store the outgoing Floor Request message in the passive floor request queue."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.3]

"Upon receiving a Floor Release message from one floor participant interface, the floor control server interface:

NOTE: A Floor Release message can be received from the permitted floor participant and from any participant that is queued in the floor control server.

1. *shall forward a Floor Release message to the controlling MCPTT function. The Floor Release message:*

    a. *shall include all fields included by the floor participant in the Floor Release message;*

    b. *if a Track Info field is included, shall include the temporary identifier at the end of the <Floor Participant Reference> value item; and*

    c. *if a Track Info field is not included, shall include a Track Info field as follows:*

        i. *shall include the "mc_queueing" fmtp attribute value negotiated as specified in clause 14 in the <Queueing Capability> value; and*

        ii. *shall include the temporary identifier as the first <Floor Participant Reference> value; and*

2. *if a Floor Request message received from this floor participant is in the passive floor request queue, shall remove the floor request from the passive floor request queue."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.5]

"Upon receiving a Floor Ack message from one floor participant interface the floor control server interface:

1. *shall send the Floor Ack message towards the controlling MCPTT function. The Floor Ack message:*

   a. *shall include all fields included by the floor participant in the Floor Ack message;*

   b. *if Track Info field is included, shall include the temporary identifier at the end of the <Floor Participant Reference> value item; and*

   c. *if a Track Info field is not included, shall include a Track Info field with temporary identifier as the first <Floor Participant Reference>."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.6]

"Upon receiving a Floor Granted message sent from the controlling MCPTT function, the floor control server interface:

1. *shall send the Floor Granted to the floor participant interface identified by the <Participant Reference> value at the end of the Track Info field. The Floor Granted message:*

   a. *shall include the fields as received with the following exceptions:*

      i. *if the Track Info field only contains one <Participant Reference> value, shall remove the Track Info field from the outgoing Floor Granted message; and*

      ii. *if the Track Info field contains more than one <Participant Reference> value, shall remove the last <Participant Reference> value from the Track Info field from the outgoing Floor Granted message; and*

   b *if the Floor Indicator field is included in the Floor Granted message and the G-bit is set to '1' (Dual floor), shall store the SSRC of granted floor participant and associate the stored value with dual floor;*

2. *if:*

   a. *a SSRC of granted floor participant and associate the stored value with dual floor is not stored, shall send a Floor Taken message populated as specified in step d. below to all participant interfaces with the exception of the floor participant interface to which the Floor Granted message is sent;*

   b. *a SSRC of granted floor participant associate with dual floor is stored and if the Floor Indicator field is not included in the Floor Granted message or if the Floor Indicator field is included in the Floor Granted message and the G-bit is set to '0' (Not dual floor), shall send a Floor Taken message populated as specified in step d. below to all participant interfaces with the exception of:*

      i. *the floor participant interface to which the Floor Granted message is sent; and*

      ii. *the floor participants only listening to the overriding floor participant;*

   c. *the Floor Indicator field is included in the Floor Granted message and the G-bit is set to '1' (Dual floor):*

      i. *shall send a Floor Taken message populated as specified in step d. below to floor participants that will listen to the RTP media from the overriding MCPTT client according to local policy;*

NOTE 1: The MCPTT client overridden by the overriding MCPTT client is still sending voice (overridden). The list of floor participants that receive the overriding, overridden, or both transmissions is based on configuration.

   d. *The Floor Taken message:*

      i. *shall include the granted MCPTT user's MCPTT ID in the Granted Party's Identity field and may include the functional alias of the granted MCPTT user in the Functional Alias field, if privacy is not requested by the granted floor participant when the MCPTT client was invited to the session;*

NOTE 2: The privacy request was stored for each invited MCPTT client when the MCPTT client accepted the invitation as specified in subclause 6.5.2.

ii. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1;

iii. shall include the Permission to Request Floor field to '0', if the group call is a broadcast group call;

iv. may include the Permission to Request the Floor field set to '1', if the group call is not a broadcast group call; and

v. shall set the first bit in the subtype of the Floor Taken message to '0' (acknowledgement is not required); and

NOTE 3: A Floor Taken message sent to all participants does not require acknowledgement.

e. if the Floor Indicator field was included in the Floor Granted message, shall include the received Floor Indicator field; and

3. if the Floor Request message received from the floor participant is in the passive floor request queue, shall remove the floor request from the passive floor request queue; and

4 if the Floor Indicator field is included in the Floor Granted message and the G-bit is set to '1' (Dual floor), shall send a Floor Idle message to those floor participants that will only listen to RTP media from the overriding MCPTT client. The Floor Idle message:

a. shall include the Floor Indicator field as received in the Floor Granted message with the G-bit set to '0' (Not dual floor); and

b. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.8]

"Upon receiving a Floor Idle message sent from the controlling MCPTT function, the floor control server interface:

NOTE 1: The Floor Idle message can be either destined to floor participants in all MCPTT clients or is sent to the floor participant in a specific MCPTT client. In the latter case the Floor Idle message contains the Track Info field.

1. if the Floor Idle message contains a Track Info field;

a. shall use the <Participant Reference> value at the end of the Track Info field to identify the floor participant interface;

b. if:

i. the Track Info field only contains one <Participant Reference> value:

A. shall remove the Track Info field from the outgoing Floor Idle message;

B. shall increase the stored message sequence number value with 1; and

C. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and

ii. if the Track Info field contains more than one <Participant Reference> value, shall remove the last <Participant Reference> value from the Track Info field; and

c. shall send the Floor Idle message to the floor participant interface;

2. if the Floor Idle message does not contain a Track Info field;

a. shall set the first bit in the subtype of the Floor Idle message to '0' (acknowledgement is not required);

NOTE 2: A Floor Idle message sent to all participants does not require acknowledgement.

b if:

i. the Floor Indicator field is not included in the Floor Idle message or if the Floor Indicator field is included and the G-bit is set to '0' (Not dual floor), shall send the Floor Idle message to all floor participant interfaces. The Floor Idle message:

A. shall include received fields; and

B. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and

ii. the Floor Indicator field is included in the Floor Idle message and the G-bit is set to '1' (Dual floor),

A. shall send the Floor Idle message to floor participants listening to the overriding MCPTT client according to local policy. The Floor Idle message:

- *shall include received fields; and*
- *shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and*

B. shall discard the stored SSRC of granted floor participant associated with a dual floor;

c. shall send a Floor Ack message towards the controlling MCPTT function if the first bit in the subtype of the received Floor Idle message is set to '1' (acknowledgement is required) as specified in subclause 8.2.2. The Floor Ack message:

i. shall include the Source field set to '3' (the non-controlling MCPTT function is the source); and

ii. shall include the Message Type field set to '5' (Floor Idle); and

3. shall empty the passive floor request queue."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.9]

"Upon receiving a Floor Taken message sent from the controlling MCPTT function, the floor control server interface:

NOTE 1: The Floor Taken message can be either destined to floor participants in all MCPTT clients or is sent to the floor participant in a specific MCPTT client. In the latter case the Floor Taken message contains the Track Info field.

1. if the Floor Taken message contains a Track Info field;

a. shall use the <Participant Reference> value at the end of the Track Info field to identify the floor participant interface;

b. if the Track Info field only contains one <Participant Reference> value:

A. shall remove the Track Info field from the outgoing Floor Taken message; and

B. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1;

c. if the Track Info field contains more than one <Participant Reference> value, shall remove the last <Participant Reference> value from the Track Info field; and

d. shall send the Floor Taken message to the floor participant interface;

2. if the Floor Taken message does not contain a Track Info field:

a. shall set the first bit in the subtype of the Floor Taken message to '0' (acknowledgement is not required);

NOTE 2: A Floor Taken message sent to all participants does not require acknowledgement.

b. if the Floor Indicator field is included in the Floor Taken message and the G-bit is set to '1' (Dual floor), shall store the SSRC of granted floor participant and associate the stored SSRC with dual floor;

c. if there is no stored SSRC of granted floor participant associated with a dual floor, shall send the Floor Taken message to all floor participant interfaces. The Floor Taken message:

i. shall include all received fields with the exception of the Message Sequence Number field;

ii. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1;

d. if there is a stored SSRC of granted floor participant associated with a dual floor:

i. if the Floor Indicator field is not included in the Floor Taken message or if the Floor Indicator field is included in the Floor Taken message and the G-bit is set to '0' (Not dual floor), shall send the Floor Taken message to all floor participant interfaces with the exception of the floor participants only listening to RTP media from the overriding floor participant. The Floor Idle message:

A. shall include all received fields; and

B. shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and

ii. if the Floor Indicator field is included in the Floor Taken message and the G-bit is set to '1' (Dual floor):

A. shall send the Floor Taken message to all floor participants listening to the RTP media from the overriding MCPTT client according to local policy. The Floor Taken message:

- *shall include all received fields; and*
- *shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and*

B. shall send a Floor Idle message to those floor participants that will only listen to RTP media from the overriding MCPTT client according to local policy. The Floor Idle message:

- *shall include all received fields;*
- *shall include in the Message Sequence Number field the local <Message Sequence Number> value increased with 1; and*
- *shall update the received Floor Indicator field and set the G-bit to '0' and;*

e. shall send a Floor Ack message towards the controlling MCPTT function if the first bit in the subtype of the received Floor Taken message is set to '1' (acknowledgement is required) as specified in subclause 8.2.2. The Floor Ack message:

i. shall include the Source field set to '3' (the non-controlling MCPTT function is the source); and

ii. shall include the Message Type field set to '2' (Floor Taken);"

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.14]

"Upon receiving an MCPTT call release step 1 request from the application and signalling plane e.g. when the session is going to be released, the floor control interface:

1. *shall ignore floor control message from the floor control server;*
2. *shall request the media distributor to stop distributing RTP media packets to the network media interface of the MCPTT clients; and*
3. *shall ignore any floor control messages received from the floor participant interfaces."*

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.4.15]

"Upon receiving an MCPTT call release step 2 request from the application and signalling plane, the floor control server interface:

1. shall release all resources associated with this session."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 6.5.3]

"When an MCPTT client leaves an MCPTT call and the MCPTT call remains ongoing with the other MCPTT clients, the non-controlling MCPTT function of an MCPTT group follows a two-step procedure:

Step 1 The floor participant interface stops sending floor control messages and RTP media packets to the MCPTT client leaving the MCPTT call and the floor participant interface discards floor control messages and RTP media packets received from the MCPTT client leaving the MCPTT call; and

Step 2 When the application and signalling plane has determined that the session with this floor participant has been released, the corresponding instance of the 'floor participant state transition' state machine is released.

When an MCPTT call is released, the floor control server interface follows a two-step procedure:

Step 1 The floor control server interface stops sending floor control messages and RTP media packets to MCPTT clients in the MCPTT call; and

Step 2 When the application and signalling plane has determined that the MCPTT call has been released, resources in the floor control server interface are released, along with all 'floor participant state transition' state machines.

The non-controlling MCPTT function of an MCPTT group can initiate an MCPTT call release depending on the release policy specified in 3GPP TS 24.379 [2]."

[ETSI TS 124 380 [5] (3GPP TS 24.380), clause 14.3.4]

"If the mc_granted" fmtp attribute is included in an SDP offer, the controlling MCPTT function:

1. if the MCPTT call is not a temporary group session, may use the SIP 200 (OK) response to indicate that the implicit floor request is granted; and

2. if the MCPTT call is a temporary group session, shall not indicate that the implicit floor request is granted.

NOTE: A MCPTT group call is a temporary group session when the <on-network-temporary> element is present in the <list-service> element as specified in 3GPP TS 24.481 [12].

If the controlling MCPTT function grants the implicit request and decide to use the SIPP 200 (OK) response to grant the implicit request, the controlling MCPTT function shall include the "mc_granted" fmtp attribute in the SDP answer.

If the controlling MCPTT function decides not to grant the implicit request, the controlling MCPTT function shall not include the "mc_granted" fmtp attribute in the SDP answer."

5.2.2.4.3 Test description

5.2.2.4.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the participating MCPTT functions
  - SS (MCPTT Server) provides the controlling MCPTT function
- SS-UE1 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.2.4.3.1-1.

- SS-UE2 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.2.4.3.1-1.
- SS-UE3 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.2.4.3.1-1.
- SS-UE4 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 5.2.2.4.3.1-1.

IUT:

- IUT (MCPTT Server)
  - IUT (MCPTT Server) provides the non-controlling MCPTT function
- The IUT (MCPTT Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCPTT Server) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (MCPTT Server) is connected to SS (MCPTT Server) defined in Annex A, Figures A.4.2 and A:4.5.
- The SS-UE1 (MCPTT client), the SS-UE2 (MCPTT client), the SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are registered and authorized to use the MCPTT system.
- Group A and Group B have been regrouped in Group T
- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.
- The SS-UE3 (MCPTT client) and the SS-UE4 (MCPTT client) are affiliated with Group B and are authorized to initiate group calls.
- Group B is controlled by the non-controlling MCPTT function.
- There are no ongoing calls in Group A and Group B.
- There is not a trusted mutual aid relationship between the partner MCPTT system and the primary MCPTT system.

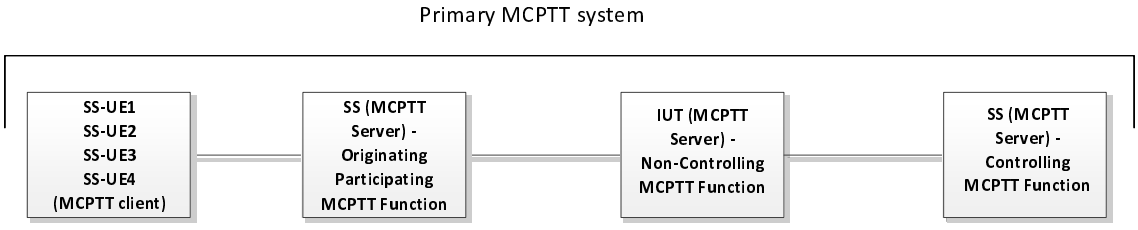


Figure 5.2.2.4.3.1-1: Functions of the testing components

5.2.2.4.3.2 Test procedure sequence

Table 5.2.2.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE3 (MCPTT client) (member of constituent Group B) to start a chat group call.	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P
3	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK) message?	-->	SIP 200 (OK)	1	P
4	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message.	<--	SIP ACK	-	-
5	Check: Does the IUT (MCPTT Server) send a SIP INVITE to the SS (MCPTT Server) providing the controlling MCPTT function?	-->	SIP INVITE	1	P
6	The SS (MCPTT Server) providing the controlling MCPTT function responds with a SIP 200 (OK) message.	<--	SIP 200 (OK)	-	-
7	Check: Does the IUT (MCPTT Server) send a SIP ACK to the SS (MCPTT Server) providing the controlling MCPTT function?	-->	SIP ACK	1	P
8	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE4 (MCPTT client) (member of constituent Group B) to join the chat group call.	<--	SIP INVITE	-	-
9	Check: Does the IUT (MCPTT Server) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P
10	Check: Does the IUT (MCPTT Server) respond with a SIP 200 (OK) message?	-->	SIP 200 (OK)	1	P
11	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message.	<--	SIP ACK	-	-
12	The SS (MCPTT Server) providing the controlling MCPTT function sends a Floor Taken to the IUT (MCPTT Server) informing that the floor is taken by SS-UE1 (MCPTT client)?	<--	Floor Taken	-	-
13	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the participating MCPTT function informing SS-UE3 that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
14	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the participating MCPTT function informing SS-UE4 that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
15	The SS (MCPTT Server) providing the controlling MCPTT function sends a Floor Idle message to the IUT (MCPTT Server)	<--	Floor Idle		-
16	Check: Does the IUT (MCPTT Server) send a Floor Idle message for SS-UE3 to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	2	P
17	Check: Does the IUT (MCPTT Server) send a Floor Idle message for SS-UE4 to the SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	Floor Idle	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
18	The SS (MCPTT Server) providing the participating MCPTT function sends a Floor Request message from SS-UE3	<--	Floor Request		-
19	Check: Does the IUT (MCPTT Server) send a Floor Request message from SS-UE3 including a Track-Info field with the participant reference?	-->	Floor Request	2	P
20	The SS (MCPTT Server) providing the controlling MCPTT function sends a Floor Granted message for SS-UE3 including the Track-Info field sent in the Floor Request.	<--	Floor Granted		-
21	Check: Does the IUT (MCPTT Server) send a Floor Granted message to the SS (MCPTT Server) providing the participating MCPTT function for SS-UE3 without the Track-Info field?	-->	Floor Granted	2	P
22	Check: Does the IUT (MCPTT Server) send a Floor Taken to the SS (MCPTT Server) providing the terminating MCPTT function informing SS-UE4 that the floor is taken by SS-UE3 (MCPTT client)?	-->	Floor Taken	2	P
23	The SS (MCPTT Server) providing the participating MCPTT function sends a Floor Release with an acknowledgement required message from SS-UE3 to release the floor.	<--	Floor Release		-
24	Check: Does the IUT (MCPTT Server) send a Floor Release message from SS-UE3 with an acknowledgement required and including a Track-Info field with the participant reference?	-->	Floor Release	2	P
25	The SS (MCPTT Server) providing the controlling MCPTT function sends a Floor Ack for the Floor Release send by the IUT (MCPTT Server).	<--	Floor Ack		-
26	Check: Does the IUT (MCPTT Server) send a Floor Ack message to the SS (MCPTT Server) providing the participating MCPTT function for the Floor Release?	-->	Floor Ack	2	P
27	The SS (MCPTT Server) providing the controlling MCPTT function sends a Floor Idle message to the IUT (MCPTT Server)-	<--	Floor Idle		-
28	Check: Does the IUT (MCPTT Server) send a Floor Idle message for SS-UE3 to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	2	P
29	Check: Does the IUT (MCPTT Server) send a Floor Idle message for SS-UE4 to the SS (MCPTT Server) providing the non-controlling MCPTT function?	-->	Floor Idle	2	P
30	The SS (MCPTT Server) providing the participating MCPTT function sends a SIP BYE request from SS-UE3 to the IUT (MCPTT Server)	<--	SIP BYE		-
31	Check: Does the IUT (MCPTT Server) providing the non-controlling MCPTT function respond with a SIP 200 (OK) message?	-->	SIP 200 (OK)	3	P
32	The SS (MCPTT Server) providing the participating MCPTT function sends a SIP BYE request from SS-UE4 to the IUT (MCPTT Server)	<--	SIP BYE		-
33	Check: Does the IUT (MCPTT Server) providing the non-controlling MCPTT function respond with a SIP 200 (OK) message?	-->	SIP 200 (OK)	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
34	Check: Does the IUT (MCPTT Server) providing the non-controlling MCPTT function send a SIP BYE request to the SS (MCPTT Server) providing the controlling MCPTT function?	-->	SIP BYE	3	P
35	The SS (MCPTT Server) providing the controlling MCPTT function sends a SIP 200 (OK) response.	<--	SIP 200 (OK)		-

5.2.2.4.3.3 Specific message contents

Table 5.2.2.4.3.3-1: SIP INVITE from the SS (MCPTT-Server) (Step 1, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server providing the non-controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.4.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.4.3.3-1B			

Table 5.2.2.4.3.3-1A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.2.4.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions INITIAL_SDP_OFFER

Table 5.2.2.4.3.3-1B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 5.2.2.4.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions CHAT-GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_C			
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_B_ID			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 5.2.2.4.3.3-2: SIP 200 (OK) from the IUT (MCPTT Server) (Steps 3 and 10, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	any allowed value			
refresher	"uac"			
Message-body				
SDP message	SDP message as described in Table 5.2.2.4.3.3-2A			

Table 5.2.2.4.3.3-2A: SDP in SIP 200 (OK) from the IUT (MCPTT Server) (Table 5.2.2.4.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the MCPTT Server			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the MCPTT Server			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the MCPTT Server (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the MCPTT Server			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the MCPTT Server being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCPTT Server			

Table 5.2.2.4.3.3-3: SIP INVITE from the IUT (MCPTT-Server) (Step 5, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server which is simulated by the SS, providing controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server providing the non-controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the MCPTT Server which is simulated by the SS, providing controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_C	The URI of the MCPTT Server providing the non-controlling function.		
feature-param	"isfocus" not present			
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			
Session-Expires				
generic-param	any allowed value			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Max-Forwards				
value	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 5.2.2.4.3.3-3A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 5.2.2.4.3.3-3B			

Table 5.2.2.4.3.3-3A: SDP in SIP INVITE from the IUT (MCPTT-Server) (Table 5.2.2.4.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions INITIAL_SDP_OFFER				
Media description[2]		Media description for media control		
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"10"			
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.2.7.3.3-3B: MCPTT-Info in SIP INVITE from IUT (MCPTT Server) (Table 5.2.2.7.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-calling-group-id> with mcpttURI set to px_MCPTT_Group_T_ID	The URI of the temporary group		
mcptt-calling-user-id	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_Group_B_ID	The URI of the group		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 5.2.2.4.3.3-4: SIP 200 (OK) from the SS (MCPTT Server) (Step 6, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	any allowed value			
refresher	"uac"			
Message-body				
SDP message	SDP message as described in Table 5.2.2.7.3.3-4A			

Table 5.2.2.4.3.3-4A: SDP in SIP 200 (OK) from the SS (MCPTT Server) (Table 5.2.2.4.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]		Media description for media control		
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"10"			
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 5.2.2.4.3.3-5: SIP BYE from the SS (MCPTT-Server) (Steps 30 and 32, 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the MCPTT Server has sent earlier in the Contact header of a response within the same dialog	Contact URI of the MCPTT Server		
From				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Local URI of the dialog (from the SS' point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Local tag of the dialog (from the SS' point of view)		
To				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Remote URI of the dialog (from the SS' point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Remote tag of the dialog ID (from the SS' point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	70	The recommended initial value is 70 in IETF RFC 3261 [17].		

Table 5.2.2.4.3.3-6: SIP BYE from the IUT (MCPTT-Server) (Step 34, 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server) providing the controlling function.		
From				
addr-spec	Same URI of the MCPTT Server as used earlier in the dialog	Local URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the MCPTT Server as used earlier in the dialog	Local tag of the dialog (from the MCPTT Server's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the MCPTT Server's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the MCPTT Server's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

Table 5.2.2.4.3.3-7: Floor Taken (Steps 12, 13 and 14, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 5.2.2.4.3.3-8: Floor Taken (Step 22, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_C			

Table 5.2.2.4.3.3-9: Floor Idle (Steps 15-17 and 27-29, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 5.2.2.4.3.3-10: Floor Request (Step 18, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1

Table 5.2.2.4.3.3-11: Floor Granted (Step 21, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

Table 5.2.2.4.3.3-12: Floor Request (Step 19, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Track Info			
Queueing Capability	"1"	The client supports queueing	
Participant Type Value	px_MCX_User_C_ParticipantType	ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1	
Floor Participant Reference	Any allowed value	The selected value is randomly chosen - 32 bit value	

Table 5.2.2.4.3.3-13: Floor Granted (Step 20, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Track Info	The same value as received in Floor Request		

Table 5.2.2.4.3.3-14: Floor Release (Step 23, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 5.2.2.4.3.3-15: Floor Release (Step 24, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	
Track Info			
Queueing Capability	"1"	The client supports queueing	
Participant Type Value	px_MCX_User_C_ParticipantType	ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1	
Floor Participant Reference	Any allowed value	The selected value is randomly chosen - 32 bit value	

Table 5.2.2.4.3.3-16: Floor Ack (Step 25, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	
Track Info	The same value as received in Floor Release		

Table 5.2.2.4.3.3-17: Floor Ack (Step 26, Table 5.2.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

6 IWF - MCPTT Server operation

6.1 Group Calls

6.1.1 Pre-arranged Group Call

6.1.1.1 IWF - MCPTT Server / Pre-arranged Group Call / Automatic Commencement Mode / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / IWF Controlling Role

6.1.1.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the SS-UE1 (MCPTT client) initiates an on-demand pre-arranged group call with automatic
commencement mode and implicit floor control }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the call by sending SIP
messages }
            }
```

(2)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
automatic commencement mode and implicit floor control }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Ack, Floor Idle, Floor
Granted, Floor Revoke, Floor Deny, Floor Queue Position Info ) by sending messages to the SS (MCPTT
Server) }
            }
```

(3)

```
with { IUT (MCPTT Server) having established an MCPTT On-demand Pre-arranged Group Call with
automatic commencement mode and implicit floor control }
ensure that {
  when { the SS-UE1 (MCPTT client) upgrades the call to an emergency group call with implicit floor
request }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the emergency call by
sending SIP messages }
            }
```

(4)

```
with { IUT (MCPTT Server) having established an MCPTT emergency group call }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Granted, Floor Idle, Floor
Ack ) by sending messages to the SS (MCPTT Server) }
            }
```

(5)

```
with { IUT (MCPTT Server) having established an MCPTT emergency group call }
ensure that {
  when { the SS-UE1 (MCPTT client) downgrades the call to a normal group call }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the normal group call by
sending SIP messages }
            }
```

(6)

```
with { IUT (MCPTT Server) having established an MCPTT group call }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Idle, Floor Idle, Floor
Granted ) by sending messages to the SS (MCPTT Server) }
            }
```

(7)

```
with { IUT (MCPTT Server) having established an MCPTT group call }
ensure that {
  when { the SS-UE1 (MCPTT client)upgrades the call to imminent-peril group call }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the imminent-peril group
call by sending SIP messages }
            }
```

(8)

```
with { IUT (MCPTT Server) having established an MCPTT imminent-peril group call }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Idle, Floor Ack, Floor
Granted ) by sending messages to the SS (MCPTT Server) }
            }
```

(9)

```
with { IUT (MCPTT Server) having established an MCPTT imminent-peril group call }
ensure that {
  when { the SS-UE1 (MCPTT client) downgrades the call to a normal group call }
    then { IUT (MCPTT Server) interacts with SS (MCPTT Server) to set up the normal group call by
sending SIP messages }
            }
```

(10)

```
with { IUT (MCPTT Server) having established an MCPTT group call }
ensure that {
  when { the SS-UE1 (MCPTT client) and SS-UE2 (MCPTT client) engage in communication }
    then { IUT (MCPTT Server) forwards floor control (Floor Taken, Floor Idle, Floor Ack, Floor
Granted ) by sending messages to the SS (MCPTT Server) }
            }
```

(11)

```
with { IUT (MCPTT Server) having established an MCPTT group call }
ensure that {
  when { the SS-UE1 (MCPTT client) ends the pre-arranged group call }
    then { IUT (MCPTT Server) responds via the SS (MCPTT Server) by sending a SIP 200 (OK) message
to the client ending the call and sends a SIP BYE message towards SS-UE2 (MCPTT client) }
            }
```

6.1.1.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 129 379 [30] (3GPP TS 29.379), clauses 10.1.4.1.1, 6.6.3.1.1, 6.6.3.1.12, 10.1.4.3, 6.6.3.2.1, 6.6.7.1, 6.6.3.1.2, 10.1.4.4.2, 10.1.4.4.3, 10.1.4.7, 10.1.4.8, 6.6.3.1.3, 6.6.3.1.4, 6.6.3.1.7, 6.6.3.1.10, 6.3.3.1.11, 6.6.3.5.2, 6.4.1.1, 6.4.1.2, 6.4.1.3, 6.4.1.6, 6.4.1.7, 6.4.1.8, 6.4.1.10 and 6.4.1.11, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 5.3, 6.3.2.2 and 6.3.3. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.4.1.1]

"This clause describes the procedures for inviting an MCPTT user to an MCPTT session. The procedure is initiated by the IWF performing the controlling role as the result of a request from the LMR system or an action in clause 10.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The IWF performing the controlling role:

1) shall generate a SIP INVITE request as specified in clause 6.6.3.1.1;

2) shall set the Request-URI to the public service identity of the terminating participating MCPTT function associated to the MCPTT user to be invited;

NOTE 1: How the IWF performing the controlling role finds the address of the terminating MCPTT participating function is out of the scope of the current release.

NOTE 2: If the terminating MCPTT user is part of a partner MCPTT system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCPTT function.

3) shall set the P-Asserted-Identity header field to the public service identity of the IWF performing the controlling role;

4) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcptt-request-uri> element set to the MCPTT ID of the terminating user; and

b) the <mcptt-calling-group-id> element set to the group identity;

NOTE 3: The <mcptt-calling-user-id> is already included in the MIME body as a result of calling clause 6.6.3.1.1 in step 1).

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in 3GPP TS 24.379 [29], clause 6.3.3.1.1, with the IWF acting as the controlling MCPTT function;

6) if the in-progress emergency state of the group is set to a value of "true" the IWF performing the controlling role:

a) shall include a Resource-Priority header field populated with the values for an MCPTT emergency group call as specified in clause 6.6.3.1.12.

b) if the IWF needs to set the group state to emergency:

i) shall include in the outgoing SIP INVITE request in the application/vnd.3gpp.mcptt-info+xml MIME body an <emergency-ind> element set to a value of "true"; and

ii) if the IWF needs to set an emergency alert and the MCPTT group is authorised for the initiation of MCPTT emergency alerts as determined by the procedures of clause 6.6.3.1.8.1, shall populate the application/vnd.3gpp.mcptt-info+xml MIME body and the application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause 6.6.3.1.7. Otherwise, shall set the <alert-ind> element to a value of "false"; and

c) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the IWF performing the controlling role:

a) shall include a Resource-Priority header field populated with the values for an MCPTT imminent peril group call as specified in clause 6.6.3.1.12; and

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true"; and

8) shall send the SIP INVITE request towards the terminating network in accordance with 3GPP TS 24.229 [3].

Upon receiving a SIP 183 (Session Progress) response containing a Require header field with the option tag "100rel" and containing a P-Answer-State header field with the value "Unconfirmed" in response to the SIP INVITE request the IWF performing the controlling role:

1) shall send a SIP PRACK request towards the MCPTT client according to 3GPP TS 24.229 [3].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the IWF performing the controlling role:

1) shall interact with the media plane as specified in 3GPP TS 29.380 [31] clause 6.3;

2) shall send a SIP NOTIFY request to all MCPTT participants with a subscription to the conference event package as specified in clause 10.3; and

3) shall increment the local counter of the number of SIP 200 (OK) responses received from invited members, by 1."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.3.1.1]

"This clause is referenced from other procedures.

The IWF performing the controlling role shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [3].

The IWF performing the controlling role:

1) shall include in the Contact header field an MCPTT session identity for the MCPTT session with the g.3gpp.mcptt media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" according to IETF RFC 3840 [9];

2) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [5];

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [3]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [7] in the SIP INVITE request;

4) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with parameters "require" and "explicit" according to IETF RFC 3841 [5];

5) shall include a Referred-By header field with the public service identity of the IWF;

6) should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [6]. The refresher parameter shall be omitted;

7) shall include the Supported header field set to "timer";

9) may include an unmodified Priv-Answer-Mode header field;

10) if the request will not contain a Priv-Answer-Mode header field, shall include an Answer-Mode header field; and

11) may include an application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing INVITE request."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.3.1.12]

"This clause is referenced from other procedures.

The IWF performing the controlling role may populate the Resource-Priority header field to assist interworked MC systems in setting bearer priorities. The IWF can set emergency, imminent peril and normal priorities for private and group calls. The priority values and namespaces are as specified in IETF RFC 8101 [23].

NOTE: How the IWF obtains the values for Resource-Priority header fields is out of scope of the present document."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.4.3]

"Upon receiving a SIP BYE request the IWF performing the controlling role shall follow the procedures as specified in clause 6.6.3.2.1."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.3.2.1]

"Upon receiving a SIP BYE request the IWF performing the controlling role:

1) shall interact with the media plane as specified in clause 6.3 in 3GPP TS 29.380 [31] for releasing the media plane resource associated with the SIP session towards the MCPTT client;

NOTE: The IWF performing the non-controlling role is also regarded as a MCPTT client in a temporary MCPTT group session.

2) shall generate a SIP 200 (OK) response and send the SIP response towards the MCPTT client according to 3GPP TS 24.229 [3];

3) shall check the MCPTT session release policy as specified in clause 6.6.7.1 and clause 6.6.7.2 whether the MCPTT session needs to be released for each participant of the MCPTT session;

4) if release of the MCPTT session is required:

a) shall perform the procedures as specified in the clause 6.6.3.1.2 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcptt-info+xml MIME body, copy the application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and

5) if a release of the MCPTT session is not required:

a) shall send a SIP NOTIFY request to all remaining MCPTT clients in the MCPTT session with a subscription to the conference event package as specified in 3GPP TS 24.379 [29], clause 10.1.3.4.2 with the IWF acting as the controlling MCPTT function.

Upon receiving a SIP 200 (OK) response to the SIP BYE request the IWF performing the controlling shall interact with the media plane as specified in clause 6.3 in 3GPP TS 29.380 [31] for releasing media plane resources associated with the SIP session with the MCPTT participant."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.7.1]

"If:

1) the call is a pre-arranged group call and if the IWF performing the controlling role receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 29.380 [31] expired and if there is at least one participant of the prearranged group call that is an MCPTT user;

2) there are only one or no participants in the call, including both MCPTT participants and participants homed in the IWF;

3) if the call is a pre-arranged group call and if it is according to local policy, the initiator of the group call leaves the call;

4) less than the minimum number of affiliated group members is present;

5) timer TNG3 (group call timer) expires; or

6) the call is a broadcast group call and if the controlling MCPTT function receives an indication from the media plane that the T4 (Inactivity) timer specified in 3GPP TS 29.380 [31] expired;

then the IWF performing the controlling role;

1) shall release the MCPTT session for the group call if any participants are MCPTT users."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.3.1.2]

"When the IWF performing the controlling role needs to remove an MCPTT participant:

1) shall interact with the media plane as specified in 3GPP TS 29.380 [31] for the MCPTT session release;

2) shall generate a SIP BYE request according to 3GPP TS 24.229 [3]; and

3) shall send the SIP BYE request to the MCPTT participants according to 3GPP TS 24.229 [3].

If timer TNG3 (group call timer) has not expired, then when the last participant is removed from the MCPTT session, the IWF performing the controlling role shall stop timer TNG3 (group call timer).

When the MCPTT group session needs to be released, the IWF performing the controlling role shall send SIP BYE requests as described in this clause, to all MCPTT participants of the group session.

Upon receiving a SIP 200 (OK) response to a SIP BYE request the IWF performing the controlling role shall interact with the media plane as specified in clause 6.3 in 3GPP TS 29.380 [31] for releasing media plane resources associated with the session with the MCPTT clients."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.4.4.2]

"When the MCPTT session for group call needs to be released as specified in clause 6.6.7.1, the IWF performing the controlling role, shall follow the procedures in clause 6.6.3.1.2."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.4.4.3]

"When an MCPTT client needs to be removed from the MCPTT session (e.g. due to de-affiliation or admitting a higher priority user), the IWF performing the controlling role shall follow the procedures in clause 6.6.3.1.2.

After successful removing the MCPTT client from the MCPTT session, the IWF performing the controlling role may generate a notification to the MCPTT clients, which have subscribed to the conference event package that an MCPTT user has been removed from the MCPTT session, as specified in clause 6.6.3.4 and send the SIP NOTIFY request to the MCPTT client according to 3GPP TS 24.229 [3]."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.4.7]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a SIP re-INVITE request for an MCPTT session identity identifying an on-demand prearranged MCPTT group session, the IWF performing the controlling role:

…

2) if received SIP re-INVITE request includes an application/vnd.3gpp.mcptt-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in 3GPP TS 24.379 [29], clause 6.3.3.1.17, with the IWF acting as the controlling MCPTT function;

…

6) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "true" and is an authorised request to initiate an MCPTT emergency group call as determined by clause 6.6.3.1.8.2, the IWF performing the controlling role:

i) shall cache the MCPTT ID of the MCPTT user that has initiated an MCPTT emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCPTT emergency alert meeting the conditions specified in clause 6.6.3.1.8.1, shall cache the MCPTT ID of the MCPTT user that has initiated an MCPTT emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "true":

A) for each of the other affiliated MCPTT members of the group generate a SIP MESSAGE request notification of the MCPTT user's emergency indication as specified in 3GPP TS 24.379 [29], clause 6.3.3.1.11, with the IWF acting as the controlling MCPTT function, setting the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true";

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [3]; and

C) if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false"; and

iv) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.6.3.1.10;

NOTE 2: The interactions of TNG2 with the TNG3 (group call timer) are explained in clause 6.6.3.5.2.

C) shall generate SIP re-INVITE requests for the MCPTT emergency group call to the other MCPTT participants of the MCPTT group call as specified in clause 6.6.3.1.3;

D) shall send the SIP re-INVITEs towards the other MCPTT participants of the MCPTT group call; and

E) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the IWF performing the controlling role shall interact with the media plane as specified in 3GPP TS 29.380 [31];

…

8) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is determined to be an authorised request for an MCPTT emergency call cancellation as specified in clause 6.6.3.1.10 and the in-progress emergency state of the group to is set to a value of "true" the IWF performing the controlling role:

a) shall set the in-progress emergency group state of the group to a value of "false";

b) shall clear the cache of the MCPTT ID of the MCPTT user as having an outstanding MCPTT emergency group call;

c) if an <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body is included and set to "false" and is determined to be an authorised request for an MCPTT emergency alert cancellation as specified in clause 6.6.3.1.8.3 and there is an outstanding MCPTT emergency alert for the MCPTT user shall:

i) if the received SIP re-INVITE request contains an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, clear the cache of the MCPTT ID of the MCPTT user identified by the <originated-by> element as having an outstanding MCPTT emergency alert; or

ii) if the received SIP re-INVITE request does not contain an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, clear the cache of the MCPTT ID of the sender of the SIP re-INVITE request as having an outstanding MCPTT emergency alert;

d) shall generate SIP re-INVITE requests to the MCPTT participants in the group call as specified in clause 6.6.3.1.3. The MCPTT controlling function:

i) for each of the other MCPTT participants in the group call shall send the SIP re-INVITE request towards the MCPTT client as specified in 3GPP TS 24.229 [3]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the IWF performing the controlling role shall interact with the media plane as specified in 3GPP TS 29.380 [31];

NOTE 3: Clause 6.6.3.1.3 will inform the group call (MCPTT) participants of the cancellation of the MCPTT group's in-progress emergency state and the cancellation of the MCPTT emergency alert if applicable.

e) shall stop timer TNG2 (in-progress emergency group call timer); and

NOTE 4: The interactions of TNG2 with the TNG3 (group call timer) are explained in clause 6.6.3.5.2;

f) for each of the affiliated MCPTT members of the group that are not participating in the call:

i) generate a SIP MESSAGE request notification of the cancellation of the MCPTT user's emergency call as specified in 3GPP TS 24.379 [29], clause 6.3.3.1.11, with the IWF acting as the controlling MCPTT function;

ii) set the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "false";

iii) if indicated above in step 8) c), set the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "false"; and

iv) send the SIP MESSAGE request according to 3GPP TS 24.229 [3];

9) if the received SIP re-INVITE request contains an imminent peril indication and the in-progress emergency group state of the group is set to a value of "false", shall perform the procedures specified in clause 10.1.4.8 and skip the rest of the steps.

Upon receiving a SIP 200 (OK) response to a SIP re-INVITE request the IWF performing the controlling role shall interact with the media plane as specified in 3GPP TS 29.380 [31];

1) shall generate a SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [3];

2) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [3] with the clarifications specified in 3GPP TS 24.379 [29], clause 6.3.3.2.1, with the IWF acting as the controlling MCPTT function;

3) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [12];

4) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [13];

…

7) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true", this is an authorised request for an MCPTT imminent peril group call and the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in 3GPP TS 24.379 [29], clause 4.4, with the IWF acting as the controlling MCPTT function;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCPTT emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

8) shall interact with media plane as specified in 3GPP TS 29.380 [31]; and

9) shall send the SIP 200 (OK) response towards the MCPTT client according to 3GPP TS 24.229 [3].

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCPTT client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in 3GPP TS 24.379 [29], clause 4.4, with the IWF acting as the controlling MCPTT function, the IWF performing the controlling role shall follow the procedures in clause 6.6.3.1.11.

Upon receipt of a SIP 2xx response for an outgoing SIP MESSAGE request, shall handle according to 3GPP TS 24.229 [3]."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.4.8]

"This procedure is initiated by the IWF performing the controlling role as the result of an action in clause 10.1.4.7.

In the procedures in this clause:

1) imminent peril indication in an incoming SIP re-INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

When the IWF performing the controlling role receives a SIP re-INVITE request with an imminent peril indication set to "true", the IWF performing the controlling role:

1) if the in-progress emergency state of the group is set to a value of "false" and if the SIP re-INVITE request contains an imminent peril indication set to a value of "true" or the in-progress imminent peril state of the group to "true", the IWF performing the controlling role shall:

NOTE 1: The calling procedure has already determined that this is not an unauthorised request for an MCPTT imminent peril call, therefore that check does not need to be repeated in the current procedure.

a) if the in-progress imminent peril state of the group is set to a value of "true" and the MCPTT user is indicating a new imminent peril indication:

i) for each of the other affiliated MCPTT member of the group generate a SIP MESSAGE request notification of the MCPTT user's imminent peril indication as specified in 3GPP TS 24.379 [29], clause 6.3.3.1.11, with the IWF acting as the controlling MCPTT function, with the following clarifications;

A) set the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [3];

b) if the in-progress imminent peril state of the group is set to a value of "false";

i) set the value of the in-progress imminent peril state of the group to "true";

ii) generate SIP re-INVITE requests for the MCPTT imminent peril group call to MCPTT participants in the MCPTT group call as specified in 3GPP TS 24.379 [29], clause 6.3.3.1.15, with the IWF acting as the controlling MCPTT function;

iii) send the SIP re-INVITES to all of the other MCPTT participants in the MCPTT group call;

iv) for each of the affiliated MCPTT members of the group not participating in the group call, generate a SIP MESSAGE request notification of the MCPTT user's imminent peril indication as specified in 3GPP TS 24.379 [29], clause 6.3.3.1.11, with the IWF acting as the controlling MCPTT function, with the following clarifications;

A) set the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [3]; and

c) cache the information that the MCPTT user has initiated an MCPTT imminent peril call;

…

3) if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "false" and is determined to be an authorised request for an MCPTT imminent peril call cancellation as specified in clause 6.6.3.1.8.6 and the in-progress imminent peril state of the group to is set to a value of "true" the IWF performing the controlling role shall:

a) set the in-progress imminent peril state of the group to a value of "false";

b) cache the information that the MCPTT user no longer has an outstanding MCPTT imminent peril group call;

c) generate SIP re-INVITES requests to the other MCPTT participants in the MCPTT group call as specified in 3GPP TS 24.379 [29], clause 6.3.3.1.15, with the IWF acting as the controlling MCPTT function. The MCPTT controlling function:

i) for each MCPTT participant shall send the SIP re-INVITE request towards the MCPTT client as specified in 3GPP TS 24.229 [3]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the IWF performing the controlling role interact with the media plane as specified in 3GPP TS 29.380 [31]; and

NOTE 2: 3GPP TS 24.379 [29], clause 6.3.3.1.15, with the IWF acting as the controlling MCPTT function, will inform the affiliated and joined MCPTT members of the cancellation of the MCPTT group's in-progress emergency state and the cancellation of the MCPTT emergency alert if applicable.

d) for each of the affiliated MCPTT members of the group not participating in the call shall:

i) generate a SIP MESSAGE request notification of the cancellation of the MCPTT user's imminent peril call as specified in 3GPP TS 24.379 [29], clause 6.3.3.1.11, with the IWF acting as the controlling MCPTT function;

ii) set the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to a value of "false"; and

iii) send the SIP MESSAGE request according to 3GPP TS 24.229 [3];

4) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [3] with the clarifications specified in 3GPP TS 24.379 [29], clause 6.3.3.2.1, with the IWF acting as the controlling MCPTT function;

5) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [12];

6) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [13];

7) shall interact with media plane as specified in 3GPP TS 29.380 [31]; and

8) shall send the SIP 200 (OK) response towards the MCPTT client according to 3GPP TS 24.229 [3].

Upon receipt of a SIP 2xx response for an outgoing SIP MESSAGE request, shall handle according to 3GPP TS 24.229 [3]."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.3.1.3]

"This clause is referenced from other procedures.

The IWF performing the controlling role shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [3].

The IWF performing the controlling role:

1) shall include an SDP offer with the media parameters as currently established with the terminating MCPTT client according to 3GPP TS 24.229 [3];

2) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcptt-calling-user-id> element set to the MCPTT ID of the initiating MCPTT user;

3) if the in-progress emergency group state of the group is set to a value of "true" the IWF performing the controlling role:

a) shall include a Resource-Priority header field with the namespace populated with the values for an MCPTT emergency group call as specified in clause 6.6.3.1.12;

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body the <emergency-ind> element set to a value of "true";

c) if the received request included a request for an emergency alert and MCPTT emergency alerts are authorised for this group and MCPTT user as determined by the procedures of clause 6.6.3.1.8.1, shall populate the application/vnd.3gpp.mcptt-info+xml MIME body and application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause 6.6.3.1.7. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcptt-info+xml MIME body; and

d) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <imminentperil-ind> element set to a value of "false"; and

NOTE: If the imminent peril state of the group is true at this point, the controlling function will be setting it to false as part of the calling procedure. This is in effect an upgrade of an MCPTT imminent peril group call to an MCPTT emergency group call.

4) if the in-progress emergency group state of the group is set to a value of "false":

a) shall include a Resource-Priority header field populated with the values for a normal MCPTT group call as specified in clause 6.6.3.1.12; and

b) if the received SIP re-INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "false" and this is an authorised request to cancel an MCPTT emergency group call as determined by the procedures of clause 6.6.3.1.8.4:

i) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with the <emergency-ind> element set to a value of "false"; and

ii) if the received SIP re-INVITE request contained an application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "false" and this is an authorised request to cancel an MCPTT emergency alert as determined by the procedures of 3GPP TS 24.379 [29], clause 6.3.3.1.15 with the IWF acting as the controlling function, shall:

A) include in the application/vnd.3gpp.mcptt-info+xml MIME body an <alert-ind> element set to a value of "false"; and

B) if the received SIP request contains an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, copy the contents of the received <originated-by> element to an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body in the outgoing SIP re-INVITE request."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.3.1.4]

"This clause is referenced from other procedures.

This clause describes the procedures for inviting an MCPTT user to an MCPTT session associated with an MCPTT emergency group call or MCPTT imminent peril group call. The procedure is initiated by the IWF performing the controlling role as the result of an action in clause 10.2.3.1.1.

The IWF performing the controlling role:

1) shall generate a SIP INVITE request as specified in clause 6.6.3.1.1;

2) shall set the Request-URI to the address of the terminating participating MCPTT function associated with the MCPTT ID of the targeted MCPTT user;

3) shall include an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element populated as follows:

a) the <mcptt-request-uri> element set to the value of the MCPTT ID of the targeted MCPTT user;

b) the <mcptt-calling-user-id> element set to the value of the MCPTT ID of the calling user; and

c) the <mcptt-calling-group-id> element set to the value of the MCPTT group ID of the emergency group call.

4) shall include in the P-Asserted-Identity header field the public service identity of the IWF performing the controlling role;

5) shall include in the SIP INVITE request an SDP offer according to the procedures specified in 3GPP TS 24.379 [29], clause 6.3.3.1.1, with the IWF acting as the controlling function; and

6) if the in-progress emergency group state of the group is set to a value of "true" the IWF performing the controlling role:

a) shall include a Resource-Priority header field populated with the values for an MCPTT emergency group call as specified in clause 6.6.3.1.12;

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <emergency-ind> element set to a value of "true";

c) if

i) the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCPTT user and MCPTT group are authorised for the initiation of MCPTT emergency alerts as determined by the procedures of clause 6.6.3.1.8.1, shall populate the application/vnd.3gpp.mcptt-info+xml MIME body and the application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause 6.6.3.1.7. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcptt-info+xml MIME body; or

ii) the call request originated from the IWF, and the IWF decides to indicate an emergency alert, shall populate the application/vnd.3gpp.mcptt-info+xml MIME body and the application/vnd.3gpp.mcptt-location-info+xml MIME body as specified in clause 6.6.3.1.7. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcptt-info+xml MIME body;

d) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <imminentperil-ind> element set to a value of "false"; and

NOTE: If the imminent peril state of the group is true at this point, the controlling function will set it to false as part of the calling procedure.

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the IWF performing the controlling role:

a) shall include a Resource-Priority header field populated with the values for an MCPTT imminent peril group call as specified in clause 6.6.3.1.12; and

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true"."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.3.1.7]

"This clause is referenced from other procedures.

This clause describes the procedures for populating the application/vnd.3gpp.mcptt-info+xml and application/vnd.3gpp.mcptt-location-info+xml MIME bodies for an MCPTT emergency alert. The procedure is initiated by the IWF performing the controlling role when it has received request initiating an MCPTT emergency alert and generates a message containing the MCPTT emergency alert information required by 3GPP TS 23.379 [2].

The IWF performing the controlling role:

1) shall include, if not already present, an application/vnd.3gpp.mcptt-info+xml MIME body as specified in 3GPP TS 24.379 [29], clause F.1, and set the <alert-ind> element to a value of "true";

2) shall determine the value of the user's Mission Critical Organization identity.

NOTE: How the IWF determines the user's Mission Critical Organization identity is out of scope of the present document;

3) shall include in the <mcpttinfo> element containing the <mcptt-Params> element containing an <mc-org> element set to the value of the user's Mission Critical Organization identity; and

4) shall include an application/vnd.3gpp.mcptt-location-info+xml MIME body in the outgoing SIP request."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.3.1.10]

"Upon expiry of timer TNG2 (in-progress emergency group call timer) for an MCPTT group, the IWF performing the controlling role:

1) shall set the in-progress emergency state of the group to a value of "false";

2) shall, if an MCPTT group call or MCPTT group session is in progress on the indicated group, for each of the participating members:

a) generate a SIP re-INVITE request as specified in clause 6.6.3.1.6; and

b) send the SIP re-INVITE request towards the MCPTT client according to 3GPP TS 24.229 [3]; and

3) shall for each affiliated but non-participating member of the group:

a) generate a SIP MESSAGE request according to 3GPP TS 24.379 [29] clause 6.3.3.1.11, with the IWF acting as the controlling function and include in the application/vnd.3gpp.mcptt-info+xml MIME body an <emergency-ind> element set to a value of "false";

b) shall include in the P-Asserted-Identity header field the public service identity of the controlling MCPTT function;

c) include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [3]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [7]; and

d) send the SIP MESSAGE request towards the MCPTT client according to rules and procedures of 3GPP TS 24.229 [3].

Upon receiving a SIP 200 (OK) response to a re-SIP INVITE request the IWF performing the controlling role shall interact with the media plane as specified in 3GPP TS 29.380 [31]."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.3.3.1.11]

"This clause is referenced from other procedures and describes how the IWF performing the controlling role generates a SIP INFO request due to the receipt of a SIP request for a priority call.

The IWF performing the controlling role:

1) shall generate a SIP INFO request according to rules and procedures of 3GPP TS 24.229 [3] and IETF RFC 6086 [26];

2) shall include the Info-Package header field set to g.3gpp.mcptt-info in the SIP INFO request;

3) shall include an application/vnd.3gpp.mcptt-info+xml MIME body in the SIP INFO request and:

a) if the received SIP request contained application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "true" and this is an unauthorised request for an MCPTT emergency alert as specified in clause 6.6.3.1.8.1, shall set the <emergency-ind> element to a value of "true" and the <alert-ind> element to a value of "false";

b) if the received SIP request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <alert-ind> element set to a value of "false" and if this is an unauthorised request for an MCPTT emergency alert cancellation, shall set <alert-ind> element to a value of "true"; and

c) if the received SIP request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <imminentperil-ind> element set to a value of "true", this is an authorised request for an MCPTT imminent peril group call and the in-progress emergency state of the group is set to a value of "true", shall set the <imminentperil-ind> element to a value of "false" and the <emergency-ind> element set to a value of "true"; and

4) shall send the SIP INFO request towards the inviting MCPTT client in the dialog created by the SIP request from the inviting MCPTT client, as specified in 3GPP TS 24.229 [3]."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.3.5.2]

"If the IWF performing the controlling role starts timer TNG2 (in-progress emergency group call timer), it shall not start timer TNG3 (group call timer).

If timer TNG3 (group call timer) is running and the MCPTT group call is upgraded to an MCPTT emergency group call, then the IWF performing the controlling role shall stop timer TNG3 (group call timer) and shall start timer TNG2 (in-progress emergency group call timer). If timer TNG2 (in-progress emergency group call timer) is running and the MCPTT emergency group call is cancelled, then the IWF performing the controlling role shall stop timer TNG2 (in-progress emergency group call timer) and shall start timer TNG3 (group call timer).

NOTE 1: How the IWF determines the value of the TNG2 and TNG3 timers is out of scope of the present document.

If timer TNG2 (in-progress emergency group call timer) is running and sequentially expires, then the controlling MCPTT function shall start timer TNG3 (group call timer).

NOTE 2: The above conditions for starting timer TNG2 (in-progress emergency group call timer) and timer TNG3 (group call timer) also apply in the case that these timers are re-started. For example: the case where the timer TNG3 was initially running, the MCPTT group call is upgraded to an MCPTT emergency group call and then the MCPTT emergency group call is cancelled."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.4.1.1]

"This clause is referenced from other procedures.

When the MCPTT emergency state is set, the IWF performing the participating role:

1) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the SIP INVITE request, an <emergency-ind> element set to "true" and if the MCPTT emergency group call state is set to "MEGC 1: emergency-gc-capable", shall set the MCPTT emergency group call state to "MEGC 2: emergency-call-requested";

2) if the IWF has determined that an MCPTT emergency alert is to be sent, and the MCPTT emergency alert state is set to "MEA 1: no-alert", shall:

a) set the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to "true" and set the MCPTT emergency alert state to "MEA 2: emergency-alert-confirm-pending"; and

b) include in the SIP INVITE request the specific location information for MCPTT emergency alert as specified in clause 6.5.1;

3) if the IWF has determined that an MCPTT emergency alert is not to be sent and the MCPTT emergency alert state is set to "MEA 1: no-alert", shall set the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to "false"; and

4) if the MCPTT client emergency group state of the user homed in the IWF of the group is set to a value other than "MEG 2: in-progress" set the MCPTT client emergency group state of the user homed in the IWF of the MCPTT group to "MEG 4: confirm-pending".

NOTE 1: This is the case of a user homed in the IWF already being in the MCPTT emergency state it initiated previously while originating an MCPTT emergency group call or MCPTT emergency alert. All group calls the user homed in the IWF originates while in MCPTT emergency state will be MCPTT emergency group calls.

When the MCPTT emergency state is clear and the MCPTT emergency group call state is set to "MEGC 1: emergency-gc-capable" and the IWF determines that the request for MCPTT emergency group call is authorized by local policy, the IWF performing the participating role shall set the MCPTT emergency state and perform the following actions:

1) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the SIP INVITE request an <emergency-ind> element set to "true" and set the MCPTT emergency group call state to "MEGC 2: emergency-call-requested" state;

2) if the user homed in the IWF has also requested an MCPTT emergency alert to be sent, shall:

a) include in the application/vnd.3gpp.mcptt-info+xml MIME body the <alert-ind> element set to "true" and set the MCPTT emergency alert state to "MEA 2: emergency-alert-confirm-pending"; and

b) include in the SIP INVITE request the specific location information, if available, for MCPTT emergency alert as specified in clause 6.5.1;

3) if the IWF has determined that an MCPTT emergency alert is not to be sent, shall set the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to "false"; and

4) if the MCPTT client emergency group state of the user homed in the IWF of the group is set to a value other than "MEG 2: in-progress" shall set the MCPTT client emergency group state of the user homed in the IWF of the MCPTT group to "MEG 4: confirm-pending".

NOTE 2: This is the case of an initial MCPTT emergency group call and optionally an MCPTT emergency alert being sent. As the MCPTT emergency state is not sent, there is no MCPTT emergency alert outstanding.

NOTE 3: An MCPTT group call originated by an affiliated member of an MCPTT group which is in an in-progress emergency state (as tracked on the MCPTT client by the MCPTT client emergency group state) but is not in an MCPTT emergency state of their own will also be an MCPTT emergency group call. The <emergency-ind> and <alert-ind> elements of the application/vnd.3gpp.mcptt-info+xml MIME body do not need to be included in this case and hence no action needs to be taken in this clause."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.4.1.2]

"This clause is referenced from other procedures.

If the MCPTT emergency group call state is set to either "MEGC 2: emergency-call-requested" or "MEGC 3: emergency-call-granted", or the MCPTT client emergency group state of the group is set to "MEG 2: in-progress", the IWF performing the participating role shall include in the SIP INVITE request a Resource-Priority header field populated with the values for an MCPTT emergency group call as specified in clause 6.4.1.11.

NOTE: The IWF performing the participating role ideally would not need to maintain knowledge of the in-progress emergency state of the group (as tracked for the client by the MCPTT client emergency group state by the IWF performing the participating role) but can use this knowledge to provide a Resource-Priority header field set to emergency level priority, which starts the infrastructure priority adjustment process sooner than otherwise would be the case.

If the MCPTT client emergency group state of the group is "no-emergency" or "cancel-pending", the IWF performing the participating role shall include in the SIP INVITE request a Resource-Priority header field populated with the values for a normal MCPTT group call as specified in clause 6.4.1.11."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.4.1.3]

"This clause is referenced from other procedures.

If the MCPTT emergency group call state is set to "MEGC 3: emergency-call-granted" and the MCPTT emergency alert state is set to "MEA 1: no-alert", the IWF performing the participating role shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [3] with the clarifications given below.

The IWF performing the participating role:

1) shall include in the SIP re-INVITE request an application/vnd.3gpp.mcptt-info+xml MIME body as defined in 3GPP TS 24.379 [29], clause F.1 with the <emergency-ind> element set to "false";

2) shall clear the MCPTT emergency state; and

3) shall set MCPTT emergency group state of the MCPTT group to "MEG 3: cancel-pending".

NOTE 1: This is the case of a user homed in the IWF who has initiated an MCPTT emergency group call and wants to cancel it.

If the MCPTT emergency group call state is set to "MEGC 3: emergency-call-granted" and the MCPTT emergency alert state is set to a value other than "MEA 1: no-alert" and only the MCPTT emergency group call is to be cancelled, the IWF performing the participating role:

1) shall include in the SIP re-INVITE request an application/vnd.3gpp.mcptt-info+xml MIME body as defined in 3GPP TS 24.379 [29], clause F.1 with the <emergency-ind> element set to "false"; and

2) shall set the MCPTT emergency group state of the MCPTT group to "MEG 3: cancel-pending".

NOTE 2: This is the case of a user homed in the IWF that has initiated both an MCPTT emergency group call and an MCPTT emergency alert and wishes to only cancel the MCPTT emergency group call. This leaves the MCPTT emergency state set.

If the MCPTT emergency group call state is set to "MEGC 3: emergency-call-granted" and the MCPTT emergency alert state is set to a value other than "MEA 1: no-alert" and the user homed in the IWF has indicated that the MCPTT emergency alert on the MCPTT group should be cancelled in addition to the MCPTT emergency group call, the IWF performing the participating role:

1) shall include in the SIP re-INVITE request an application/vnd.3gpp.mcptt-info+xml MIME body as defined in 3GPP TS 24.379 [29], clause F.1 with the <emergency-ind> element set to "false";

2) shall:

a) include in the application/vnd.3gpp.mcptt-info+xml MIME body an <alert-ind> element set to "false";

b) set the MCPTT emergency alert state to "MEA 4: Emergency-alert-cancel-pending"; and

c) clear the MCPTT emergency state; and

3) shall set the MCPTT emergency group state of the MCPTT group to "MEG 3: cancel-pending".

NOTE 3: This is the case of a user homed in the IWF that has initiated both an MCPTT emergency group call and an MCPTT emergency alert and wishes to cancel both."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.4.1.6]

"This clause is referenced from other procedures.

When the IWF performing the participating role determines to originate an MCPTT imminent peril group call, the IWF performing the participating role:

1) if the client homed in the IWF's imminent peril group state is set to "MIGC 1: imminent-peril-gc-capable" and the in-progress emergency state of the group is set to a value of "false":

a) shall include in the SIP request an application/vnd.3gpp.mcptt-info+xml MIME body as defined in 3GPP TS 24.379 [29], clause F.1 with the <imminentperil-ind> element set to "true" and set the MCPTT emergency group call state to "MIGC 2: imminent-peril-call-requested" state; and

b) if the client homed in the IWF's imminent peril group state of the group is set to a value other than "MIG 2: in-progress" shall set the client homed in the IWF's emergency group state of the MCPTT group to "MIG 4: confirm-pending".

NOTE: An MCPTT group call originated by an affiliated member of an MCPTT group which is in an in-progress imminent peril state (as tracked on the client by the client imminent peril group state) will also have the priority associated with MCPTT imminent peril group calls. The <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info MIME body does not need to be included in this case, nor do any state changes result and hence no action needs to be taken in this clause."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.4.1.7]

"This clause is referenced from other procedures.

If the MCPTT imminent peril group call state is set to "MIGC 3: imminent-peril-call-granted" or the MCPTT imminent peril group state of the MCPTT group is set to "MIG 2: in-progress", the IWF shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [3] with the clarifications given below.

The IWF:

1) shall include in the SIP re-INVITE request an application/vnd.3gpp.mcptt-info+xml MIME body as defined in clause F.1 with the <imminentperil-ind> element set to "false"; and

2) shall set MCPTT imminent peril group state of the MCPTT group to "MIG 3: cancel-pending".

NOTE: This is the case of a user who has initiated an MCPTT imminent peril group call and wants to cancel it, or another authorised member of the group who wishes to cancel the in-progress imminent peril state of the group."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.4.1.8]

"This clause is referenced from other procedures.

When the MCPTT imminent peril group call state is set "MIGC 2: imminent-peril-call-requested" or "MIGC 3: imminent-peril-call-granted", or the client homed in the IWF's imminent peril state of the group is set to "MIG 2: in-progress", the IWF performing the participating role:

1) shall include in the SIP INVITE request a Resource-Priority header field populated with the values for an MCPTT imminent peril group call as specified in clause 6.4.1.11.

NOTE: The IWF performing the participating role ideally would not need to maintain knowledge of the in-progress imminent peril state of the group (as tracked for the client by the client imminent peril group state) but can use this knowledge to provide a Resource-Priority header field set to imminent peril level priority, which starts the infrastructure priority adjustment process sooner than otherwise would be the case.

When the MCPTT imminent peril group call state is set to "MIGC 1: imminent-peril-gc-capable" and the IWF performing the participating role determines that cancellation of the MCPTT imminent peril group call is authorized, or the MCPTT client imminent peril group state of the group is "MIG 1: no-imminent-peril" or "MIG 3: cancel-pending", the IWF performing the participating role:

1) shall include in the SIP INVITE request a Resource-Priority header field populated with the values for a normal MCPTT group call as specified in clause 6.4.1.11."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.4.1.10]

"This clause is referenced from other procedures.

Upon the need to cancel an in-progress emergency group state of a group, the IWF performing the participating role shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [3] with the clarifications given below.

The IWF performing the participating role:

1) shall include in the SIP re-INVITE request an application/vnd.3gpp.mcptt-info+xml MIME body as defined in 3GPP TS 24.379 [29], clause F.1 with the <emergency-ind> element set to "false";

2) shall set MCPTT emergency group state of the MCPTT group to "MEG 3: cancel-pending"; and

3) if the MCPTT emergency alert on the MCPTT group originated by a MCPTT user is to be cancelled:

a) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <alert-ind> element set a value of "false"; and

b) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body an <originated-by> element set to the MCPTT ID of the MCPTT user who originated the MCPTT emergency alert.

NOTE: When an MCPTT emergency alert is cancelled by a user other than its originator, the <originated-by> element is needed to identify which MCPTT emergency alert is being cancelled, as more than one MCPTT user could have originated emergency alerts to the same group."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.4.1.11]

"This clause is referenced from other procedures.

The IWF performing the participating role may populate the Resource-Priority header as described for the IWF performing the controlling role in clause 6.6.3.1.12."

[ETSI TS 129 380 [31] (3GPP TS 29.380), clause 5.3]

"An IWF performing the controlling role shall:

1. support the role of a controlling MCPTT function as specified in 3GPP TS 23.379 [5];

2. support negotiating media plane control channel media level attributes as specified in clause 4.3;

3. provide floor control arbitration logic with an interface to MCPTT clients as described in clause 4.2.1 and support the related procedures as specified in clause 6.3;

4. provide a network media interface as described in clause 4.2.1 and support associated procedures as specified in clause 6.3;

5. provide a media distributor with a floor control interface towards the MCPTT clients as described in 3GPP TS 24.380 [25] clause 4.2.1 and support associated procedures in clause 6.3;

6. support the on-network mode floor control protocol elements as specified in clause 8: and

7. support media plane security as specified in clause 13.

An IWF performing the controlling role may:

1. *provide a floor request queue as described in clause 4.2.1 and support the related procedures as specified in clause 6.3 and clause 4.3."*

[ETSI TS 129 380 [31] (3GPP TS 29.380), clause 6.3.2.2]

"When an MCPTT call is established a new instance of the floor control server state machine for 'general floor control operation' is created.

For each MCPTT media endpoint added to the MCPTT call, a new instance of the floor control server state machine for 'basic floor control operation towards the floor participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the IWF floor control server could queue the implicit floor control request for the MCPTT media endpoint.

…"

[ETSI TS 129 380 [31] (3GPP TS 29.380), clause 6.3.3]

"When an MCPTT media endpoint leaves an MCPTT call and the MCPTT call remains ongoing with the other media endpoints, the IWF floor control server follows a two-step procedure.

Step 1 The IWF stops sending floor control messages and RTP media packets to the MCPTT media endpoint leaving the MCPTT call and, the IWF discards floor control messages and RTP media packets received from the MCPTT media endpoint leaving the MCPTT call.

Step 2 When the application and signalling plane has determined that the MCPTT call with this MCPTT floor participant has been released, the corresponding instance of the IWF floor control server state machine for 'basic floor control operation towards the floor participant' is released.

When an MCPTT call is released, the IWF floor control server follows a two-step procedure.

Step 1 The IWF stops sending floor control messages and RTP media packets to all floor participants in the MCPTT call.

Step 2 When the application and signalling plane has determined that the MCPTT call has been released, the corresponding instance of the floor control server state machine for 'general floor control operation' are also terminated, along with any 'basic floor control operation towards the floor participant' state machines for the floor participants of this call.

The floor control server state machine for general floor control operation initiates the MCPTT call release depending on the release policy specified in 3GPP TS 29.379 [26]."

6.1.1.1.3 Test description

6.1.1.1.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the participating MCPTT functions
- SS-UE1 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 6.1.1.1.3.1-1.
- SS-UE2 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 6.1.1.1.3.1-1.

IUT:

- IUT (IWF)

- IUT (IWF) provides the interworking function (IWF) performing the controlling role

- The IUT (IWF) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (IWF) also consists of all sub-systems of the MCPTT Server, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (IWF) is connected to SS (MCPTT Server) defined in Annex A, Figure A.4.2, and to the SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) as defined in Annex A, Figure A.4.1.

- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are registered and authorized to use the MCPTT system.

- The SS-UE1 (MCPTT client) and the SS-UE2 (MCPTT client) are affiliated with Group A and are authorized to initiate group calls.

- Group A is controlled by the IUT (IWF).

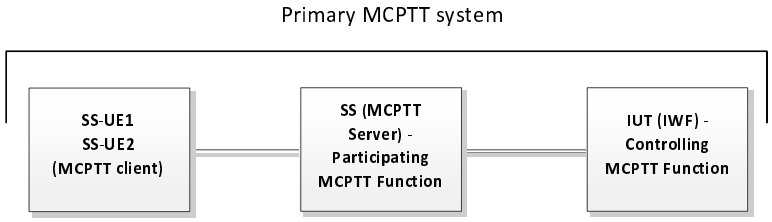


Figure 6.1.1.1.3.1-1: Functions of the testing components

6.1.1.1.3.2 Test procedure sequence

Table 6.1.1.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) to initiate an on-demand pre-arranged group call with automatic commencement mode and implicit floor control	<--	SIP INVITE	-	-
2	Check: Does the IUT (IWF) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	1	P
3	Check: Does the IUT (IWF) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP INVITE	1	P
4	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 183 (Session Progress) message sent reliably.	<--	SIP 183 (Session Progress)		-
5	Check: Does the IUT (IWF) send a SIP PRACK message to the SS (MCPTT Server) providing the terminating participating MCPTT function in response to the SIP 183 (Session Progress)?	-->	SIP PRACK	1	P
6	The SS (MCPTT Server) responds to the SIP PRACK with a SIP 200 (OK) message.	<--	SIP 200 (OK)		-
7	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Depending on the IWF implementation, steps 8-9 and steps 10-11 might be performed in parallel.	-	-	-	-
8	Check: Does the IUT (IWF) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	1	P
9	Check: Does the IUT (IWF) send a Floor Taken to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
10	Check: Does the IUT (IWF) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	1	P
11	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message.	<--	SIP ACK		-
12	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor.	<--	Floor Release		-
13	Check: Does the IUT (IWF) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P
14	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	2	P
15	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	2	P
16	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
17	Check: Does the IUT (IWF) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	2	P
18	Check: Does the IUT (IWF) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
19	The SS (MCPTT Server) providing the terminating participating MCPTT function sends a Floor Request message from SS-UE2 with a higher priority than SS-UE1 (MCPTT client).	<--	Floor Request		-
20	Check: Does the IUT (IWF) send a Floor Revoke message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Revoke	2	P
21	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1.	<--	Floor Release		-
22	Check: Does the IUT (IWF) send a Floor Granted message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Granted	2	P
23	Check: Does the IUT (IWF) send a Floor Taken message to the SS (MCPTT Server) providing the originating participating MCPTT function informing that the floor is taken by SS-UE2 (MCPTT client)?	-->	Floor Taken	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
24	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1 with a lower priority than SS-UE2 (MCPTT client) and the Floor Indicator F-bit set to 0.	<--	Floor Request		-
25	Check: Does the IUT (IWF) send a Floor Deny message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Deny	2	P
26	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1 with a lower priority than SS-UE2 (MCPTT client) and the Floor Indicator F-bit set to 1.	<--	Floor Request		-
27	Check: Does the IUT (IWF) send a Queue Position Info message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Queue Position Info	2	P
28	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Queue Position Request message from SS-UE1.	<--	Floor Queue Position Request		-
29	Check: Does the IUT (IWF) send a Queue Position Info message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Queue Position Info	2	P
30	The SS (MCPTT Server) providing the terminating participating MCPTT function sends a Floor Release message from SS-UE2 with an acknowledgement required to release the floor	<--	Floor Release		-
31	Check: Does the IUT (IWF) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P
32	Check: Does the IUT (IWF) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	2	P
33	Check: Does the IUT (IWF) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	2	P
34	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
35	Check: Does the IUT (IWF) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	2	P
36	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	2	P
37	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	2	P
38	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) to upgrade the call to an emergency group call with implicit floor request.	<--	SIP INVITE		-
39	Check: Does the IUT (IWF) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	3	P
40	Check: Does the IUT (IWF) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP INVITE	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Depending on the IWF implementation, steps 41-43 and steps 44-46 might be performed in parallel.	-	-	-	-
41	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
42	Check: Does the IUT (IWF) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	3	P
43	Check: Does the IUT (IWF) send a Floor Taken to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	4	P
44	Check: Does the IUT (IWF) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	3	P
45	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
46	Check: Does the IUT (IWF) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	4	P
47	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
48	Check: Does the IUT (IWF) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	4	P
49	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	4	P
50	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	4	P
51	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
52	Check: Does the IUT (IWF) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	4	P
53	Check: Does the IUT (IWF) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	4	P
54	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
55	Check: Does the IUT (IWF) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	4	P
56	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	4	P
57	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	4	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
58	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) to downgrade the call to a normal group call without implicit floor request.	<--	SIP INVITE		-
59	Check: Does the IUT (IWF) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	5	P
60	Check: Does the IUT (IWF) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP INVITE	5	P
-	EXCEPTION: Depending on the IWF implementation, steps 61-62 and steps 63-64 might be performed in parallel.	-	-	-	-
61	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
62	Check: Does the IUT (IWF) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	5	P
63	Check: Does the IUT (IWF) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	5	P
64	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
65	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	6	P
66	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	6	P
67	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
68	Check: Does the IUT (IWF) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	6	P
69	Check: Does the IUT (IWF) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	6	P
70	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
71	Check: Does the IUT (IWF) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	6	P
72	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	6	P
73	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	6	P
74	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) to upgrade the call to an imminent-peril group call with implicit floor request.	<--	SIP INVITE		-
75	Check: Does the IUT (IWF) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	7	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
76	Check: Does the IUT (IWF) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP INVITE	7	P
-	EXCEPTION: Depending on the IWF implementation, steps 77-79 and steps 80-82 might be performed in parallel.	-	-	-	-
77	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
78	Check: Does the IUT (IWF) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	7	P
79	Check: Does the IUT (IWF) send a Floor Taken to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	8	P
80	Check: Does the IUT (IWF) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	7	P
81	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
82	Check: Does the IUT (IWF) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	8	P
83	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
84	Check: Does the IUT (IWF) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	8	P
85	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	8	P
86	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle		P
87	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
88	Check: Does the IUT (IWF) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	8	P
89	Check: Does the IUT (IWF) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	8	P
90	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
91	Check: Does the IUT (IWF) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	8	P
92	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	8	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
93	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	8	P
94	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP INVITE request from SS-UE1 (MCPTT client) to downgrade the call to a normal group call without implicit floor request.	<--	SIP INVITE		-
95	Check: Does the IUT (IWF) send a SIP 100 Trying message to the SS (MCPTT Server)?	-->	SIP 100 Trying	9	P
96	Check: Does the IUT (IWF) send a SIP INVITE to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP INVITE	9	P
-	EXCEPTION: Depending on the IWF implementation, steps 97-98 and steps 99-100 might be performed in parallel.	-	-	-	-
97	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
98	Check: Does the IUT (IWF) send a SIP ACK to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	SIP ACK	9	P
99	Check: Does the IUT (IWF) send a SIP 200 (OK) to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)	9	P
100	The SS (MCPTT Server) providing the originating participating MCPTT function responds with a SIP ACK message	<--	SIP ACK		-
101	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	10	P
102	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	10	P
103	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Request message from SS-UE1	<--	Floor Request		-
104	Check: Does the IUT (IWF) send a Floor Granted message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Granted	10	P
105	Check: Does the IUT (IWF) send a Floor Taken message to the SS (MCPTT Server) providing the terminating participating MCPTT function informing that the floor is taken by SS-UE1 (MCPTT client)?	-->	Floor Taken	10	P
106	The SS (MCPTT Server) providing the originating participating MCPTT function sends a Floor Release message from SS-UE1 with an acknowledgement required to release the floor	<--	Floor Release		-
107	Check: Does the IUT (IWF) respond with a Floor Ack to the Floor Release message sent by the SS (MCPTT Server)?	-->	Floor Ack	10	P
108	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the originating participating MCPTT function?	-->	Floor Idle	10	P
109	Check: Does the IUT (IWF) send a Floor Idle message to the SS (MCPTT Server) providing the terminating participating MCPTT function?	-->	Floor Idle	10	P
110	The SS (MCPTT Server) providing the originating participating MCPTT function sends a SIP BYE request from SS-UE1	<--	SIP BYE		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
111	Check: Does the IUT (IWF) send a SIP 200 (OK) response to SS (MCPTT Server) providing the originating participating MCPTT function?	-->	SIP 200 (OK)		P
112	Check: Does the IUT (IWF) send a SIP BYE message to the SS (MCPTT Server) providing the terminating participating MCPTT function to end the call?	-->	SIP BYE		P
113	The SS (MCPTT Server) providing the terminating participating MCPTT function responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-

6.1.1.1.3.3 Specific message contents

Table 6.1.1.1.3.3-1: SIP INVITE from the SS (MCPTT-Server) (Step 1, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the IWF providing the controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the IWF providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing originating participating function.		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 6.1.1.1.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 6.1.1.1.3.3-1B			

Table 6.1.1.1.3.3-1A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions INITIAL_SDP_OFFER, IMPLICIT_GRANT_REQUESTED, IMPLICIT_FLOOR_GRANTED

Table 6.1.1.1.3.3-1B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 6.1.1.1.3.3-2: SIP INVITE from the IUT (IWF) (Step 3, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing terminating participating function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the IWF providing the controlling function.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the MCPTT Server which is simulated by the SS, providing terminating participating function.		
Call-ID				
callid	any allowed value			
CSeq				
value	any allowed value			
method	"INVITE"			
Session-Expires				
generic-param	any allowed value			
Answer-Mode	not present			
Priv-Answer-Mode	not present			
Max-Forwards				
value	any allowed value			
Message-body				
MIME body part		**SDP message**		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME-part-body	SDP message as described in Table 6.1.1.1.3.3-2A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 6.1.1.1.3.3-2B			

6.1.1.1.3.3-2A: SDP in SIP INVITE from the IUT (IWF) (Table 6.1.1.1.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the IWF performing the controlling role			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IWF performing the controlling role			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IWF performing the controlling role			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IWF performing the controlling role			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 6.1.1.1.3.3-2B: MCPTT-Info in SIP INVITE from the IUT (IWF) (Table 6.1.1.1.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 6.1.1.1.3.3-3: SIP 183 (Session Progress) from the SS (MCPTT-Server) (Step 4, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.1-1, conditions MCPTT, 100rel

Table 6.1.1.1.3.3-4: SIP PRACK from the IUT (IWF) (Step 5, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.10.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Request-URI	same URI as the SS has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server), providing terminating participating function.		
Max-Forwards				
value	any allowed value			

Table 6.1.1.1.3.3-5: SIP 200 (OK) to PRACK from the SS (MCPTT-Server) (Step 6, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Content-Type	not present			
Content-Length	not present			

Table 6.1.1.1.3.3-6: SIP 200 (OK) from the SS (MCPTT Server) (Step 7, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	3600			
refresher	"uas"			
Message-body				
SDP message	SDP message as described in Table 6.1.1.1.3.3-6A			

Table 6.1.1.1.3.3-6A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value		pc_MCPTT_FloorRequestQueueing
mc_priority	"2"	"2" is the value of the <user-priority> element for user B in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table 6.1.1.1.3.3-7: SIP 200 (OK) from the IUT (IWF) (Step 10, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Session-Expires				
generic-param	any allowed value			
refresher	"uac"			
Message-body				
SDP message	SDP message as described in Table 6.1.1.1.3.3-7A			

Table 6.1.1.1.3.3-7A: SDP in SIP 200 (OK) from the IUT (IWF) (Table 6.1.1.1.3.3-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_FLOOR_GRANTED, IMPLICIT_GRANT_REQUESTED				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
username	any allowed value			
session-id	any allowed value			
sess-version	any allowed value			
unicast-address	IP address of the IWF			
Session Name	at least one UTF-8-encoded character, or if no name is given, a single empty space	s= line		
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IWF			
Media description[1]		**Media description for audio**		
media description				
port	port number assigned by the IWF (even integer)			
Connection Data	present if session description does not contain a c=line; optional otherwise	**c= line**		
connection-address	IP address of the IWF			
Media description[2]		**Media description for media control**		
media description				
port	port number assigned by the IWF being different than the port number of the audio channel (RTP) and it associated control channel (RTCP)			
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IWF			

Table 6.1.1.1.3.3-8: SIP INVITE from the SS (MCPTT-Server) (Step 38, Table 6.1.1.1.3.2-1)

Derivation Path: Table 6.1.1.1.3.3-1 conditions MCPTT, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IWF has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IWF		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-8A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-8B			

Table 6.1.1.1.3.3-8A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-8)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_OFFER, IMPLICIT_GRANT_REQUESTED

Table 6.1.1.1.3.3-8B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-8)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 6.1.1.1.3.3-9: SIP INVITE from the IUT (IWF) (Step 40, Table 6.1.1.1.3.2-1)

Derivation Path: Table 6.1.1.1.3.3-2, conditions MCPTT, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-9A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-9B			

Table 6.1.1.1.3.3-9A: SDP in SIP INVITE from the IUT (IWF) (Table 6.1.1.1.3.3-9)

Derivation Path: Table 6.1.1.1.3.3-2A, condition SDP_OFFER instead of INITIAL_SDP_OFFER

Table 6.1.1.1.3.3-9B: MCPTT-Info in SIP INVITE from the IUT (IWF) (Table 6.1.1.1.3.3-9)

Derivation Path: Table 6.1.1.1.3.3-2B, conditions EMERGENCY-CALL

Table 6.1.1.1.3.3-10: SIP 200 (OK) from the SS (MCPTT-Server) (Steps 41, 61, 77 and 97, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 6.1.1.1.3.3-10A			

Table 6.1.1.1.3.3-10A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-10)

Derivation Path: Table 6.1.1.1.3.3-6A

Table 6.1.1.1.3.3-11: SIP 200 (OK) from the IUT (IWF) (Steps 44 and 80, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 6.1.1.1.3.3-11A			

Table 6.1.1.1.3.3-11A: SDP in SIP 200 (OK) from the IUT (IWF) (Table 6.1.1.1.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_GRANT_REQUESTED

Table 6.1.1.1.3.3-12: SIP 200 (OK) from the IUT (IWF) (Steps 63 and 99, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 6.1.1.1.3.3-12A			

Table 6.1.1.1.3.3-12A: SDP in SIP 200 (OK) from the IUT (IWF) (Table 6.1.1.1.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER

Table 6.1.1.1.3.3-13: SIP INVITE from the SS (MCPTT-Server) (Step 58, Table 6.1.1.1.3.2-1)

Derivation Path: Table 6.1.1.1.3.3-1 conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IWF has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IWF		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-13A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-13B			

Table 6.1.1.1.3.3-13A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-13)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_OFFER

Table 6.1.1.1.3.3-13B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-13)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
emergency-ind	Encrypted (see note) <emergency-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 6.1.1.1.3.3-14: SIP INVITE from the IUT (IWF) (Step 60, Table 6.1.1.1.3.2-1)

Derivation Path: Table 6.1.1.1.3.3-2, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-14A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-14B			

Table 6.1.1.1.3.3-14A: SDP in SIP INVITE from the IUT (IWF) (Table 6.1.1.1.3.3-14)

Derivation Path: Table 6.1.1.1.3.3-9A

Table 6.1.1.1.3.3-14B: MCPTT-Info in SIP INVITE from the IUT (IWF) (Table 6.1.1.1.3.3-14)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
emergency-ind	Encrypted (see note) <emergency-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 6.1.1.1.3.3-15: SIP INVITE from the SS (MCPTT-Server) (Step 74, Table 6.1.1.1.3.2-1)

Derivation Path: Table 6.1.1.1.3.3-1, conditions MCPTT, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IWF has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IWF		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-15A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-15B			

Table 6.1.1.1.3.3-15A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-15)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_OFFER, IMPLICIT_GRANT_REQUESTED,

Table 6.1.1.1.3.3-15B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-15)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 6.1.1.1.3.3-16: SIP INVITE from the IUT (IWF) (Step 76, Table 6.1.1.1.3.2-1)

Derivation Path: Table 6.1.1.1.3.3-2, conditions MCPTT, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-16A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-16B			

Table 6.1.1.1.3.3-16A: SDP in SIP INVITE from the IUT (IWF) (Table 6.1.1.1.3.3-16)

Derivation Path: Table 6.1.1.1.3.3-9A

Table 6.1.1.1.3.3-16B: MCPTT-Info in SIP INVITE from the IUT (IWF) (Table 6.1.1.1.3.3-16)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, INVITE_REFER, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 6.1.1.1.3.3-17: SIP INVITE from the SS (MCPTT-Server) (Step 94, Table 6.1.1.1.3.2-1)

Derivation Path: Table 6.1.1.1.3.3-1, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IWF has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IWF		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-17A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-17B			

Table 6.1.1.1.3.3-17A: SDP in SIP INVITE from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-17)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_OFFER

Table 6.1.1.1.3.3-17B: MCPTT-Info in SIP INVITE from the SS (MCPTT-Server) (Table 6.1.1.1.3.3-17)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
imminentperil-ind	Encrypted (see note) <imminentperil-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 6.1.1.1.3.3-18: SIP INVITE from the IUT (IWF) (Step 96, Table 6.1.1.1.3.2-1)

Derivation Path: Table 6.1.1.1.3.3-2, conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table 6.1.1.1.3.3-18A			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.1.3.3-18B			

Table 6.1.1.1.3.3-18A: SDP in SIP INVITE from the IUT (IWF) (Table 6.1.1.1.3.3-18)

Derivation Path: Table 6.1.1.1.3.3-9A

Table 6.1.1.1.3.3-18B: MCPTT-Info in SIP INVITE from the IUT (IWF) (Table 6.1.1.1.3.3-18)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
imminentperil-ind	Encrypted (see note) <imminentperil-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table 6.1.1.1.3.3-19: Floor Taken (Steps 9, 18, 33, 69 and 105, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 6.1.1.1.3.3-20: Floor Release (Steps 12, 30, 34, 70 and 106, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 6.1.1.1.3.3-21: Floor Ack (Steps 13, 31, 35, 48, 55, 71, 84, 91 and 107, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 6.1.1.1.3.3-22: Floor Idle (Steps 14, 15, 36, 37, 65, 66, 72, 73, 101, 102, 108 and 109, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 6.1.1.1.3.3-23: Floor Request (Steps 16, 26, 67 and 103, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 6.1.1.1.3.3-24: Floor Request (Step 24, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	
Floor Indicator			
Floor Indicator	"1000000000000000"	bit A=1 (Normal call) bit F=0 (Queueing not supported)	

Table 6.1.1.1.3.3-25: Floor Granted (Steps 17, 32, 68 and 104, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

Table 6.1.1.1.3.3-26: Floor Request (Step 19, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"2"	The highest priority for the user	

Table 6.1.1.1.3.3-27: Floor Revoke (Step 20, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.8-1

Table 6.1.1.1.3.3-28: Floor Release (Step 21, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1

Table 6.1.1.1.3.3-29: Floor Granted (Step 22, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"2"	The highest priority for the user	

Table 6.1.1.1.3.3-30: Floor Taken (Step 23, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1

Table 6.1.1.1.3.3-31: Floor Deny (Step 25, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.4-1

Table 6.1.1.1.3.3-32: Floor Queue Position Info (Steps 27 and 29, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.10-1

Table 6.1.1.1.3.3-33: Floor Queue Position Request (Step 28, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.9-1

Table 6.1.1.1.3.3-34: Floor Taken (Steps 43 and 53, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1, condition EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 6.1.1.1.3.3-35: Floor Idle (Steps 49, 50, 56 and 57, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1, condition EMERGENCY-CALL

Table 6.1.1.1.3.3-36: Floor Request (Step 51, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1 condition EMERGENCY-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 6.1.1.1.3.3-37: Floor Granted (Steps 46 and 52, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition EMERGENCY-CALL

Table 6.1.1.1.3.3-38: Floor Release (Steps 47 and 54, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition EMERGENCY-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 6.1.1.1.3.3-39: Floor Taken (Steps 79 and 89, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1 condition IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Granted Party's Identity	px_MCPTT_ID_User_A			

Table 6.1.1.1.3.3-40: Floor Idle (Steps 85, 86, 92 and 93, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1 condition IMMPERIL-CALL

Table 6.1.1.1.3.3-41: Floor Request (Step 87, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1 condition IMMPERIL-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 6.1.1.1.3.3-42: Floor Granted (Steps 82 and 88, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition IMMPERIL-CALL

Table 6.1.1.1.3.3-43: Floor Release (Steps 83 and 90, Table 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition IMMPERIL-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 6.1.1.1.3.3-44: SIP BYE from the SS (MCPTT-Server) (Step 110, 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the IWF has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IWF		
From				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Local URI of the dialog (from the SS' point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Local tag of the dialog (from the SS' point of view)		
To				
addr-spec	Same URI of the IWF as used earlier in the dialog	Remote URI of the dialog (from the SS' point of view)		
tag	Same tag of the IWF as used earlier in the dialog	Remote tag of the dialog ID (from the SS' point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	70	The recommended initial value is 70 in IETF RFC 3261 [17].		

Table 6.1.1.1.3.3-45: SIP BYE from the IUT (IWF) (Step 112, 6.1.1.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
From				
addr-spec	Same URI of the IWF as used earlier in the dialog	Local URI of the dialog (from the IWF's point of view)		
tag	Same tag of the IWF as used earlier in the dialog	Local tag of the dialog (from the IWF's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the IWF's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the IWF's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

6.1.1.2 IWF - MCPTT Server / Pre-arranged Group Call / Automatic Commencement Mode / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / IWF Originating Participating Role

6.1.1.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCPTT Server) connected to SS (MCPTT Server)}
ensure that {
  when { the IUT (IWF) initiates an on-demand pre-arranged group call with automatic commencement
mode and implicit floor control }
    then { IUT (IWF) interacts with SS (MCPTT Server) to set up pre-arranged group call by sending
SIP messages }
           }
```

(2)

```
with { SS (MCPTT Server) having established an MCPTT Pre-arranged Group Call with Automatic
Commencement Mode }
ensure that {
  when { the IUT (IWF) engage in communication }
    then { IUT (IWF) forwards floor control (Floor Request, Floor Release, Floor Queue Position
Request ) by forwarding messages to the SS (MCPTT Server) }
           }
```

(3)

```
with { SS (MCPTT Server) having established an MCPTT Pre-arranged Group Call with Automatic
Commencement Mode }
ensure that {
  when { the IUT (IWF) upgrades the call to an emergency group call with implicit floor request }
    then { IUT (IWF) interacts with SS (MCPTT Server) to set up an emergency group call by sending
SIP messages }
           }
```

(4)

```
with { SS (MCPTT Server) having established an MCPTT emergency group call with implicit floor
request }
ensure that {
  when { the IUT (IWF) engage in communication }
    then { IUT (IWF) forwards floor control (Floor Request, Floor Release ) by forwarding messages
via the SS (MCPTT Server) }
            }
```

(5)

```
with { SS (MCPTT Server) having established an MCPTT emergency group call with implicit floor
request }
ensure that {
  when { the IUT (IWF)  downgrades the call to a normal group call without implicit floor request }
    then { IUT (IWF)  interacts with SS (MCPTT Server) to set up the normal group call by sending
SIP messages }
            }
```

(6)

```
with { SS (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the IUT (IWF) engage in communication }
    then { IUT (IWF) forwards floor control (Floor Request, Floor Release ) by sending messages via
the SS (MCPTT Server) }
            }
```

(7)

```
with { SS (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the IUT (IWF) upgrades the call to imminent-peril group call with implicit floor request }
    then { IUT (IWF) interacts with SS (MCPTT Server) to set up the imminent-peril group call by
sending SIP messages }
            }
```

(8)

```
with { SS (MCPTT Server) having established an MCPTT imminent-peril group call with implicit floor
request }
ensure that {
  when { the IUT (IWF) engage in communication }
    then { IUT (IWF) forwards floor control (Floor Request, Floor Release ) by sending messages via
the SS (MCPTT Server) }
            }
```

(9)

```
with { SS (MCPTT Server) having established an MCPTT imminent-peril group call with implicit floor
request }
ensure that {
  when { the IUT (IWF) downgrades the call to a normal group call without implicit floor request }
    then { IUT (IWF) interacts with SS (MCPTT Server) to set up the normal group call by sending SIP
messages }
            }
```

(10)

```
with { SS (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the IUT (IWF) engage in communication }
    then { IUT (IWF) forwards floor control (Floor Request, Floor Release ) by sending messages via
the SS (MCPTT Server) }
            }
```

(11)

```
with { SS (MCPTT Server) having established an MCPTT group call without implicit floor request }
ensure that {
  when { the IUT (IWF) ends the pre-arranged group call }
    then { IUT (IWF) forwards a SIP BYE message to SS (MCPTT Server) }
            }
```

6.1.1.2.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 379 [4] (3GPP TS 24.379), clauses 10.1.3.1.1, 10.1.3.1.2, 10.1.3.2, 10.1.3.3.1, 10.1.3.4.1, 10.1.3.6, 10.1.2.1, 10.1.2.2, 10.1.2.3, 10.1.2.4, 10.1.2.5, 10.1.2.6, 6.6.2.1.1.1, 6.6.2.1.2, 6.6.2.1.3, 6.6.2.2.2.1, 6.4.1.1 and 6.4.1.6, ETSI TS 124 380 [5] (3GPP TS 24.380), clauses 6.4.2.1, 6.4.2.2 and 6.4.2.3. Unless otherwise stated, these are Rel-13 requirements.

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.3.1.1]

"Editor's Note: Behaviour for cases where the IWF affiliates on behalf of its LMR users is FFS.

In this clause, the IWF originates a prearranged group session on behalf of an LMR user.

NOTE 1: How the IWF determines the MCPTT ID of the calling user is out of scope of the present document.

The IWF, performing the originating participating function:

1) if the user identified by the MCPTT ID is not affiliated to the group as determined by clause 9.2.1.2.8 and this is an authorised request for originating a priority call as determined by clause 6.6.2.1.4.2, shall perform the actions specified in clause 9.2.1.2.9 for implicit affiliation;

2) shall determine the public service identity of the controlling MCPTT function associated with the group identity of the group on which the call is to be originated;

NOTE 2: How the IWF discovers the public service identity of the controlling MCPTT function associated with the group identity is out of scope of the present document.

3) shall generate a SIP INVITE request as specified in clause 10.1.2.1;

4) shall modify the SIP INVITE request as specified in clause 6.6.2.1.2;

5) may insert the calling user's location information into an application/vnd.3gpp.mcptt-location-info+xml MIME body to be included in the outgoing SIP request;

6) shall set the Request-URI to the public service identity of the controlling MCPTT function associated with the group identity;

7) shall set the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP INVITE request to the MCPTT ID of the calling user;

8) shall update the SDP as specified in clause 6.6.2.1.1.1; and

9) shall send the SIP INVITE request to the controlling MCPTT function as specified in 3GPP TS 24.229 [3].

Upon receipt of a SIP 302 (Moved Temporarily) response to the above SIP INVITE request, the participating IWF function:

1) shall generate a SIP INVITE request as specified in clause 6.6.2.1.5;

2) shall include an SDP offer based upon the SDP offer in the SIP INVITE request generated by the IWF in the step above; and

3) shall forward the SIP INVITE request according to 3GPP TS 24.229 [3].

Upon receipt of a SIP 2xx response in response to the above SIP INVITE request, the IWF performing the participating role:

NOTE 3: If an <MKFC-GKTPs> element is received, the IWF ignores that element.

1) if the procedures of clause 9.2.1.2.9 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 9.2.1.2.10;

2) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [6].

3) shall perform the steps for SIP 2xx as specified in clause 10.1.2.1.

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request, the participating IWF function:

1) if the implicit affiliation procedures of clause 9.2.1.2.9 were invoked in this procedure, shall perform the procedures of clause 9.2.1.2.11; and

2) shall perform the steps for the received SIP 4xx, 5xx or 6xx as specified in clause 10.1.2.1.

On receiving a SIP INFO request where the Request-URI contains an MCPTT session ID identifying an ongoing group session, the IWF shall perform the steps for the received SIP INFO in clause 10.1.2.1."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.3.1.2]

"Upon a need to send a SIP re-INVITE request for an MCPTT session identifying an on-demand prearranged MCPTT group session, the IWF performing the participating role:

1) if the request is for an upgrade to an in-progress emergency or an imminent peril, shall perform the steps in clause 10.1.2.3;

2) if the request is for a cancellation of an in-progress emergency, shall perform the steps in clause 10.1.2.4;

3) if the request is for a cancellation of an in-progress imminent peril, shall perform the steps in clause 10.1.2.5;

4) shall include the MCPTT ID of the originating user in <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP re-INVITE request;

NOTE 1: How the IWF determines the MCPTT ID of a user homed in the IWF is out of scope of the present document.

5) shall include in the SIP re-INVITE request an SDP offer as specified in clause 6.6.2.1.1;

6) if the SIP re-INVITE requires a Resource-Priority header field, shall include a Resource-Priority header field according to 6.4.1.11; and

7) shall send the SIP re-INVITE request according to 3GPP TS 24.229 [3].

Upon receipt of a SIP 2xx response to the above SIP re-INVITE request, the IWF performing the participating role:

1) shall interact with the media plane as specified in 3GPP TS 29.380 [31];

2) if the request in the present clause above is for an upgrade for emergency or imminent peril, shall follow the procedures for SIP 200 (OK) response as specified in clause 10.1.2.3;

3) if the request in the present clause above is for an in-progress emergency cancel, shall follow the procedures for SIP 200 (OK) response as specified in clause 10.1.2.4; and

4) if the request in the present clause above is for an in-progress imminent peril cancel, shall follow the procedures for SIP 200 (OK) response as specified in clause 10.1.2.5.

Upon receipt of a SIP 403 (Forbidden) response to the above SIP re-INVITE request, the IWF performing the participating role shall interact with the media plane as specified in 3GPP TS 29.380 [31].

On receiving a SIP 4xx response, a SIP 5xx response or a SIP 6xx response to the SIP re-INVITE request, the IWF performing the participating role:

1) if the request in the present clause above is for an upgrade for emergency or imminent peril, shall follow the procedures for SIP 4xx, SIP 5xx and SIP 6xx responses as specified in clause 10.1.2.3;

2) if the request in the present clause above is for an in-progress emergency cancel, shall follow the procedures for SIP 4xx, SIP 5xx and SIP 6xx responses as specified in clause 10.1.2.4; or

3) if the request in the present clause above is for an in-progress imminent peril cancel, shall follow the procedures for SIP 4xx, SIP 5xx and SIP 6xx responses as specified in clause 10.1.2.5.

Upon receiving a SIP INFO request where the Request-URI contains an MCPTT session ID identifying an ongoing session, the IWF performing the participating role:

1) if the SIP re-INVITE request in the present clause above is for an upgrade for emergency or imminent peril, shall follow the procedures for SIP INFO as specified in clause 10.1.2.3;

2) if the SIP re-INVITE request in the present clause above is for an in-progress emergency cancel, shall follow the procedures for SIP INFO as specified in clause 10.1.2.4; or

3) if the SIP re-INVITE request in the present clause above is for an in-progress imminent peril cancel, shall follow the procedures for SIP INFO as specified in clause 10.1.2.5."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a "SIP INVITE request for terminating participating MCPTT function", the IWF performing the participating role:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The IWF performing the participating role may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [14], and shall not continue with the rest of the steps;

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the IWF performing the participating role shall reject the request with a SIP 403 (Forbidden) response with the warning text set to "104 isfocus not assigned" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

3) may reject the request with a SIP 480 (Temporarily Unavailable) response with the warning text set to "146 T-PF unable to determine the service settings for the called user" in a Warning header field as specified in 3GPP TS 24.379 [29], clause 4.4, with the IWF acting as the MCPTT server and shall not continue with the rest of the steps; and

NOTE: If an <MKFC-GKTPs> element is received, the IWF ignores it.

4) shall perform the steps in 3GPP TS 24.379 [29], clause 10.1.2.1.2."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.3.3.1]

"When the IWF performing the participating role determines it has to send a SIP BYE request, the IWF shall follow the procedures as specified in clause 6.6.2.1.3."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.3.4.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the IWF performing the participating role shall follow the procedures as specified in clause 6.6.2.2.2.1."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.3.6]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

Upon receipt of a SIP re-INVITE request for an MCPTT group containing an emergency indication or imminent peril indication, the IWF performing the participating role:

1) shall perform the steps in clause 10.1.2.6;

2) shall interact with the media plane as specified in 3GPP TS 29.380 [31]; and

3) shall send the SIP 200 (OK) response according to 3GPP TS 24.229 [3]."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.2.1]

"This clause is referred to by other clauses.

To establish an MCPTT prearranged group session, the IWF performing the participating role shall generate an initial SIP INVITE request by following the UE originating session procedures specified in 3GPP TS 24.229 [3], with the clarifications given below.

The IWF performing the participating role:

1) if originating an MCPTT emergency group call or originating an MCPTT prearranged group call and the MCPTT emergency state is already set, the IWF performing the participating role shall comply with the procedures in clause 6.4.1.1;

2) if originating an MCPTT imminent peril group call, the IWF performing the participating role shall comply with the procedures in clause 6.4.1.6;

3) if originating a broadcast group call, the IWF performing the participating role shall include in the application/vnd.3gpp.mcptt-info+xml MIME body the <broadcast-ind> element set to "true" as defined in 3GPP TS 24.379 [29], clause F.1;

4) shall include the g.3gpp.mcptt media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field of the SIP INVITE request according to IETF RFC 3840 [9];

5) shall include an Accept-Contact header field containing the g.3gpp.mcptt media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [5];

6) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [5];

7) should include the "timer" option tag in the Supported header field;

8) should include the Session-Expires header field according to IETF RFC 4028 [6]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

9) if the emergency group state for this group is set to "MEG 2: in-progress" or "MEG 4: confirm-pending", the IWF performing the participating role shall include the Resource-Priority header field and comply with the procedures in clause 6.4.1.2;

10) if the imminent peril group state for this group is set to "MIG 2: in-progress" or "MIG 4: confirm-pending" shall include the Resource-Priority header field and comply with the procedures in clause 6.4.1.8;

11) shall contain an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <session-type> element set to a value of "prearranged";

b) the <mcptt-request-uri> element set to the group identity;

c) the <mcptt-client-id> element set to a value determined by the IWF; and

NOTE 1: How the IWF determines the value of the <mcptt-client-id> element is out of scope of the present document.

d) if the group identity can be determined to be a TGI and if the IWF performing the participating role can associate the TGI with a MCPTT group ID, the <associated-group-id> element set to the MCPTT group ID;

NOTE 2: The MCPTT ID will be inserted into the body of the SIP INVITE request by the referring clause.

NOTE 3: The text "can associate the TGI with a MCPTT group ID" means that the IWF performing the participating role is able to determine that there is a constituent group of the temporary group that it is a member of.

NOTE 4: The IWF performing the participating role is informed about temporary groups and regrouping of MCPTT groups that the users homed in the IWF are members of as specified in 3GPP TS 24.481 [16].

NOTE 5: If the TGI has several MCPTT groups as constituent groups, where the user homed in the IWF is a member, the IWF performing the participating role selects one of those MCPTT groups.

12) shall include an SDP offer according to 3GPP TS 24.229 [3] with the clarifications given in clause 6.1.1; and

13) if an implicit floor request is required, shall indicate this as specified in clause 6.7; and shall skip the rest of the steps.

On receiving a SIP 2xx response to the SIP INVITE request, the IWF performing the participating role:

1) shall interact with the user plane as specified in 3GPP TS 29.380 [31];

2) if the MCPTT emergency group call state is set to "MEGC 2: emergency-call-requested" or "MEGC 3: emergency-call-granted" or the MCPTT imminent peril group call state is set to "MIGC 2: imminent-peril-call-requested" or "MIGC 3: imminent-peril-call-granted", the IWF performing the participating role shall perform the actions specified in clause 6.4.1.4; and

3) may subscribe to the conference event package as specified in clause 10.3.

On receiving a SIP 4xx response, a SIP 5xx response or a SIP 6xx response to the SIP INVITE request:

1) if the MCPTT emergency group call state is set to "MEGC 2: emergency-call-requested" or "MEGC 3: emergency-call-granted"; or

2) if the MCPTT imminent peril group call state is set to "MIGC 2: imminent-peril-call-requested" or "MIGC 3: imminent-peril-call-granted";

the IWF performing the participating role shall perform the actions specified in clause 6.4.1.5.

On receiving a SIP INFO request where the Request-URI contains an MCPTT session ID identifying an ongoing group session, the IWF performing the participating role shall follow the actions specified in clause 6.4.1.9."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.2.2]

"This clause is referenced from other procedures.

In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body.

The IWF performing the participating role:

1) may reject the SIP INVITE request,

NOTE: The conditions under which the IWF rejects the request are out of scope of the present document.

2) if the SIP INVITE request is rejected in step 1), shall respond towards the controlling MCPTT function either with an appropriate reject code as specified in 3GPP TS 24.229 [3] and warning texts as specified in clause 4.2.2 or with a SIP 480 (Temporarily unavailable) response and skip the rest of the steps of this clause;

3) if the SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with the <emergency-ind> element set to a value of "true":

a) shall set the MCPTT emergency group state to "MEG 2: in-progress";

b) shall set the MCPTT imminent peril group state to "MIG 1: no-imminent-peril"; and

c) shall set the MCPTT imminent peril group call state to "MIGC 1: imminent-peril-gc-capable"; otherwise

4) if the SIP INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with the <imminentperil-ind> element set to a value of "true", shall set the MCPTT imminent peril group state to "MIG 2: in-progress";

5) shall perform the automatic commencement procedures specified in clause 6.2.1 if one of the following conditions are met:

a) if the SIP INVITE request contains an Answer-Mode header field with the value "Auto" and IWF is configured for automatic commencement mode for receiving the call;

b) if the SIP INVITE request contains an Answer-Mode header field with the value "Auto" and IWF is configured to allow automatic commencement mode for receiving the call; or

c) SIP INVITE request contains an Answer-Mode header field with the value "Manual" and IWF is not configured to allow manual commencement mode for receiving the call;

6) shall perform the manual commencement procedures specified in clause 6.2.2 if one of the following conditions are met:

a) if the SIP INVITE request contains an Answer-Mode header field with the value "Manual" and IWF is configured for manual commencement mode for receiving the call;

b) if the SIP INVITE request contains an Answer-Mode header field with the value "Manual" and IWF is configured to allow manual commencement mode for receiving the call; or

c) SIP INVITE request contains an Answer-Mode header field with the value "Automatic" and IWF is not configured to allow automatic commencement mode for receiving the call; and

7) when the SIP 200 (OK) response to the SIP INVITE request is sent, may subscribe to the conference event package as specified in clause 10.3."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.2.3]

"This clause is referenced from other procedures.

To upgrade the MCPTT group session to an emergency condition or an imminent peril condition on an MCPTT prearranged group, the IWF performing the participating role shall perform the steps below.

1) if the request is to upgrade the MCPTT group session to an MCPTT emergency call, the IWF performing the participating role:

a) shall include an application/vnd.3gpp.mcptt-info+xml MIME body populated as specified in clause 6.4.1.1; and

b) shall include a Resource-Priority header field and comply with the procedures in clause 6.4.1.2.

2) if the MCPTT user has requested to upgrade the MCPTT group session to an MCPTT imminent peril call, the IWF performing the participating role:

a) shall include an application/vnd.3gpp.mcptt-info+xml MIME body populated as specified in clause 6.4.1.6; and

b) shall include a Resource-Priority header field and comply with the procedures in clause 6.4.1.8;

3) if the SIP re-INVITE request is to be sent within an on-demand session, shall include in the SIP re-INVITE request an SDP offer according to 3GPP TS 24.229 [3] with the clarifications specified in clause 6.1.1;

NOTE: The SIP re-INVITE request can be sent within an on-demand session.

4) if an implicit floor request is required, shall indicate this as specified in clause 6.7; and

5) shall exit the procedure in the present clause.

On receiving a SIP 2xx response to the SIP re-INVITE request the IWF performing the participating role:

1) shall perform the actions specified in clause 6.4.1.4.

On receiving a SIP INFO request where the Request-URI contains an MCPTT session ID identifying an ongoing group session, the IWF performing the participating role shall follow the actions specified in clause 6.4.1.9.

On receiving a SIP 4xx response, SIP 5xx response or a SIP 6xx response to the SIP re-INVITE request the IWF performing the participating role shall perform the actions specified in clause 6.4.1.5."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.2.4]

"This clause is referenced from other procedures.

To cancel the in-progress emergency condition on a prearranged MCPTT group, the IWF performing the participating role shall generate a SIP re-INVITE request by following the UE originating session procedures specified in 3GPP TS 24.229 [3], with the clarifications given below.

The IWF performing the participating role:

1) shall, if cancelling an in-progress emergency condition and optionally an MCPTT emergency alert originated by the user homed in the IWF, include an application/vnd.3gpp.mcptt-info+xml MIME body populated as specified in clause 6.4.1.3;

2) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <session-type> element set to a value of "prearranged"; and

b) the <mcptt-request-uri> element set to the group identity;

3) shall set the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP re-INVITE request to the MCPTT ID of the calling user;

4) shall include the g.3gpp.mcptt media feature tag in the Contact header field of the SIP re-INVITE request according to IETF RFC 3840 [9];

5) shall include in the SIP re-INVITE request an SDP offer according to 3GPP TS 24.229 [3] with the clarifications specified in clause 6.1.1;

6) shall include a Resource-Priority header field and comply with the procedures in clause 6.4.1.2; and

7) shall exit the procedure in the present clause.

On receiving a SIP 2xx response to the SIP re-INVITE request, the IWF performing the participating role:

1) shall interact with the user plane as specified in 3GPP TS 29.380 [31];

2) shall set the MCPTT emergency group state of the group to "MEG 1: no-emergency";

3) shall set the MCPTT emergency group call state of the group to "MEGC 1: emergency-gc-capable"; and

4) if the MCPTT emergency alert state is set to "MEA 4: Emergency-alert-cancel-pending", the sent SIP re-INVITE request did not contain an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body and the SIP 2xx response to the SIP request for a priority group call does not contain a Warning header field as specified in clause 4.4 with the warning text containing the mcptt-warn-code set to "149", shall set the MCPTT emergency alert state to "MEA 1: no-alert".

On receiving a SIP INFO request where the Request-URI contains an MCPTT session ID identifying an ongoing group session, the IWF performing the participating role shall follow the actions specified in clause 6.4.1.9 with the IWF acting as the MCPTT client.

On receiving a SIP 4xx response, SIP 5xx response or SIP 6xx response to the SIP re-INVITE request:

1) shall set the MCPTT emergency group state as "MEG 2: in-progress";

2) if the SIP 4xx response, SIP 5xx response or SIP 6xx response contains an application/vnd.3gpp.mcptt-info+xml MIME body with an <alert-ind> element set to a value of "true" and the sent SIP re-INVITE request did not contain an <originated-by> element in the application/vnd.3gpp.mcptt-info+xml MIME body, the IWF performing the participating role shall set the MCPTT emergency alert state to "MEA 3: emergency-alert-initiated"; and

3) if the SIP 4xx response, SIP 5xx response or SIP 6xx response did not contain an application/vnd.3gpp.mcptt-info+xml MIME body with an <alert-ind> element and did not contain an <originated-by> element, the MCPTT emergency alert (MEA) state shall revert to its value prior to entering the current procedure.

NOTE: If the in-progress emergency group state cancel request is rejected, the state of the session does not change, i.e. continues with MCPTT emergency group call level priority."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.2.5]

"This clause is referenced from other procedures.

To cancel the in-progress imminent peril condition on a prearranged MCPTT group, the IWF performing the participating role shall generate a SIP re-INVITE request by following the procedures specified in 3GPP TS 24.229 [3], with the clarifications given below.

The IWF performing the participating role:

1) shall include an application/vnd.3gpp.mcptt-info+xml MIME body populated as specified in clause 6.4.1.7;

2) shall include a Resource-Priority header field and comply with the procedures in clause 6.4.1.8;

3) shall include in the application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with:

a) the <session-type> element set to a value of "prearranged"; and

b) the <mcptt-request-uri> element set to the group identity;

4) shall set the <mcptt-calling-user-id> element of the application/vnd.3gpp.mcptt-info+xml MIME body of the SIP re-INVITE request to the MCPTT ID of the calling user;

5) shall include the g.3gpp.mcptt media feature tag in the Contact header field of the SIP re-INVITE request according to IETF RFC 3840 [9];

6) shall include in the SIP re-INVITE request an SDP offer according to 3GPP TS 24.229 [3] with the clarifications specified in clause 6.1.1; and

7) shall exit the procedure in the present clause.

On receiving a SIP 2xx response to the SIP re-INVITE request, the IWF performing the participating role:

1) shall interact with the user plane as specified in 3GPP TS 29.380 [31];

2) shall set the MCPTT imminent peril group state of the group to "MIG 1: no-imminent-peril"; and

3) shall set the MCPTT imminent peril group call state of the group to "MIGC 1: imminent-peril-gc-capable".

On receiving a SIP 4xx, SIP 5xx response or SIP 6xx response to the SIP re-INVITE request:

1) if the SIP 4xx response, SIP 5xx response or SIP 6xx response:

a) contains an application/vnd.3gpp.mcptt-info+xml MIME body with an <imminentperil-ind> element set to a value of "true"; or

b) does not contain an application/vnd.3gpp.mcptt-info+xml MIME body with an <imminentperil-ind> element;

then the IWF performing the participating role shall set the MCPTT imminent peril group state as "MIG 2: in-progress".

NOTE: This is the case where the IWF performing the participating role requested the cancellation of the MCPTT imminent peril in-progress state and was rejected."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 10.1.2.6]

"This clause is referred to by other clauses.

The IWF:

1) *if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with the <emergency-ind> element set to a value of "true":*

   a) *shall set the MCPTT emergency group state to "MEG 2: in-progress";*

   b) *shall set the MCPTT imminent peril group state to "MIG 1: no-imminent-peril"; and*

   c) *shall set the MCPTT imminent peril group call state to "MIGC 1: imminent-peril-gc-capable";*

2) *if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with the <imminentperil-ind> element set to a value of "true":*

   a) *shall set the MCPTT imminent peril group state to "MIG 2: in-progress";*

3) *if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with the <emergency-ind> element set to a value of "false":*

   a) *if the <mcpttinfo> element containing the <mcptt-Params> element contains an <alert-ind> element set to "false":*

      i) *if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body including an <originated-by> element:*

         A) *if the MCPTT ID contained in the <originated-by> element is the MCPTT ID of the receiving user homed in the IWF, shall set the MCPTT emergency alert state to "MEA 1: no-alert";*

   b) *shall set the MCPTT emergency group state to "MEG 1: no-emergency"; and*

   c) *if the MCPTT emergency group call state of the group is set to "MEGC 3: emergency-call-granted", shall set the MCPTT emergency group call state of the group to "MEGC 1: emergency-gc-capable";*

4) *if the SIP re-INVITE request contains an application/vnd.3gpp.mcptt-info+xml MIME body with the <mcpttinfo> element containing the <mcptt-Params> element with the <imminentperil-ind> element set to a value of "false":*

   a) *shall set the MCPTT imminent peril group state to "MIG 1: no-imminent-peril"; and*

   b) *shall set the MCPTT imminent peril group call state to "MIGC 1: imminent-peril-gc-capable";*

5) *shall accept the SIP re-INVITE request and generate a SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [3];*

6) *shall include the g.3gpp.mcptt media feature tag in the Contact header field of the SIP 200 (OK) response;*

7) *shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field of the SIP 200 (OK) response; and*

8) *shall include an SDP answer in the SIP 200 (OK) response to the SDP offer in the incoming SIP re-INVITE request according to 3GPP TS 24.229 [3] with the clarifications given in clause 6.1.2."*

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer generated by the IWF performing the participating role:

1) shall contain only one SDP media-level clause for speech; and

2) shall contain an SDP media-level clause for one media-floor control entity.

When composing the SDP offer according to 3GPP TS 24.229 [3], the IWF performing the participating role:

1) shall insert the IP address and port number selected by the IWF performing the participating role for the media stream in the SDP offer;

NOTE 1: Requirements can exist for the IWF performing the participating role to be always included in the path of the offered media stream, for example: for the support of features such as lawful interception and recording. Other examples can exist.

2) shall insert the IP address and port number selected by the IWF performing the participating role for the offered media floor control entity, if any, in the received SDP offer; and

3) shall include an "m=audio" media-level clause for the media stream consisting of:

a) the port number for the media stream selected; and

b) the codec(s) and media parameters and attributes with the following clarification:

i) if a call is being initiated to a group identity; and

ii) if the IWF performing the participating role determines one or more preferred codecs;

then the IWF:

i) shall include the name of the chosen codec in the <encoding name> field of the "a=rtpmap" attribute as defined in IETF RFC 4566 [8];

c) "i=" field set to "speech" according to 3GPP TS 24.229 [3];

4) if floor control shall be used during the session, shall include an "m=application" media-level clause as specified in 3GPP TS 29.380 [31] clause 12 for a media-floor control entity, consisting of:

a) the port number for the media-floor control entity selected as specified in 3GPP TS 29.380 [31]; and

b) the 'fmtp' attributes as specified in 3GPP TS 29.380 [31] clause 14;

5) if security between the IWF and the MCPTT system is required for a private call, shall include the MIKEY-SAKKE I_MESSAGE in an "a=key-mgmt" attribute as a "mikey" attribute value in the SDP offer as specified in IETF RFC 4567 [22]; and

6) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value.

NOTE: End-to-end security of voice traffic is supported through the use of Interworking Security Data messages specified in 3GPP TS 29.582 [33]. Through the use of these messages, an LMR system can communicate security information to the LMR functionality of a device on the MCPTT system. This will allow the use of LMR algorithms and keys for encrypting and decrypting voice packets within the device that can be transmitted to and received from the LMR system."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.2.1.2]

"This clause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [3] the IWF performing the participating role:

1) *should include the Session-Expires header field according to IETF RFC 4028 [6]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";*

2) *shall include the option tag "timer" in the Supported header field;*

3) *shall include in the P-Asserted-Identity header field the public service identity of the IWF performing the participating role;*

4) *shall include the g.3gpp.mcptt media feature tag into the Contact header; and*

5) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [3]), into the P-Asserted-Service header field."*

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.2.1.3]

"When the IWF is ending participation in an MCPTT session and decides to send a SIP BYE request, the IWF:

1) *shall interact with the media plane as specified in clause 6.4 in 3GPP TS 29.380 [31];*

2) *shall generate a SIP BYE request as specified in 3GPP TS 24.229 [3];*

3) *shall set the Request-URI to the MCPTT session identity of the MCPTT session;*

4) *shall set the P-Asserted-Identity header field of the outgoing SIP BYE request to the public service identity of the IWF;*

5) *may insert an application/vnd.3gpp.mcptt-info+xml MIME body into the outgoing SIP BYE request; and*

6) *shall send the SIP BYE request towards the controlling MCPTT function, according to 3GPP TS 24.229 [3].*

Upon receiving a SIP 200 (OK) response to the SIP BYE request the IWF should complete any further actions necessary to dissociate the LMR user from the MCPTT session and shall interact with the media plane to release any resources as specified in clause 6.4 in 3GPP TS 29.380 [31] for releasing media plane resources associated with the SIP session with the controlling MCPTT function."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.6.2.2.1]

"Upon receiving a SIP BYE request from the controlling MCPTT function, the IWF performing the participating role:

1) *shall interact with the media plane as specified in clause 6.4 in 3GPP TS 29.380 [31] for releasing media plane resource associated with the SIP session; and*

2) *shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCPTT function according to 3GPP TS 24.229 [3]."*

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.4.1.1]

"This clause is referenced from other procedures.

When the MCPTT emergency state is set, the IWF performing the participating role:

1) *shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the SIP INVITE request, an <emergency-ind> element set to "true" and if the MCPTT emergency group call state is set to "MEGC 1: emergency-gc-capable", shall set the MCPTT emergency group call state to "MEGC 2: emergency-call-requested";*

2) *if the IWF has determined that an MCPTT emergency alert is to be sent, and the MCPTT emergency alert state is set to "MEA 1: no-alert", shall:*

   a) *set the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to "true" and set the MCPTT emergency alert state to "MEA 2: emergency-alert-confirm-pending"; and*

   b) *include in the SIP INVITE request the specific location information for MCPTT emergency alert as specified in clause 6.5.1;*

3) *if the IWF has determined that an MCPTT emergency alert is not to be sent and the MCPTT emergency alert state is set to "MEA 1: no-alert", shall set the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to "false"; and*

4) *if the MCPTT client emergency group state of the user homed in the IWF of the group is set to a value other than "MEG 2: in-progress" set the MCPTT client emergency group state of the user homed in the IWF of the MCPTT group to "MEG 4: confirm-pending".*

NOTE 1: This is the case of a user homed in the IWF already being in the MCPTT emergency state it initiated previously while originating an MCPTT emergency group call or MCPTT emergency alert. All group calls the user homed in the IWF originates while in MCPTT emergency state will be MCPTT emergency group calls.

When the MCPTT emergency state is clear and the MCPTT emergency group call state is set to "MEGC 1: emergency-gc-capable" and the IWF determines that the request for MCPTT emergency group call is authorized by local policy, the IWF performing the participating role shall set the MCPTT emergency state and perform the following actions:

1) *shall include in the application/vnd.3gpp.mcptt-info+xml MIME body in the SIP INVITE request an <emergency-ind> element set to "true" and set the MCPTT emergency group call state to "MEGC 2: emergency-call-requested" state;*

2) *if the user homed in the IWF has also requested an MCPTT emergency alert to be sent, shall:*

   a) *include in the application/vnd.3gpp.mcptt-info+xml MIME body the <alert-ind> element set to "true" and set the MCPTT emergency alert state to "MEA 2: emergency-alert-confirm-pending"; and*

   b) *include in the SIP INVITE request the specific location information, if available, for MCPTT emergency alert as specified in clause 6.5.1;*

3) *if the IWF has determined that an MCPTT emergency alert is not to be sent, shall set the <alert-ind> element of the application/vnd.3gpp.mcptt-info+xml MIME body to "false"; and*

4) *if the MCPTT client emergency group state of the user homed in the IWF of the group is set to a value other than "MEG 2: in-progress" shall set the MCPTT client emergency group state of the user homed in the IWF of the MCPTT group to "MEG 4: confirm-pending".*

NOTE 2: This is the case of an initial MCPTT emergency group call and optionally an MCPTT emergency alert being sent. As the MCPTT emergency state is not sent, there is no MCPTT emergency alert outstanding.

NOTE 3: An MCPTT group call originated by an affiliated member of an MCPTT group which is in an in-progress emergency state (as tracked on the MCPTT client by the MCPTT client emergency group state) but is not in an MCPTT emergency state of their own will also be an MCPTT emergency group call. The <emergency-ind> and <alert-ind> elements of the application/vnd.3gpp.mcptt-info+xml MIME body do not need to be included in this case and hence no action needs to be taken in this clause."

[ETSI TS 129 379 [30] (3GPP TS 29.379), clause 6.4.1.6]

"This clause is referenced from other procedures.

When the IWF performing the participating role determines to originate an MCPTT imminent peril group call, the IWF performing the participating role:

1) if the client homed in the IWF's imminent peril group state is set to "MIGC 1: imminent-peril-gc-capable" and the in-progress emergency state of the group is set to a value of "false":

a) shall include in the SIP request an application/vnd.3gpp.mcptt-info+xml MIME body as defined in 3GPP TS 24.379 [29], clause F.1 with the <imminentperil-ind> element set to "true" and set the MCPTT emergency group call state to "MIGC 2: imminent-peril-call-requested" state; and

b) if the client homed in the IWF's imminent peril group state of the group is set to a value other than "MIG 2: in-progress" shall set the client homed in the IWF's emergency group state of the MCPTT group to "MIG 4: confirm-pending".

NOTE: An MCPTT group call originated by an affiliated member of an MCPTT group which is in an in-progress imminent peril state (as tracked on the client by the client imminent peril group state) will also have the priority associated with MCPTT imminent peril group calls. The <imminentperil-ind> element of the application/vnd.3gpp.mcptt-info MIME body does not need to be included in this case, nor do any state changes result and hence no action needs to be taken in this clause."

[ETSI TS 129 380 [31] (3GPP TS 29.380), clause 6.4.2.1]

"Based on the negotiations during the call establishment specified in 3GPP TS 29.379 [26], a new instance of a media session shall be created for this call.

The SIP INVITE request sent by the application and signalling plane:

1. shall be regarded an implicit floor request when an implicit floor request is negotiated; and

2. shall not be regarded as an implicit floor request in case of a rejoin to an already on-going group call.

NOTE: The IWF floor participant can negotiate the use of prioritization of the Floor Request message. In that case, the floor participant can request permission to send media at a priority level that is either the same as or lower than the highest priority that was permitted to the participant in the MCPTT call initialization. If a floor participant is authorized for pre-emptive priority in the MCPTT call it is good practise to always request permission to send RTP media packets at a priority level that is lower than pre-emptive priority unless the IWF explicitly needs to pre-empt the current RTP media packets' sender. In any case pre-emptive priority will have no effect for audio cut-in floor control."

[ETSI TS 129 380 [31] (3GPP TS 29.380), clause 6.4.2.2]

"When a call is initiated as described in 3GPP TS 29.379 [26], the IWF floor participant:

1. shall create a media session;

2. if the originating IWF floor participant receives a floor control message before it receives the SIP 200 (OK) response, shall store the floor control message; and

NOTE: The originating IWF floor participant might receive a floor control message before the SIP 200 (OK) response when initiating, joining or rejoining a call because of processing delays of the SIP 200 (OK) response in the SIP core.

3. if a floor control message is received after a SIP INVITE request with implicit floor request and before the reception of the SIP 200 (OK), shall act upon the floor control message."

[ETSI TS 129 380 [31] (3GPP TS 29.380), clause 6.4.2.3]

"When an MCPTT call is established, the terminating IWF floor participant shall create an instance of a media session.

NOTE: From an IWF floor participant perspective the MCPTT call is established when the application and signalling plane sends the SIP 200 (OK) response."

6.1.1.2.3 Test description

6.1.1.2.3.1 Pre-test conditions

System Simulator:

- SS (MCPTT Server)
  - SS (MCPTT Server) provides the controlling MCPTT function
- SS-UE2 (MCPTT client)
  - Simulated behind the SS (MCPTT Server) as defined in Figure 6.1.1.2.3.1-1.

IUT:

- IUT (IWF)
  - IUT (IWF) provides the interworking function (IWF) performing the originating participating MCPTT role
- The IUT (IWF) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (IWF) also consists of all sub-systems of the IWF, including the Media Distribution Function, the MCPTT User Database, the SIP AS, the HTPP Server, the HTTP Client, and the Floor Control Server.

Preamble:

- The IUT (IWF) is connected to SS (MCPTT Server) defined in Annex A, Figures A.4.2 and A.4.3.
- The SS-UE2 (MCPTT client) is registered and authorized to use the MCPTT system.
- The SS-UE2 (MCPTT client) is affiliated with Group A and are authorized to initiate group calls.
- Group A is controlled by the controlling MCPTT function.

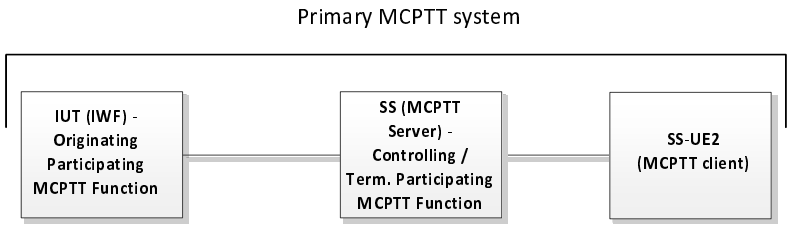


Figure 6.1.1.2.3.1-1: Functions of the testing components

6.1.1.2.3.2 Test procedure sequence

Table 6.1.1.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Make the IWF initiate an on-demand pre-arranged group call with automatic commencement mode and implicit floor control on behalf of an LMR user.	-	-	-	-
2	Check: Does the IUT (IWF) send a SIP INVITE request to the SS (MCPTT Server) to initiate a pre-arranged group call with automatic commencement mode and implicit floor control?	-->	SIP INVITE	1	P
3	The SS (MCPTT Server) responds with a 100 Trying message to the IUT (IWF).	<--	SIP 100 Trying		-
4	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
-	EXCEPTION: Depending on the IWF implementation, steps 5-7 might be performed in parallel.	-	-	-	-
5	Check: Does the IUT (IWF) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	1	P
6	Make the IWF release the floor with an acknowledgement required on behalf of an LMR user.	-	-	-	-
7	Check: Does the IUT (IWF) send a Floor Release message to the SS (MCPTT Server) with an acknowledgement required?	-->	Floor Release	2	P
8	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		
9	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
10	Make the IWF request the floor on behalf of an LMR user.	-	-	-	-
11	Check: Does the IUT (IWF) send a Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	2	P
12	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
13	The SS (MCPTT Server) sends a Floor Revoke message with no acknowledgment required.	<--	Floor Revoke		-
14	Check: Does the IUT (IWF) send the Floor Release message to the SS (MCPTT Server)?	-->	Floor Release	2	P
15	The SS (MCPTT Server) sends a Floor Taken message to the IUT (IWF) informing that the floor is taken by SS-UE2 (MCPTT client)	<--	Floor Taken		-
16	Make the IWF request the floor on behalf of an LMR user with a lower priority than SS-UE2 (MCPTT client) and without queueing support.	-	-	-	-
17	Check: Does the IUT (IWF) send the Floor Request message to the SS (MCPTT server) with a lower priority than SS-UE2 (MCPTT client) and the Floor Indicator F-bit set to 0?	-->	Floor Request	2	P
18	The SS (MCPTT Server) sends a Floor Deny message to the IUT (IWF)	<--	Floor Deny		-
19	Make the IWF request the floor on behalf of an LMR user with a lower priority than SS-UE2 (MCPTT client) and with queueing support.	-	-	-	-
20	Check: Does the IUT (IWF) send the Floor Request message to the SS (MCPTT server) with a lower priority than SS-UE2 (MCPTT client) and the Floor Indicator F-bit set to 1?	-->	Floor Request	2	P
21	The SS (MCPTT Server) sends a Floor Queue Position Info message to the IUT (IWF)	<--	Floor Queue Position Info		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
22	Make the IUT (IWF) request the position in the queue on behalf of the LMR user.	-	-	-	-
23	Check: Does the IUT (IWF) send a Floor Queue Position Request message to the SS (MCPTT Server)?	-->	Floor Queue Position Request	2	P
24	The SS (MCPTT Server) sends a Floor Queue Position Info message to the IUT (IWF)	<--	Floor Queue Position Info		-
25	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
26	Make the IWF release the floor with an acknowledgement required on behalf of an LMR user.	-	-	-	-
27	Check: Does the IUT (IWF) send a Floor Release message to the SS (MCPTT Server) with an acknowledgement required?	-->	Floor Release	2	P
28	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		-
29	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
30	Make the IUT (IWF) request the modification of the call to upgrade the call to an emergency group call with implicit floor request on behalf of the LMR user.	-	-	-	-
31	Check: Does the IUT (IWF) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE	3	P
32	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
33	Check: Does the IUT (IWF) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	3	P
34	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
35	Make the IWF release the floor with an acknowledgement required on behalf of an LMR user.	-	-	-	-
36	Check: Does the IUT (IWF) send a Floor Release message to the SS (MCPTT Server) with an acknowledgement required?	-->	Floor Release	4	P
37	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		-
38	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
39	Make the IWF request the floor on behalf of an LMR user.	-	-	-	-
40	Check: Does the IUT (IWF) send a Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	4	P
41	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
42	Make the IWF release the floor with an acknowledgement required on behalf of an LMR user.	-	-	-	-
43	Check: Does the IUT (IWF) send a Floor Release message to the SS (MCPTT Server) with an acknowledgement required?	-->	Floor Release	4	P
44	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		-
45	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
46	Make the IUT (IWF) request the modification of the call to downgrade the call to a normal group call without implicit floor request on behalf of the LMR user.	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
47	Check: Does the IUT (IWF) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE	5	P
48	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
49	Check: Does the IUT (IWF) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	5	P
50	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
51	Make the IWF request the floor on behalf of an LMR user.	-	-	-	-
52	Check: Does the IUT (IWF) send a Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	6	P
53	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
54	Make the IWF release the floor with an acknowledgement required on behalf of an LMR user.	-	-	-	-
55	Check: Does the IUT (IWF) send a Floor Release message to the SS (MCPTT Server) with an acknowledgement required?	-->	Floor Release	6	P
56	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		-
57	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
58	Make the IUT (IWF) request the modification of the call to upgrade the call to an imminent-peril group call with implicit floor request on behalf of the LMR user.	-	-	-	-
59	Check: Does the IUT (IWF) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE	7	P
60	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
61	Check: Does the IUT (IWF) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	7	P
62	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
63	Make the IWF release the floor with an acknowledgement required on behalf of an LMR user.	-	-	-	-
64	Check: Does the IUT (IWF) send a Floor Release message to the SS (MCPTT Server) with an acknowledgement required?	-->	Floor Release	8	P
65	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		-
66	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
67	Make the IWF request the floor on behalf of an LMR user.	-	-	-	-
68	Check: Does the IUT (IWF) send a Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	8	P
69	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
70	Make the IWF release the floor with an acknowledgement required on behalf of an LMR user.	-	-	-	-
71	Check: Does the IUT (IWF) send a Floor Release message to the SS (MCPTT Server) with an acknowledgement required?	-->	Floor Release	8	P
72	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		-
73	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
74	Make the IUT (IWF) request the modification of the call to downgrade the call to a normal group call without implicit floor request on behalf of the LMR user.	-	-	-	-
75	Check: Does the IUT (IWF) send a SIP INVITE to the SS (MCPTT Server)?	-->	SIP INVITE	9	P
76	The SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-
77	Check: Does the IUT (IWF) send a SIP ACK to the SS (MCPTT Server)?	-->	SIP ACK	9	P
78	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
79	Make the IWF request the floor on behalf of an LMR user.	-	-	-	-
80	Check: Does the IUT (IWF) send a Floor Request message to the SS (MCPTT Server)?	-->	Floor Request	10	P
81	The SS (MCPTT Server) sends a Floor Granted message with no acknowledgment required.	<--	Floor Granted		-
82	Make the IWF release the floor with an acknowledgement required on behalf of an LMR user.	-	-	-	-
83	Check: Does the IUT (IWF) send a Floor Release message to the SS (MCPTT Server) with an acknowledgement required?	-->	Floor Release	10	P
84	The SS (MCPTT server) sends a Floor Ack message	<--	Floor Ack		-
85	The SS (MCPTT server) sends a Floor Idle message	<--	Floor Idle		-
86	Make the IUT (IWF) request the release of the group call on behalf of the LMR user.	-	-	-	-
87	Check: Does the IUT (IWF) forward the SIP BYE message to the SS (MCPTT Server) to end the call?	-->	SIP BYE	11	P
88	SS (MCPTT Server) responds with a SIP 200 (OK) message	<--	SIP 200 (OK)		-

6.1.1.2.3.3 Specific message contents

Table 6.1.1.2.3.3-1: SIP INVITE from the IUT (IWF) (Step 2, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCPTT_PublicServiceId_B	The URI of the IWF which is simulated by the SS, providing the controlling function.		
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the IUT (IWF)			
Record-Route	not present			
From				
addr-spec	tsc_MCPTT_PublicServiceId_A	The URI of the IWF performing the originating participating role.		
To				
addr-spec	tsc_MCPTT_PublicServiceId_B	The URI of the IWF which is simulated by the SS, providing the controlling function.		
Contact				
user-info and host	tsc_MCPTT_PublicServiceId_A	The URI of the IWF performing the originating participating role.		
feature-param	"isfocus" not present			
Priv-Answer-Mode	not present			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 6.1.1.2.3.3-1A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info message as described in Table 6.1.1.2.3.3-1B			

Table 6.1.1.2.3.3-1A: SDP in SIP INVITE from the IUT (IWF) (Table 6.1.1.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions INITIAL_SDP_OFFER, IMPLICIT_GRANT_REQUESTED				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the IUT (IWF)			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (IWF)			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (IWF)			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (IWF)			

Table 6.1.1.2.3.3-1B: MCPTT-Info in SIP INVITE from the IUT (IWF) (Table 6.1.1.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 6.1.1.2.3.3-2: SIP 200 (OK) from the SS (MCPTT Server) (Step 4, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 6.1.1.2.3.3-2A			

Table 6.1.1.2.3.3-2A: SDP in SIP 200 (OK) from the SS (MCPTT Server) (Table 6.1.1.2.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER, IMPLICIT_FLOOR_GRANTED, IMPLICIT_GRANT_REQUESTED

Table 6.1.1.2.3.3-3: SIP INVITE from the IUT (IWF) (Step 31, Table 6.1.1.2.3.2-1)

Derivation Path: Table 6.1.1.2.3.3-2 conditions MCPTT, re_INVITE, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.2.3.3-3A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.2.3.3-3B			

Table 6.1.1.2.3.3-3A: SDP in SIP INVITE from the IUT (IWF) (Table 6.1.1.2.3.3-3)

Derivation Path: Table 6.1.1.2.3.3-2A, condition SDP_OFFER instead of INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin	Same o=line as in the previous SDP message sent by the IUT (IWF) except that sess-version is incremented by one			

Table 6.1.1.2.3.3-3B: MCPTT-Info in SIP INVITE from the IUT (IWF) (Table 6.1.1.2.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, EMERGENCY-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 6.1.1.2.3.3-4: SIP 200 (OK) from the SS (MCPTT Server) (Steps 32 and 60, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 6.1.1.2.3.3-4A			

Table 6.1.1.2.3.3-4A: SDP in SIP 200 (OK) from the SS (MCPTT Server) (Table 6.1.1.2.3.3-4)

Derivation Table 6.1.1.2.3.3-2A

Table 6.1.1.2.3.3-5: SIP INVITE from the IUT (IWF) (Step 47, Table 6.1.1.2.3.2-1)

Derivation Path: Table 6.1.1.2.3.3-2 conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.2.3.3-5A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.2.3.3-5B			

Table 6.1.1.2.3.3-5A: SDP in SIP INVITE from the IUT (IWF) (Table 6.1.1.2.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1 conditions SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin				
unicast-address	IP address of the IUT (IWF)			
Connection Data	not required if included in all media	c= line		
connection-address	IP address of the IUT (IWF)			
Media description[1]		**Media description for audio**		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (IWF)			
Media description[2]		Media description for media control		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the IUT (IWF)			

Table 6.1.1.2.3.3-5B: MCPTT-Info in SIP INVITE from the IUT (IWF) (Table 6.1.1.2.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
emergency-ind	Encrypted (see note) <emergency-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 6.1.1.2.3.3-6: SIP INVITE from the IUT (IWF) (Step 59, Table 6.1.1.2.3.2-1)

Derivation Path: Table 6.1.1.2.3.3-2 conditions MCPTT, re_INVITE, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.2.3.3-6A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.2.3.3-6B			

Table 6.1.1.2.3.3-6A: SDP in SIP INVITE from the IUT (IWF) (Table 6.1.1.2.3.3-6)

Derivation Path: Table 6.1.1.2.3.3-3A

Table 6.1.1.2.3.3-6B: MCPTT-Info in SIP INVITE from the IUT (IWF) (Table 6.1.1.2.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER, IMMPERIL-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 6.1.1.2.3.3-7: SIP INVITE from the IUT (IWF) (Step 75, Table 6.1.1.2.3.2-1)

Derivation Path: Table 6.1.1.2.3.3-2 conditions MCPTT, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.1.2.3.3-7A			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table 6.1.1.2.3.3-7B			

Table 6.1.1.2.3.3-7A: SDP in SIP INVITE from the IUT (IWF) (Table 6.1.1.2.3.3-14)

Derivation Path: Table 6.1.1.2.3.3-5A

Table 6.1.1.2.3.3-7B: MCPTT-Info in SIP INVITE from the IUT (IWF) (Table 6.1.1.2.3.3-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1 conditions GROUP-CALL, INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A			
imminentperil-ind	Encrypted (see note) <imminentperil-ind> with mcpttBoolean set to false			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table 6.1.1.2.3.3-8: Floor Release (Steps 7, 27, 55 and 83, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Subtype	10100	Acknowledgement is required	

Table 6.1.1.2.3.3-9: Floor Ack (Steps 8, 28, 56 and 84, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 6.1.1.2.3.3-10: Floor Release (Step 13, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1

Table 6.1.1.2.3.3-11: Floor Request (Steps 11, 20, 52 and 80, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 6.1.1.2.3.3-12: Floor Request (Step 17, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	
Floor Indicator			
Floor Indicator	"1000000000000000"	bit A=1 (Normal call) bit F=0 (Queueing not supported)	

Table 6.1.1.2.3.3-13: Floor Taken (Step 15, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1

Table 6.1.1.2.3.3-14: Floor Granted (Steps 12, 25, 53 and 81, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1

Table 6.1.1.2.3.3-15: Floor Deny (Step 18, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.4-1

Table 6.1.1.2.3.3-16: Floor Queue Position Info (Steps 21 and 24, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.10-1

Table 6.1.1.2.3.3-17: Floor Queue Position Request (Step 23, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.9-1

Table 6.1.1.2.3.3-18: Floor Idle (Steps 9, 29, 50, 57, 78 and 85, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1

Table 6.1.1.2.3.3-19: Floor Revoke (Step 13, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.8-1

Table 6.1.1.2.3.3-20: Floor Release (Steps 36 and 43, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition ACK, EMERGENCY-CALL

Table 6.1.1.2.3.3-21: Floor Ack (Steps 37 and 44, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition DOWNLINK, EMERGENCY-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 6.1.1.2.3.3-22: Floor Request (Step 40, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1 condition EMERGENCY-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 6.1.1.2.3.3-23: Floor Granted (Steps 34 and 41, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition EMERGENCY-CALL

Table 6.1.1.2.3.3-24: Floor Idle (Steps 38 and 45, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.9-1 condition EMERGENCY-CALL

Table 6.1.1.2.3.3-25: Floor Release (Steps 64 and 71, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1 condition ACK, IMMPERIL-CALL

Table 6.1.1.2.3.3-26: Floor Ack (Steps 65 and 72, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.11-1 condition DOWNLINK, IMMPERIL-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Message Type			
Message Type	"00010100"	Acknowledgement for Floor Release message	

Table 6.1.1.2.3.3-27: Floor Request (Step 68, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.2-1 condition IMMPERIL-CALL			
Information Element	**Value/remark**	**Comment**	**Condition**
Floor priority	"0"	The lowest priority	

Table 6.1.1.2.3.3-28: Floor Granted (Steps 62 and 69, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition IMMPERIL-CALL

Table 6.1.1.2.3.3-29: Floor Idle (Steps 66 and 73, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.9-1 condition IMMPERIL-CALL

Table 6.1.1.2.3.3-30: SIP BYE from the IUT (IWF) (Step 87, 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	same URI as the SS (MCPTT Server) has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS (MCPTT Server)		
From				
addr-spec	Same URI of the IWF as used earlier in the dialog	Local URI of the dialog (from the IWF's point of view)		
tag	Same tag of the IWF as used earlier in the dialog	Local tag of the dialog (from the IWF's point of view)		
To				
addr-spec	Same URI of the SS (MCPTT Server) as used earlier in the dialog	Remote URI of the dialog (from the IWF's point of view)		
tag	Same tag of the SS (MCPTT Server) as used earlier in the dialog	Remote tag of the dialog ID (from the IWF's point of view)		
Max-Forwards			IETF RFC 3261 [17]	
value	any allowed value	Non-zero value		

Table 6.1.1.2.3.3-39: SIP 200 (OK) from the IUT (IWF) (Steps 72 and 114, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 6.1.1.2.3.3-39A			

Table 6.1.1.2.3.3-39A: SDP in SIP 200 (OK) from the IUT (IWF) (Table 6.1.1.2.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER

Table 6.1.1.2.3.3-40: SIP 200 (OK) from the SS (MCPTT-Server) (Steps 70 and 112, Table 6.1.1.2.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 conditions MCPTT, INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Record-Route	not present			
Message-body				
SDP message	SDP message as described in Table 6.1.1.2.3.3-40A			

Table 6.1.1.2.3.3-40A: SDP in SIP 200 (OK) from the SS (MCPTT-Server) (Table 6.1.1.2.3.3-11)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1 conditions SDP_ANSWER

Annex A (normative): Common Test Environment

A.1 Introduction

This annex specifies additional generic test procedures to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), to be used by the test cases wherever applicable.

A.2 Generic test procedures for Server-Client MCS operation

A.2.1 General

The purpose of the procedures specified in the following clauses is to facilitate test description by providing procedure sequences which can be referred from the relevant server-to-client TCs specified in the present document.

The procedures specified are required to ensure that any MC service can take place between a MC client and a MC server or specific MC relevant pre-conditions are met before a test case can be executed.

A.2.2 Void

A.2.3 MCX Server Originated notifications regarding temporary group creation or tear down

A.2.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.3.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.3.3 Procedure

Table A.2.3.3-1: MCX Server originated notifications regarding temporary group creation or tear down

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP NOTIFY to the SS-UE informing about change of group A's configuration document?	<--	SIP NOTIFY	-	P
2	The SS-UE (MCX Client) sends a SIP 200 (OK) message to the MCX Server.	-->	SIP 200 (OK)	-	-
3	The SS-UE (MCX Client) sends an HTTP GET Request message to the MCX Server that contains the access token and the XCAP-URI of the Group Configuration document.	-->	HTTP GET	-	-
4	Check: Does the MCX Server send a HTTP 200 (OK) message including the updated Group Document?	<--	HTTP 200 (OK)	-	P
5	Check: Does the MCX Server send a SIP NOTIFY message to the SS-UE (MCX Client) containing the group key transport payloads (GKTP) document including the group keys?	<-	SIP NOTIFY	-	P
6	The SS-UE (MCX Client) sends a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.3.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).
- Clarifications specified in clause 5.3.22.4 shall also be followed.

Table A.2.3.4-1: SIP NOTIFY (Step 1, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-1

Table A.2.3.4-2: SIP 200 (OK) (Steps 2, 6, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-2

Table A.2.3.4-3: HTTP GET (Step 3, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-3

Table A.2.3.4-4: HTTP 200 (OK) (Step 4, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-4

Table A.2.3.4-5: SIP NOTIFY (Step 5, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-6

A.2.4 MCX Server Terminated Group Creation

A.2.4.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.4.2 Definition of system information messages

N/a (out of E-UTRA coverage)

A.2.4.3 Procedure

Table A.2.4.3-1: MCX Server Terminated Group Creation procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP PUT to the Group Management Server for requesting the creation of the new group. (see note)	-->	HTTP PUT	-	-
2	Check: Does the MCX Server respond with a HTTP 201 (Created) response message?	<--	HTTP 201 (Created)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.4.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.4.4-1: HTTP PUT (Step 1, Table A.2.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.4-1, condition GROUPCREATE

A.2.5 MCX Server Terminated Temporary Group Creation

A.2.5.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.5.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.5.3 Procedure

Table A.2.5.3-1: MCX Participating Terminated Temporary Group Creation procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP POST to the MCX Server to request for creation of a temporary group. (see note)	-->	HTTP POST	-	-
2	Check: Does the MCX Server send a HTTP 200 (OK) containing the GMOP group-regroup-creation-response?	<--	HTTP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.5.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.5.4-1: HTTP POST (Step 1, Table A.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition TEMPGROUP

Table A.2.5.4-2: HTTP 200 (OK) (Step 2, Table A.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition TEMPGROUP

A.2.6 MCX Server Terminated Temporary Group Tear Down

A.2.6.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.6.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.6.3 Procedure

Table A.2.6.3-1: MCX Server Terminated Temporary Group Tear Down procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP DELETE to the MCX Server to request for tear down of a temporary group. (see note)	-->	HTTP DELETE	-	-
2	Check: Does the MCX Server send a HTTP 200 (OK) response message?	<--	HTTP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.6.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.6.4-1: HTTP DELETE (Step 1, Table A.2.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.5-1, condition TEMPGROUP

A.2.7 MCX Subscription and Notification

A.2.7.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.7.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.7.3 Procedure

Table A.2.7.3-1: MCX Subscription and Notification

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP SUBSCRIBE message request.	-->	SIP SUBSCRIBE	-	-
2	Check: Does the MCX Server respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCX Server send a SIP NOTIFY message?	<--	SIP NOTIFY	-	P
4	The SS-UE (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)		
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.7.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.7.4-1: SIP SUBSCRIBE (Step 1, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
value	"4294967295"	to receive the current status and later notification	ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3; ETSI TS 124 281 [11] (3GPP TS 24.281) clause 8.2.1.3; ETSI TS 124 282 [10] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
Message-body				
MIME body part		**MCPTT Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.1.3 and 9A.2.1.3	MCPTT
MIME-part-body	MCPTT-Info as described in Table A.2.7.4-2			
MIME body part		**MCVIDEO Info**	ETSI TS 124 281 [11] (3GPP TS 24.281) clause 8.2.1.3	MCVIDEO
MIME-part-body	MCVIDEO-Info as described in Table A.2.7.4-3			
MIME body part		**MCDATA Info**	ETSI TS 124 282 [10] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	MCDATA
MIME-part-body	MCDATA-Info as described in Table A.2.7.4-4			

Table A.2.7.4-2: MCPTT-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table A.2.7.4-3: MCVideo-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
mcvideo-request-uri	encrypted (see note) <mcvideo-request-uri> with mcvideoURI set to px_MCVIDEO_ID_User_A		ETSI TS 124 281 [11] (3GPP TS 24.281) clause 8.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2A.				

Table A.2.7.4-4: MCData-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	encrypted (see note) <mcdata-request-uri> with mcdataURI set to px_MCDATA_ID_User_A		ETSI TS 124 282 [10] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3A.				

Table A.2.7.4-5: SIP 200 (OK) (Step 2, Table A.2A.7.3-1)

Derivation Path: Table ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), 5.5.2.17.1.2-1 with condition SUBSCRIBE-RSP

A.2.8 MCX Participating Terminated SIP MESSAGE

A.2.8.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.8.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.8.3 Procedure

Table A.2.8.3-1: MCX Participating Terminated SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) send a SIP MESSAGE message (see note)	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server responds with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.8.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.9 MCX Participating Originated SIP MESSAGE

A.2.9.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData

A.2.9.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.9.3 Procedure

Table A.2.9.3-1: MCX Participating Originated SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP MESSAGE message?	<--	SIP MESSAGE	-	P
2	The SS-UE (MCX Client) responds with a SIP 200 (OK) message (see note)	→	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.9.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.10 MCX E2E SIP MESSAGE

A.2.10.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.10.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.10.3 Procedure

Table A.2.10.3-1: MCX E2E SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP MESSAGE message	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server send a SIP MESSAGE to SS-UE2?	<--	SIP MESSAGE	-	P
3	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	-	P

A.2.10.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.11 MCX E2E SIP MESSAGE Request - Accept

A.2.11.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.11.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.11.3 Procedure

Table A.2.11.3-1: MCX E2E SIP MESSAGE Request - Accept Participating Originated

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP MESSAGE message.	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server respond with a SIP 200 (OK) message	<--	SIP 200 (OK)	-	P
3	Check: Does the MCX server send a SIP MESSAGE accepting the request to SS-UE1?	<--	SIP MESSAGE	-	P
4	The SS-UE1 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
-	EXCEPTION: Depending on the MCX Server implementation, the following steps may be performed in parallel to steps 3-4.	-	-	-	-
5	Check: Does the MCX Server send a SIP MESSAGE message to SS-UE2?	<--	SIP MESSAGE	-	P
6	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-

A.2.11.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.12 MCX Presence Event Status Change

A.2.12.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.12.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.12.3 Procedure

Table A.2.12.3-1: MCX Presence Event Status Change

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP PUBLISH message.	-->	SIP PUBLISH	-	P
2	Check: Does the MCX Server respond to the SIP PUBLISH message with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	-
3	Check: Does the MCX Server send a SIP NOTIFY message informing about the status change progress?	<--	SIP NOTIFY	-	P
4	The SS-UE (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
5	Check: Does the MCX Server send a SIP NOTIFY informing about the final status?	<--	SIP NOTIFY	-	P
6	The SS-UE responds with a SIP 200 (OK).	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.12.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.12.4-1: SIP PUBLISH (Step 1, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.11-1, condition PRESENCE-EVENT

Table A.2.12.4-2: MCPTT-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [4] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table A.2.12.4-3: MCVideo-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
mcvideo-request-uri	encrypted (see note) <mcvideo-request-uri> with mcvideoURI set to px_MCVIDEO_ID_User_A		ETSI TS 124 281 [11] (3GPP TS 24.281) clause 8.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2A.				

Table A.2.12.4-4: MCData-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	encrypted (see note) <mcdata-request-uri> with mcdataURI set to px_MCDATA_ID_User_A		ETSI TS 124 282 [10] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3A.				

Table A.2.12.4-5: SIP 200 (OK) (Step 2, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, condition PUBLISH-RSP

Table A.2.12.4-3: SIP NOTIFY (Steps 3 and 5, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition PRESENCE-EVENT

Table A.2.12.4-4: SIP 200 (OK) (Steps 4 and 6, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1

A.2.13 MCX Participating Terminated pre-established session establishment

A.2.13.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.13.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.13.3 Procedure

Table A.2.13.3-1: MCX Participating Terminated pre-established session establishment

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP INVITE message in order to create a pre-established session	-->	SIP INVITE	-	-
-	EXCEPTION: Step 2a1 describes behaviour that depends on the MCX Server implementation; the "lower case letter" identifies a step sequence that take place if the MCX Server responds to a SIP INVITE with a SIP 100 (Trying)	-	-	-	-
2a1	Check: Does the MCX Server respond with a SIP 100 (Trying) message?	<--	SIP 100 Trying	-	P
3	Check: Does the MCX Server respond with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
4	The SS-UE (MCX Client) responds with a SIP ACK message.	-->	SIP ACK	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.13.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.13.4-1: SIP INVITE (Step 1, Table A.2.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.3.4-1

Table A.2.13.4-2: SIP 200 (OK) (Step 3, Table A.2.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.3.4-2

A.2.14 MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying

A.2.14.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.14.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.14.3 Procedure

Table A.2.14.3-1: MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call (see note)	-->	SIP INVITE	-	-
-	EXCEPTION: Step 2a1 describes behaviour that depends on the MCX Server implementation; the "lower case letter" identifies a step sequence that take place if the MCX Server responds to a SIP INVITE with a SIP 100 (Trying)	-	-	-	-
2a1	Check: Does the MCX Server respond with a SIP 100 Trying?	<--	SIP 100 (Trying)	-	P
3	Check: Does the MCX Server respond with a SIP 200 (OK)?	<--	SIP 200 (OK)	-	P
4	The SS-UE (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification (see note)	-->	SIP ACK	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.14.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.14.4-1: SIP INVITE (Step 1, Table A.2.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (see note)
NOTE: With condition re_INVITE for session modifications.

Table A.2.14.4-1: SIP 200 (OK) (Step 3, Table A.2.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.15 MCX Participating Terminated call release

A.2.15.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.15.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.15.3 Procedure

Table A.2.15.3-1: MCX Participating Terminated call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP BYE request to terminate the MCX session (see note)	-->	SIP BYE	-	-
2	Check: Does the MCX Server respond with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.15.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.16 MCX Participating Originated session establishment/modification without provisional responses other than 100 Trying

A.2.16.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.16.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.16.3 Procedure

Table A.2.16.3-1: MCX Participating Originated session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP INVITE requesting the establishment/modification of a call?	<--	SIP INVITE	-	P
2	The SS-UE (MCX Client) sends a SIP 100 Trying (see note)	-->	SIP 100 (Trying)	-	-
3	The SS-UE (MCX Client) responds with a SIP 200 (OK) (see note)	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification?	<--	SIP ACK	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.16.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.16.4-1: SIP INVITE (Step 1, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (see note).
NOTE: With condition re_INVITE for session modifications.

Table A.2.16.4-2: SIP 200 (OK) (Step 3, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

A.2.17 MCX Participating Originated call release

A.2.17.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.17.2 Definition of system information messages

N/a (out of E-UTRA coverage)

A.2.17.3 Procedure

Table A.2.17.3-1: MCX Participating Originated call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP BYE request to terminate the MCX session?	<--	SIP BYE	-	P
2	The SS-UE (MCX Client) responds with a SIP 200 (OK) message (see note).	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.17.4 Specific message contents

All message contents are as specified in clause 5.5. and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.18 MCX E2E session establishment/modification without provisional responses other than 100 Trying

A.2.18.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.18.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.18.3 Procedure

Table A.2.18.3-1: MCX E2E session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
6	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
7	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
8	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.18.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.18.4-1: SIP INVITE (Step 1, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (see note).
NOTE: With condition re_INVITE for session modifications.

Table A.2.18.4-2: SIP INVITE (Step 2, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (see note).
NOTE: With condition re_INVITE for session modifications.

Table A.2.18.4-1: SIP 200 (OK) (Step 5, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.18.4-1: SIP 200 (OK) (Step 6, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.19 MCX E2E call release

A.2.19.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.19.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.19.3 Procedure

Table A.2.19.3-1: MCX E2E call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP BYE request to terminate the MCX session	-->	SIP BYE	-	-
2	Check: Does the MCX Server respond a SIP BYE request to terminate the MCX session to SS-UE2 (MCX Client)?	<--	SIP BYE	-	P
3	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX Server respond a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	-	P

A.2.19.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.19.4-1: SIP BYE from the SS-UE1 (Step 1, Table A.2.19.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1

Table A.2.19.4-2: SIP BYE from the MCPTT Server (Step 2, Table A.2.19.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1

A.2.20 MCX E2E group call establishment, manual commencement

A.2.20.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.20.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.20.3 Procedure

Table A.2.20.3-1: MCX E2E group call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) sends a SIP 183 (Session Progress) reliably.	-->	SIP 183 (Session Progress)	-	-
6	Check: Does the MCX Server acknowledge the receipt of SIP 183 (Session Progress) from SS-UE2?	<--	PRACK	-	P
7	The SS-UE2 (MCX Client) responds a SIP 200 (OK) to PRACK message	-->	SIP 200 (OK)	-	-
8	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
9	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
10	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
11	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.20.4 Specific message contents

All message contents are as specified in clause 5.5 with conditions GROUP-CALL and MANUAL; and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

For chat group calls, CHAT-GROUP-CALL condition applies instead of GROUP-CALL.

Table A.2.20.4-1: SIP 200 (OK) (Step 8, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.20.4-2: SIP 200 (OK) (Step 9, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.21 MCX E2E private call establishment, manual commencement

A.2.21.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.21.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.2.21.3 Procedure

Table A.2.21.3-1: MCX E2E private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) sends a SIP 180 (Ringing) unreliably.	-->	SIP 180 (Ringing)	-	-
6	Check: Does the MCX Server respond with a unreliable SIP 180 (Ringing) to SS-UE1 (MCX Client)?	<--	SIP 180 (Ringing)	-	P
7	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
8	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
9	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
10	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.21.4 Specific message contents

All message contents are as specified in clause 5.5 with conditions PRIVATE-CALL and MANUAL; and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.21.4-1: SIP 200 (OK) (Step 7, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.21.4-1: SIP 200 (OK) (Step 8, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.3 Generic test procedures for MCPTT Server-Client operation

A.3.0 General

The purpose of the procedures specified in the following clauses is to facilitate test description by providing MCPTT procedure sequences which can be referred from the relevant TCs specified in the present document.

The procedures specified are required to ensure that MCPTT service can take place between a MCPTT Client and a MCPTT server or specific MCPTT relevant pre-conditions are met before a test case can be executed.

A.3.1 MCPTT E2E session establishment/modification without provisional responses other than 100 Trying

A.3.1.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.1.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.1.3 Procedure

Table A.3.1.3-1: MCPTT E2E session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT server correctly perform test procedure MCX E2E session establishment/modification without provisional responses other than 100 Trying' for as described in Table A.2.18.3-1 with MCPTT as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit floor control in the SIP INVITE request but it is not implicitly granted (NOTE1)	-	-	-	-
2a1	Check: Does the MCPTT server send a Floor Granted message to SS-UE1?	<--	Floor Granted	-	P
-	EXCEPTION: Step 3a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is granted to SS-UE1, either implicit or explicitly. (NOTE1)	-	-	-	-
3a1	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<--	Floor Taken	-	P
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is not granted to SS-UE1. (NOTE1)	-	-	-	-
4a1	Check: Does the MCPTT server send a Floor Idle message to SS-UE1?	<--	Floor Idle	-	P
4a2	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<--	Floor Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:

a. SS-UE1 sends SDP offer with media description for floor control but without implicit floor request.
b. SS-UE1 sends SDP offer with media description for floor control and with implicit floor request:
   i. SDP answer from MCPTT Server contains "mc_implicit_request" and "mc_granted" (Floor is implicitly granted).
   ii. SDP answer from MCPTT Server contains "mc_implicit request" and but no "mc_granted" (Floor needs to be explicitly granted at step 2a1).
   iii. SDP answer from MCPTT Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the floor).
c. SS-UE1 sends SDP offer without media description for floor control.

A.3.1.4 Specific message contents

All message contents are as specified in clause 5.5 with condition MCPTT and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.1.4-1: SIP INVITE (Step 1, Table A.3.1.3-1; Step 1, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (see note), with condition MCPTT NOTE: With condition re_INVITE for session modifications.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [25]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [25].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-2			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table A.3.1.4-3			

Table A.3.1.4-2: SDP in SIP INVITE (Table A.3.1.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [13] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [9]	
"RR"	2000		IETF RFC 3556 [9]	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_FLOORCONTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note 3	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
NOTE 1: If the SS-UE requests a call with media floor control and implicit floor (Option b of note on Table A.3.1.3-1), apply condition IMPLICIT_GRANT_REQUESTED.				
NOTE 2: If the SS-UE requests a call without media floor (Option c of note on Table A.3.1.3-1), apply condition WITHOUT_FLOORCONTROL.				
NOTE 3: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1.)				

Table A.3.1.4-3: MCPTT-Info in SIP INVITE (Table A.3.1.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition INVITE_REFER

Table A.3.1.4-4: SIP INVITE from the MCPTT-Server (Step 1, Table A.3.1.3-1; Step 2 Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (see note), with condition MCPTT NOTE: With condition re_INVITE for session modifications.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [23] IETF RFC 7134 [18] IETF RFC 8101 [21]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
Session-Expires			IETF RFC 4028 [25]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.1.4-6			

Table A.3.1.4-5: SDP in SIP INVITE (Table A.3.1.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER See note 1.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note 2	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	Any value lower or equal to the priority specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
NOTE 1: If the SS-UE1 requests a call without media floor control (Option c of NOTE 1 on Table A.3.1.3-1), apply condition WITHOUT_FLOORCONTROL. NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.1.4-6: MCPTT-Info in SIP INVITE (Table A.3.1.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table A.3.1.4-7: SIP 200 (OK) (Step 1, Table A.3.1.3-1; Step 5, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [16]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-8			

Table A.3.1.4-8: SDP in SIP 200 (OK) (Table A.3.1.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [13] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [9]	
"RR"	2000		IETF RFC 3556 [9]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_ FLOORCO NTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.1.4-9: SIP 200 (OK) (Step 1, Table A.3.1.3-1; Step 6, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-10			

Table A.3.1.4-10: SDP in SIP 200 (OK) (Table A.3.1.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER
NOTE 1: If the SS-UE requests an implicit floor (Option b of note on Table A.3.1.3-1), more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.
NOTE 2: If the SS-UE requests a call without media floor (Option c of note on Table A.3.1.3-1) apply condition WITHOUT_FLOORCONTROL.
NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Information Element	Value/remark	Comment	Reference	Condition
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_FLOOR_GRANTED
mc_implicit_request	not present		TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER
NOTE 1: If the SS-UE requests an implicit floor (Option b of note on Table A.3.1.3-1), more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.
NOTE 2: If the SS-UE requests a call without media floor (Option c of note on Table A.3.1.3-1) apply condition WITHOUT_FLOORCONTROL.
NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Information Element	Value/remark	Comment	Reference	Condition
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

A.3.2 MCPTT E2E group call establishment, manual commencement

A.3.2.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.2.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.2.3 Procedure

Table A.3.2.3-1: MCPTT E2E group call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT server correctly perform test procedure MCX E2E group call establishment, manual commencement' as described in Table A.2.20.3-1 with MCPTT as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit floor control in the SIP INVITE request but it is not implicitly granted (see note)	-	-	-	-
2a1	Check: Does the MCPTT server send a Floor Granted message to SS-UE1?	<--	Floor Granted	-	P
-	EXCEPTION: Step 3a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is granted to SS-UE1, either implicit or explicitly (see note)	-	-	-	-
3a1	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<--	Floor Taken	-	P
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is not granted to SS-UE1 (see note)	-	-	-	-
4a1	Check: Does the MCPTT server send a Floor Idle message to SS-UE1?	<--	Floor Idle	-	P
4a2	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<--	Floor Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:
a. SS-UE1 sends SDP offer with media description for floor control but without implicit floor request.
b. SS-UE1 sends SDP offer with media description for floor control and with implicit floor request:
i. SDP answer from MCPTT Server contains "mc_implicit_request" and "mc_granted" (Floor is implicitly granted).
ii. SDP answer from MCPTT Server contains "mc_implicit request" and but no "mc_granted" (Floor needs to be explicitly granted at step 2a1).
iii. SDP answer from MCPTT Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the floor).

A.3.2.4 Specific message contents

All message contents are as specified in clause 5.5 with condition MCPTT and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.2.4-1: SIP INVITE from the SS-UE1 (Step 1, Table A.3.2.3-1; Step 1, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1, with condition MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [25]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [25].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-2			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.2.4-3			

Table A.3.2.4-2: SDP in SIP INVITE (Table A.3.2.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and IMPLICIT_GRANT_REQUESTED if the SS-UE1 requests implicit floor				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [13] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [9]	
"RR"	2000		IETF RFC 3556 [9]	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and IMPLICIT_GRANT_REQUESTED if the SS-UE1 requests implicit floor				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
NOTE: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1.)				

Table A.3.2.4-3: MCPTT-Info in SIP INVITE (Table A.3.2.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition GROUP-CALL, INVITE_REFER

Table A.3.2.4-4: SIP INVITE from the MCPTT-Server (Step 1, Table A.3.2.3-1; Step 2, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, with condition MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [23] IETF RFC 7134 [18] IETF RFC 8101 [21]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
Session-Expires			IETF RFC 4028 [25]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.2.4-6			

Table A.3.2.4-5: SDP in SIP INVITE (Table A.3.2.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
NOTE: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1.)				

Table A.3.2.4-6: MCPTT-Info in SIP INVITE (Table A.3.2.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1, condition GROUP-CALL (see note 1)				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note 2) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note 2) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE 1: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL. NOTE 2: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table A.3.2.4-7: SIP 200 (OK) from the SS-UE2 (Step 1, Table A.3.2.3-1; Step 8, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1, with condition MCPTT, INVITE-RSP and GROUP-CALL (see note)				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [16]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-8			
NOTE: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.				

Table A.3.2.4-8: SDP in SIP 200 (OK) (Table A.3.2.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [13] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [9]	
"RR"	2000		IETF RFC 3556 [9]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]		Media description for media control		
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE1			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.2.4-9: SIP 200 (OK) from the MCPTT Server (Step 1, Table A.3.2.3-1; Step 9, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-10			

Table A.3.2.4-10: SDP in SIP 200 (OK) (Table A.3.2.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER. NOTE 1: If the SS-UE requests an implicit floor (Option b of NOTE 1 on Table A.3.2.3-1), more conditions may apply: i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied. ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies. NOTE 2: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description [2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_ssrc	any value		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.2.4-11: SIP 183 (Session Progress) from the SS-UE2 (Step 1, Table A.3.2.3-1; Step 5, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.2-1, with condition MCPTT and 100rel

A.3.3 MCPTT E2E private call establishment, manual commencement

A.3.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.3.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.3.3 Procedure

Table A.3.3.3-1: MCPTT E2E private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT server correctly perform test procedure MCX E2E private call establishment, manual commencement' as described in Table A.2.21.3-1 with MCPTT as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit floor control in the SIP INVITE request but it is not implicitly granted (see note)	-	-	-	-
2a1	Check: Does the MCPTT server send a Floor Granted message to SS-UE1?	<--	Floor Granted	-	P
-	EXCEPTION: Step 3a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is granted to SS-UE1, either implicit or explicitly (see note)	-	-	-	-
3a1	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<--	Floor Taken	-	P
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is not granted to SS-UE1 (see note)	-	-	-	-
4a1	Check: Does the MCPTT server send a Floor Idle message to SS-UE1?	<--	Floor Idle	-	P
4a2	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<--	Floor Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:
a. SS-UE1 sends SDP offer with media description for floor control but without implicit floor request.
b. SS-UE1 sends SDP offer with media description for floor control and with implicit floor request:
   i. SDP answer from MCPTT Server contains "mc_implicit_request" and "mc_granted" (Floor is implicitly granted).
   ii. SDP answer from MCPTT Server contains "mc_implicit request" and but no "mc_granted" (Floor needs to be explicitly granted at step 2a1).
   iii. SDP answer from MCPTT Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the floor).
c. SS-UE1 sends SDP offer without media description for floor control.

A.3.3.4 Specific message contents

All message contents are as specified in clause 5.5 with condition MCPTT and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.3.4-1: SIP INVITE from the SS-UE1 (Step 1, Table A.3.3.3-1; Step 1, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1, with condition MCPTT, PRIVATE-CALL and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [25]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [25].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-2			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.3.4-3			

Table A.3.3.4-2: SDP in SIP INVITE (Table A.3.3.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and PRIVATE-CALL NOTE 1: If the SS-UE requests a call with media floor control and implicit floor (Option b of note on Table A.3.3.3-1), apply condition IMPLICIT_GRANT_REQUESTED. NOTE 2: If the SS-UE requests a call without media floor (Option c of note on Table A.3.3.3-1), apply condition WITHOUT_FLOORCONTROL.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [13] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and PRIVATE-CALL
NOTE 1: If the SS-UE requests a call with media floor control and implicit floor (Option b of note on Table A.3.3.3-1), apply condition IMPLICIT_GRANT_REQUESTED.
NOTE 2: If the SS-UE requests a call without media floor (Option c of note on Table A.3.3.3-1), apply condition WITHOUT_FLOORCONTROL.

Information Element	Value/remark	Comment	Reference	Condition
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [9]	
"RR"	2000		IETF RFC 3556 [9]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_FLOORCONTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and PRIVATE-CALL
NOTE 1: If the SS-UE requests a call with media floor control and implicit floor (Option b of note on Table A.3.3.3-1), apply condition IMPLICIT_GRANT_REQUESTED.
NOTE 2: If the SS-UE requests a call without media floor (Option c of note on Table A.3.3.3-1), apply condition WITHOUT_FLOORCONTROL.

Information Element	Value/remark	Comment	Reference	Condition
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	

Table A.3.3.4-3: MCPTT-Info in SIP INVITE (Table A.3.3.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition PRIVATE-CALL, INVITE_REFER

Table A.3.3.4-4: SIP INVITE from the MCPTT-Server to SS-UE2 (Step 1, Table A.3.3.3-1; Step 2, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, with condition MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [23] IETF RFC 7134 [18] IETF RFC 8101 [21]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in Table 5.5.8.8-1 for MCVideo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
Session-Expires			IETF RFC 4028 [25]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.3.4-6			

Table A.3.3.4-5: SDP in SIP INVITE (Table A.3.3.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER NOTE: If the SS-UE1 requests a call without media floor control (Option c of note on Table A.3.3.3-1), apply condition WITHOUT_FLOORCONTROL.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing

Table A.3.3.4-6: MCPTT-Info in SIP INVITE (Table A.3.3.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table A.3.3.4-7: SIP 200 (OK) from the SS-UE2 to MCPTT Server (Step 1, Table A.3.3.3-1; Step 7, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1, with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [16]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-8			

Table A.3.3.4-8: SDP in SIP 200 (OK) (Table A.3.3.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [13] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [9]	
"RR"	2000		IETF RFC 3556 [9]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]		Media description for media control		NOT WITHOUT_ FLOORCO NTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [**5**] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.3.4-9: SIP 200 (OK) from the MCPTT Server to SS-UE1 (Step 1, Table A.3.3.3-1; Step 8, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-10			

Table A.3.3.4-10: SDP in SIP 200 (OK) (Table A.3.3.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER.
NOTE 1: If the SS-UE1 request an implicit floor (Option b of note on Table A.3.3.3-1), more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.
NOTE 2: If the SS-UE1 request a call without media floor (Option c of note on Table A.3.3.3-1) apply condition WITHOUT_FLOORCONTROL.
NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Information Element	Value/remark	Comment	Reference	Condition
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_FLOOR_GRANTED
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	not present		TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

**Table A.3.3.4-11: SIP 180 (Ringing) response
(Step 1, Table A.3.3.3-1; Steps 5, Table A.2.21.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.2.1-1, with condition MCPTT

**Table A.3.3.4-12: SIP 180 (Ringing) response
(Step 1, Table A.3.3.3-1; Steps 6, Table A.2.21.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.2.2-1, with condition MCPTT

A.3.4 MCPTT E2E call establishment using a pre-established session, manual commencement

A.3.4.1 Initial conditions

As specified in the test case which calls the procedure.

A.3.4.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.4.3 Procedure

Table A.3.4.3-1: MCPTT E2E call establishment using a pre-established session, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT Client) sends a SIP REFER message to request the establishment of an MCPTT call using a pre-established session	-->	SIP REFER	-	-
2	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1 indicating that the MCPTT call has been properly requested?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCPTT Server send a SIP re-INVITE requesting the establishment of a call to SS-UE2 (MCPTT Client)?	<--	SIP re-INVITE	-	P
4	The SS-UE2 (MCPTT Client) responds a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
5	The SS-UE2 (MCPTT Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
6	Check: Does the MCPTT server send a SIP ACK to acknowledge the session establishment to SS-UE2?	<--	SIP ACK	-	P
-	EXCEPTION: Depending on the MCPTT Server implementation, the following steps might be performed in parallel.	-	-	-	-
7	Check: Does the MCPTT server send a Connect message to SS-UE1 (MCPTT Client)?	<--	Connect	-	P
8	The SS-UE1 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-
9	Check: Does the MCPTT server send a Connect message to SS-UE2 (MCPTT Client)?	<--	Connect	-	P
10	The SS-UE2 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-

A.3.4.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.4.4-1: SIP REFER (Step 1, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1 with conditions MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body			IETF RFC 3261 [17]	
MIME body part		**Resource list**	IETF RFC 5366 [28]	
MIME-part-body	Resource-lists as described in Table A.3.4.4-2			

Table A.3.4.4-2: Resource-lists in SIP REFER (Table A.3.4.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1, condition PRE-ESTABLISH with the uri attribute of the entry extended with the SIP URI header fields as specified in Table A.3.4.4-3A

Table A.3.4.4-2A: SIP header fields extending the uri attribute of the resource-lists' single entry (Table A.3.4.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-2, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
body				
MIME body part		**SDP Message**		
MIME-part-headers				
Content-Type	"application/sdp"			
MIME-part-body	SDP Message as described in Table A.3.4.4-2B			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.4.4-2C			

Table A.3.4.4-2B: SDP in SIP header fields (Table A.3.4.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION NOTE: If the SS-UE requests an implicit floor condition IMPLCIT_GRAN_REQUESTED also applies.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [13] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION
NOTE: If the SS-UE requests an implicit floor condition IMPLCIT_GRAN_REQUESTED also applies.

Information Element	Value/remark	Comment	Reference	Condition
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [9]	
"RR"	2000		IETF RFC 3556 [9]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTP		
foundation	1234	arbitrarily selected		
priority	2130706431	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTCP		
foundation	1234	Same as for RTP		
priority	2130706430	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]		Media description for media control		
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION NOTE: If the SS-UE requests an implicit floor condition IMPLCIT_GRAN_REQUESTED also applies.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_implicit_request	not present			
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	any value if present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION NOTE: If the SS-UE requests an implicit floor condition IMPLCIT_GRAN_REQUESTED also applies.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for Media Control messages		
foundation	4321	arbitrarily selected; different than for RTP/RTCP		
priority	2130706430	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
NOTE: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1.)				

Table A.3.4.4-2C: MCPTT-Info in SIP header fields (Table A.3.4.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition INVITE-REFER

Table A.3.4.4-3: SIP 200 (OK) (Step 2, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with conditions MCPTT and REFER-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type				
media-type	"application/sdp"			
Message-body				
SDP message	SDP message as described in Table A.3.4.4-3A			

Table A.3.4.4-3A: SDP in SIP 200 (OK) (Table A.3.4.4-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

NOTE 1: If the SS-UE requests an implicit floor more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.

NOTE 2: If the SS-UE requests a call without media floor condition WITHOUT_FLOORCONTROL shall be applied.

NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Information Element	Value/remark	Comment	Reference	Condition
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTP		
foundation	any value			
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTCP		
foundation	any value			
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_FLOORCONTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

NOTE 1: If the SS-UE requests an implicit floor more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.

NOTE 2: If the SS-UE requests a call without media floor condition WITHOUT_FLOORCONTROL shall be applied.

NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Information Element	Value/remark	Comment	Reference	Condition
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_FLOOR_GRANTED
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION
NOTE 1: If the SS-UE requests an implicit floor more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.
NOTE 2: If the SS-UE requests a call without media floor condition WITHOUT_FLOORCONTROL shall be applied.
NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Information Element	Value/remark	Comment	Reference	Condition
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for Media Control messages		
foundation	any value			
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		

Table A.3.4.4-4: SIP INVITE (Step 3, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 with conditions MCPTT, MANUAL and re_INVITE

Information Element	Value/remark	Comment	Reference	Condition
Resource-Priority			IETF RFC 4412 [23] IETF RFC 7134 [18] IETF RFC 8101 [21]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 with conditions MCPTT, MANUAL and re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
Session-Expires			IETF RFC 4028 [25]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.4.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.4.4-6			

Table A.3.4.4-5: SDP in SIP INVITE (Table A.3.4.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER See note 1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note 2	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	Any value lower or equal to the priority specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
NOTE 1: If the SS-UE requests a call without media floor control condition WITHOUT_FLOORCONTROL shall be applied. NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1.)				

Table A.3.4.4-6: MCPTT-Info in SIP INVITE (Table A.3.4.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (see note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (see note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table A.3.4.4-7: SIP 200 (OK) (Step 4, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with conditions MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [16]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.4.4-8			

Table A.3.4.4-8: SDP in SIP 200 (OK) (Table A.3.4.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [13] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [9]	
"RR"	2000		IETF RFC 3556 [9]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_FLOORCONTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_FLOORCONTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.4.4-9: Connect (Step 7, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Session Identity field			
Session Type	"00000000"	No session type	
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	
Inviting MCPTT User Identity field	not present		

Table A.3.4.4-10: Connect (Step 9, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	

A.3.5 MCPTT E2E call establishment using a pre-established session, automatic commencement

A.3.5.1 Initial conditions

As specified in the test case which calls the procedure.

A.3.5.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.5.3 Procedure

Table A.3.5.3-1: MCPTT E2E call establishment using a pre-established session, automatic commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT Client) sends a SIP REFER message to request the establishment of an MCPTT call using a pre-established session	-->	SIP REFER	-	-
2	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1 indicating that the MCPTT call has been properly requested?	<--	SIP 200 (OK)	-	P
-	EXCEPTION: Depending on the MCPTT Server implementation, the following steps might be performed in parallel.	-	-	-	-
3	Check: Does the MCPTT server send a Connect message to SS-UE1 (MCPTT Client)?	<--	Connect	-	P
4	The SS-UE1 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-
5	Check: Does the MCPTT server send a Connect message to SS-UE2 (MCPTT Client)?	<--	Connect	-	P
6	The SS-UE2 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-

A.3.5.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.5.4-1: SIP REFER (Step 1, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1 with conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body			IETF RFC 3261 [17]	
MIME body part		**Resource list**	IETF RFC 5366 [28]	
MIME-part-body	Resource-lists as described in Table A.3.5.4-2			

Table A.3.5.4-2: Resource-lists in SIP REFER (Table A.3.5.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1, condition PRE-ESTABLISH with the uri attribute of the entry extended with the SIP URI header fields as specified in Table A.3.5.4-3A

Table A.3.5.4-2A: SIP header fields extending the uri attribute of the resource-lists' single entry (Table A.3.5.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-2, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
body				
MIME body part		**SDP Message**		
MIME-part-headers				
Content-Type	"application/sdp"			
MIME-part-body	SDP Message as described in Table A.3.5.4-2B			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.5.4-2C			

Table A.3.5.4-2B: SDP in SIP header fields (Table A.3.5.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION See note 1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [13] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [12] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [9]	
"RR"	2000		IETF RFC 3556 [9]	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION See note 1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTP		
foundation	1234	arbitrarily selected		
priority	2130706431	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTCP		
foundation	1234	Same as for RTP		
priority	2130706430	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]		Media description for media control		
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_FLOORCONTROL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION See note 1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1) See note 2	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_implicit_request	not present			
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	any value if present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION See note 1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for Media Control messages		
foundation	4321	arbitrarily selected; different than for RTP/RTCP		
priority	2130706430	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
NOTE 1: If the SS-UE requests an implicit floor condition IMPLCIT_GRAN_REQUESTED also applies. NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1.)				

Table A.3.5.4-2C: MCPTT-Info in SIP header fields (Table A.3.5.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition INVITE-REFER

Table A.3.5.4-3: SIP 200 (OK) (Step 2, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with conditions MCPTT and REFER-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type				
media-type	"application/sdp"			
Message-body				
SDP message	SDP message as described in Table A.3.5.4-3A			

Table A.3.5.4-3A: SDP in SIP 200 (OK) (Table A.3.5.4-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION NOTE 1: If the SS-UE requests an implicit floor more conditions may apply: i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied. ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies. NOTE 2: If the SS-UE requests a call without media floor condition WITHOUT_FLOORCONTROL shall be applied. NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [15]	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTP		
foundation	any value			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

NOTE 1: If the SS-UE requests an implicit floor more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.

NOTE 2: If the SS-UE requests a call without media floor condition WITHOUT_FLOORCONTROL shall be applied.

NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Information Element	Value/remark	Comment	Reference	Condition
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for RTCP		
foundation	any value			
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.1-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_F LOOR_GR ANTED
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_ GRANT_RE QUESTED

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

NOTE 1: If the SS-UE requests an implicit floor more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.

NOTE 2: If the SS-UE requests a call without media floor condition WITHOUT_FLOORCONTROL shall be applied.

NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Information Element	Value/remark	Comment	Reference	Condition
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [5] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [7]	
candidate		candidate for Media Control messages		
foundation	any value			
priority	any value	IETF RFC 5245 [7] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		

Table A.3.5.4-4: Connect (Step 3, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Session Identity field			
Session Type	"00000000"	No session type	
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	
Inviting MCPTT User Identity field	not present		

Table A.3.5.4-5: Connect (Step 5, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	
PCK I_MESSAGE field	MIKEY-SAKKE I_MESSAGE as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-2A		PRIVATE-CALL

A.3.6 MCPTT E2E call release keeping the pre-established session

A.3.6.1 Initial conditions

As specified in the test case which calls the procedure.

A.3.6.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.6.3 Procedure

Table A.3.6.3-1: MCPTT E2E call release keeping the pre-established session

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT Client) sends a SIP REFER message with method "BYE" to release the MCPTT session and keep the pre-established session.	-->	SIP REFER	-	-
2	Check: Does the MCPTT Server respond with a SIP 200 (OK) to SS-UE1?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCPTT Server release the call by sending a Disconnect message to SS-UE2 (MCPTT Client)?	<--	Disconnect	-	P
4	The SS-UE2 (MCPTT Client) sends an Acknowledgement to accept the release of the call	-->	Acknowledge	-	-

A.3.6.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.6.4-1: SIP REFER (Step 1, Table A.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1 with conditions MCPTT and METHOD-BYE

Table A.3.6.4-2: SIP 200 (OK) (Step 2, Table A.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with conditions MCPTT and REFER-RSP

Table A.3.6.4-3: Disconnect (Steps 3, Table A.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.13-1 with condition ACK

A.3.7 MCPTT Group Affiliation Status Determination and Subscription

A.3.7.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.7.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.7.3 Procedure

Table A.3.7.3-1: MCPTT Group affiliation status determination and subscription

Same procedure as defined in: Table A.2.7.3-1

A.3.7.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.7.4-1: SIP SUBSCRIBE (Step 1, Table A.3.7.3-1)

Derivation Path: Table A.2.7.4-1, conditions MCPTT and PRESENCE-EVENT

Table A.3.7.4-2: SIP NOTIFY (Step 3, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions MCPTT and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.7.4-3			

Table A.3.7.4-3: PIDF in SIP NOTIFY (Table A.3.7.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1 (see note)
NOTE: PIDF document contains tuple with empty <status> element.

A.3.8 MCPTT Group Affiliation Status Change

A.3.8.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.8.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.8.3 Procedure

Table A.3.8.3-1: MCPTT Group Affiliation Status Change

Same procedure as defined in Table A.2.12.3-1

A.3.8.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.8.4-1: SIP PUBLISH (Step 1, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-1, condition MCPTT

Table A.3.8.4-2: PIDF in SIP PUBLISH (Table A.3.8.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.1-1, condition AFFILIATION

Table A.3.8.4-3: SIP 200 (OK) (Step 2, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			IETF RFC 3261 [17] IETF RFC 3903 [22]	
value	same value as in the request or lower			

Table A.3.8.4-4: SIP NOTIFY (Step 3, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-3, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.8.4-5			

Table A.3.8.4-5: PIDF in SIP NOTIFY (Table A.3.8.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.3.5	
..p-id	same value as received in the SIP PUBLISH message.			

Table A.3.8.4-6: SIP NOTIFY (Step 5, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-3, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.8.4-7			

Table A.3.8.4-7: PIDF in SIP NOTIFY (Table A.3.8.4-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliated"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9.2.2.3.5	
..p-id	same value as received in the SIP PUBLISH message.			

A.3.9 MCPTT Functional Alias Status Determination and Subscription

A.3.9.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.9.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.9.3 Procedure

Table A.3.9.3-1: MCPTT Functional Alias Status Determination and Subscription

Same procedure as defined in Table A.2.7.3-1

A.3.9.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.9.4-1: SIP SUBSCRIBE (Step 1, Table A.3.9.3-1)

Derivation Path: Table A.2.7.4-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT Info**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.1.3	
MIME-part-body	MCPTT-Info as described in Table A.2.7.4-2			

Table A.3.9.4-2: MCPTT-Info in SIP SUBSCRIBE (Table A.3.9.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
anyExt				
request-type	functional-alias-status-determination		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.1.3	

Table A.3.9.4-3: SIP NOTIFY (Step 3, Table A.3.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1 with conditions MCPTT and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.9.4-4			

Table A.3.9.4-4: PIDF in SIP NOTIFY (Table A.3.9.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1 (see note)
NOTE: PIDF document contains tuple with empty <status> element.

A.3.10 MCPTT Functional Alias Status Change

A.3.10.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.10.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.10.3 Procedure

Table A.3.10.3-1: MCPTT Functional Alias Status Change

Same procedure as defined in Table A.2.12.3-1

A.3.10.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.10.4-1: SIP PUBLISH (Step 1, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-1, condition MCPTT

Table A.3.10.4-2: PIDF in SIP PUBLISH (Table A.3.10.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.1-1, condition FUNCTIONAL_ALIAS_STATUS_CHANGE

Table A.3.10.4-3: SIP 200 (OK) (Step 2, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			IETF RFC 3261 [17] IETF RFC 3903 [22]	
value	same value as in the request or lower			

Table A.3.10.4-4: SIP NOTIFY (Step 3, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-3, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.10.4-5			

Table A.3.10.4-5: PIDF in SIP NOTIFY (Table A.3.10.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, conditions FUNCTIONAL_ALIAS_ACTIVATED and NOTIFY_FOR_PUBLISH				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
functionalAlias				
status	"activating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.3.5	

Table A.3.10.4-6: SIP NOTIFY (Step 5, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-3, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.10.4-7			

Table A.3.10.4-7: PIDF in SIP NOTIFY (Table A.3.10.4-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, conditions FUNCTIONAL_ALIAS_ACTIVATED and NOTIFY_FOR_PUBLISH				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
functionalAlias				
status	"activated"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [4] (3GPP TS 24.379) clause 9A.2.2.3.5	

A.3.11 MCPTT Floor Request - Floor Granted

A.3.11.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.11.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.11.3 Procedure

Table A.3.11.3-1: MCPTT Floor Request - Floor Granted

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Request message. (see note)	-->	Floor Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Granted message with acknowledgement required?	<--	Floor Granted	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Granted message. (see note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Granted message with no acknowledgement required?	<--	Floor Granted	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.11.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.11.4-1: Floor Granted (Step 2a1, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition ACK

A.3.12 MCPTT Floor Request - Floor Queue Position Info

A.3.12.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.12.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.12.3 Procedure

Table A.3.12.3-1: MCPTT Floor Request - Floor Queue Position Info

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Request message. (see note)	-->	Floor Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with acknowledgement required?	<--	Floor Queue Position Info	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Queue Position Info message. (see note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with no acknowledgement required?	<--	Floor Queue Position Info	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.12.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.12.4-1: Floor Queue Position Info (Step 2a1, Table A.3.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.10-1 condition ACK

A.3.13 MCPTT Queuing Position Request

A.3.13.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.13.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.13.3 Procedure

Table A.3.13.3-1: MCPTT Queuing Position Request

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Queue Position Request message. (see note)	-->	Floor Queue Position Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with acknowledgement required?	<--	Floor Queue Position Info	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Queue Position Info message. (see note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with no acknowledgement required?	<--	Floor Queue Position Info	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.13.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

Subclauses referring to messages from SS relate to the IUT (MCX Server).

- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.13.4-1: Floor Queue Position Info (Step 2a1, Table A.3.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.10-1 condition ACK

A.3.14 MCPTT Floor Request - Floor Deny

A.3.14.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.14.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.14.3 Procedure

Table A.3.14.3-1: MCPTT Floor Request - Floor Deny

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Request message. (see note)	-->	Floor Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Deny message indicating that the Floor Request message has been rejected, with acknowledgement required?	<--	Floor Deny	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Deny message. (see note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Deny message indicating that the Floor Request message has been rejected, with no acknowledgement required?	<--	Floor Deny	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.14.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.14.4-1: Floor Deny (Step 2a1, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.4-1 condition ACK

A.3.15 MCPTT Floor Release - Floor Idle

A.3.15.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.15.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.15.3 Procedure

Table A.3.15.3-1: MCPTT Floor Release - Floor Idle

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Release message with acknowledgment required. (see note)	-->	Floor Release	-	-
2	Check: Does the MCPTT Server send a Floor Ack message in response to the Floor Release message?	<--	Floor Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
3a1	Check: Does the MCPTT Server send a Floor Idle message with acknowledgement required?	<--	Floor Idle	-	P
3a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Idle message. (see note)	-->	Floor Ack	-	-
3b1	Check: Does the MCPTT Server send a Floor Idle message with no acknowledgement required?	<--	Floor Idle	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.15.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.15.4-1: Floor Release (Step 1, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
RTCP header				
Subtype	10100			

Table A.3.15.4-2: Floor Idle (Step 3a1, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1 condition ACK

A.3.16 MCPTT Floor Release - Floor Taken

A.3.16.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.16.2 Definition of system information messages

N/a (out of E-UTRA coverage).

A.3.16.3 Procedure

Table A.3.16.3-1: MCPTT Floor Release - Floor Taken

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Release message with acknowledgment required. (see note)	-->	Floor Release	-	-
2	Check: Does the MCPTT Server send a Floor Ack message in response to the Floor Release message?	<--	Floor Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
3a1	Check: Does the MCPTT Server send a Floor Taken message with acknowledgement required?	<--	Floor Taken	-	P
3a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Taken message. (see note)	-->	Floor Ack	-	-
3b1	Check: Does the MCPTT Server send a Floor Taken message with no acknowledgement required?	<--	Floor Taken	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.16.4 Specific message contents

All message contents are as specified in clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.16.4-1: Floor Release (Step 1, Table A.3.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
RTCP header				
Subtype	10100			

Table A.3.16.4-2: Floor Taken (Step 3a1, Table A.3.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1 condition ACK

A.4 MCPTT Conformance testing test point models

Based on the test points models shown in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 4.1, the following section extends the test environment implementation examples provided in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 4.2 for Server-Client and Server-Server conformance testing.

Figure A.4.1 shows test configuration where the IUT and the system simulator, simulating MCPTT Clients, communicate, one with the other, over the LTE radio interface (test points (1)).

Figures A.4.2, A.4.3, A.4.4, A.4.5 and A.4.6 show test configuration where the IUT and the System Simulator communicate, one with the other, over the MCPTT-3 or IWF-1 interfaces, as defined by ETSI TS 123 179 [3] (3GPP TS 23.179), clause 7.5.2.4, and ETSI TS 123 283 [33] (3GPP TS 23.283), clause 7.4.1, respectively (test points (4)).

Figure A.4.7 shows test configuration where the IUT and the System Simulator communicate, one with the other, over the CSC-16 or IWF-3 interface, as defined by ETSI TS 123 280 [32] (3GPP TS 23.280), clause 7.5.2.16, and ETSI TS 123 283 [33] (3GPP TS 23.283), clause 7.4.3, respectively (test points (4)).

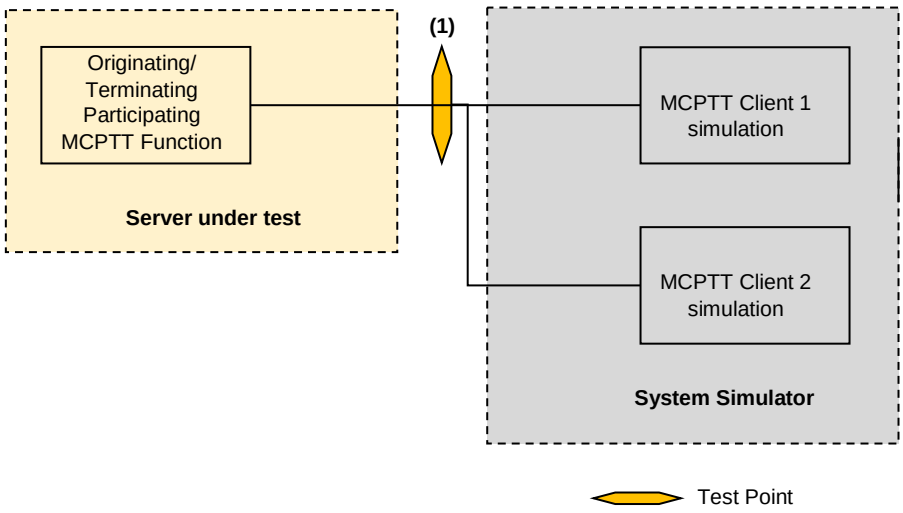


Figure A.4.1: Testing the MCPTT Server (server-to-client)

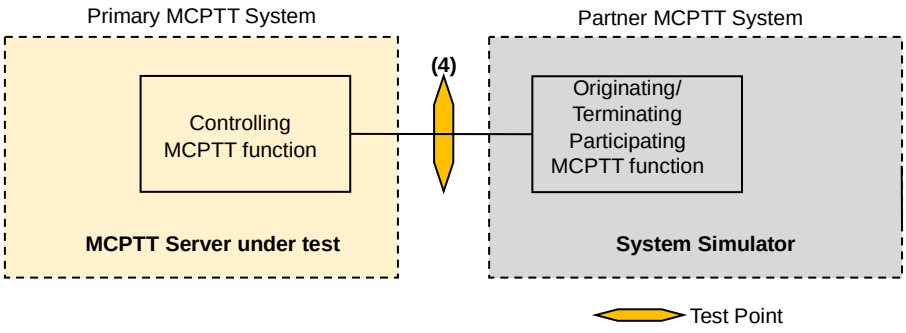


Figure A.4.2: Testing the MCPTT Server (server-to-server), Controlling function

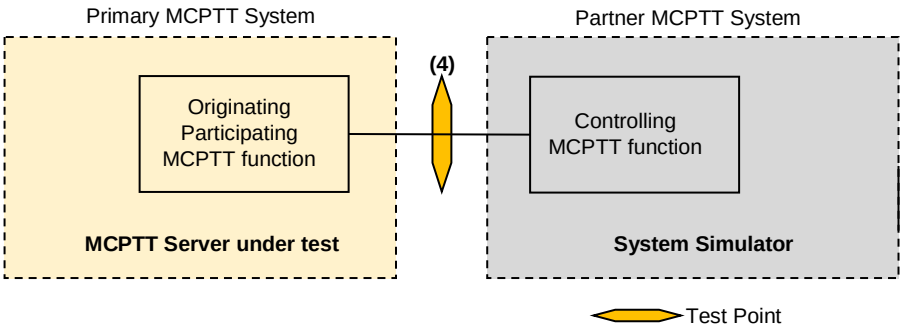


Figure A.4.3: Testing the MCPTT Server (server-to-server), Originating function

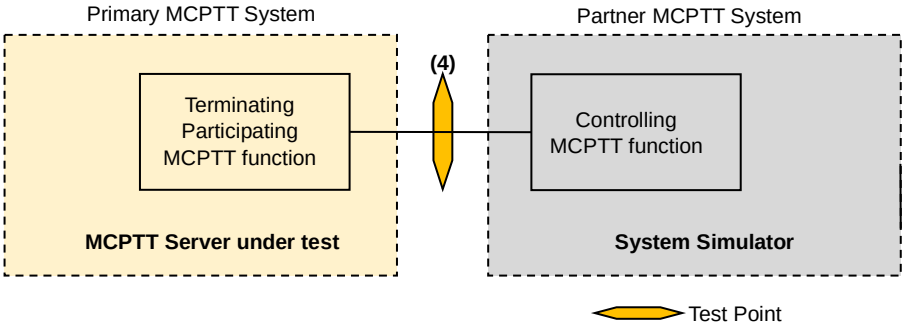


Figure A.4.4: Testing the MCPTT Server (server-to-server), Terminating function

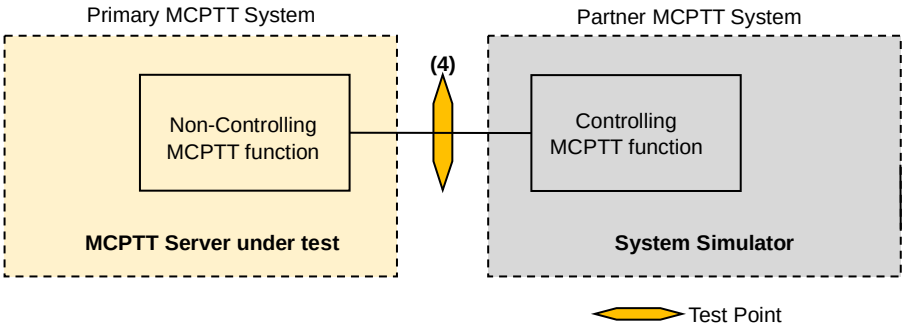


Figure A.4.5: Testing the MCPTT Server (server-to-server), Non-controlling function

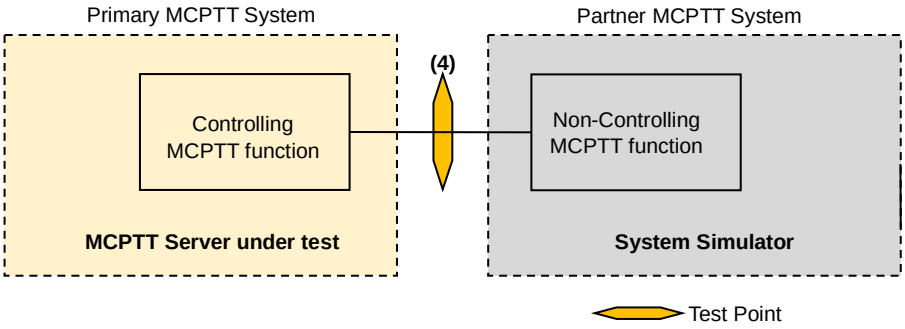


Figure A.4.6: Testing the MCPTT Server (server-to-server), Controlling function; trusted and untrusted modes

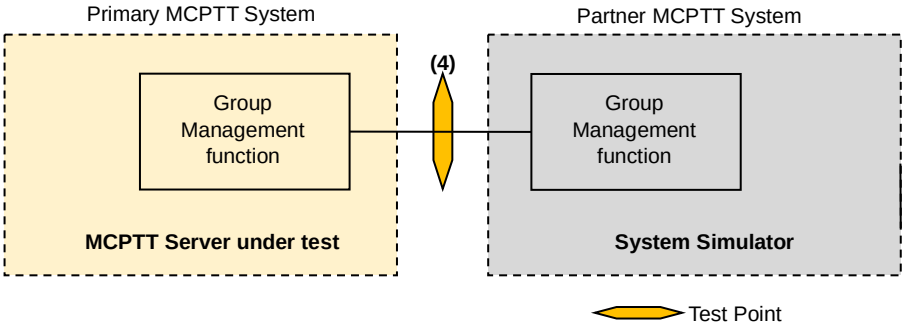


Figure A.4.7: Testing the MCPTT Server (server-to-server), Group Management function

Annex B (informative): Bibliography

- IETF RFC 3830: "MIKEY: Multimedia Internet KEYing".
- IETF RFC 3840: "Indicating User Agent Capabilities in the Session Initiation Protocol (SIP)".
- IETF RFC 3841: "Caller Preferences for the Session Initiation Protocol (SIP)".
- IETF RFC 4488: "Suppression of Session Initiation Protocol (SIP) REFER Method Implicit Subscription".
- IETF RFC 4538: "Request Authorization through Dialog Identification in the Session Initiation Protocol (SIP)".
- IETF RFC 4825: "The Extensible Markup Language (XML) Configuration Access Protocol (XCAP)".
- IETF RFC 5875: "An Extensible Markup Language (XML) Configuration Access Protocol (XCAP) Diff Event Package".
- IETF RFC 6043: "MIKEY-TICKET: Ticket-Based Modes of Key Distribution in Multimedia Internet KEYing (MIKEY)".
- IETF RFC 6509: "MIKEY-SAKKE: Sakai-Kasahara Key Encryption in Multimedia Internet KEYing (MIKEY)".
- IETF RFC 7614: "Explicit Subscriptions for the REFER Method".
- 3GPP TS 24.482: "Mission Critical Services (MCS) identity management; Protocol specification".
- 3GPP TS 24.484: "Mission Critical Services (MCS) configuration management; Protocol specification".
- 3GPP TS 33.180: "Security of the Mission Critical (MC) service".

History

Version	Date	Status
V1.1.1	January 2026	Publication